BUREAU OF
RECLAMATION

San Juan Lateral – Turnout Buildings

Navajo-Gallup Water Supply Project, New Mexico

Upper Colorado Basin Region

Solicitation No. 140R4026B0002

U.S. Department of the Interior **JULY 2026**

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CONTINUATION OF
SECTION B – SUPPLIES OR SERVICES AND PRICES

SAN JUAN LATERAL – TURNOUT BUILDINGS
NAVAJO-GALLUP WATER SUPPLY PROJECT, NEW MEXICO

B.1 PRICE SCHEDULES

(a) Offers will be considered for award on the following Price Schedules, but no offer will be considered for award on only part of the Price Schedules. Offers are subject to the terms and conditions of this solicitation.

(b) Quantities in Price Schedules are estimated quantities for comparison of offers only. Except as provided with the provision at H.4, Variation in Estimated Quantity, no claim shall be made against Government for overruns or underruns. Contract clause at H.4, Variation in Estimated Quantity does not apply to lump sum items.

(c) See contract clause at WBR 1452.232-81, Payment for Mobilization and Preparatory Work, for CLINs 1-1 and 2-1.

(d) Definitions:

(1) CLIN – Contract Line Item Number

(e) Note: In accordance with Section 7-9-3.3 of New Mexico Statutes Annotated (NMSA) 1978 and Section 402 of Navajo Nation's Business Activity Tax Statute, it is Contractor's responsibility to ensure that all applicable New Mexico Gross Receipts Tax and Navajo Nation Gross Receipts Tax are included in their price proposal.

(f) In accordance with FAR 52.232-5 (Payments under Fixed-Price Construction Contracts) (b)(2), payment may be approved for materials delivered to Contractor at locations other than the site if Contractor obtains CO approval prior to making arrangements for obtaining the materials. Contractor shall also be required to complete a "Materials on Hand Certificate" which will certify:

(1) Materials described for use on identified contract.

(2) Materials are clearly marked to identify that they belong to contract cited.

(3) Full title for those materials vested in Prime Contractor sited.

(4) Materials are free of liens and encumbrances.

(5) Materials are properly stored, secured, and protected against loss or damage.

PRICE SCHEDULE 1 (Sanostee)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
1-1	WBR 1452.232-81	Mobilization and Preparatory Work	For the lump sum of		$
1-2	01 31 30	Contract Document Management System (CDMS)	For the lump sum of		$
1-3	01 56 15	Utility Crossing Investigations	For the lump sum of		$
1-4	01 78 30	As-built Drawings	For the lump sum of		$10,000
1-5	01 78 30	Final Drawings	For the lump sum of		$5,000
1-6	13 34 00	Turnout Buildings	For the lump sum of		$
1-7	25 00 01	System Control and Monitoring	For the lump sum of		$
1-8	26 05 02	Complete Electrical System for Outdoor Equipment Only	For the lump sum of		$
1-9	26 05 33	Electrical Conduit	450 ft. Per location	$	$
1-10	26 05 33	Fabricated Sheet Steel Boxes, Wiring Troughs, and Precast Handholes	For the lump sum of		$
1-11	26 42 00	Corrosion System(s) for Turnout Yard Pipe	For the lump sum of		$
1-12	26 42 11	Buried Galvanic Anode Cathodic Protection for Turnout Yard Pipe	For the lump sum of		$
1-13	27 30 31	Telephone System	For the lump sum of		$
1-14	28 10 01	Security and Surveillance System	For the lump sum of		$
1-15	31 02 10	Water for Dust Abatement	15 M (1,000) Gal	$	$
1-16	31 02 30	Dust Palliative	0.19 acre	$	$
1-17	31 37 00	Riprap	50 yd^3	$	$

PRICE SCHEDULE 1 (Sanostee)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
1-18	32 15 10	Gravel Surfacing	90 yd^3	$	$
1-19	32 31 15	Ornamental Fence	220 ft.	$	$
1-20	32 92 20	Seeding	0.06 acre	$	$
1-21	33 13 00	Disinfection of Water Utility Distribution	For the lump sum of		$
1-22	35 21 95	Steel Manifold Piping	For the lump sum of		$
1-23	35 22 14	Valves and Equipment	For the lump sum of		$

TOTAL FOR PRICE SCHEDULE 1 $_____

PRICE SCHEDULE 2 (Newcomb)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
2-1	WBR 1452.232-81	Mobilization and Preparatory Work	For the lump sum of		$
2-3	01 56 15	Utility Crossing Investigations	For the lump sum of		$
2-4	01 78 30	As-built Drawings	For the lump sum of		$10,000
2-5	01 78 30	Final Drawings	For the lump sum of		$5,000
2-6	13 34 00	Turnout Buildings	For the lump sum of		$
2-7	25 00 01	System Control and Monitoring	For the lump sum of		$
2-8	26 05 02	Complete Electrical System for Outdoor Equipment Only	For the lump sum of		$
2-9	26 05 33	Electrical Conduit	250 ft.	$	$
2-10	26 05 33	Fabricated Sheet Steel Boxes, Wiring Troughs, and Precast Handholes	For the lump sum of		$

PRICE SCHEDULE 2 (Newcomb)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
2-11	26 42 00	Corrosion System(s) for Turnout Yard Pipe	For the lump sum of		$
2-12	26 42 11	Buried Galvanic Anode Cathodic Protection for Turnout Yard Pipe	For the lump sum of		$
2-13	27 30 31	Telephone System	For the lump sum of		$
2-14	28 10 01	Security and Surveillance System	For the lump sum of		$
2-15	31 02 10	Water for Dust Abatement	9.2 M (1,000) Gal	$	$
2-16	31 02 30	Dust Palliative	0.13 acre	$	$
2-17	31 23 50	Access Road to Window Rock	For the lump sum of		$
2-19	31 37 00	Riprap	32 yd^3	$	$
2-20	32 15 10	Gravel Surfacing	50 yd^3	$	$
2-21	32 31 15	Ornamental Fence	170 ft.	$	$
2-22	32 92 20	Seeding	0.05 acre	$	$
2-23	33 13 00	Disinfection of Water Utility Distribution	For the lump sum of		$
2-24	35 21 95	Steel Manifold Piping	For the lump sum of		$
2-25	35 22 14	Valves and Equipment	For the lump sum of		$

TOTAL FOR PRICE SCHEDULE 2 $_____

PRICE SCHEDULE 3 (Naschitti)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
3-1	WBR 1452.232-81	Mobilization and Preparatory Work	For the lump sum of		$

PRICE SCHEDULE 3 (Naschitti)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
3-3	01 56 15	Utility Crossing Investigations	For the lump sum of		$
3-4	01 78 30	As-built Drawings	For the lump sum of		$10,000
3-5	01 78 30	Final Drawings	For the lump sum of		$5,000
3-6	13 34 00	Turnout Buildings	For the lump sum of		$
3-7	25 00 01	System Control and Monitoring	For the lump sum of		$
3-8	26 05 02	Complete Electrical System for Outdoor Equipment Only	For the lump sum of		$
3-9	26 05 33	Electrical Conduit	250 ft.	$	$
3-10	26 05 33	Fabricated Sheet Steel Boxes, Wiring Troughs, and Precast Handholes	For the lump sum of		$
3-11	26 42 00	Corrosion System(s) for Turnout Yard Pipe	For the lump sum of		$
3-12	26 42 11	Buried Galvanic Anode Cathodic Protection for Turnout Yard Pipe	For the lump sum of		$
3-13	27 30 31	Telephone System	For the lump sum of		$
3-14	28 10 01	Security and Surveillance System	For the lump sum of		$
3-15	31 02 10	Water for Dust Abatement	19 M (1,000) Gal	$	$
3-16	31 02 30	Dust Palliative	0.25 acre	$	$
3-18	31 37 00	Riprap	125 yd^3	$	$
3-19	32 15 10	Gravel Surfacing	135 yd^3	$	$
3-20	32 31 15	Ornamental Fence	220 ft.	$	$
3-21	32 92 20	Seeding	0.03 acre	$	$
3-22	33 11 12	Line Pipe for BBN9 Turnout	For the lump sum of		$

PRICE SCHEDULE 3 (Naschitti)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
3-23	33 13 00	Disinfection of Water Utility Distribution	For the lump sum of		$
3-24	3-24	Steel Manifold Piping	For the lump sum of		$
3-25	35 22 14	Valves and Equipment	For the lump sum of		$

TOTAL FOR PRICE SCHEDULE 3 $_____

PRICE SCHEDULE 4 (BBN9)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
4-1	WBR 1452.232-81	Mobilization and Preparatory Work	For the lump sum of		$
4-3	01 56 15	Utility Crossing Investigations	For the lump sum of		$
4-4	01 78 30	As-built Drawings	For the lump sum of		$10,000
4-5	01 78 30	Final Drawings	For the lump sum of		$5,000
4-6	13 34 00	Turnout Buildings	For the lump sum of		$
4-7	25 00 01	System Control and Monitoring	For the lump sum of		$
4-8	26 05 02	Complete Electrical System for Outdoor Equipment Only	For the lump sum of		$
4-9	26 05 33	Electrical Conduit	250 ft.	$	$
4-10	26 05 33	Fabricated Sheet Steel Boxes, Wiring Troughs, and Precast Handholes	For the lump sum of		$
4-11	26 42 00	Corrosion System(s) for Turnout Yard Pipe	For the lump sum of		$
4-12	26 42 11	Buried Galvanic Anode Cathodic Protection for Turnout Yard Pipe	For the lump sum of		$

PRICE SCHEDULE 4 (BBN9)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
4-13	27 30 31	Telephone System	For the lump sum of		$
4-14	28 10 01	Security and Surveillance System	For the lump sum of		$
4-15	31 02 10	Water for Dust Abatement	19 M (1,000) Gal	$	$
4-16	31 02 30	Dust Palliative	0.25 acre	$	$
4-17	31 23 50	Low Water Crossing	For the lump sum of		$
4-19	31 37 00	Riprap	52 yd^3	$	$
4-20	32 15 10	Gravel Surfacing	81 yd^3	$	$
4-21	32 31 15	Ornamental Fence	180 ft.	$	$
4-22	32 92 20	Seeding	0.13 acre	$	$
4-23	33 11 12	Line Pipe for BBN9 Turnout	For the lump sum of		$
4-24	33 13 00	Disinfection of Water Utility Distribution	For the lump sum of		$
4-25	35 21 95	Steel Manifold Piping	For the lump sum of		$
4-26	35 22 14	Valves and Equipment	For the lump sum of		$

TOTAL FOR PRICE SCHEDULE 4 $_____

PRICE SCHEDULE 5 (Coyote Canyon)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
5-1	WBR 1452.232-81	Mobilization and Preparatory Work	For the lump sum of		$
5-3	01 56 15	Utility Crossing Investigations	For the lump sum of		$
5-4	01 78 30	As-built Drawings	For the lump sum of		$10,000
5-5	01 78 30	Final Drawings	For the lump sum of		$5,000

PRICE SCHEDULE 5 (Coyote Canyon)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
5-6	13 34 00	Turnout Buildings	For the lump sum of		$
5-7	25 00 01	System Control and Monitoring	For the lump sum of		$
5-8	26 05 02	Complete Electrical System for Outdoor Equipment Only	For the lump sum of		$
5-9	26 05 33	Electrical Conduit	250 ft.	$	$
5-10	26 05 33	Fabricated Sheet Steel Boxes, Wiring Troughs, and Precast Handholes	For the lump sum of		$
5-11	26 42 00	Corrosion System(s) for Turnout Yard Pipe	For the lump sum of		$
5-12	26 42 11	Buried Galvanic Anode Cathodic Protection for Turnout Yard Pipe	For the lump sum of		$
5-13	27 30 31	Telephone System	For the lump sum of		$
5-14	28 10 01	Security and Surveillance System	For the lump sum of		$
5-15	31 02 10	Water for Dust Abatement	12 M (1,000) Gal	$	$
5-16	31 02 30	Dust Palliative	0.16 acre	$	$
5-17	31 23 50	Access Road to Coyote Canyon	For the lump sum of		$
5-19	31 37 00	Riprap	53 yd^3	$	$
5-20	32 15 10	Gravel Surfacing	67 yd^3	$	$
5-21	32 31 15	Ornamental Fence	165 ft.	$	$
5-22	32 92 20	Seeding	0.05 acre	$	$
5-23	33 11 12	Line Pipe for Coyote Canyon Turnout	For the lump sum of		$
5-24	33 13 00	Disinfection of Water Utility Distribution	For the lump sum of		$
5-25	4-25	Steel Manifold Piping	For the lump sum of		$

PRICE SCHEDULE 5 (Coyote Canyon)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
5-26	35 22 14	Valves and Equipment	For the lump sum of		$

TOTAL FOR PRICE SCHEDULE 5 $_____

PRICE SCHEDULE 6 (Rock Springs)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
6-1	WBR 1452.232-81	Mobilization and Preparatory Work	For the lump sum of		$
6-3	01 56 15	Utility Crossing Investigations	For the lump sum of		$
6-4	01 78 30	As-built Drawings	For the lump sum of		$10,000
6-5	01 78 30	Final Drawings	For the lump sum of		$5,000
6-6	13 34 00	Turnout Buildings	For the lump sum of		$
6-7	25 00 01	System Control and Monitoring	For the lump sum of		$
6-8	26 05 02	Complete Electrical System for Outdoor Equipment Only	For the lump sum of		$
6-9	26 05 33	Electrical Conduit	250 ft.	$	$
6-10	26 05 33	Fabricated Sheet Steel Boxes, Wiring Troughs, and Precast Handholes	For the lump sum of		$
6-11	26 42 00	Corrosion System(s) for Turnout Yard Pipe	For the lump sum of		$
6-12	26 42 11	Buried Galvanic Anode Cathodic Protection for Turnout Yard Pipe	For the lump sum of		$
6-13	27 30 31	Telephone System	For the lump sum of		$
6-14	28 10 01	Security and Surveillance System	For the lump sum of		$

PRICE SCHEDULE 6 (Rock Springs)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
6-15	31 02 10	Water for Dust Abatement	13.5 M (1,000) Gal	$	$
6-16	31 02 30	Dust Palliative	0.18 acre	$	$
6-18	31 37 00	Riprap	515 yd^3	$	$
6-19	32 15 10	Gravel Surfacing	97 yd^3	$	$
6-20	32 31 15	Ornamental Fence	175 ft.	$	$
6-21	32 92 20	Seeding	0.03 acre	$	$
6-22	33 13 00	Disinfection of Water Utility Distribution	For the lump sum of		$
6-23	35 21 95	Steel Manifold Piping	For the lump sum of		$
6-24	35 22 14	Valves and Equipment	For the lump sum of		$

TOTAL FOR PRICE SCHEDULE 6 $_____

PRICE SCHEDULE 7 (Window Rock)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
7-1	WBR 1452.232-81	Mobilization and Preparatory Work	For the lump sum of		$
7-3	01 56 15	Utility Crossing Investigations	For the lump sum of		$
7-4	01 78 30	As-built Drawings	For the lump sum of		$10,000
7-5	01 78 30	Final Drawings	For the lump sum of		$5,000
7-6	13 34 00	Turnout Buildings	For the lump sum of		$
7-7	25 00 01	System Control and Monitoring	For the lump sum of		$
7-8	26 05 02	Complete Electrical System for Outdoor Equipment Only	For the lump sum of		$

PRICE SCHEDULE 7 (Window Rock)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
7-9	26 05 33	Electrical Conduit	250 ft.	$	$
7-10	26 05 33	Fabricated Sheet Steel Boxes, Wiring Troughs, and Precast Handholes	For the lump sum of		$
7-11	26 42 00	Corrosion System(s) for Turnout Yard Pipe	For the lump sum of		$
7-12	26 42 11	Buried Galvanic Anode Cathodic Protection for Turnout Yard Pipe	For the lump sum of		$
7-13	27 30 31	Telephone System	For the lump sum of		$
7-14	28 10 01	Security and Surveillance System	For the lump sum of		$
7-15	31 02 10	Water for Dust Abatement	10 M (1,000) Gal	$	$
7-16	31 02 30	Dust Palliative	0.14 acre	$	$
7-17	31 23 50	Access Road to Window Rock	For the lump sum of		$
7-19	31 37 00	Riprap	64 yd^3	$	$
7-20	32 15 10	Gravel Surfacing	59 yd^3	$	$
7-21	32 31 15	Ornamental Fence	135 ft.	$	$
7-22	32 92 20	Seeding	0.03 acre	$	$
7-23	33 13 00	Disinfection of Water Utility Distribution	For the lump sum of		$
7-24	35 21 95	Steel Manifold Piping	For the lump sum of		$
7-25	35 22 14	Valves and Equipment	For the lump sum of		$

TOTAL FOR PRICE SCHEDULE 7 $_____

END OF SUPPLIES OR SERVICES AND PRICES

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SUMMARY OF WORK

PART 1 GENERAL

1.01 LOCATION

A. Work is located at 7 sites in northwest New Mexico between Window Rock and Shiprock, approximate latitude and longitude coordinates for each are:

1. Sanostee: 36.429389, -108.720139 in San Juan County.
2. Newcomb: 36.288306, -108.704806 in San Juan County.
3. Naschitti: 36.068778, -108.682833 in San Juan County.
4. BBN9: 35.869694, -108.657278 in McKinley County.
5. Coyote Canyon: 35.746306, -108.741778 in McKinley County.
6. Rock Springs: 35.636472, -108.869389 in McKinley County.
7. Window Rock: 35.663306, -109.045278 in McKinley County.

1.02 PRINCIPAL COMPONENTS OF WORK

A. Overview: Work consists of constructing a potable water delivery turnout building and associated improvements at each site.

B. Install at each site:

1. Sitework, including buried piping, valves, and accessories; access road (Coyote Canyon and Window Rock only); steel fencing with gate; gravel surfacing; and drainage.
2. Steel piping connection to existing steel lateral (supply) pipe.
3. Steel piping connection to existing steel distribution (delivery) pipe.
4. Buried thrust blocks.
5. Fiberoptic cable infrastructure from existing lateral alignment to building.
6. Precast concrete building and foundation, including geotechnical investigation, foundation design, and building design.
7. Interior exposed piping, valves, and accessories.
8. Building mechanical, electrical, controls, instrumentation, and security.
9. Controls integration with existing San Juan Lateral SCADA system.

C. Flush, test, and disinfect installed piping, including furnishing and disposal of water.

D. NTUA will provide a transformer to each turnout location to step-down power for each location's power needs. At each site, a transformer will feed into a meter then disconnect before powering up the building. The meter socket will delineate ownership with equipment upstream from the meter, including the meter, will be owned and maintained by NTUA. FCCO will own and maintain from the meter socket and downstream.

1.03 SPECIFICATIONS REQUIREMENTS

A. Requirements in Division 1, General Requirements, apply to Divisions 2 through 53.

B. Imperative statements in these specifications are Contractor requirements unless otherwise stated.

C. Where specifications are written in streamlined form, the words "shall be" are included by inference where a colon (:) is used within a sentence or phrase.

1.04 DEFINITIONS

A. When specifications use a word or term defined in Federal Acquisition Regulations (FAR), definition of the word or term shall be in accordance with FAR sections in effect at the time solicitation was issued.

1.05 REQUIREMENTS

A. Design and construct per National Fire Protection Association (NFPA) 101, Life Safety Code.

1.06 ACRONYMS

A. The following acronyms apply to specifications Divisions 1 through 53:

1. CO: Contracting Officer.
2. COR: Contracting Officer's Representative.
3. NTUA: Navajo Tribal Utility Authority.
4. OGR: Onsite Government Representative.
5. FCCO: Four Corners Construction Office.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 01 12 10
OTHER CONTRACTS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work.

1.02 OTHER WORK

A. During work under this contract, other work may be performed in the vicinity of the work by Government or other contractors, including: San Juan Lateral Water Treatment Plant.

B. Cooperate and coordinate with other contractors on access to and use of work areas and contractor use areas. Comply with clause at FAR 52.236-8, Other Contracts.

C. Coordinate work with COR so to not interfere with work of other contractors or Government in vicinity of work.

D. Cooperate and coordinate with other contractors on traffic control and use of temporary utilities.

E. Access roads are not for sole use of any one contractor.

F. Cooperate and coordinate with other contractors on maintaining dust-free operations.

G. Cooperate and coordinate work with surrounding residents.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 01 14 10
USE OF SITE

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work.

1.02 REFERENCE STANDARDS

A. Bureau of Reclamation (Reclamation)

1.	Cleaning Manual	Inspection and Cleaning Manual for Equipment and Vehicles to Prevent the Spread of Invasive Species, available at: https://www.usbr.gov/mussels/prevention/docs/EquipmentInspectionandCleaningManual2021.pdf

1.03 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 14 10-1, Land Use and Landscape Rehabilitation Plan:

1. Use site on Right-of-Way (ROW):

   a. Show use location and extent of impact. Uses include but are not limited to the following:

      1) Buildings and service areas including onsite offices, shops, warehouses, storage areas, fuel and oil storage areas, and fabrication yards.

      2) Parking areas, temporary roads, and haul routes.

      3) Utilities including gas, air, power, and water lines; fire hydrants; and compressor station.

      4) First-aid and medical facilities.

      5) Areas for processing, storing, and disposing of waste materials from construction operations.

      6) Temporary fences.

   b. Describe methods to preserve, protect, and repair, vegetation (such as trees, shrubs, and grass) and other landscape features on or adjacent to jobsite, which are not to be removed and which do not interfere with work

required under this contract. Include methods to mark work area limits, protect disturbed areas, and prevent erosion.

c. Describe methods to reseed disturbed pipe laydown areas.

d. Describe methods to preserve, protect, identified cultural resources adjacent to jobsite, which are not to be removed and which do not interfere with work required under this contract. Include methods to mark work area limits, protect sensitive areas, and prevent erosion.

e. Describe methods to protect and repair, existing improvements and utilities at or near jobsite.

f. Describe methods for removing temporary structures and facilities, cleanup, and rehabilitating site after completion of construction activities.

g. Describe methods for cleaning construction equipment.

C. Submit revised drawings of changes in use of Government land made during design and erection stages or after use of Government land is in operation.

D. RSN 01 14 10-2, Optional Onsite Security Plan:

1. Schedule for single individual to live on job site to provide full-time inhabitance including estimated of fixed dates.

2. Residence facilities, indicating:

a. If existing job trailer will be used of if a new trailer will be placed onsite.

b. Utilities for residence facility to be used and if modifications are required:

1) Electric.

2) Water.

3) Waste management.

c. Any changes required to existing site security lighting

3. Map showing residence facility in relation to overall site, including plant, construction trailer, etc.

4. Other items not listed pertinent to additional site use.

1.04 PROJECT CONDITIONS

A. Government land, ROW, and Easements as shown on drawings may be used for required construction facilities: Refer to Contractor Use Areas as shown on drawings.

B. When private land is used for construction facilities, or other construction purposes, make necessary arrangements associated with use of private land. Prior to using land outside of ROW, obtain cultural and environmental approvals in accordance with Section 01 31 19 – Project Management and Coordination and Section 01 57 90 – Preservation of Historical and Archeological Data.

C. Location, construction, operation, maintenance, and removal of construction facilities on ROW or Easements will be subject to approval of COR.

D. Do not interfere with work of other contractors, Navajo Nation or Government in vicinity, or with reservations made by Government for use of such land.

E. Government may backcharge Contractor for costs incurred by the Government from contractors failing to cooperate with Government or other Contractors.

F. Blasting is not allowed.

G. See Section 01 51 00 – Temporary Utilities.

H. Natural Conditions, Contractor to verify (https://wrcc.dri.edu/cgi-bin/cliMAIN.pl?nm3340):

1. Ambient Temperature Range: Averages 12 to 94 degrees F with extremes of minus 21 to 110 degrees F possible.
2. Annual Average Precipitation: 6 inches.
3. Altitude: 5,865 to 6,853 feet.
4. Wind:
   a. Maximum sustained: 9.5 mph.
   b. Maximum gust: 64 mph.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 CLEANING

A. Construction Equipment:

1. Before bringing on site, clean construction equipment, to remove dirt, vegetation, and other organic material to prevent introduction of noxious weeds, and invasive plant and animal species.
2. Contractor cleaning procedures shall result in equipment and vehicles being cleaned as well or better than procedures described in Cleaning Manual.
3. OGR will inspect construction equipment following procedures described in Cleaning Manual before allowing equipment onsite.

3.02 MAINTENANCE

A. Clean and service portable toilets weekly or more frequently as directed by COR.

B. Keep approach walks free of mud, water, ice, and snow.

3.03 RESTORATION

A. Restore Contactor Use Areas to preconstruction condition.

B. Restore temporary construction roads to original contours and make impassable to vehicular traffic when no longer required.

C. After completion of work, regrade and scarify Government land, easements, and ROWs used for construction purposes and not required for completed installation so that surfaces blend with natural terrain and are in a condition that facilitates revegetation, provides drainage, and prevents erosion.

D. Seed disturbed areas of Government land used for construction purposes and not required for completed installation in accordance with Section 32 92 20 - Seeding with species of native plants, or other approved perennial species, as recommended by an experienced local horticulturist.

END OF SECTION

SECTION 01 14 12
WORK RESTRICTIONS AND LIMITATIONS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work.

1.02 WORK RESTRICTIONS AND LIMITATIONS

A. Allowable Work Hours: Sunrise to sunset, unless otherwise authorized by COR.

B. Comply with noise and light restrictions listed in Section 01 57 20 – Environmental Controls and 01 35 20 – Safety and Health.

C. Perform work in roadways in accordance with Section 01 55 20 – Traffic Control.

D. Complete work on San Juan Lateral by January 1, 2028, to accommodate flushing and disinfection of the lateral.

1. Install tee and buried valve at Coyote Canyon.
2. Install valves at BBN9. Refer to Drawing 1695-D-62548.
3. Replace blind flanges on existing turnouts with valves.
4. Limit outage to San Juan Lateral to no more than 28 days. Coordinate outage with COR. Notify COR 28 days before outage.

E. Complete BBN9, Coyote Canyon, and Naschitti by September 1, 2028.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 01 31 04
PARTNERING

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work.

1.02 MEETINGS

A. Team scoping/partnering workshop will be held immediately following Notice to Proceed to discuss the specific elements related to the preconstruction services phase of the project. Discussion shall include submittal requirements and process, reconciliation of submittal review comments, project goals, project communication and issue escalation plan, schedule management, risk analysis, construction partnering.

B. Traditional construction Partnering workshops will be held in addition to the initial team scoping/partnering workshop.

1.03 INTENT

A. Discuss effectively and successfully implementing work throughout construction. Form an informal management partnership with the Contractor to develop a cohesive building team.

B. Form a bilateral relationship striving for mutual trust, dedication to common goals and an understanding of individual expectations and values.

C. Develop a cooperative management team drawing on strengths of each team member. Expected benefits include improved efficiency, cost effectiveness and innovation between parties to ensure a quality deliverable completed on time and within budget.

D. Anticipated partnership between the primary contractor, subcontractors, and Reclamation.

E. Partnering Sessions:

1. Facilitated by a third party selected by the Contractor with approval by Reclamation.
2. Held in Farmington, New Mexico.
3. One half-day initial partnering session.
4. Quarterly half day sessions during construction.
5. Half-day final lessons learned session after construction is completed.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

SECTION 01 31 19
PROJECT MANAGEMENT AND COORDINATION

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedule for other items of work.

1.02 REFERENCE STANDARDS

A. Bureau of Reclamation (Reclamation)

1.	RSHS	Reclamation Safety and Health Standards, including revisions, available at https://www.usbr.gov/safety/rshs/index.html

1.03 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 31 19-1, Written Summary:

1. Written weekly summary of upcoming work, traffic plans and progress meeting minutes. Provide work schedule for next 3 weeks including list of RSNs and RFIs.

C. RSN 01 31 19-2, Off ROW Land Use Coordination:

1. Show expanded use location and extent of impact.
2. Request cultural approvals:
   a. In accordance with Section 01 57 90 – Preservation of Historical and Archeological Data.
   b. Minimum information required for cultural analysis: Map and UTM data in NAD83 format of the location, narrative of intended use of the new area, summary of intended access to the new area, and a need by date.
3. Request or revise Environmental Permits in accordance with:
   a. Section 01 57 20 – Environmental Controls.
   b. Section 01 57 30 – Water Pollution Control.
   c. Section 01 57 50 – Tree and Plant Protection.
   d. Section 01 57 60 – Protected Species.
4. Do not use proposed land until approved by COR:
   a. Approvals may require up to 6 months.

5. Requirement does not apply to existing commercial facilities.

D. RSN 01 31 19-3, Key Personnel:

1. Submit list of key personnel:
   a. Project Manager.
   b. Site Superintendent.
   c. Contractor's Onsite Safety Representative.
   d. Quality Control Supervisor.
2. Include resume of Project Manager and Site Superintendent.
3. CO approval of key personnel required prior to work being performed.
4. Personally Identifiable Information (PII) shall be encrypted using WINZip or Adobe Acrobat before transmitting electronically.

E. RSN 01 31 19-4, Replacement of Key Personnel:

1. Submit key personnel changes with a minimum notice of 48 hours to CO.
2. CO approval of changes in key personnel required prior to work being performed by new personnel.

1.04 QUALIFICATIONS

A. Project Manager

1. Comparable role on three projects of similar nature, size, and complexity.
2. Employed by prime contractor.

B. Site Superintendent:

1. Comparable role on three projects of similar nature, size, and complexity.
2. Employed by prime contractor.

C. Contractor's Onsite Safety Representative: Meet requirements of Section 01 35 30 – Contractor's Onsite Safety Personnel.

D. Quality Control Supervisor: Meet requirements of Section 01 46 00 – Quality Procedures.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 PROJECT COORDINATION

A. Coordinate work with COR, other Bureau of Reclamation (Reclamation) staff, Bureau of Indian Affairs, and other Federal, Tribal, State, and local agencies, as needed.

B. Facilitate onsite work with above listed agencies.

C. 48-Hour Look Ahead Notification:

1. Provide COR, OGR, Cultural Contract COR, and Cultural Resource Contractor via email, and phone with a 48-hour notice when Cultural Resource Contractor monitoring services are required.

2. Contact information including Cultural Contract COR and Cultural Resource Contractor will be provided at preconstruction meeting.

3.02 MEETINGS

A. The following meetings are considered significant, but do not relieve Contractor from responsibility of other meetings required by contract:

1. Preconstruction Meeting:

a. Meet with Government prior to start of construction. Meeting to occur after approval of RSN 01 35 20-1, Safety Program.

b. Provide, as a minimum, Project Manager, Project Superintendent, and Project Engineer for Preconstruction Meeting.

c. Review will include, but may not be limited to:

1) Use of premises by Contractor, Government, and public:

a) Off ROW land use coordination.

2) Construction facilities and controls.

3) Access to work and haul routes.

4) Temporary utilities.

5) Survey layout and installation of temporary fencing of the entire project work area prior to surveying.

6) Security.

7) Housekeeping procedures.

8) Schedules and sequence of work.

9) Procedures for testing.

10) Procedures for maintaining record documents.

11) Special site requirements:

a) Archeological: Refer to Section 01 57 90 – Preservation of Historical and Archaeological Data.

b) Biological: Refer to Section 01 57 60 – Protected Species.

c) Environmental: Refer to:

i. Section 01 57 20 – Environmental Controls

ii. Section 01 57 30 – Water Pollution Control.

12) Roles and responsibilities:

a) Government organization and personnel.

b) Contractor organization and personnel.

c) Subcontractors.

d) Proposed work schedule.

13) Contract requirements:

a) Progress payments.

b) Invoices.

c) Differing site conditions.

d) Changes.

e) Superintendence by Contractor.

f) Payment for mobilization and preparatory work.

g) Submittal procedures.

h) Emails and telephone calls.

2. Cultural Resource Sensitivity Awareness Training:

a. Refer to Section 01 57 90 – Preservation of Historical and Archaeological Data.

b. Complete Government-provided training before initial groundbreaking. Schedule for training will be mutually agreed upon.

3. Preconstruction Safety Meeting:

a. Meet with COR prior to start of construction. Meeting to occur after approval of RSN 01 35 20-1, Safety Program.

b. Review will include:

1) Safety requirements.

2) Protection of cultural resources and cultural sensitivity training requirements.

3) Monthly joint safety policy meetings.

4) Weekly toolbox safety meetings.

5) Safety program.

6) Job hazard analysis.

7) Required safety inspections.

8) Government will monitor Contractor's operations for compliance with RSHS.

4. Progress Meetings:

a. Meet weekly with Government to review:

1) Progress made.

2) Review as-built drawings.

3) Difficulties in performing work.

4) Safety performance.

5) Resolution necessary to meet specifications requirements.

6) Status of contract documents (e.g., RFI's, submittals).

7) Three-week look ahead, including installing temporary safety fencing and upcoming cultural monitoring areas discussed.

8) Updating of progress reports.

9) Other items as determined by COR and OGR.

b. Develop meeting agenda and minutes, may be used as Written Summary.

END OF SECTION

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SECTION 01 31 30
CONTRACT DOCUMENT MANAGEMENT SYSTEM

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Contract Document Management System:

1. Payment: Lump-sum price offered in Price Schedule 1.

1.02 DEFINITIONS

A. Contract Document Management System (CDMS):

1. Contract documents include, but are not limited to:
    a. Specifications.
    b. Contract Drawings.
    c. Submittals.
    d. RFCs: Requests for Change (Contractor generated documents).
    e. RFIs: Requests for Information (Contractor generated documents).
    f. RFPs: Requests for Proposal (Government generated documents).
    g. CMs: Contract Modifications.
    h. Contractor Proposals.
    i. Value Engineering Proposals.
    j. Invoices and Progress Payments.
    k. Contract Schedules.
    l. Meeting Agendas and Meeting Minutes.
    m. Letters and Memos.

1.03 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 31 30-1, Approval Data:

1. System provider experience, number of systems operations, references.
2. Documentation of system capabilities.
3. Instructions for system use.

C. RSN 01 31 30-2, Final Data:

1. Documents uploaded to CDMS on a Reclamation provided USB drive.

1.04 QUALIFICATIONS

A. System Provider: At least 5 years successful performance in providing required services.

1.05 CDMS REQUIREMENTS

A. CDMS shall be capable of generating, storing, tracking, categorizing, and managing Contract Documents.

B. Access:

1. Web-based:
    a. Contract Documents are stored and accessed by authorized individuals via an internet site.
    b. Compatible with web browsers Microsoft Edge, Apple Safari, Mozilla Firefox, and Google Chrome.
2. Ability to upload Contract Documents and make available for user download in the following supported file formats, minimum:
    a. Adobe Acrobat.
    b. MS Word.
    c. MS Excel.
    d. MS Project.
    e. Oracle Primavera P6 Project Manager.
    f. AutoCAD Civil 3D Version 2018 or newer.
    g. Contract Manager.
3. Allow Contract Documents to be prepared by Contractor or by Government:
    a. Preparing organization will control access of documents. See Document Security and Backup paragraph below.
4. Contractor and Government shall each have a project manager for CDMS:
    a. Each project manager shall have capabilities to assign users within their organization and to assign user rights which control access to documents based on user class and document type.
    b. Each organization shall be capable of defining its access hierarchy.
    c. Inclusion of users will be prerogative of organizational project manager.
    d. Government shall be able to create and modify project properties (i.e., Contractor, project location, description, bid amount, and project directory).

5. Email notifications shall be automatically sent to selectable users when new documents are submitted and available for viewing, or alerts are generated. Examples include:
    a. Updates to project information (e.g., entry of a new submittal).
    b. Changes to project information (e.g., approval of a submittal).
    c. Alerts (e.g., submittal under review for greater than 15 days without action).
6. Ability to link files; examples include:
    a. RFP linked to a CM.
    b. Submittal linked to a Specification or Drawing.

C. Reports:
1. CDMS shall:
    a. Generate reports that list and sort documents by status; examples include:
        1) Submittals that have been approved.
        2) Invoices paid to date.
        3) RFIs under review.
    b. Generate alerts when documents requiring action approach or exceed allowable time.
    c. Generate customizable summary reports; examples include:
        1) Submittal Turn-around.
        2) RFI Turn-Around.
        3) CM Processing Turn-Around.
        4) Total CM Costs.
        5) Total CM Request Exposure.
    d. Reports shall be printable, exportable as searchable .pdf or printer friendly .html, and exportable to Excel (.xls, .xml, or .csv).
    e. Ability to link from an item in a report directly to item by clicking on it:
        1) For example: In submittal report, clicking on submittal that is 15 days old takes you directly to information on that submittal.
    f. Capable of producing complete and logically organized set of documents within CDMS in both .pdf and .xml formats.

D. Document Security and Backup:
1. Secure Server Language (SSL) encryption for secured data exchange between browser and server (Secure access to documents and information).

2. Allow document access security until authoring party chooses to share it with others:

    a. Show parties that have access to document.

    b. Extent of document sharing shall be determined by authoring party.

    c. Once documents are shared, track documents so that changes cannot occur without record of changes.

3. Backup/archiving to FTP site of documents to which an organization has access on a regular (minimum monthly) basis. Backups shall be in .pdf format.

4. Documents shall be downloadable to FTP sites where Government and other organizations can access their own data confidentially.

E. General Tracking and Control:

1. Documents shall be grouped into categories (Submittals, RFI, RFP, CM, Payments, Daily Inspection Reports, and Meeting Minutes).

2. CDMS shall track:

    a. When documents were received or returned, as well as status of documents (e.g., Under Review; Returned, Approved or Rejected).

    b. Document changes including who entered or changed document and date change was made.

    c. When a reviewer has opened an item.

F. Submittal Tracking and Control:

1. Ability to create master list of submittals ("Schedule of Submittals").

2. Supply standard submittal form for entering information. Allow customization of standard submittal form. Contractor shall be able to:

    a. Enter required submittal number (RSN), title, description, due date, and specification section.

    b. Upload submittal data in electronic format.

3. Track date received, date response due, date returned, and status:

    a. Status designations may be modified to fit Government standard designations – A/AAN/PAR/NA/ACK (Approved/Approved As Noted/Partially Approved - Resubmit/Not Approved/Acknowledged).

4. Supply Submittal Tracking form for Government to enter review comments and action taken. Allow customizing of submittal tracking form.

    a. Allow for Government to route submittal to third parties (e.g., consultant, internal design or operations and maintenance groups) by generating design-review transmittals with each submittal package item, to track status of individual sub-items within the submittal package, and to route each sub-item to appropriate reviewers.

5. Ability to generate automatic notification after specified days without response.

G. RFI Tracking and Control:

1. Provide standard RFI form to enter information.
2. Ability to create or attach documents to the RFI form or links to other documents within CDMS.
3. Ability for Reclamation to route RFI to appropriate reviewers and for reviewers to enter comments into standard response form.

H. Contract Modifications:

1. Supply standard Contractor Proposal form with ability to attach documents and link to other documents within CDMS.
2. Supply ability to upload Government's standard CM form and RFP forms, or create an acceptable alternative, to enter information.
3. Ability to attach documents to these forms, and link to other documents within CDMS.
4. Ability for Contractor to respond to RFP and CM form with Cost and Time impacts.
5. Ability to track date CM issued and date responded, with automatic notification after specified days without response.

I. Cost Tracking and Payment Request Management:

1. Ability to import payment bid items cost from a MS Excel spreadsheet.
2. Ability to edit percent complete and payment amount each month. Format to be spreadsheet; for example, items available for editing at once, rather than needing to edit one item at a time, save, open next item, and edit that.

J. Correspondence:

1. Ability to generate letters and memos:
   a. Ability to attach documents to letters and memos, and to reference other documents within CDMS.
   b. CDMS shall automatically assign sequential numbers to letters and memos.
2. Ability to generate responses to letters and memos:
   a. System automatically generates memo suffixes (e.g., the 1st response to Memo 32 to be 32.1 and 2nd response to be 32.2).
3. Official correspondence from Government to Contractor shall contain Government's official letterhead and corresponding logos.
4. Ability to generate meeting agendas and meeting minutes and to attach documents to the minutes:

a. CDMS shall automatically assign sequential meeting numbers.

K. Miscellaneous Tracking and Controls. Ability to:

1. Generate logs tracking RFIs, CORs, COs, Submittals, Payments, Memos, daily inspection reports, survey requests, plant inspection reports, and testing reports.

2. Filter logs based on criteria:

a. Submittals: Specification Section, Days Out, Status (A/AAN/PAR/NA/ACK).

b. RFIs: Days out, Status (Open/Closed), Specification Section/Drawing Number, Responsibility/Ball-In-Court.

3. Sort logs based on a criteria (e.g., for Submittal, Specification Section, Days Out, Status, or other criteria determined by COR).

4. Supply forms to match Government's CM Form and Progress Payment Form or other forms when needed.

5. Track Cost Exposure with links to RFI and CM requests.

6. Measure productivity benchmarks on a project such as computing average review days for RFIs and submittals, categorize CMs, and track total CM percentages by category.

7. Keyword search of documents in database.

8. Allow online Payment Requests (electronic invoicing) by Contractor.

L. Budget and Funding Source Tracking:

1. Ability to enter funding sources and assign contract line items to multiple funding sources.

PART 2 PRODUCTS

2.01 SOFTWARE

A. Procore, as manufactured by Procore, www.procore.com; or equal, having the following essential characteristics: Capable of performing requirements of Part 1 and Part 3.

PART 3 EXECUTION

3.01 IMPLEMENTATION

A. CDMS shall be in place and operating before Contractor transmits Contract Documents to Government, except RSN 01 31 30-1, Approval Data.

1. CDMS operation includes COR, or other authorized Government representative, having program control to assign users and user rights to Government personnel to access appropriate areas of system.

B. Input Schedule of Submittals before making submittals. Include:

1. RSNs listed in Table 01 33 00A – List of Submittals.
2. Scheduled submittal date.

C. Provide a report of submittal status monthly or as requested by COR. Submittal status report shall be in a table format and include:

1. Required Submittal Number (RSN).
2. Section Title.
3. Submittal Title.
4. Date Submitted.
5. Date of Response.
6. Approval Status: A/AAN/PAR/NA/ACK (Approved/Approved As Noted/Partially Approved - Resubmit/Not Approved/Acknowledged).
7. Days in Review.
8. Due Date.

D. CDMS shall be operational until final project closeout.

END OF SECTION

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SECTION 01 32 10
CONSTRUCTION PROGRAM

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Costs:

1. Developing Baseline Schedule: Include in lump-sum prices offered in Price Schedules for mobilization and preparatory work.
2. Updating and Using Construction Program: Include in prices offered in Price Schedules for other items of work.

1.02 DEFINITIONS

A. Schedule: Critical Path Method (CPM) of planning and scheduling a construction project where activities are arranged based on activity relationships and network calculations determine when activities can be performed and critical path of project.

B. Project Calendar(s): Cross reference of numerical work days with calendar days. Project calendar serves as basis for day/date conversion and assigns work days and non-workdays.

C. Resources: Equipment, labor or crews, materials, subcontractors, fabricators, manufacturers, and consultants.

D. Out of Sequence Work: An activity that starts before its predecessor activities are completed.

1.03 REFERENCE STANDARDS

A. Associated General Contractors of America (AGC)

1. AGC Manual-04 — Construction Planning and Scheduling Manual, 2nd Edition

1.04 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals:

1. Furnish database files in format compatible with Oracle Primavera P6 Professional Project Management.
2. Furnish schedule and analysis via CDMS.
3. Upon request, provide information and data used to develop and maintain Construction Program to COR.

B. RSN 01 32 10-1, Representative Information:

1. Designation of authorized representative to develop and maintain Construction Program. Include resume with training and experience.

C. RSN 01 32 10-2, Baseline Schedule:

1. Include:

a. Construction Program/Schedule database.

b. Definition of project calendars.

c. Primavera layout file (.plf).

d. Gantt chart (bar chart) for project on 11- by 17-inch prints and in portable document format (.plf).

e. Activity report including logic constraints consisting of predecessors, successors, and constraint dates.

f. Tabular listings of resources and resource limitations used to produce baseline schedule. Correlate resources with schedule assumptions and offered price. Resource listing may be independent of schedule database.

g. Narrative explanation of project schedule development. Include impacts or resource limitations and weather conditions on project schedule development. Include planned number of work hours per day and hours per day for subcontractors.

D. RSN 01 32 10-3, Updated Schedule Reports:

1. This submittal is required for a proper invoice for progress payments in accordance with the clauses at FAR 52.232-27, Prompt Payment for Construction Contracts.

2. Include:

a. Construction Program database with updated activity and milestone data.

b. Definition of project calendars if revised from baseline calendars.

c. Primavera layout file (.plf).

d. Gantt chart for project on 11- by 17-inch prints and in portable document format (.plf).

e. Narrative report specifically stating status of project:

1) When negative float exists, cite specific actions and conditions which caused "behind schedule" condition and provide proposed course of action to complete project within specified delivery time. When float changed from previous month’s updated schedule, explain reason for change.

2) List of Contractor-initiated changes to current schedule stating reason for action taken and any unresolved issues relating to Construction Program. Government reserves right to reject Contractor-initiated changes to current schedule which negatively impact any Government action which was initiated on basis of current schedule.

3) Discuss contract milestones and significant activities that were started, continued, or completed during update period or scheduled during next update period.

f. Submittals Register:

1) List submittals as listed in Table 01 33 00A – List of Submittals.

2) Update Submittal Register monthly to include submittals and revised submittal. Register fields shall include RSN name and number, planned submittal date, date submitted, date of response, status (e.g., approved, approved as noted/partially approved - resubmit/not approved/ acknowledged).

3) This register document is separate from scheduling database.

E. RSN 01 32 10-4, Time Impact Analysis:

1. Include:

a. Construction Program database with proposed revised activity and milestone data.

b. Proposed revised schedule due to change or delay with added, changed, or deleted activities.

c. Primavera layout file (.plf).

d. Gantt chart for project on 11- by 17-inch print(s) and in portable document format (.plf).

e. Narrative report explaining results and conclusions.

1.05 QUALIFICATIONS

A. Representative: Experienced in developing and maintaining construction schedules and knowledgeable of activities and progress onsite to develop and maintain accurate and reliable schedules.

1.06 GENERAL

A. Develop, maintain, and use Construction Program to plan, monitor, evaluate, and report accomplishment of work.

1.07 SCHEDULES

A. Prepare construction schedules using Critical Path Method, outlined in AGC Manual. Use computer software program to perform mathematical analysis of scheduling data:

1. Prepare schedule based on required sequence and interdependence of activities. Logically sequence activities.

2. Prepare detailed activity network for accomplishing required work organized by Work Breakdown Structure (WBS). Utilize WBS in lieu of activity codes for organization of schedule. If activity codes are utilized in addition to WBS, use unique activity code names assigned as project codes rather than global codes.

3. Activities except "Award" shall have predecessor activities and activities except "Contract Complete" shall have successor activities.

4. Meet contract requirements; milestone(s) in accordance with the provision at Section H.6, Commencement, Prosecution, and Completion of Work; and relevant clauses and specification sections. Include interim milestone dates, Government interface dates, contract completion date, and other time or seasonal constraints specified in contract documents.

5. Include work of subcontractors, Government interfaces, and contract milestones.

6. Adjust Construction Program/schedule for seasonal weather conditions. Provide a narrative explaining expected weather and plan for incorporation in schedule. Consider work influenced by high or low ambient temperatures or precipitation. Utilize Primavera P6 calendars to depict historical weather days that impact work. Assign weather calendars to work impacted by weather.

7. Define activities to level of detail resulting in durations being no greater than 20 workdays:

    a. Durations for administrative activities (e.g., submittals and reviews) fabrication, manufacturing, and typical level of effort activities (e.g., dewatering, storm water management) will not be subject to workday duration limitations.

8. Include separate activities for submittal preparation and submittal reviews when submittal addresses work that involves significant quantities, long lead times, is on the critical path or near critical path, or as requested by COR. Assign Government submittal review activities to a seven-day calendar with durations as specified in Section 01 33 00 – Submittals. Include required Submittal Number (RSN) in activity description.

9. Include separate activities for fabrication or manufacturing when work involves significant quantities, long lead times, is on the critical path or near critical path, or as requested by COR.

10. Include contract title, contract number, and Contractor's name on transmittal cover sheet and each sheet of Gantt chart.

11. For each activity on Gantt chart, display activity identification number, activity description planned or original duration, start date, finish date, total float, and calendar identification. Do not display early start and late start fields.

12. Include table of abbreviations used in schedule, listed and defined alphabetically.

13. Use finish to start logic relationships between activities. Do not utilize start to start, finish to finish, or start to finish logic relationships. Do not use negative lead or lag times.

14. Use durations in units of whole workdays.

15. Provide best estimate of time required to complete activity considering quantity of work, work conditions, location of work, and planned resources for activity.

16. Equate durations of Government reviews and other identified actions to maximum number of calendar days specified in their respective paragraphs.

17. Establish workday calendar(s), and use these in schedule to translate activity's workday duration into calendar dates. Use unique calendar names unique; do not use software default calendar names. Save as project calendars, not global calendars.

B. Baseline Schedule:

1. Represents Contractor’s as-planned approach to accomplishing work. Do not include actual start dates, percent completes, or actual finish dates.

C. Updated Schedule:

1. Meet monthly with COR at Government’s project office, or at location approved by COR, to review progress made to end date of progress payment period. Establish dates that activities were started and completed and remaining duration for each activity started but not completed during period:

   a. Discuss and mutually agree upon changes to schedule.

   b. Out-of-sequence activities are not allowed. Revise logic to reflect change in work plan.

   c. Finalize updated Construction Program database with mutually agreed upon changes.

2. Following receipt of executed contract modification, incorporate activity data stipulated in modification into current schedule for inclusion in next scheduled progress update. Provide appropriate logic relationships for revised activities.

3. Assign unique project file name for each schedule update.

1.08 TIME IMPACT ANALYSIS

A. Provide time impact analysis for contract changes, (e.g., a change order, proposed modification, or value engineering change proposal) to support a claim or request for equitable adjustment to contract which involves a delay or accelerated schedule.

B. CO may use time impact analysis to determine if a time extension or reduction to contract milestone dates are justified:

1. Time impact analysis is required for contract changes whether Contractor's current schedule milestone dates are same as, earlier than, or later than, those required under contract.

2. Changes, additions, or deletions to activities; activity durations; or activity time frames; or activity predecessors and successors will not automatically determine that an extension or reduction of contract time is warranted or due Contractor.

3. Time extensions for performance will be considered only to extent that Contractor's current scheduled milestone dates exceed contract milestone dates.

C. Float is not for exclusive use by or benefit of either Government or Contractor.

D. Perform time impact analyses using data in most recent approved schedule update prior to change or delay event:

1. Prepare proposed revised schedule and narrative description describing and highlighting changes or delays.

2. Prepare summary comparing results of two schedule analyses: One using current schedule data from last approved updated schedule prior to event requiring analysis, and one using proposed schedule data incorporating changes or delays:

a. Show contract milestones and activities whose periods of performance have shifted as result of change which affects production and/or manufacture schedules, material orders, construction seasons, and labor and/or equipment utilization.

b. For activities directly affected by change or delay, include current and proposed items:

1) Activity description.

2) Types and quantities or major pieces of equipment, principal manpower, and pacing materials (materials that affect activity start, duration, or finish).

3) Activity duration.

4) Narrative containing rationale used in developing proposed logic relationships and activity data.

c. Data date for impacted schedule used in comparison shall be same as approved updated schedule data date.

d. Base schedule comparisons on status or work and available float at time CO directs or proposes change to work, Contractor submits a value engineering change proposal, or when a delay occurs as shown in approved updated schedule.

1.09 REVIEW AND EVALUATION

A. Baseline Schedule:

1. Within 21 calendar days after receipt of baseline schedule:

   a. CO will accept or not accept proposed baseline schedule.

   b. Upon request from CO, meet with COR for joint review of proposed baseline schedule.

   c. If schedule is not accepted, revise and resubmit within 7 calendar days following date of rejection letter.

2. Do not proceed with onsite work, except mobilization and surveying, until baseline schedule has been approved by Government.

B. Updated Schedules:

1. Government will require 21 calendar days after receipt of each monthly updated schedule to review and accept or not accept updated schedule.

2. Submit updated schedule at time of submission of each invoice request for progress payment.

3. If updated schedule is not accepted, revise and resubmit updated schedule within 7 calendar days following date of rejection letter.

C. Failure to include any element of work in schedules will not release Contractor from completing required work under contract.

D. Performance will be evaluated by Government using approved CPM schedules.

1.10 FAILURE TO COMPLY

A. Failure to comply with requirements of this Section shall be grounds for a determination by CO that Contractor is not prosecuting work with sufficient diligence to ensure completion within specified time.

B. CO may terminate Contractor's right to proceed with work, or any separable part of it, in accordance with default terms of this contract.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

SECTION 01 33 00
SUBMITTALS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work.

1.02 REFERENCE STANDARDS

A. American Society of Mechanical Engineers (ASME)

1.	ASME Y14.1-2020	Decimal Inch Drawing Sheet Size and Format

B. National Institute of Building Sciences (NIBS)

1.	NIBS NCS-2014	United States National CAD Standards, Version 6

1.03 DEFINITIONS

A. Days: Calendar days.

B. Required Submittal Number (RSN): Identifies items to be submitted together as a complete submittal.

C. Submittal Types, as listed in Table 01 33 00A – List of Submittals:

1. A – Action:
    a. Government will respond as to adequacy of submittal.
    b. Action submittals: Considered to be “shop drawings” within terms of the clause at FAR 52.236-21, Specifications and Drawings for Construction.
2. I – Informational:
    a. Government will acknowledge receipt of Informational submittals.
    b. Government may reject an Informational submittal when submittal does not comply with contract. Contractor shall correct mistakes or deficiencies in rejected Informational submittals and resubmit.
    c. Informational submittals: Considered to be “shop drawings” within terms of the clause at FAR 52.236-21, Specifications and Drawings for Construction, except that approval by Government is not required.

1.04 SUBMITTAL REQUIREMENTS

A. In case of conflict between requirements of this section and requirements included elsewhere in these specifications, requirements included elsewhere take precedence.

B. General:

1. Prepare in English.
2. Label with contract number, title, and RSN.
3. Measurement units: US Customary Units.
4. Provide unique transmittal number for each submittal.

C. Drawings:

1. Minimum identification in title block:
   a. Contract number and title.
   b. Contractor's or supplier's title and drawing number.
   c. Date.
2. Reserve 3- by 3-inch space next to title block for review stamps.
3. Print Size: D size (22- by 34-inches) as defined by ASME Y14.1.
4. Draw to scale using computer drafting or drafting equipment, unless otherwise specified:
   a. Computer drafted drawings:
      1) In accordance with NIBS NCS.
      2) Electronic file format: Compatible with AutoCAD, Version 2018 or newer.
      3) Compile using "eTransmit" utility in AutoCAD.
   b. Drawing prepared with drafting equipment, when allowed: Lettering shall be neat.
5. Drawings designated as "Government Format" in specifications:
   a. Computer drafted.
   b. Government will provide electronic AutoCAD format template.
   c. Title block and sheet format:
      1) As shown on Standard Drawing 40-D-7102.
      2) Government will provide template.
      3) Government will provide specific title block information to be used.

6. Final Drawings:
    a. Computer drafted.
    b. Government will provide electronic AutoCAD format template.
    c. Show as-built changes, including revision dates, made during installation. Indicate changes by clouding.
    d. Electronic Files: On Reclamation issued USB Drive.

D. Product Data:
1. Mark manufacturer's data for commercial products or equipment, such as catalog cut sheets:
    a. Identify manufacturer's name, type, model, size, and characteristics.
    b. Illustrate that product or equipment meets requirements of specifications.
    c. Mark items to be furnished in a manner that will photocopy (no highlighter).
    d. Strike through items that do not apply.

E. Certifications:
1. Certifications by a registered professional: Signed and sealed by registered professional:
    a. Not required for non-design work.
2. Manufacturer’s certifications: Signed by authorized representative of manufacturer.

F. Manuals:
1. Copies:
    a. Printed copies: Bound and indexed.
    b. Electronic copies: Searchable Adobe .pdf on Reclamation issued USB Drive:
        1) Bookmark longer files to assist in navigating file.
        2) Electronic files may be submitted as separate parts of a manual. After each separate electronic file is approved, compile each approved electronic file to assemble an entire manual which shall be consistent with final manual submittal.
2. Contents:
    a. Parts identification lists, lists of special tools, and accessories.
    b. Schematics and wiring diagrams.
    c. Detailed instructions for installing, operating, lubricating, and maintaining equipment.

d. As-built drawings, photographs, and test records or reports if required by specifications.

G. Photographs:

1. Digital files: In .jpeg or similar format.

1.05 SUBMITTALS PROCEDURES

A. Submit only checked submittals. Submittals without evidence of Contractor's approval will be returned for resubmission.

B. Submit complete sets of required materials for each RSN as specified in "Submittals Required" column in Table 01 33 00A – List of Submittals. A complete set includes all listed items for RSNs with multiple parts.

C. Submit sets specified in "Sets to be sent:" columns in Table 01 33 00A – List of Submittals:

1. Submittals identified with "CDMS" in "Sets to be sent" column shall be submitted electronically in accordance with Section 01 31 30 – Contract Document Management System.

2. Submittals identified with "CDMS" in the top row of the "Sets to be sent" column require approval (Action) or acknowledgement (Informational) before sending required sets of hardcopy submittals.

3. Submittals identified with "CDMS" in the bottom row of the "Sets to be sent" column require submittal of hardcopy submittals before finalizing submittal in CDMS.

D. Include the following information in transmittal letters:

1. Contract number and title.

2. RSN for each attached submittal.

3. Responsible code.

4. Number of sets for each RSN.

5. Identify submittal as initial or resubmittal.

E. Resubmittal of submittals not approved:

1. Mark changes such that they are readily identifiable and show revision date.

2. Describe reasons for significant changes in transmittal letter.

3. Resubmit returned submittals within 28 days after receiving the comments, unless otherwise directed.

4. Requirements for initial submittals apply to resubmittals.

1.06 REVIEW OF SUBMITTALS

A. Time Required:

1. Submittal review will require 28 days for review of each submittal or resubmittal, unless otherwise specified.
2. Time required for review of each submittal or resubmittal begins when complete sets of materials required for a particular RSN are received and extends through digital signature date stamp and upload date in CDMS.

B. Time in Excess of Specified:

1. CO may extend contract completion date to allow additional time for completing work affected by excess review time:
   a. Time extension will be to extent that excess review time caused delay to contract completion date.
   b. Time extension will not exceed time used in excess of specified number of days for review of submittals or resubmittals.
   c. Concurrent days of excess review time resulting from review of two or more separate submittals or resubmittals will be counted only once in extending contract completion date.
2. No time extension will be allowed if Contractor fails to make complete action submittals in sequence and within time periods specified.
3. Adjustment for delay will be made only to extent that:
   a. Approval was required under contract, and
   b. Requests for approval were properly and timely submitted and were approved.
4. Adjustment will be subject to terms of paragraphs (b) and (c) of the clause at FAR 52.242-14, Suspension of Work; however, no such delay shall be deemed to be a "suspension order" as the term is used in that clause.

C. Return of Submittals:

1. Return of submittals will be by CDMS response or hard copy, as applicable.
2. Action Submittals: One set of submittals required for action will be returned either approved, partially approved, approved subject to identified changes and resubmit, or not approved:
   a. Submittals that are not approved shall be revised and resubmitted.
   b. Do not change designs without approval of CO after drawings, documentation, and technical data have been approved.
   c. Exception: Schedules will be either accepted, partially accepted or not accepted.

3. Informational Submittals: Government will acknowledge Informational submittals:

   a. Informational submittals will not be returned when they comply with specifications.

   b. Informational submittals that do not comply with specifications may be returned for resubmittal or additional information may be requested.

1.07 HARD COPY OF TRANSMITTALS

A. Addresses for codes listed in Table 01 33 00A – List of Submittals:

   1. Contracting Officer, Bureau of Reclamation, Attn: UC-840, 125 South State St., Room 6107, Salt Lake City, UT 84138-1147.

   2. Construction Engineer, Bureau of Reclamation, 1235 La Plata Hwy, Farmington, New Mexico, 87401.

   3. Technical Service Center, Bureau of Reclamation, Attn: 86-68160, PO Box 25007, Denver CO 80225-0007; Express Mail: Sixth and Kipling, Building 67, Room 152.

   4. Navajo Tribal Utility Authority (NTUA), Attn: Operations Manager, PO Box 1749, Shiprock, NM 87420.

B. Send original transmittal letter with appropriate number of sets to office listed in "Responsible Code" column in Table 01 33 00A – List of Submittals:

   1. Responsible codes starting with "86-6" are located in the Technical Service Center. Send these submittals to the TSC address shown above.

C. Send copy of transmittal letter with appropriate number of sets to offices that are not the responsible code but show "No. of sets to be sent to" in Table 01 33 00A – List of Submittals.

D. Provide copy of transmittal letter on CDMS.

E. When "No. of sets to be sent to" is 0, send a copy of transmittal letter to that office.

F. Submittals required by the specifications, but not listed in Table 01 33 00A – List of Submittals:

   1. Submit in accordance with this section.

   2. Submit to CE, unless otherwise specified.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

Table 01 33 00A – List of Submittals

* Submittal types: A – Action, I – Information

** CO indicates Contracting Officer, CE indicates Construction Engineer, TSC indicates Technical Service Center, and NTUA indicates the City of Gallup New Mexico.

RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
01 14 10-1	Use of Site	Land Use and Landscape Rehabilitation Plan	At least 63 days before use of ROW	A	CE	CDMS			
01 14 10-2	Use of Site	Optional Onsite Security Plan	At least 60 days before use of site	A	CE	CDMS			
01 31 19-1	Project Management and Coordination	Written Summary	Monthly once onsite construction begins	A	CE	CDMS			
01 31 19-2	Project Management and Coordination	Off ROW Land Use Coordination	At least 6 months before use of OFF ROW land	I	CE	CDMS			
01 31 19-3	Project Management and Coordination	Key Personnel	Within 7 days after receipt of Notice of Award	A	CE	CDMS			
01 31 19-4	Project Management and Coordination	Replacement of Key Personnel	A minimum of 2 business days prior to work being performed by new personnel	A	CE	CDMS			
01 31 30-1	Contract Document Management System	Approval Data	Within 21 days of Award	A	CE	0	1	0	0
						CDMS			
01 31 30-2	Contract Document Management System	Final Data	Within 14 days of completion of work	A	CE	0	2	2	0
						CDMS			
01 32 10-1	Construction Program	Representative Information	Within 7 days after receipt of Notice of Award	I	CE	CDMS			
01 32 10-2	Construction Program	Baseline Schedule	Within 21 days after receipt of initial Notice to Proceed	A	CE	CDMS			
01 32 10-3	Construction Program	Updated Schedule Reports	Monthly or with Progress Payments	A	CE	CDMS			

Table 01 33 00A – List of Submittals

* Submittal types: A – Action, I – Information

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RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
01 32 10-4	Construction Program	Time Impact Analysis	Within 28 days after CO directs a contract change, with any proposal for a future modification, with any value engineering proposal, or with any request or claim for an equitable adjustment to the contract.	A	CE	CDMS			
01 35 10-1	Safety Data Sheets	Complete LHM and SDS	At least 14 days before jobsite delivery of hazardous material	I	CE	CDMS			
01 35 10-2	Safety Data Sheets	Updated LHM and SDS	At least 14 days before jobsite delivery of hazardous material not previously listed	I	CE	CDMS			
01 35 20-1	Safety and Health (WBR 1452.223-81)	Safety Program	Submitted and accepted before commencing onsite work. Submit at least 45 days before beginning onsite work. See Section 3 of RSHS	A	CE	CDMS			
01 35 20-2	Safety and Health	Job Hazard Analyses	At least 14 days before commencing onsite activities for each phase of work	A	CE	CDMS			
01 35 20-3	Safety and Health	Monthly Accident Summary Report	First day of each month. See paragraph 3.8 of RSHS	A	CE	CDMS			

Table 01 33 00A – List of Submittals

* Submittal types: A – Action, I – Information

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RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
01 35 30-1	Contractor's Onsite Safety Personnel	Resumes	At least 45 days before beginning onsite construction; resubmit upon personnel change	A	CE	CDMS			
01 35 30-2	Contractor's Onsite Safety Personnel	Safety Inspection Reports	At least once each week	I	CE	CDMS			
01 46 00-1	Quality Procedures	Definable Features of Work List	Within 21 days after receipt of initial Notice to Proceed	A	CE	CDMS			
01 46 00-2	Quality Procedures	Contractor Quality Control (QC) Plan	At least 45 days before beginning onsite work	A	CE	CDMS			
01 46 00-3	Quality Procedures	Contractor's Quality Daily Report	Weekly	I	CE	CDMS			
01 46 00-4	Quality Procedures	Quality Control Supervisor (QCS) Resume	At least 45 days before beginning onsite work. Resubmit upon personnel change.	A	CE	CDMS			
01 46 00-5	Quality Procedures	Contractor's Quality Testing Plan	At least 45 days before beginning onsite work	A	CE	CDMS			
01 46 00-6	Quality Procedures	Summary of Monthly Test Results	Monthly, on or before 5th day of each month	A	CE	CDMS			
01 46 20-1	Testing Agency Services	Testing Agencies Services Plan	At least 45 days before testing is required	A	CE	CDMS			
01 46 25-1	Special Inspections	Preconstruction Submittals	At least 45 days before special inspection is required	A	CE	CDMS			
01 46 25-2	Special Inspections	Test Reports	Daily and Biweekly	A	CE	CDMS			
01 46 25-3	Special Inspections	Certificates	At least 45 days before special inspection is required	A	CE	CDMS			

Table 01 33 00A – List of Submittals

* Submittal types: A – Action, I – Information

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RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
01 46 25-4	Special Inspections	Closeout Submittals	Within 14 days of completion of work	A	CE	CDMS			
01 55 00-1	Vehicular Access and Parking	Pre-Construction Digital Recording	At least 21 days before beginning onsite work	A	CE	0	1	0	0
01 55 00-2	Vehicular Access and Parking	Right of Way (ROW) Access Plan	At least 21 days before beginning onsite work.	A	CE	CDMS			
01 55 00-3	Vehicular Access and Parking	Post-Construction and Post-Repair Digital Recording	At least 21 days before leaving site.	A	CE	0	1	0	0
01 55 20-1	Traffic Control	Information of Traffic Control Plan (TCP)	At least 70 days before affecting public traffic.	A	CE	CDMS			
01 55 20-2	Traffic Control	Approved NMDOT Permit(s)	At least 45 days before affecting public traffic.	A	CE	CDMS			
01 56 10-1	Protection of Existing Installations	Plan for Protecting Existing Installations	At least 45 days before start of onsite construction work	A	CE	CDMS			
01 56 15-1	Protection of Existing Utilities	Utility Owner Acknowledgement	At least 45 days before start of onsite construction work	A	CE	CDMS			
01 56 15-2	Protection of Existing Utilities	Utility Crossing Investigation	At least 45 days before start of onsite construction work	A	CE	CDMS			
01 56 15-3	Protection of Existing Utilities	Work Plan within Utility Easements	At least 45 days before start of onsite construction work	A	CE	CDMS			
01 57 20-1	Environmental Controls	Copy of Applicable Air Quality Permit	At least 14 days before beginning onsite work	I	CE	CDMS			
01 57 30-1	Water Pollution Control	Stormwater Pollution Prevention Plan (SWPPP)	At least 28 days before start of onsite construction work	A	CE	CDMS			

Table 01 33 00A – List of Submittals

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RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
01 57 30-2	Water Pollution Control	Spill Prevention, Control, and Countermeasure (SPCC) Plan	At least 28 days before delivery or storage of oil	A	CE	CDMS			
01 57 40-1	Pesticides	Qualifications	At least 45 days before beginning onsite work	A	CE	CDMS			
01 57 40-2	Pesticides	Pesticide Use Plan	At least 45 days before application of first pesticide	A	CE	CDMS			
01 57 50-1	Tree and Plant Protection	Protection Plan	At least 45 days before beginning onsite work	A	CE	CDMS			
01 57 90-1	Preservation of Historical and Archeological Data	Alternate Use Area	At least 45 days before beginning offsite work, if commercial site; if non-commercial site then at least 6-months prior to use of land	A	CE	CDMS			
01 61 10-1	Buy American	Buy American Certificates	Prior to project closeout	I	CE	CDMS			
01 71 20-1	Surveying	Surveying Plan	Within 45 days after Notice to Proceed, and approved before beginning survey work	A	CE	CDMS			
01 71 20-2	Surveying	Resume(s)	At least 45 days before beginning survey work; resubmit upon personnel change	I	CE	CDMS			
01 71 20-3	Surveying	Accuracy Check Results	At least 45 days before beginning survey work	I	CE	CDMS			
01 74 00-1	Cleaning and Waste Management	Waste Production and Disposal Plan	At least 45 days before beginning onsite work	I	CE	CDMS			

Table 01 33 00A – List of Submittals

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RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
01 74 00-2	Cleaning and Waste Management	Waste Production and Disposal Records	Within 7 days of hazardous waste disposal	I	CE	CDMS			
01 74 00-3	Cleaning and Waste Management	Environmental Consultant Resume	At least 45 days before beginning environmental assessment	I	CE	CDMS			
01 74 00-4	Cleaning and Waste Management	Environmental Site Assessment	Within 14 days of completion of work	I	CE	CDMS			
01 78 30-1	Project Record Documents	Electronic Media Release Form, Upon Request	With request for Electronic Drawings	A	CE	CDMS			
01 78 30-2	Project Record Documents	Progress As-Built Drawings	Monthly once onsite construction begins.	A	CE	CDMS			
01 78 30-3	Project Record Documents	Final As-Built Drawings	Within 28 days of Government issued substantial completion of work	A	CE	CDMS			
01 78 30-4	Project Record Documents	Final Drawings	Within 14 days of approval of RSN 01 78 30-3	A	CE	0	2	0	2
						CDMS			
						CDMS			
03 11 10-1	Concrete Forming	Formwork Design and Drawings	At least 45 days prior to installation work	A	CE	CDMS			
03 20 00-1	Concrete Reinforcement	Reinforcement Diagrams and Lists	At least 45 days before placing concrete	A	CE	CDMS			
03 20 60-1	Concrete Anchors	Instructions	At least 45 days before placing concrete	A	CE	CDMS			
03 20 60-2	Concrete Anchors	Test Data and Results	At least 7 days before testing	A	86-8530	CDMS			

Table 01 33 00A – List of Submittals

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RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
03 30 00-1	Cast-In-Place Concrete	Approval Data	At least 45 days before placing concrete	A	86-8530	CDMS			
03 30 00-2	Cast-In-Place Concrete	Concrete Placement Schedule	At least 45 days before placing concrete	A	86-8530	CDMS			
03 30 00-3	Cast-In-Place Concrete	Certifications	At least 45 days before placing concrete	A	86-8530	CDMS			
03 30 00-4	Cast-In-Place Concrete	Test Reports	At least 45 days before placing curing compound	A	86-8530	CDMS			
03 48 00-1	Precast Concrete Manholes	Certificates	At least 45 days before placing concrete	A	86-8530	CDMS			
03 48 00-2	Precast Concrete Manholes	Shop Drawings	At least 45 days before placing concrete	A	86-8530	CDMS			
03 48 00-3	Precast Concrete Manholes	Product Data	At least 45 days before placing concrete	A	86-8530	CDMS			
03 62 20-1	Nonshrink Grout for Equipment and Metalwork	Packaged Nonshrink Grout Data	At least 45 days before placing concrete	A	86-8530	CDMS			
03 63 00-1	Epoxy Grout	Approval Data	At least 45 days before beginning grouting	A	86-8530	CDMS			
09 96 20-1	Coatings	Approval Data	At least 28 days before beginning coating work	A	86-8540	CDMS			
09 96 20-2	Coatings	Final Approval Data	At least 28 days before beginning coating work	A	86-8540	CDMS			
09 96 20-3	Coatings	Paint Chip Samples	At least 28 days before beginning coating work	A	86-8540	CDMS			
						0	1	1	0
09 96 20-4	Coatings	Qualifications	At least 28 days before beginning coating work	A	86-8540	CDMS			
09 96 20-5	Coatings	Quality Control Plan	At least 28 days before beginning coating work	A	86-8540	CDMS			

Table 01 33 00A – List of Submittals

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RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
09 96 20-6	Coatings	Contractor Quality Testing Report	At least 14 days after completion of test	A	86-8540	CDMS			
10 14 00-1	Signage	Shop Drawings and Approval Data	At least 28 days before procurement	A	86-8120	CDMS			
10 14 26-1	Utility Markers	Utility Marker Plan	At least 28 days before procurement	A	CE	CDMS			
10 44 20-1	Fire Extinguishers	Approval Data	At least 28 days before procurement	A	CE	CDMS			
13 34 00-1	Prefabricated Engineered Structures	Approval Drawings and Data	At least 28 days before procurement	A	CE	CDMS			
13 34 00-2	Prefabricated Engineered Structures	Sample and Color Sample Kit	At least 28 days before procurement	A	CE	CDMS			
						0	1	1	0
13 34 00-3	Prefabricated Engineered Structures	Geotechnical Report	At least 28 days before procurement	A	86-68314	CDMS			
13 34 00-4	Prefabricated Engineered Structures	Factory Test Reports	Within 14 days after completion of test	A	86-68460	CDMS			
13 34 00-5	Prefabricated Engineered Structures	Field Quality Test Reports	Within 14 days after completion of test	A	86-68460	CDMS			
13 34 00-6	Prefabricated Engineered Structures	Final Data	At least 28 days before procurement	A	86-68460	CDMS			
25 00 01-1	System Control and Monitoring	Qualifications	At least 28 days before beginning work	A	CE	CDMS			
25 00 01-2	System Control and Monitoring	System Description Documentation	At the same time as equipment is shipped to the site	A	CE	CDMS			
25 00 01-3	System Control and Monitoring	Approval Drawings and Data	Within 150 days after Notice to Proceed, and at least 45 days prior to installation	A	CE	CDMS			

Table 01 33 00A – List of Submittals

* Submittal types: A – Action, I – Information

** CO indicates Contracting Officer, CE indicates Construction Engineer, TSC indicates Technical Service Center, and NTUA indicates the City of Gallup New Mexico.

RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
25 00 01-4	System Control and Monitoring	Layout Drawings	Within 150 days after Notice to Proceed, and at least 45 days prior to installation	A	CE	CDMS			
25 00 01-5	System Control and Monitoring	Software Development Documentation	Within 150 days after Notice to Proceed, and at least 45 days prior to installation	I	CE	CDMS			
25 00 01-6	System Control and Monitoring	Final Drawings, Documentation, and Operations and Maintenance Manuals	Within 14 days of completion of work	I	CE	CDMS			
25 08 10-1	Automation System Testing and Acceptance	SPAT Procedure	At least 40 days before beginning test	A	CE	CDMS			
25 08 10-2	Automation System Testing and Acceptance	SPAT Start Date	At least 40 days before beginning test	I	CE	CDMS			
25 08 10-3	Automation System Testing and Acceptance	SPAT Test Report	At least 14 days after completion of test	A	CE	CDMS			
25 08 10-4	Automation System Testing and Acceptance	Lateral Integration Test Start Date	At least 40 days before beginning test	I	CE	CDMS			
25 08 10-5	Automation System Testing and Acceptance	Lateral Integration Test Report	At least 40 days before beginning test	A	CE	CDMS			
26 05 10-1	Conductors and Cables	Approval Data	Submitted and approved at least 28 days before procurement	A	86-68460	CDMS			
26 05 10-2	Conductors and Cables	Field Test Reports	No later than 14 days after completing testing	A	86-68460	CDMS			
26 05 20-1	Grounding and Bonding	Approval Drawings and Data	Submitted and approved at least 28 days before procurement	A	86-68460	CDMS			

Table 01 33 00A – List of Submittals

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RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
26 05 20-2	Grounding and Bonding	As-built Drawings	No later than 28 days after completing work	A	86-68460	CDMS			
26 05 20-3	Grounding and Bonding	Test Reports	No later than 14 days after completing testing	A	86-68460	CDMS			
26 05 33-1	Raceways and Boxes	Approval Drawings and Data	Submitted and approved at least 28 days before procurement	A	86-68460	CDMS			
26 05 33-2	Raceways and Boxes	As-built Drawings	No later than 28 days after completing work	A	86-68460	CDMS			
26 27 10-1	Meter Socket	Approval Drawings and Data	Submitted and approved at least 28 days before procurement	A	86-68460	CDMS			
26 28 20-1	Service Disconnect Switch	Approval Data	Submitted and approved at least 28 days before procurement	A	86-68460	CDMS			
26 42 00-1	Cathodic Protection Systems	Certification and Data	At least 42 days before procuring materials	A	86-68540	CDMS			
26 42 11-1	Buried Galvanic Anode Cathodic Protection	Certification and Data	At least 42 days before procuring materials	A	86-68540	CDMS			
26 42 11-2	Buried Galvanic Anode Cathodic Protection	Final Data	Within 28 days of final testing	A	86-68540	CDMS			
27 30 01-1	Telephone System	Communication System Design	Within 150 days after Notice to Proceed, and at least 45 days prior to installation	A	CE	CDMS			
27 30 01-2	Telephone System	Approval Data	Within 150 days after Notice to Proceed, and at least 45 days prior to installation	A	CE	CDMS			

Table 01 33 00A – List of Submittals

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RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
27 30 01-3	Telephone System	Test Procedure and Test Schedule	At least 28 days before beginning test	I	CE	CDMS			
27 30 01-4	Telephone System	Operation and Maintenance Manual	At least 28 days before beginning test	I	CE	CDMS			
27 30 01-5	Telephone System	Field Operational Checkout Test Report	At least 14 days after completion of test	A	CE	CDMS			
28 10 01-1	Security and Surveillance System	Approval Drawings and Data	Within 150 days after Notice to Proceed, and at least 45 days prior to installation	A	CE	CDMS			
28 10 01-2	Security and Surveillance System	Test Procedure and Test Schedule	At least 28 days before beginning test	I	CE	CDMS			
28 10 01-3	Security and Surveillance System	Test Report	At least 14 days after completion of test	A	CE	CDMS			
28 10 01-4	Security and Surveillance System	Final Drawings and Operations and Maintenance Manuals	Within 14 days of completion of work	I	CE	CDMS			
31 02 10-1	Water for Dust Abatement	Meter Calibration	At least 21 days prior to use of equipment	A	CE	CDMS			
31 02 30-1	Dust Palliative	Dust Palliative	At least 28 days before beginning onsite work	A	CE	CDMS			
31 03 33-1	Removal and Control of Water	Removal and Control of Water Plan	At least 28 days before beginning onsite work	A	CE	CDMS			
31 23 02-3	Compacting Earth Materials	Test Results	Within 7 days after completion of tests	I	CE	CDMS			
31 23 10-1	Earthwork for Site and Structures	Processing Plan for Soils under Structures	At least 42 days before beginning onsite work	A	CE	CDMS			
31 23 10-2	Earthwork for Site and Structures	Test Results	Within 7 days after completion of tests	I	CE	CDMS			

Table 01 33 00A – List of Submittals

* Submittal types: A – Action, I – Information

** CO indicates Contracting Officer, CE indicates Construction Engineer, TSC indicates Technical Service Center, and NTUA indicates the City of Gallup New Mexico.

RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
31 23 22-1	Pipe Trench Earthwork	Processing Plan	At least 28 days before beginning onsite work	A	CE	CDMS			
31 23 22-2	Pipe Trench Earthwork	Test Results	Within 1 day of testing, immediately if test failed	I	CE	CDMS			
31 23 70-1	Controlled Low Strength Material (CLSM)	Approval Data for CLSM Mix	At least 28 days before beginning onsite work	A	CE	CDMS			
31 23 70-2	Controlled Low Strength Material (CLSM)	Cold Weather Protection Plan	At least 28 days before beginning onsite work	A	CE	CDMS			
31 23 70-3	Controlled Low Strength Material (CLSM)	Quality Control Test Results	Within 1 day of testing, immediately if test failed	I	CE	CDMS			
31 31 30-1	Soil-Applied Herbicide	Use Plan	At least 28 days before beginning onsite work	I	CE	CDMS			
31 31 30-2	Soil-Applied Herbicide	Applicator Certification	At least 28 days before beginning onsite work	I	CE	CDMS			
31 37 00-1	Riprap	Approval Data and Test Results	At least 45 days before installing rip rap	A	CE	CDMS			
32 15 10-1	Gravel Surfacing	Gravel Certification	At least 14 days before placing surfacing	A	CE	CDMS			
32 15 10-2	Gravel Surfacing	Geotextile Installation	At least 14 days before placing surfacing	A	CE	CDMS			
32 15 10-3	Gravel Surfacing	Geotextile Certification	At least 14 days before placing surfacing	A	CE	CDMS			
32 31 19-1	Steel Ornamental Fence	Certifications	At least 14 days before installing fence	A	CE	CDMS			
32 31 19-2	Steel Ornamental Fence	Installation Plans	At least 14 days before installing fence	A	CE	CDMS			
32 92 20-1	Seeding	Seeding Plan	At least 14 days before seeding	A	CE	CDMS			

Table 01 33 00A – List of Submittals

* Submittal types: A – Action, I – Information

** CO indicates Contracting Officer, CE indicates Construction Engineer, TSC indicates Technical Service Center, and NTUA indicates the City of Gallup New Mexico.

RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
32 92 20-2	Seeding	Certifications	At least 14 days before seeding	A	CE	CDMS			
33 11 12-1	Steel Line Pipe	Shop Drawings	At least 28 days before beginning onsite work	A	86-68141	CDMS			
33 11 12-2	Steel Line Pipe	Ventilation Plan	At least 28 days before beginning onsite work	I	86-68141	CDMS			
33 11 12-3	Steel Line Pipe	Field Welder Qualifications	At least 28 days before beginning onsite work	A	86-68141	CDMS			
33 11 12-4	Steel Line Pipe	Filling and Testing Plan	At least 28 days before beginning onsite work	A	86-68141	CDMS			
33 11 12-5	Steel Line Pipe	Flotation Prevention Plan	At least 28 days before beginning onsite work	A	86-68141	CDMS			
33 11 12-6	Steel Line Pipe	Pipe Deflection Measurements	At least 28 days before beginning onsite work	A	86-68141	CDMS			
33 13 00-1	Disinfection of Water Utility Distribution	Flushing, Testing, Disinfection, and Verification Plan	At least 28 days before beginning onsite work	A	86-68141	CDMS			
33 13 00-2	Disinfection of Water Utility Distribution	Flushing, Testing, Disinfection, and Verification Report	Within 14 days of test completion	A	86-68141	CDMS			
35 21 95-1	Steel Manifold Piping	Shop Drawings	At least 42 days before construction	A	86-68420	CDMS			
35 21 95-2	Steel Manifold Piping	Commercial Products Data	At least 42 days before construction	A	86-68420	CDMS			
35 21 95-3	Steel Manifold Piping	Final Data	Within one week after testing	A	86-68420	CDMS			
35 21 95-4	Steel Manifold Piping	Final Drawings	Within 28 days of installation	I	86-68420	CDMS			
35 22 14-1	Valves and Equipment	Commercial Products Data	At least 42 days before fabrication or construction	A	86-68420	CDMS			
35 22 14-2	Valves and Equipment	Field Installation Procedures	At least 42 days before fabrication or procurement.	A	86-68420	CDMS			

Table 01 33 00A – List of Submittals

* Submittal types: A – Action, I – Information

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RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
35 22 14-3	Valves and Equipment	Shop Testing Certifications, Reports, and Data	Within 28 days after testing.	A	86-68420	CDMS			
35 22 14-4	Valves and Equipment	Service Manuals and Final Data	At least 28 days before installation	I	86-68420	CDMS			

END OF SECTION

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SECTION 01 33 26
ELECTRICAL DRAWINGS AND DATA

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in Price Schedules for Complete Electrical System for Outdoor Equipment Only

1.02 REFERENCE STANDARDS

A. American Society of Mechanical Engineers (ASME)

1.	ASME Y14.1-2020	Decimal Inch Drawing Sheet Size and Format

B. Institute of Electrical and Electronic Engineers (IEEE)

1.	IEEE 315-1975(1993)	Graphic Symbols For Electrical and Electronics Diagrams (Including Reference Designation Letters)
2.	IEEE C37.2-2022	Electrical Power System Device Function Numbers, Acronyms, and Contact Designations
3.	IEEE Y32.9-1972(1989)	Graphic Symbols for Electrical Wiring and Layout Diagrams Used in Architecture and Building Construction

C. National Electrical Manufacturers Association (NEMA)

1.	NEMA ICS 1-2000(2008)	Industrial Control and Systems: General Requirements

1.03 SUBMITTAL PROCEDURES

A. In addition to the requirements in Section 01 33 00 – Submittals, prepare electrical drawings and data submittals in accordance with the following.

B. Drawings, general:

1. Device designations and symbols:
   a. Conform to IEEE 315, IEEE C37.2, IEEE Y32.9, and NEMA ICS-1.
   b. As indicated on Standard Drawings 104-D-757 and 104-D-1150.
2. Size: D-size prints as defined in ASME Y14.1 made from original drawings.

3. Indicate changes on revised drawings to distinguish them from previous submittals. Describe reasons for significant changes in submittal letters.

C. Approval Drawings in Government Format:

1. General:

a. Furnish with Government title block and in form as indicated on Standard Drawing 40-D-7102.

b. Government will provide specific title block information when drawings are submitted.

c. Government may revise conductor or device designations on Contractor's drawings when drawings are approved at no change in contract price or completion time.

2. Schematic Diagrams:

a. In accordance with Standard Drawing 104-D-1150.

b. Format:

1) Vertical ladder diagram form.

2) Label rungs with sequential numbers starting with number 1.

3) Locate rung numbers adjacent and to left of associated rung.

4) Provide unique rung numbers for each circuit.

5) On each rung, indicate related rungs.

c. Conductor designations: As indicated on drawings.

d. Indicate device terminal designations.

e. Indicate functional operation.

f. Describe unusual or nonstandard operation.

g. Indicate ratings and values of devices.

h. Indicate contacts, including spare contacts, for motor contactors, auxiliary relays, timers, and protective relays.

i. Indicate cross-referencing between other schematic and wiring diagrams.

j. Do not include current and potential transformer circuits on drawing with direct-current control and protective relaying circuits.

k. Conductor designations: As indicated on drawings or as developed by Contractor. Contractor-developed conductor designations shall consist of:

1) Circuit prefix.

2) Rung number.

3) Unique rung wire letter such as 5P15C (“5”-Unit Number, “P”-Protection Circuit, “15”-Rung Number, “C”-Unique Rung Wire Letter).

l. Base instrument current and potential transformer circuit conductor designations upon primary conductor designations indicated on single-line diagram drawings.

m. Show switch developments for control, selector, and limit switches.

n. Show device and terminal block designations.

3. Wiring Diagram Drawings:

a. Match with schematic diagram drawings.

b. Indicate point-to-point wiring in accordance with Standard Drawing 104-D-1165.

c. Indicate equipment as mounted on back, side, swing, and door panels. Show panel as viewed from wiring side of panel.

d. Identify each item of equipment with designation indicated on schematic diagram.

e. Identify control and protective devices with (IEEE) device number as indicated on specifications drawings or on Standard Drawing 104-D-1150.

f. Identify components such as fuses and resistors by value.

g. Indicate wiring of devices and items of equipment including terminal number.

h. Indicate cables, cable and individual wire designations, and connections to external circuits.

i. Cable designations: Similar to designations on Standard Drawing 104-D-1165.

j. Show no more than one panel per drawing.

4. Sample Wiring Diagram: Sample of actual wiring diagram that will be submitted as check print and final drawing.

D. Approval Drawings and Data in Manufacturer’s Format:

1. General: Furnish with manufacturer’s standard format, except as noted below.

2. Equipment Layout Drawings:

a. Indicate dimensions of equipment.

b. Indicate location of devices and items of equipment including nameplates, terminal blocks, wiring ducts, bus, conduit entries, and other features in their relative physical location.

c. Indicate method of securing equipment to deck, floor, or wall.

d. Identify each device and item of equipment with a bill of material reference number.

3. Bill of Material (BOM) List:

a. Provide information on manufacturer, style, type, rating, quantity, and other identifying information for each device or item of equipment.

b. Provide unique reference number for each device or item of equipment listed on bill of material.

4. Nameplate Lists: Provide information on type of material, size, and engraved lettering.

5. Manufacturer's Technical Catalog Data:

a. Provide technical data for each device or item of equipment.

b. Include manufacturer's name and address; catalog number, type, style, or model number; electrical ratings; and dimensions.

c. Where several items are listed on same sheet, indicate items being submitted for approval.

d. Demonstrate proposed device or item of equipment meets specifications requirements.

e. Mark catalog data sheet with appropriate BOM item number.

f. Assemble catalog data sheet into one enclosing cover with BOM index.

6. Time-Current Characteristic and Equipment Coordination Curves:

a. Provide time-current characteristic curves for adjustable and non-adjustable protective devices.

b. Provide equipment coordination curves for:

1) Transformer inrush current.

2) Transformer full load current.

3) Transformer thermal and mechanical damage.

4) Motor inrush current.

5) Motor full load current.

c. Original plot on log-log graph paper.

d. Illustrate degree of selectivity being provided.

7. Job Hazard Analysis (JHA):

a. Provide when Government will be performing factory inspection.

b. Prepare JHA in accordance with 29 CFR 1910, Subpart I.

c. As a minimum include the following in JHA:

1) Factory address and name of site visit contact person.

2) Date of JHA.

3) Purpose of JHA, for example "Visitor Safety Procedures."

4) Hazards:

a) Types of hazards at factory.

b) Safety procedures in or around high-voltage test labs.

c) Safety procedures on factory floor.

5) Personal Protective Equipment:

a) List of equipment required.

b) List of equipment provided by factory.

6) Emergency response:

a) Name and phone number of Factory Safety Officer.

b) Employees trained in emergency conditions (CPR and First Aid).

c) Factory fire protection/evacuation procedures.

d) Nearest hospital or emergency room.

7) Name, title, and signature of factory official approving JHA.

E. Check Prints:

1. Schematic and wiring diagrams which indicate "As-built" condition of equipment at time of shipment.

2. Mark as changes and revisions are made during installation and checkout of equipment.

3. Keep drawings at the jobsite and mark as work progresses.

4. Maintain separate, identical set of prints for use in preparing final drawings.

5. Furnish copy of Size D check prints to Government for use in checkout and commissioning.

F. As-built Drawings:

1. Provide for grounding, conduit, and lighting systems.

a. Indicate as-built changes made during construction.

b. Indicate dimensions of:

1) Location of grounding electrodes.

2) Embedded and direct buried grounding cables.

3) Embedded and direct buried conduit.

2. Include revision dates.

3. Mark the drawings in the following colors:
    a. Red: Additions to original drawings.
    b. Green: Deletions to original drawings.
    c. Blue: Notations necessary for explanation of as-built markings.

G. Final Drawings:
1. Revised to reflect approval comments and as-built condition of installed equipment at time of contract completion.
2. Provide original plot of schematic and wiring diagrams, equipment layout drawings, bill of materials, and nameplate drawings/lists.
3. Provide computer drawing files on Reclamation issued USB Drive in AutoCAD (*.dwg).

H. Test Reports:
1. Typed, 8 1/2- by 11-inch sheets.
2. Certified.
3. Submit multi-page reports in bound folder or three-ring binder.

I. Operation and Maintenance Instructions Books:
1. Print copies:
    a. Assemble each set of material into one or more books with enclosing covers.
2. Electronic copies:
    a. Adobe Acrobat (.pdf) format on Reclamation issued USB Drive.
    b. Conform to print copies.
    c. Bookmark file(s) to aid in navigating.
3. Provide following items:
    a. Index sheet at front of each book which provides page or index tab number information for each device or item of equipment in book.
    b. Manufacturer’s operation and maintenance procedures; installation details, as necessary; and catalog data sheets for each device or item of equipment.
    c. List of recommended spare parts and components.
    d. Parts lists for replacement parts.
    e. Manufacturer’s full-size circuit breaker and fuse time-current curves.
    f. Copies of factory test reports.
    g. Copies of bills of materials

h. Copies of drawings.

1) Revised to reflect approval comments.

2) Maximum size: 11- by 17-inch.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 01 35 10
SAFETY DATA SHEETS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work.

1.02 DEFINITIONS

A. LHM: List of Hazardous Materials.

B. SDS: Safety Data Sheet: Referred to as Material Safety Data Sheets in clause at FAR 52.223-3, Hazardous Material Identifications and Material Safety Data - Alternate 1.

1.03 APPLICATION

A. For purpose of this contract, "delivered under this contract" in paragraph (b) of clause at FAR 52.223-3, Hazardous Material Identification and Material Safety Data - Alternate 1, includes materials:

1. Delivered to Government.
2. Incorporated into work.
3. Materials used by Contractor during contract performance at jobsite.

1.04 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 35 10-1, Complete LHM and SDS.

C. RSN 01 35 10-2, Updated LHM and SDS:

1. Comply with paragraph (e) of clause at FAR 52.223-3, Hazardous Material Identification and Material Safety Data - Alternate 1.

1.05 DELIVERY

A. Do not deliver hazardous materials to jobsite which are not included on original or previously updated LHM and SDS before receipt of updated LHM and SDS by Construction Engineer (CE).

B. Maintain hard copy of complete updated LHM and SDS on site for use by Government and Contractor personnel.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

SECTION 01 35 20
SAFETY AND HEALTH

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in Price Schedules for other items of work.
2. Costs for complying with updates to RSHS, FIST 1-1, and 29 CFR 1926 promulgated prior to contract completion are Contractor's responsibility.

1.02 REFERENCE STANDARDS

A. Bureau of Reclamation (Reclamation)

1.	RSHS	Reclamation Safety and Health Standards, including revisions, available at https://www.usbr.gov/safety/rshs/index.html
2.	FIST 1-1 (2019)	Hazardous Energy Control Program, available at: https://www.usbr.gov/power/data/fist_pub.html
3.	FIST 4-1 (2024)	Hazardous Energy Control Program, available at: https://www.usbr.gov/power/data/fist_pub.html

B. Code of Federal Regulations (CFR)

1.	29 CFR 1926.32	Definitions, Construction
2.	29 CFR 1926.103	Respiratory Protection

C. Applicable Tribal and State Safety and Health Regulations for Construction.

1.03 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 35 20-1, Safety Program:

1. Written safety program in accordance with RSHS.
2. Cover aspects of onsite and applicable off-site operations and activities associated with this contract.
3. Follow outline in Appendix 1.03C of RSHS.

4. Will not be accepted for review by COR unless it addresses, in order, lettered and numbered per Appendix 1.03C, a comprehensive narrative for each applicable item in outline. For items in outline that do not apply to this contract write "Not applicable to Contract" next to number/letter.
5. A generic company safety program is not acceptable. Site-Specific Safety Program required for this contract.
6. Submitted and accepted prior to commencing onsite work, including mobilization.

C. RSN 01 35 20-2, Job Hazard Analyses:

1. Submitted and accepted prior to commencing onsite activities for each phase of work.
2. Include appropriate Safety Data Sheet(s) for activities involving hazardous materials.

D. RSN 01 35 20-3, Monthly Accident Summary Report:

1. In accordance with RSHS.

1.04 DOCUMENTATION AND RECORDS

A. Prepare and retain all safety plans, programs, training content, and training records applicable to scope of work and make available to COR unless already included in written safety program. Some examples include but are not limited to:

1. Confined Space and when deemed necessary by existing or introduced hazards of Permit Required Confined Space through Written Program and Training Records.
2. Forklifts and Other Industrial Trucks Training Records.
3. Electrical Safety Requirement Training Records.
4. Emergency Action Plans Written Program and Training Records.
5. Fall Protection Written Program and Training Records.
6. Fire Protection and Prevention Written Program and Training Records.
7. Flammable and Combustible Liquids Written Spill Control Plan.
8. General Safety and Health Provisions, Written Safety Program, and Regular Safety Inspection Records.
9. Hazard Communication Written Program and Training Records.
10. Ladder Written Compliance Program and Training Records.
11. Lockout/Tagout: Hazardous Energy Control Program (HECP) Training Records. Refer to RSHS Sections 1.11 and 1.13.
12. Occupational Noise Exposure Written Program and Training Audiometric Testing Records.
13. Personal Protective Equipment (PPE) Hazard Assessments and Training Records.

14. Power Operated Hand Tools Training Records.
15. Respiratory Protection Written Program and Training Records.
16. Safety Training and Education Written Program and Training Records.
17. Steel Erection Training Records, if applicable.
18. Welding, Cutting, and Brazing Written Program and Training Records.

1.05 SAFETY AND HEALTH

A. Provide and maintain work environment and procedures that:

1. Safeguard public, Government personnel, and Contractor employees exposed to Contractor operations and activities.
2. Avoid interruptions of site operations and delays in project completion dates.
3. Control costs in contract performance.

B. Do not begin onsite work, including mobilization, until CE accepts Safety Program.

C. Participate in Contractor Safety Program Review meeting prior to mobilization.

D. Minimum work crew shall consist of no less than 2 people, unless approved by COR.

E. Develop JHA for each distinct phase of work under the contract.

1. Do not begin phase of work until JHA is acceptable to COR and shared with construction employees. At a minimum JHA to include elements in RSHS 1.05.
2. Activities involving hazardous materials shall have appropriate Safety Data Sheet(s) attached to JHA.

F. Comply with RSHS and 29 CFR 1926. Include fatigue management plan as required by RSHS.

G. Minimum light requirements:

1. Perform on-site work between sunrise to sunset, unless approved by COR.
2. If COR approves work outside of daylight hours, submit after-hours plan to COR that includes light plants/equipment to provide a minimum of 20 foot-candles to illuminate all working areas.
    a. Provide additional lighting as necessary to comply with County requirements for night work within County Right of Way.

H. Confined Space:

1. Provide rescue team when entering confined space.
2. Complete a mock rescue prior to entering confined space.

I. Correct safety and health violations identified by CO or COR. Correct safety and health violations identified by COR within 2 days for minor violations and immediately for serious violations.

J. When Contractor fails or refuses to correct a compliance directive, CO may issue order to stop all or part of the work:

1. When satisfactory corrective action is taken, order to resume work will be issued.
2. Contractor shall not be entitled to extension of time, claim for damage or additional compensation by reason of either directive or stop order.
3. Failure of CO to order discontinuance of Contractor's operations shall not relieve Contractor of responsibility for safety of personnel and property.

K. Maintain accurate records of and report to CO, COR, and as required in RSHS.

L. Rights and remedies of Government provided in this section are in addition to any other rights and remedies provided by law or under this contract:

1. In the event of a conflict between requirements contained in RSHS, specification paragraphs, Contractor's approved Safety Program, referenced safety and health codes and standards, or U.S. Department of Labor Construction Safety and Health Standards, promulgated under Section 107 of Contract Work Hours and Safety Standards Act (40 U.S.C. 327 et seq.), as amended, the most stringent requirement shall prevail.

M. Provide appropriate safety barricades, signs, and warnings.

N. Provide medical and emergency assistance in timely manner consistent with recognized emergency protocols.

O. In instances where local emergency responders are not readily available, responsibility to provide emergency services shall default to Contractor.

P. Perform all training as required by Federal, Tribal, State, and local regulations prior to any activity requiring training:

1. Submit training records to COR upon request.
2. In no case shall an employee perform work until all required training is complete.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 01 35 30
CONTRACTOR'S ONSITE SAFETY PERSONNEL

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work.

1.02 REFERENCE STANDARDS

A. Bureau of Reclamation (Reclamation)

1.	RSHS	Reclamation Safety and Health Standards, including revisions, available at https://www.usbr.gov/safety/rshs/index.html

1.03 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 35 30-1, Resumes:

1. Resumes verifying qualifications listed in 1.04 Qualifications.
2. Include resumes for alternate Onsite Safety Representative, and Onsite Safety and Health Professional.
3. Contractor's Onsite Safety Representative.
    a. Include current telephone numbers of references, description of safety responsibilities, and copies of training certifications.
    b. Submit alternate safety manager as needed should have all qualifications as primary safety manager.
4. Contractor's Onsite Safety and Health Professional
    a. Include resume with current telephone numbers of references, description of safety responsibilities, and copies of OSHA 3015 and other training certifications.

C. RSN 01 35 30-2, Safety Inspection Reports:

1. List noted deficiencies, digital photos, abatement dates, and follow-up action for jobsite activities.
2. Base inspection report on findings of jobsite walk-through with Government personnel.

1.04 QUALIFICATIONS

A. Contractor's Onsite Safety Representative:

1. Competent supervisory employee with current safety and health related training, at least 5 years' experience in pipeline construction as an onsite safety representative, and duties on at least three projects of similar nature, size, and complexity.

2. OSHA Construction 30-Hour Training Course including applicable elective subjects, for example Concrete and Masonry Construction; Confined Space Entry; Cranes, Derricks, Hoists, Elevators, and Conveyors; Ergonomics; Excavations; Fire Protection and Prevention; Materials Handling, Storage, Use, and Disposal; Motor Vehicles, Mechanized Equipment, and Marine Operations; Rollover Protective Structures and Overhead Protection; Signs; Signals, and Barricades; Powered Industrial Vehicles; Safety and Health Programs; Scaffolds; Steel Erection; Tools - Hand and Power; Welding and Cutting.

   a. Provide copy of OSHA Construction 30-Hour training certificate issued within the last 3 years required.

3. Current first aid and CPR certification. OSHA 3015 trained.

4. Knowledgeable of current safety trends and industry practices. Exercises technical expertise.

B. Contractor's Onsite Safety and Health Professional:

1. The Contractor shall identify a full-time Onsite Safety and Health Professional that is acceptable to Reclamation. The individual shall be qualified by virtue of education, training, certification, and experience in which professional status in the construction safety field has been established.

2. Include resume with current telephone numbers of references, description of safety professional responsibilities on similar construction projects, and copies of training certifications, and degree, and registration.

3. The individual shall be qualified by virtue of education, training, certification, and experience in which professional status in the construction safety field has been established.

4. The Contractor Onsite Safety and Health Professional shall have no other duties.

5. Competent supervisory employee with current safety and health related training, experience and duties on at least three projects of similar nature, size, complexity, and work to be performed as the current project

6. OSHA Construction 30-Hour Training Course including applicable elective subjects, for example Concrete and Masonry Construction; Confined Space Entry; Cranes, Derricks, Hoists, Elevators, and Conveyors; Ergonomics; Excavations; Fire Protection and Prevention; Materials Handling, Storage, Use and Disposal; Motor Vehicles, Mechanized Equipment and Marine Operations; Rollover Protective Structures and Overhead Protection; and Signs; Signals and Barricades;

Powered Industrial Vehicles; Safety and Health Programs; Scaffolds; Steel Erection; Tools - Hand and Power; Welding and Cutting. Current first aid CPR certification.

7. Registered Professional Safety Engineer, Certified Safety Professional (CSP), or CIH certified by the Board of Certified Safety Professionals/ American Board of Industrial.

8. Knowledgeable of current safety trends, industry practices, and can exercise technical expertise.

1.05 APPLICATION

A. Contractor's Onsite Safety Representative Authorities, Duties, and Responsibilities:

1. Responsible for effectively implementing Contractor's Safety Program.

2. Coordinate all emergency response activities.

3. Recommend controls for observed or predicted hazards.

4. Full authorization to correct unsafe acts on the spot, including authority to stop work to correct safety and health problems.

5. Lead weekly safety meetings at jobsite.

6. Onsite during any and all construction activities.

7. Assemble, review, and sign each Job Hazard Analysis (JHA). Review JHA with employees.

8. Verify correct use of respiratory protection and personal protective equipment.

9. Conduct and prepare weekly safety inspections in conjunction with OGR listing noted deficiencies, abatement dates, and follow up action. Report shall be available for review onsite.

10. Provide direct oversight and assemble detailed report of all incident reviews, near-misses, accidents, and emergency response or rescue actions.

11. Contractor's Onsite Safety Representative shall have no other non-safety duties.

12. Report directly to project manager or higher. Duties to include subcontractor oversite.

B. Contractor's Onsite Safety and Health Professional Authorities, Duties, and Responsibilities:

1. The Onsite Safety and Health Professional's job shall be hazard recognition, control and accident prevention.

2. The Onsite Safety and Health Professional shall always be at the worksite during work activities.

3. Shall NOT be supervised by the onsite superintendent but shall report directly to the corporate manager/owner.

4. Shall have full authorization to correct unsafe acts on the spot, including the authority to stop work to correct safety and health problems.
5. Shall participate in the planning, preparation and execution of all activities including, but not limited to:
    a. Development and approval of JHA.
    b. Review of equipment and materials.
    c. Employee training.
    d. Review of JHA with employees.
    e. Observing and reporting on activities.
6. Duties include, but are not limited to the following:
    a. Before start of construction, identify existing and predictable unsanitary, hazardous, or dangerous conditions.
    b. Approve the site-specific Contractor safety and health plans and program prior to submission.
    c. Coordinating all elements contained in the Contractor safety and health program and these specifications, including safety training.
    d. May act as the qualified person for designing fall protection, excavation and scaffolding systems in accordance with the regulatory standards. Review and finalize all incident reviews, near-misses and accident reports in accordance with RSHS 1.03.
    e. Assemble, review, and sign each Job Hazard Analysis (JHA).
    f. Review and finalize all incident reviews, near-misses, and accident reports in accordance with RSHS
    g. Conduct and prepare weekly safety inspections in conjunction with the onsite government representative listing noted deficiencies, their abatement dates, and follow up action. The report shall be available for review onsite.
    h. Attend the Safety Program Review Meeting and Conduct the monthly Joint Safety Policy Meetings in accordance with the RSHS Section 1.03.

1.06 QUALITY ASSURANCE

A. Contractor's Onsite Safety Representative:
1. Effectiveness of Contractor's Onsite Safety Representative in prosecuting the safety program will be subject to continued review and approval by COR.
2. Should Contractor's onsite safety representative's effort be deemed insufficient, Contractor may be required to employ a full-time qualified Safety Professional at no cost to Government.

B. Contractor's Onsite Safety and Health Professional:

1. The effectiveness of the Contractor's Onsite Safety and Health Professional in prosecuting the safety program will be subject to continued review and approval by the COR.

2. Should the Contractor's safety effort be considered inadequate, the CO has the option to require the Contractor to replace the full-time qualified Safety and Health Professional, at no additional cost to the Government.

C. Safety Program:

1. Effectiveness of Contractor's Safety Program will be subject to continued review and approval by CO.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 01 42 10
REFERENCE STANDARDS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work.

1.02 REFERENCE STANDARDS

A. Referenced editions of standard specifications, codes, and manuals form a part of this specification to extent referenced.

B. These specifications take precedence when conflicting requirements occur between specifications and referenced standard.

1.03 JOBSITE REFERENCE STANDARDS

A. Maintain at fabrication site, access to referenced standard specifications, codes, and manuals required for work in progress at fabrication site. Make available for use by Government.

B. Maintain onsite, access to referenced standard specifications, codes, and manuals required for onsite work in progress. Make available for use by Government.

1.04 AVAILABILITY

A. Code of Federal Regulation (CFR):

1. Available online, authorized by the National Archives and Records Administration (NARA) and the Government Printing Office (GPO), at https://www.govinfo.gov/help/cfr.

B. Federal Specifications, Standards, and Commercial Item Descriptions; and Military Specifications:

1. Copies of Federal Specifications, Standards, and Commercial Item Descriptions may be obtained from GSA Federal Supply Service, see the provision at Section H.7, Availability of Specifications Listed in the GSA Index of Federal Specifications, Standards and Commercial Item Descriptions, FPMR Part 101-29.
2. Copies of Military Specifications may be obtained from Department of Defense, see the provision at Section H.8, Availability of Specifications, Standards, and Data Item Descriptions Listed in the Acquisition Streamlining and Standardization Information System (ASSIST).

C. Bureau of Reclamation (Reclamation) Documents:

1. Reclamation Safety and Health Standards (RSHS) may be downloaded at https://www.usbr.gov/safety/rshs/index.html.

2. Reclamation manuals and other publications including significant scientific, technical, and engineering works are available upon request to the CO.

D. Industrial and Governmental Documents:

1. When a reference has a joint designation (e.g., ANSI/IEEE) these specifications generally cite the proponent organization (e.g., IEEE).

2. Addresses for obtaining industrial and governmental (other than Federal and Reclamation specifications and standards) specifications, standards, and codes are listed in Table 01 42 10A – Addresses for Specifications, Standards, and Codes.

Table 01 42 10A – Addresses for Specifications, Standards, and Codes

Acronym	Name and Address	Telephone
AASHTO	American Association of State Highway and Transportation Officials 444 North Capitol Street, NW., Suite 249 Washington, DC 20001 www.aashto.org	202-624-5800 800-231-3475
ACI	American Concrete Institute 38800 Country Club Dr. Farmington Hills, MI 48331-3439 USA www.concrete.org	248-848-3700
AGC	Associated General Contractors of America 2300 Wilson Boulevard, Suite 300 Arlington, VA 22201 www.agc.org	703-548-3118
ASCE	American Society for Civil Engineers 1801 Alexander Bell Drive Reston, VA 20191 www.asce.org	800-548-2723 703-295-6300
ANSI	American National Standards Institute 1819 L. Street, NW. Washington, DC 20036 www.ansi.org	202-293-8020

Table 01 42 10A – Addresses for Specifications, Standards, and Codes

Acronym	Name and Address	Telephone
ASME	American Society of Mechanical Engineers Two Park Avenue New York, NY 10016-5990 www.asme.org	800-843-2763 646-616-3100
ASTM	ASTM International P.O. Box C700 100 Barr Harbor Drive West Conshohocken, PA 19428-2959 www.astm.org	610-832-9500
AWWA	American Water Works Association 6666 W. Quincy Ave. Denver, CO 80235 www.awwa.org	303-794-7711
AWS	American Welding Society 550 NW LeJeune Road Miami, FL 33126 www.amweld.org	800-443-9353 305-443-9353
ICC	International Code Council 200 Massachusetts Avenue Northwest, Suite 250 Washington, DC 20001 www.iccsafe.org	888-422-7233 202-370-1800
IEEE	Institute of Electrical and Electronics Engineers 3 Park Avenue,17th Floor New York, NY 10016-5997 www.ieee.org	212-419-7900
ISA	International Society of Automation P.O. Box 12277 Research Triangle Park, NC 27709 www.isa.org	919-549-8411
NEMA	National Electrical Manufacturers Association 1300 17th Street North, Suite 900 Arlington, VA 22209 www.nema.org	703-841-3200
NETA	International Electrical Testing Association 3050 Old Centre Road, Suite 101 Portage, MI 49024 www.netaworld.org	888-300-6382

Table 01 42 10A – Addresses for Specifications, Standards, and Codes

Acronym	Name and Address	Telephone
NIBS	National Institute of Building Sciences 1090 Vermont Avenue Northwest, Suite 700 Washington, DC 20005 www.nibs.org	202-289-7800
OSHA	U. S. Department of Labor Occupational Safety and Health Administration 200 Constitution Avenue Northwest Washington, DC 20210 www.osha.gov	800-321-OSHA 800-321-6742
UL	Underwriters Laboratories, Inc. 333 Pfingsten Road Northbrook, IL 60062 www.ul.com	847-272-8800

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

SECTION 01 46 00
QUALITY PROCEDURES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work.

1.02 DEFINITIONS

A. Quality Assurance: Inspection and tests performed by Government to ensure compliance with terms of contract.

B. Quality Control: Activities performed by Contractor to ensure work conforms to contract requirements:

1. Clause at FAR 52.246-12, Inspection of Construction, requires Contractor to establish an inspection system to ensure quality.

2. Quality Control includes activities in addition to specified Contractor Quality Testing to ensure work conforms to contract requirements.

C. Contractor Quality Testing: Specified tests shall be performed by Contractor:

1. Government may use test results for Quality Assurance.

2. Test results shall be a part of Contractor's Quality Control Program.

a. Tests do not relieve Contractor of providing adequate quality control measures in accordance with clause at FAR 52.246-12, Inspection of Construction.

D. Definable Features of Work: Task that is separate and distinct from other tasks, has separate control requirements, and may be identified by different trades or disciplines, or may be work by the same trade in a different environment.

1.03 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 46 00-1, Definable Features of Work List:

1. Provide list of Definable Features of Work that Contractor Quality Control Plan will be based on.

a. Although each section of the specifications may generally be considered as a definable feature of work, there are frequently more than one definable features under a particular section.

C. RSN 01 46 00-2, Contractor Quality Control (QC) Plan:

1. Outline Contractors Quality Control Program. As a minimum, cover construction operations, both onsite and offsite, work by subcontractors, fabricators, suppliers, and purchasing agents. Include:

a. Description of quality control organization, a chart showing lines of authority, and acknowledgement that Quality Control Supervisor (QCS) staff shall implement the three-phase control system for all aspects of work specified.

b. Identify staff including a QCS who reports to project manager:

1) An alternate QCS with equal lines of authority to act on behalf of the primary QCS should they not be present on the project site.

2) Names, qualifications (in resume format), duties, responsibilities, and authorities of each person assigned a QC function.

c. Letter to QCS, signed by an authorized official of the firm.

1) Describe responsibilities and delegate sufficient authorities to adequately perform the functions of the QCS.

2) Include authority to stop work not in compliance with the contract.

d. QCS shall issue letters of direction to other quality control representatives outlining duties, authorities, and responsibilities.

1) Furnish copy of letters to Government.

e. Procedures for:

1) Scheduling, reviewing, certifying, and managing submittals, including those of subcontractors, offsite fabricators, suppliers and purchasing agents.

2) Control, verification, and acceptance testing procedures for each specific test to include:

a) Test name.

b) Specification paragraph requiring test.

c) Feature of work to be tested.

d) Test frequency.

e) Person responsible for each test.

3) Tracking:

a) Preparatory.

b) Initial.

c) Follow-up control phases and control.

d) Verification and acceptance tests including documentation.

e) Construction deficiencies from identification through acceptable corrective action. Establish procedures to verify correction of identified deficiencies.

4) Reporting procedures, including proposed reporting formats.

f. Base program upon a mutually agreed upon list of Definable Features of Work. Include preparatory, initial, and follow-up for each definable feature.

2. Detail implementation of Quality Control Plan summarized in Article 3.01 below.

3. Include Quality Testing Plan that address methods, procedures, frequencies, and scheduling for performing quality testing as required in other specification sections.

4. Include Quality Testing Plan that address methods, procedures, and frequencies, and scheduling for performing quality testing as required in other specification sections.

D. RSN 01 46 00-3, Contractor's Quality Daily Report:

1. Submit daily reports on a weekly basis.

2. As a minimum, Contractor's Quality Daily Report shall:

a. Record Contractor's activities throughout shift regardless of how minor or complex.

b. Be started at beginning of shift, written during shift, and completed at end of shift.

c. Include such detail that a person not familiar with the day's work could determine what was done and calculate Contractor's direct cost for that day.

d. Include temperature (maximum/minimum), wind conditions, estimated humidity, sunshine, and precipitation (type and intensity), if necessary, for the type of work being performed.

e. Note name of contractor's representative on site during the shift and hours on site.

f. Document subcontractor activity.

g. Identify specific safety incidents noted during shift including action(s) taken, including:

1) Names of people involved.

2) Name of person confirming correction.

3) Record of significant discussions held between Reclamation and Contractor personnel or other parties involved with project.

h. Main body of report:

1) Locations and a detailed description of all work, including:

a) Sketches or photographs where appropriate to assist in description.

b) Description shall reflect what, where, when, why, who, and how work was being performed.

c) Provide:

i. Record of quantities installed.

ii. Specific stations or location of work.

iii. Description by CPM program activity (if applicable).

iv. Show labor classification, work labor force and equipment used during shift.

v. List equipment that is idle or being repaired.

i. List materials and equipment delivered to job site:

1) Materials installed during shift.

2) Materials refused or wasted during shift.

j. Other items to report include:

1) Changes in site conditions.

2) Modifications in design and installation.

3) Photographs taken.

4) Environmental compliance or noncompliance.

5) Hazardous waste and material concerns.

6) Landowner contacts.

7) Worker complaints.

8) Potential problems observed.

E. RSN 01 46 00-4, Quality Control Supervisor (QCS) Resume:

1. Designation of authorized individual as QCS to develop and manage quality control processes.

2. Designated QCS Resume.

3. Include alternate QCS Resume, verifying alternate meets Article 1.04 Qualifications.

F. RSN 01 46 00-5, Contractor's Quality Testing Plan:

1. Address methods, procedures, frequencies, and scheduling for performing quality testing as required in other specification sections.

G. RSN 01 46 00-6, Summary of Monthly Test Results.

1.04 QUALIFICATIONS

A. Designate an employee as Quality Control Supervisor (QCS).

B. Quality Control Supervisor (QCS):

1. Qualifications:

a. Graduate engineer, graduate architect, or graduate of construction management, with minimum 4 years' construction experience in charge of quality control for projects similar in nature to that required by this contract; or

b. Construction person with 10 years' experience in related QC work.

C. Both QCS and alternate shall have the authority to stop work if necessary.

D. QCS shall hold no other duties.

1.05 TESTING ACCEPTANCE

A. Material acceptance based on passing of both quality control testing and quality assurance testing.

B. Government quality assurance testing results will prevail in the event of a discrepancy between Contractor's test results and Government test results.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 QUALITY CONTROL PLAN

A. Base program upon definable features of work. Include three phases for each feature:

1. Preparatory Phase:

a. Performed prior to beginning of work:

1) Notify COR, in writing, 3 business days prior to beginning required action of preparatory phase.

2) Review approved submittals and identify missing information or required submittals.

3) Review each paragraph of applicable specifications and references.

4) Review contract plans or shop drawings.

5) Verify materials and equipment have been tested, submitted, and approved.

6) Verify provisions have been made to provide required inspection and testing.

7) Examine work area to verify required preliminary work has been completed.

8) Physical examination of required materials, equipment, and sample work to assure:

a) Materials are on hand and conform to approved shop drawings or submitted data.

b) Have been properly stored.

9) Review of appropriate Job Hazard Analyses (JHAs).

10) Discuss procedures for constructing work, including review of repetitive deficiencies.

11) Arrange quality control testing if required.

b. Preparatory Meeting:

1) Conducted by QCS and attended by superintendent, other quality control personnel, foreman responsible for definable feature, and Government.

2) Discuss results of preparatory phase actions. Document by separate minutes prepared by QCS and attach to daily Contractor's Quality Control Report.

3) For meeting relating to concrete batch plant or concrete placing, include the following attendees at a minimum:

a) Concrete forming, placing, and finishing on site supervisor(s); Contractor or subcontractor(s) employee as applicable.

b) Batch plant operator.

c) Concrete pump subcontractor on site supervisor, if applicable.

d) Contractor QC and Government QA testing personnel.

2. Initial Phase:

a. Notify COR 3 business days in advance of beginning any required action of initial phase.

b. Accomplished at beginning of a definable feature of work.

c. Verify that control for feature of work developed in Preparatory Meeting is implemented and feature of work is performed to the level of workmanship required.

d. Perform as detailed below:

1) Review minutes of Preparatory Meeting.

2) Check preliminary work.

3) Verify adequacy of controls to ensure full contract compliance.

4) Establish levels of workmanship.

5) Resolve all differences.

6) Check safety to include compliance with safety plan and JHA:

a) Review JHA with workers.

e. Initial Phase Meeting:

1) Conducted by QCS and attended by superintendent, other quality control personnel, foreman responsible for definable feature, and Government.

2) Discuss results of initial phase actions. Document by separate minutes prepared by QCS, and attach to daily Contractor's Quality Control Report.

3. Follow-Up Phase:

a. Perform daily checks to assure continuing compliance with contract requirements, including safety and control testing until completion of particular feature of work.

b. Make checks a matter of record in Contractor's Quality Control documentation.

c. Conduct final follow-up checks and confirm all deficiencies are corrected prior to start of additional features of work.

d. Quality control personnel shall continually refer to standards set in Preparatory Meeting and Initial Phases.

3.02 QUALITY ASSURANCE

A. During course of work, Government may perform quality assurance tests. Tests performed by Government will be used to ensure compliance with contract requirements and not as a replacement for specified Contractor quality testing:

1. Upon request, make measuring and testing devices available for use by Government for verification tests.

2. If a conflict arises between Contractor quality testing results and Government quality assurance tests, Government testing results will take precedence over Contractor testing results.

B. Contractor's QCS:

1. At project site during work activities and shall have complete understanding of all aspects of the work being performed.
2. Effectiveness of the QCS in prosecuting QC Program will be subject to continued review and approval by CO.
3. Should QCS's efforts be deemed insufficient, Contractor may be required to replace QCS at no additional cost to Government.

C. Contractor's Quality Control (QC) Program:

1. Effectiveness of Contractor's QC Program will be subject to continued review and approval by CO.

END OF SECTION

SECTION 01 46 20
TESTING AGENCY SERVICES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for items of work requiring testing agency services.

B. Progress Payments: If test reports are not submitted in a timely manner, Contractor will be considered to be in non-compliance and delaying that phase of the work to which the testing applies. CO may retain appropriate amounts of applicable progress payments.

1.02 REFERENCE STANDARDS

A. ASTM International (ASTM)

1.	ASTM D3666-16	Minimum Requirements for Agencies Testing and Inspecting Road and Paving Materials
2.	ASTM D3740-23	Minimum Requirements for Agencies Engaged in Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction
3.	ASTM E329-25	Agencies Engaged in Construction Inspection, Testing, or Special Inspection
4.	ASTM E543-21	Agencies Performing Nondestructive Testing

1.03 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 46 20-1, Testing Agency Services Plan:

1. Include:
    a. Names of agencies to perform sampling and testing.
    b. Agency accreditation to perform specified testing or agency qualifications to perform specified testing.
    c. Resumes of personnel performing tests.
    d. Samples of report forms.
2. No change in the approved plan may be made without written concurrence by COR.

1.04 QUALIFICATIONS

A. Testing agency organization:

1. Agencies testing construction materials: Meet requirements of ASTM E329.
2. Agencies engaged in nondestructive testing: Meet requirements of ASTM E543.

B. Equipment: Calibrate measuring devices, laboratory equipment, and instruments at established intervals.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 CONTRACTOR QUALITY TESTING

A. Employ accredited independent agency to perform sampling, testing, and reporting as required in the following Sections:

1. Section 03 30 00 – Cast-In-Place Concrete
2. Section 31 23 02 – Compacting Earth Materials
3. Section 31 23 70 – Controlled Low Strength Material
4. Other sections as required.

3.02 GOVERNMENT CONTRACT QUALITY ASSURANCE

A. During course of work, Government may perform quality assurance tests. Tests performed by Government will be used to ensure compliance with contract requirements and not as replacement for specified Contractor quality testing.

END OF SECTION

SECTION 01 46 25
SPECIAL INSPECTIONS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for items of work requiring special inspections.

1.02 REFERENCE STANDARDS

A. International Code Council (ICC):

1. ICC IBC-2021 International Building Code

1.03 DEFINITIONS

A. Continuous Special Inspections:

1. Constant monitoring of specific tasks by Special Inspector.
2. Carry out these inspections continuously over duration of the particular tasks:
3. In accordance with drawing 1695-D-62527.

B. DFOW: Definable Feature of Work.

C. Periodic Special Inspections: By special inspector who is intermittently present where the work to be inspected has been or is being performed.

D. Special Inspector (SI):

1. Qualified person with necessary competence to inspect construction requiring Special Inspections.
2. Independent third party hired by Contractor.

E. Associate Special Inspector (ASI):

1. Qualified person who assists SI in performing Special Inspections.
2. Performs inspection under direct supervision of SI. Cannot perform inspections without SI on site.

F. Third Party: Not employed by Contractor or subcontractor performing work requiring inspection, except in a certified manufacturing facility.

G. Special Inspector of Record (SIOR):

1. Licensed engineer in responsible charge for supervision of special inspectors for the project.

2. Independent third party hired by Contractor.

H. Statement of Special Inspections (SSI): Section identifying material, systems, components and work requiring Special Inspections.

I. Schedule of Special Inspections: List of required Special Inspections, extent to which each Special Inspection shall be performed, and required frequency for each in accordance with ICC IBC Chapter 17.

1.04 GENERAL REQUIREMENTS

A. Perform in accordance with this section and Chapter 17 of ICC IBC. For purposes of this contract, language of ICC IBC will be revised as follows:

1. Owner: Considered Government.

2. Building Official: COR or COR's representative.

3. It is Contractor's responsibility to provide Special Inspection services for listed DFOW. Special Inspection requirements as outlined in ICC IBC are required to be provided, with the exception of entity required to provide services.

B. Performed by an approved Testing Agency meeting technical requirements defined in IBC 1703.1.1 through IBC 1703.1.3 and this section.

C. Special Inspections do not replace Contractor's Quality Procedures/Quality Control (QC) Program or testing and inspections required by other sections of specifications.

D. Government will use results of Special Inspections and testing as a means of validating Contractor's Quality Control Program and are intended to ensure that work complies with Contract Documents and applicable building codes.

E. DFOW Specification Sections may contain requirements in addition to and that take precedence over the minimum requirements listed in this Section. COR will make final determination regarding requirements of Section 01 46 00 – Quality Procedures, Section 01 46 20 – Testing Agency Services, and Section 01 46 25 – Special Inspections.

1.05 STATEMENT OF SPECIAL INSPECTIONS

A. Special Inspections required for DFOW's and corresponding specification sections:

1. Cast-in-place concrete foundation for precast concrete building, if used:

a. Concrete Forming: Section 03 11 10 – Concrete Forming.

b. Concrete Reinforcing: Section 03 20 00 – Concrete Reinforcing.

c. Cast-In-Place Concrete: Section 03 30 00 – Cast-In-Place Concrete.

2. Precast Concrete Building: Section 13 34 00 – Prefabricated Engineered Structures

3. Earthwork: Section 31 23 10 – Earthwork for Site and Structures.

1.06 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 46 25-1, Preconstruction Submittals:

1. SIOR Qualifications.
2. SIO Qualifications.
3. SIOR Letter of Acceptance.
4. Special Inspections Project Manual, Special Inspections Agency's Written Practices, NDT Procedures and Equipment Calibration Records.

C. RSN 01 46 25-2, Test Reports:

1. Special Inspections Daily Reports.
2. Special Inspections Biweekly Reports.
3. Failed test or non compliant test results shall be submitted within 24 hrs after inspection or testing.

D. RSN 01 46 25-3, Certificates:

1. Fabrication Plant(s).
2. Precast Concrete Institute (PCI).
3. Certified Plant Certificate of Compliance.
4. Special Inspector of Record Qualifications.
5. Special Inspector Qualifications.
6. Qualification Records for NDT technicians.

E. RSN 01 46 25-4, Closeout Submittals:

1. Interim Final Report of Special Instructions.
2. Comprehensive Final Report of Special Instructions.

1.07 SPECIAL INSPECTOR QUALIFICATIONS

A. Meet Qualifications of Applicable Certifying Associations.

1. Partial list of certifying associations in Table 01 46 25A – Partial List of Certifying Associations.
2. Provide qualifications for each special inspector and special inspector of record.

Table 01 46 25A – Partial List of Certifying Associations

Acronym	Certifying Association
AABC	Associated Air Balance Council
ACI	American Concrete Institute
AISC	American Institute of Steel Construction
ASNT	American Society for Nondestructive Testing
AWS	American Welding Society
ICC	International Code Council
NICET	National Institute for Certification in Engineering Technologies
PCI	Precast/Prestressed Concrete Institute
UL	Underwriters Laboratories

B. Welding Structural Steel:

1. Special Inspector:

   a. ICC Structural Welding Special Inspector certificate with 1 year of related experience.

   b. AWS Certified Welding Inspector.

2. Associate Special Inspector: AWS Certified Associate Welding Inspector.

C. Nondestructive Testing of Welds:

1. Special Inspector: NDT Level III Certificate.

2. Associate Special Inspector: NDT Level II Certificate plus 1 year of related experience or equal approved by COR.

D. Precast Concrete Construction:

1. Special Inspector:

   a. ICC Precast Special Inspector Certificate with 1 year of related experience or equal approved by COR.

   b. PCI Quality Control Technician/Inspector Level II Certificate with 1 year of related experience or equal approved by COR.

E. Special Inspector of Record (SIOR): Licensed Professional Engineer in the State of New Mexico approved by COR.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 FABRICATOR SPECIAL INSPECTIONS

A. In accordance with SSI and Schedule of Special Inspections unless fabricator is certified by approved agency to perform such work without Special Inspections.

B. Provide the following certifications to COR, for information, prior to performing work in fabricator's shop without Special Inspections:

1. American Institute of Steel Construction (AISC) Certified Fabrication Plant, Category STD.
2. Truss Plate Institute (TPI) steel truss plant quality assurance program certification.
3. Steel Joist Institute Membership.
4. Precast Concrete Institute (PCI) Certified Plant, Group C.

C. At completion of fabrication, provide a certificate of compliance in comprehensive final report of Special Inspections, stating materials supplied and work performed by fabricator are in accordance with construction documents.

3.02 RESPONSIBILITIES

A. Special Inspector of Record:

1. Supervise all Special Inspectors required by specifications and IBC.
2. Provide SIOR Letter of Acceptance attesting to acceptance of duties of SIOR, signed and sealed by SIOR.
3. Verify qualifications of:
    a. Special Inspectors.
    b. Fabricators.
4. Provide Special Inspections agency's written practices for monitoring and control of agency's operations to include:
    a. Procedures for selection and administration of inspection personnel. Describe training, experience, and examination requirements for qualifications and certification of inspection personnel.
    b. Agency's inspection procedures, including general inspection, material controls, and visual welding inspection.
5. Provide qualification records for nondestructive testing (NDT) technicians designated for the project.
6. Provide NDT procedures and equipment calibration records for NDT to be performed and equipment to be used for the project.
7. Prepare Special Inspections Project Manual covering:

a. Applicable completed inspection tables from IBC Section 1705.

b. Include tasks specific to work.

c. Roles and responsibilities of SIOR, SI, General Contractor, Subcontractors, and QC Manager during Special Inspections.

d. Organizational chart or communication plan, indicating lines of communication.

e. Contractor's internal plan for scheduling inspections. Address timeliness of inspection requests, contact for inspection requests, and availability of alternate inspectors.

f. Verify Government reporting procedures.

g. Forms or templates used by SI and SIOR to document inspections.

h. Procedures for tracking nonconforming work and verification of corrections.

i. Indicate how SIOR and SI shall participate in weekly QC meetings.

j. Indicate how Special Inspections shall be handled if fabricator's shop is not certified per paragraph Fabricator Special Inspections.

k. Provide individual manual sections detailing each specific item requiring Special Inspections as indicated on Schedule of Special Inspections.

1) Provide names and qualifications of special inspector performing Special Inspections for each specific item.

2) Detail how Special Inspections shall carried out for each item and expectations for Contractor and subcontractors.

l. Make a copy of Special Inspections Project Manual available on the job site during construction. Include Special Inspector’s daily and biweekly reports.

8. Attend coordination and mutual understanding meeting to review Special Inspections Project Manual with Government, Contractor, and Special Inspectors. Verify that all parties have clear understanding of Special Inspections provisions and individual duties and responsibilities of each party.

9. Provide a copy of Special Inspector's daily reports to QC Manager.

10. Report special inspection discrepancies to QC Manager for correction. Document discrepancies in daily report if not corrected before special inspector leaves site.

11. Provide biweekly Special Inspections report until Special Inspections are complete. Include:

a. Brief summary of work performed during reporting time frame.

b. Changes and discrepancies with drawings, specifications and mechanical or electrical component certification observed during reporting period.

c. Resolution of discrepancies.

d. List nonconforming items requiring correction.

e. Applicable test results including nondestructive testing reports.

12. At completion of each DFOW requiring Special Inspections, provide interim final report documenting Special Inspections and corrections of discrepancies completed for that DFOW. Report shall be signed, dated, and sealed by SIOR.

13. At project completion, provide comprehensive final report documenting Special Inspections completed for the project and corrections of discrepancies. Report shall be signed, dated, and sealed by SIOR.

B. Special Inspectors:

1. Attend coordination and mutual understanding meeting to review Special Inspections Project Manual with Government, Contractor, and Special Inspectors. Verify that all parties have clear understanding of Special Inspections provisions and individual duties and responsibilities of each party.

2. Inspect DFOW elements for which special inspector is qualified to inspect as identified in Schedule of Special Inspections.

3. Attend preparatory phase meetings to applicable DFOW.

4. Provide Special Inspections information to SIOR for applicable DFOW.

a. Provide Special Inspections agency's written practices for monitoring and control of agency's operations to include:

1) Procedures for selection and administration of inspection personnel. Describe training, experience, and examination requirements for qualifications and certification of inspection personnel.

2) Agency’s inspection procedures, including general inspection, material controls, and visual welding inspection.

b. Provide qualification records for nondestructive testing (NDT) technicians designated for the project.

c. Provide NDT procedures and equipment calibration records for NDT to be performed and equipment to be used for the project.

5. Provide daily reports to SIOR.

6. Report special inspection discrepancies to QC Manager for correction. Document discrepancies in daily report if not corrected before special inspector leaves site.

7. Provide biweekly Special Inspections report until Special Inspections are complete. Include:

a. Brief summary of work performed during reporting time frame.

b. Changes and discrepancies with drawings, specifications and mechanical or electrical component certification observed during reporting period.

c. Resolution of discrepancies.

d. List nonconforming items requiring correction.

e. Applicable test results including nondestructive testing reports.

8. At completion of each DFOW requiring Special Inspections, provide interim final report documenting Special Inspections and corrections of discrepancies completed for that DFOW. Report shall be signed and dated by Special Inspector and shall indicate certification(s) qualifying them to conduct inspection.

9. At project completion, provide comprehensive final report documenting Special Inspections completed for the project and corrections of discrepancies. Report shall be signed and dated by Special Inspector and shall indicate certification(s) qualifying them to conduct inspection.

3.03 QUALITY ASSURANCE

A. Government reserves the right to perform Quality Assurance on the Special Inspections. Provide 48-hour notice to COR prior to performing special inspections.

3.04 DEFECTIVE WORK

A. Check work as it progresses.

B. Failure to detect defective work or materials shall not prevent later rejection if defective work or materials are discovered, nor obligate CO to accept work.

END OF SECTION

SECTION 01 51 00
TEMPORARY UTILITIES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work.

1.02 REFERENCE STANDARDS

A. Institute of Electrical and Electronics Engineers (IEEE)

1. IEEE C2-2023 National Electrical Safety Code (NESC)

1.03 TEMPORARY ELECTRICITY

A. Provide electric power required for construction and testing through contract completion and acceptance by Government.

B. Contractor is responsible for making all arrangements and payments to utility companies for temporary electricity required for construction and testing through contract completion and acceptance by Government.

C. Comply with IEEE C2 clearances and spacing for temporary communications and supply lines.

1.04 TEMPORARY WATER

A. Provide water required for construction through contract completion and acceptance by Government.

B. Potential Sources:

1. Obtain required permits.
2. Depending on availability, the following entities may have potable water:
3. Contact for access, metering, and billing arrangements:
   a. Lower Valley Water Users Cooperative Association
      4286 US-64
      Kirkland, NM 87417
      www.lowervalleywater.com
      Telephone: 505-598-5585
   b. Navajo Agricultural Products Industry

P.O. Drawer 1318
Farmington, NM 87499
www.navajopride.com
Telephone: 505-566-2636
Fax: 505-599-0572

c. Navajo Nation Department of Water Resources / Navajo Nation Water Code Administration
P.O. Box 678
Fort Defiance, AZ 86504
www.nndwr.navajo-nsn.gov, www.watercode.navajo-nsn.gov
Telephone: 928-729-4132
Fax: 928-729-4421

d. Navajo Tribal Utility Authority (NTUA)
P.O. Box 170
Fort Defiance, AZ 86504
Telephone: 928-729-5721

e. Reclamation – Four Corners Construction Office (FCCO)
2200 Bloomfield Highway
Farmington, NM 87401
Telephone: 505-325-1794
1235 La Plata Highway
Farmington, NM 87401
Telephone: 505-325-1794

4. Use water which meets specified requirements for water used in grouting and other permanent work.

C. Arrange, pay, and transport water for use during construction.

1. Provide proof of payment to water utility.

D. Provide means of conveying water to points of use.

1.05 TEMPORARY SANITARY FACILITIES

A. Provide temporary sanitary facilities.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 REMOVAL

A. Remove temporary equipment and facilities upon completion of work under this contract.

END OF SECTION

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SECTION 01 55 00
VEHICULAR ACCESS AND PARKING

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in Price Schedules for other items of work.

a. Includes maintenance, repair, and other items of work, as detailed in this section.

1.02 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 55 00-1, Pre-Construction Digital Recording:

1. Include roads, driveways, corrals, cattle guards, fences, and other areas as directed by OGR.

C. RSN 01 55 00-2, Right-of-Way (ROW) Access Plan

1. In accordance with Article 1.03 A. 1.

D. RSN 01 55 00-3, Post Construction and Post Repair Digital Recording.

1.03 SITE CONDITIONS

A. Establish rights-of-way for access to work from existing roads.

1. In accordance with clause at FAR 52.236-10, Operations and Storage Areas, use only established roadways, parking areas, and haul routes; or temporary roadways, parking areas, or haul routes constructed by Contractor when and as authorized by CO.

2. Subject to clause at FAR 52.249-10, Default (Fixed-Price Construction), unavailability of transportation facilities or limitations thereon shall not become a basis for claims for damages or extension of time for completion of work.

1.04 REGULATORY REQUIREMENTS

A. Meet requirements established by jurisdictional authority for use of existing roadways and haul routes; including seasonal or other limitations or restrictions, payment of excess size and weight fees, and posting of bonds conditioned upon repair of damage.

B. Comply with applicable regulations for haul routes over public highways, roads, or bridges.

PART 2 PRODUCTS

2.01 MATERIALS

A. Materials to maintain and repair existing roadways, parking areas, and haul routes: In accordance with requirements of jurisdictional authority.

B. Materials to construct, maintain, and repair temporary roadways, parking areas, and haul routes: As approved by OGR.

C. Materials to maintain roadways and parking areas constructed under this contract and used by Contractor for construction work: In accordance with specified requirements for construction of those roadways and parking areas.

PART 3 EXECUTION

3.01 EXAMINATION

A. Investigate condition of available public or private roads for clearances, restrictions, bridge-load limits, bond requirements, and other limitations that affect or may affect access and transportation operations to and from jobsite.

B. Refer to 01 57 90 – Preservation of Historical and Archeological Data if cultural resources are discovered or if Contractor needs to make improvements outside of Construction ROW.

3.02 RECORDS

A. Make pre-construction, post construction, and post repair digital recordings as directed by OGR:

1. Contact OGR to make arrangements with road owners for crossings.
2. Submit on Reclamation-issued USB drive.
3. Minimum 720p resolution.
4. Label:
    a. Contract number and title.
    b. Contractor’s name.
    c. Date recording is made.
    d. Preface video with this information.

B. OGR will be present during recording. Notify OGR at least 3 days prior to recording.

3.03 ESTABLISHED ROADWAYS AND PARKING AREAS

A. Established roadways and parking areas are available for Contractor's use subject to existing restrictions, Jurisdictional Authorities, and approval of OGR. Do not adversely affect established roadways and parking area conditions.

B. Any modifications to existing roadways and parking areas require cultural clearance in accordance with 01 57 90 – Preservation of Historical and Archeological Data.

3.04 TEMPORARY ROADWAYS AND PARKING AREAS

A. Roadways:

1. All temporary roadways will be restricted to already cleared rights-of-way (ROW) for the project and access/Contractor use areas already identified and cleared of cultural resources.
2. Construct temporary roadways for access from public thoroughfares to serve construction area, of a width and load-bearing capacity to provide unimpeded traffic for construction purposes.
3. Construction traffic to be free of debris prior to entry onto public thoroughfares as required by jurisdictional authorities.
4. Construct temporary bridges or culverts at stream crossings or cross-drainage channels to allow for unimpeded surface drainage.

B. Parking Areas:

1. Temporary parking areas will be restricted to already cleared ROW for the project and access/Contractor use areas already identified and cleared of cultural resources.
2. Construct temporary parking areas to accommodate use of construction personnel.
3. Provide additional offsite parking when site space is not adequate.

3.05 HAUL ROUTES

A. Perform work on rights-of-way established by Government as necessary to construct and maintain any roads, bridges, or drainage structures required for establishment and use of haul routes for construction operations.

B. Use existing available public highways, roads, or bridges as haul routes subject to applicable local regulations.

C. Minimize interference with or congestion of local traffic.

D. Provide barricades, flaggers, and other necessary precautions for safety of public where haul routes cross public highways or roads in accordance with Section 01 55 20 – Traffic Control.

E. Detours: In accordance with Section 01 55 20 – Traffic Control.

3.06 MAINTENANCE

A. Maintain roadways, parking areas, and haul routes in a sound, smooth condition in accordance with these specifications and Jurisdictional Authorities.

B. Maintain surfacing of gravel surfaced roads and parking areas until completion and acceptance of work under this contract. As approved by COR, defer until latest practicable date within specified completion period, placement of surfacing on roads or parking areas subject to heavy and deteriorating use by Contractor's construction operations or equipment in accordance with these specifications and Jurisdictional Authorities.

C. Snow removal for convenience of Contractor or to facilitate work operations of Contractor is considered to be normal required maintenance.

3.07 REPAIR

A. Promptly repair ruts, broken pavement, potholes, low areas with standing water, and other deficiencies to maintain road surfacing and drainage in original or specified condition in accordance with these Specifications and Jurisdictional Authorities.

B. Notify applicable contact(s) below prior to making improvements:

1. New Mexico 811:
   1021 Eubank Blvd NE
   Albuquerque, NM 87112
   www.nmonecall.org
   Statewide: Dial 811 or 800-321-ALERT (2537) during regular business hours.
   By Fax: 505-260-1248 or 800-727-8809 24 Hours a day.
   By Web: Request a Line Spot 24 Hours a day.

2. Navajo Nation Division of Transportation (NNDOT):

   Physical Address:
   #16 Old Coalmine Road
   Mentmore, NM 8719

   Mailing Address:
   PO Box 4620
   Window Rock, AZ 86515
   https://www.navajodot.org
   505-371-8300/8301
   Fax 505-371-8399

3. Bureau of Indian Affairs, Navajo Regional Office DOT

301 West Hill Street
Gallup, NM 87301
www.bia.gov
505-863-8314
Fax 505-863-8324

4. San Juan County Roads Department
   305 S. Oliver Drive
   Aztec, NM 87410
   https://www.sjcounty.net/government/public-works
   505-334-4520

5. New Mexico Department of Transportation (NMDOT)
   1120 Cerrillos Road
   Santa Fe, NM 87504-1149
   https://www.dot.nm.gov
   505-827-5100
   Fax 505-827-5469

6. Mckinley County Roads Department
   1980 Warehouse Lane
   Gallup, NM 87301
   https://www.co.mckinley.nm.us/164/Roads
   505-722-2303

3.08 REMOVAL

A. Remove materials used to construct temporary roadways, parking areas, and haul routes prior to contract completion. Recycle salvageable materials as approved by OGR.

END OF SECTION

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SECTION 01 55 20
TRAFFIC CONTROL

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in Price Schedules for other items of work, except as specified.
2. See Section 31 02 10 – Water for Dust Abatement for measurement and payment of water for dust abatement.

1.02 REFERENCE STANDARDS

A. Federal Highway Administration (FHWA)

1.	MUTCD, Part 6	Part 6, Temporary Traffic Control, Manual on Uniform Traffic Control Devices, available at: https://mutcd.fhwa.dot.gov/kno_11th_Edition.htm

B. New Mexico Department of Transportation (NMDOT)

1.	NMDOT Std Spec-2019	Available at: https://www.dot.nm.gov/infrastructure/plans-specifications-estimates-pse-bureau/standards/
2.	NMDOT Std Drawings-2019(r23)	Available at: https://www.dot.nm.gov/infrastructure/plans-specifications-estimates-pse-bureau/standards/
3.	NMDOT District 5	Permitting resources available at: https://www.dot.nm.gov/business-support/permits/

1.03 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 55 20-1, Information of Traffic Control Plan (TCP):

1. Provide approved copy of Traffic Control Plan.

C. RSN 01 55 20-2, Approved NMDOT Permit(s):

1. Temporary Access Control Permit Request.

1.04 PROJECT CONDITIONS

A. Project includes crossings at:

1. Various maintained roads.
2. Indian Service Routes.
3. Various unmaintained/unnamed roads.

1.05 REGULATORY REQUIREMENTS

A. Government has coordinated work with NMDOT and San Juan County. Entities have agreed to issue a permit upon NMDOT/SJC approval of Traffic Control Plan.

B. Contractor shall submit Traffic Control Plan (TCP) to New Mexico Department of Transportation (NMDOT) to obtain required permit(s) to conduct operations along state highways for daily operations of construction and entry/exit into and out of project site.

C. Traffic Control Plan must be prepared and signed by a registered professional engineer in the State of New Mexico.

D. Obtain permit from San Juan or Mckinley County as needed.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 GENERAL

A. Obtain permits from:

1. Navajo Division of Transportation (Navajo DOT)
   Physical Address:
   #16 Old Coalmine Road
   Mentmore, NM 87319
   Mailing Address:
   P.O. Box 4620
   Window Rock, AZ 86515
   https://www.navajodot.org
   505-371-8300
   Fax 505-371-8399
2. Bureau of Indian Affairs, Navajo Regional Office DOT

301 West Hill Street
Gallup, NM 87301
www.bia.gov
505-863-8314
Fax 505-863-8324

3. San Juan County Roads Department
305 S. Oliver Drive
Aztec, NM 87410
www.sjcounty.net/government/public-works
505-334-4520
Fax 505-334-3645

4. New Mexico Department of Transportation (NMDOT)
1120 Cerrillos Road
Santa Fe, NM 87504-1149
https://www.dot.state.nm.us
505-827-5100
Fax 505-827-5469

5. Mckinley County Roads Department
1980 Warehouse Lane
Gallup, NM 87301
https://www.co.mckinley.nm.us/164/Roads
505-722-2303

B. Road Closures:

1. Keep public roads open.

2. Private roads, per written agreement with owner.

3.02 TRAFFIC CONTROL

A. Meet requirements of MUTCD, Part 6, RSHS and Jurisdictional Authorities. (e.g., NMDOT and San Juan County).

B. Provide cones, delineators, concrete safety barriers, barricades, flasher lights, danger signals, signs, and other temporary traffic control devices as required to protect work and public safety.

C. Provide flaggers and guards as required to prevent accidents and damage or injury to passing traffic.

D. Do not begin work along public or private roads until proper traffic control devices for warning, channeling, and protecting motorists are in place in accordance with approved traffic control plan.

E. Maintain traffic flow and conduct construction operations to minimize obstruction and inconvenience to public traffic.

F. Provide unobstructed, smooth, and dustless passageway for two lanes of traffic through construction operations, unless otherwise approved by COR.

G. Maintain convenient access to driveways, houses, and buildings along line of work.

H. Protect roads closed to traffic with effective barricades and warning signs. Illuminate barricades and obstructions from sunset to sunrise.

I. Remove traffic control devices when no longer needed.

3.03 DETOUR

A. Obtain OGR approval prior to installing detours.

B. Construct and maintain detour to provide safe, continuous traffic flow during construction in accordance with Jurisdictional Authorities.

C. Limit detour use to 3 consecutive days without prior OGR approval or as required by permits.

D. Obtain earth materials for detour embankments from required excavation. Construct stable compacted embankments.

E. Superelevate detour curves as directed by OGR and in accordance with Jurisdictional Authorities.

F. Provide culverts and ditches as required to maintain surface drainage at detour.

G. Provide compacted gravel surfacing on detour.

H. Provide traffic control devices as required by MUTCD, Part 6.

I. Maintain detour in smooth, dust-free condition.

J. Remove the detour when no longer needed.

END OF SECTION

SECTION 01 56 10
PROTECTION OF EXISTING INSTALLATIONS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in Price Schedules for other items of work, except as specified.
2. Costs for repair of installations damaged by Contractor's operations are Contractor's responsibility.

1.02 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 56 10-1, Plan for Protecting Existing Installations.

1.03 PROJECT CONDITIONS

A. Drawings included in these specifications show existing materials, equipment, and installations but may not show all installations existing at the jobsite.

B. Obtain location of embedded conduit, pipe, cable, ground mat, and other buried items before performing drilling or excavation.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 PROTECTION

A. Provide protection for personnel and existing facilities from harm due to Contractor's operations. Protection shall be subject to approval of Government.

B. Arrange protective installations to permit operation of existing equipment and facilities by Government while work is in progress.

3.02 REMOVAL OF PROTECTIVE INSTALLATIONS

A. Remove protective installations after purpose has been served. Materials furnished by Contractor to provide protection remain property of Contractor.

3.03 REPAIR

A. Repair, at Contractor's expense, damage to existing installations due to Contractor's operations or Contractor's failure to provide proper protection. At Government's option, damage may be repaired by Government, and Contractor will be backcharged repair costs.

END OF SECTION

SECTION 01 56 15
PROTECTION OF EXISTING UTILITIES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Prices Schedules for other items of work, except for Utility Crossing Investigation.

B. Utility Crossing Investigations:

1. Payment: Lump-sum price offered in Price Schedules.

1.02 REFERENCE STANDARDS

A. Bureau of Reclamation (Reclamation)

1.	RSHS	Reclamation Safety and Health Standards, including revisions, available at https://www.usbr.gov/safety/rshs/index.html

B. Institute of Electrical and Electronics Engineers (IEEE)

1.	IEEE C2-2023	National Electrical Safety Code (NESC)

1.03 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 56 15-1, Utility Owner Acknowledgement:

1. Copy of notification acknowledgement showing underground and overhead utility agreements.

C. RSN 01 56 15-2, Utility Crossing Investigation:

1. Include the following for each utility line as applicable:
   a. Owner company name, address, and telephone number.
   b. Utility line crossing location.
   c. Utility line material.
   d. Elevation.
   e. Diameter.
   f. Service restrictions.
   g. Powerline voltage.

D. RSN 01 56 15-3, Work Plan within Utility Easements:

1. Proposed installation method including construction equipment.
2. Methods for protecting.
3. Approval of work plan from Utility owner.

1.04 PROJECT CONDITIONS

A. Drawings included in these specifications show existing utilities but may not show all utilities existing at jobsite.

B. Unknown existing utility lines may cross pipeline alignment.

C. Government does not represent that location of known existing utilities shown on drawings are exact. It is Contractor's responsibility to determine actual location of and make provision for all known and unknown utilities:

1. Verify locations, depths, and clearances of both above and below ground utilities prior to excavation.

D. Ensure that each utility line is in service as required by each utility owner.

E. Coordinate work within existing pipeline and transmission line Rights-of-Way (ROW) with all utility owners. Contractors shall comply with all Utility owner crossing requirements and shall be responsible for all costs associated with utility crossings. Obtain permission from utility owners before procuring materials to be installed in ROW.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 GENERAL

A. Have New Mexico One Call 811 locate utilities before doing underground work.

1. New Mexico 811
   1021 Eubank Blvd NE
   Albuquerque, NM 87112
   www.nm811.org
   Statewide: Dial 811 or 800-321-ALERT (2537) during regular business hours.
   By Web: Request a Line Spot 24-hours a day.

B. Utility Crossing Investigation:

1. Determine height and voltage of overhead utility lines. Clearances in accordance with RSHS and OSHA, whichever is more stringent.

2. Determine location, elevations, diameters, and materials of each underground utility line by potholing or test pits or as approved by utility owners.

3. Obtain permission from utility owner before performing physical utility investigation (potholing or test pits).

C. Notify impacted property owner at least 2 working days before disturbing waterline that serves their property. Water service to property shall not be shut off for more than 8 hours.

D. No excavation will be permitted within 10 feet of gas lines when in service without written approval from Gas Company.

E. Protect and support existing utilities that intersect work area. Before commencing work, obtain approval and necessary permits from utility owners within project ROW:

1. Protect overhead power facilities as required by utility owner if working in their ROW.

2. Protect existing waterlines as required by utility owner when working within 100 feet of their lines.

F. For abandoned electric and water lines, contact power and water companies to turn off lines prior to work.

G. Repair existing utilities damaged during construction as approved by COR and utility owner.

3.02 CLEARANCE

A. Obtain clearances required for construction operations; provide in accordance with RSHS and IEEE C2.

3.03 INTERFERENCE WITH OPERATION OR MAINTENANCE

A. Do not interfere with operation or maintenance service on utilities, existing on date offers are received:

1. Provide for access to utilities in a manner satisfactory to owners and operators and Government.

B. Provide required temporary structures; make necessary repairs, replacements, or similar operations; and furnish indemnity or other bonds.

C. Do not cross, park or travel along utility ROW without written permission from utility owner:

1. Permission shall be obtained by Contractor.

2. If permit(s) are required, Contractor shall include Government as permittee.

END OF SECTION

SECTION 01 56 20
EXISTING FENCES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work.

PART 2 PRODUCTS

2.01 MATERIALS

A. Rebuild fence with new material unless otherwise approved by COR.

B. Provide replacement materials of similar type, when necessary.

PART 3 EXECUTION

3.01 FENCE REMOVAL

A. Remove existing fences where necessary for performance of work, only when authorized by COR. Maintain fences, where designated, until work is completed or their removal is authorized.

3.02 TEMPORARY FENCES

A. Where fences are removed on rights-of-way, provide temporary fence protection for adjacent lands to prevent livestock from straying from or onto adjacent lands, complete with gates and cattle guards.

B. If Contractor does not provide necessary temporary fencing or protection within a reasonable time after need for fencing or protection arises, CO will cause work to be performed and backcharge Contractor for such work.

C. Remove temporary fences and protection as a part of cleanup operations prior to final acceptance of completed work.

3.03 FENCE REBUILDING

A. Where fences are removed to accommodate construction, immediately after construction, rebuild at original location or as close as possible with COR approval.

END OF SECTION

SECTION 01 56 32
TEMPORARY SAFETY FENCE

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work.

PART 2 PRODUCTS

2.01 SAFETY FENCE

A. Fence:
   1. High-density polyethylene grid.
   2. Minimum height: 48 inches.
   3. Color: Safety orange.
   4. Recovered Material Content:
      a. 90 to 100 percent.
   5. Postconsumer Content:
      a. 60 to 100 percent.

B. Posts: Steel fence posts.

2.02 WIRE SAFETY FENCE

A. Metal t-posts with metal grid.

PART 3 EXECUTION

3.01 INSTALLATION

A. Erect fence around work areas at location approved by COR.
   1. Install safety fence as directed by OGR.
   2. Fence along Contractor use areas.

B. Space posts 10 feet, maximum, on center.

C. Secure grid to posts.

3.02 MAINTENANCE AND REMOVAL

A. Maintain fence until work in area is complete and accepted by OGR.

B. Remove fence when no longer required.

END OF SECTION

SECTION 01 57 20
ENVIRONMENTAL CONTROLS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in Price Schedules for other items of work, except as specified.
2. Costs for damages and work stoppage are Contractor's responsibility.

B. See Section 31 02 10 – Water for Dust Abatement, for measurement and payment for applying water used for dust abatement within rights-of-way.

C. See Section 31 02 30 – Dust Palliative, for measurement and payment for applying dust palliative within rights-of-way.

1.02 REFERENCE STANDARDS

A. Bureau of Reclamation (Reclamation)

1.	RSHS	Reclamation Safety and Health Standards, including revisions, available at: https://www.usbr.gov/safety/rshs/index.html

1.03 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 57 20-1, Copy of Applicable Air Quality Permit.

1.04 REGULATORY REQUIREMENTS

A. Comply with Federal, Tribal, State, and local laws and regulations.

B. Comply with RSHS.

C. Conform to most stringent requirement in cases of conflict between specifications, regulatory requirements, and RSHS.

D. Contractor shall be responsible for damages resulting from dust originating from Contractor operations in accordance with clause at FAR 52.236-7, Permits and Responsibilities.

E. CO may stop construction activity in violation of Federal, Tribal, State, or local laws, and additional expenses resulting from work stoppage will be responsibility of Contractor.

1.05 PERMITTING

A. Develop draft environmental permit applications not covered in other sections and forward to COR for review and submission by Government to relevant authorities having jurisdiction.

1.06 DUST CONTROL

A. Provide dust control and abatement during performance of work.

B. Prevent, control, and abate dust pollution on rights-of-way provided by Government or elsewhere during performance of work.

C. Provide labor, equipment, and materials, and use efficient, environmentally acceptable methods to prevent dust nuisance or damage to persons, property, or activities, including, but not limited to, crops, cultivated fields, wildlife habitats, dwellings and residences, agricultural activities, recreational activities, traffic, and similar conditions.

1.07 AIR POLLUTION CONTROL

A. Air Quality Permits are not required for control of fugitive dust generated as result of construction-related activities on a short-term basis; however, permits are required with specific emission sources tied to aggregate processing operations releasing particulates to atmosphere.

B. Air Quality Permits are required for certain construction-related activities including, but not limited to, sandblasting, aggregate processing, or other processes which discharge pollutants into open air.

C. Air Quality Permits, and information concerning requirements, are available from: New Mexico Environmental Department (505-827-2855; https://www.env.nm.gov/air-quality/) and the USEPA.

D. Use reasonably available methods and devices to prevent, control, and otherwise minimize atmospheric emissions or discharges of air contaminants.

E. Do not operate equipment and vehicles that show excessive exhaust gas emissions until corrective repairs or adjustments reduce such emissions to acceptable levels.

1.08 NOISE CONTROL

A. Only construction activities approved by COR will be allowed between dusk and dawn.

1.09 LIGHT CONTROL

A. Shine direct stationary floodlights downward at an angle less than horizontal.

B. Shield floodlights so that floodlights will not be a nuisance to surrounding areas.

C. Direct lighting so that residences are not in direct beam of light.

D. Correct lighting control problems when they occur as approved by OGR.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 01 57 30
WATER POLLUTION CONTROL

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work.

1.02 REFERENCE STANDARDS

A. Bureau of Reclamation (Reclamation)

1.	RSHS	Reclamation Safety and Health Standards, including revisions, available at https://www.usbr.gov/safety/rshs/index.html

B. Code of Federal Regulations (CFR)

1.	40 CFR Part 110	Discharge of Oil
2.	40 CFR Part 112	Oil Pollution Prevention

C. Public Law

1.	Sections 311, 402, and 404	Clean Water Act (Public Law 92-500, as amended)

D. U.S. Environmental Protection Agency (USEPA)

1.	EPA CGP-22	Stormwater Discharge Permit associated with a Construction Site including revisions, available at: https://www.epa.gov/npdes/2022-construction-general-permit-cgp
2.	EPA Developing a SWPPP	SWPPP template and instructional resources, available at: www.epa.gov/npdes/developing-stormwater-pollution-prevention-plan-swppp

1.03 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 57 30-1, Stormwater Pollution Prevention Plan (SWPPP):

1. As required by the EPA CGP.
2. Include copy of permits.

C. RSN 01 57 30-2, Spill Prevention, Control, and Countermeasure (SPCC) Plan:

1. Submit when SPCC Plan is required in accordance with 40 CFR Part 112.

a. Required where release of oil and oil products could reasonably be expected to enter into or upon navigable waters of the United States or adjoining shorelines in quantities that may be harmful (40 CFR Part 110) and aggregate on site oil storage capacity is over 1,320 gallons. Only containers with capacity of 55 gallons and greater are included in determining onsite aggregate storage capacity.

b. Include:

1) Oil Storage Quantity.

2) Drawings of containment system.

3) Response plan to spill.

c. Reviewed and certified by a registered professional engineer in accordance with 40 CFR Part 112, as required by Section 311 of the Clean Water Act (Public Law 92-500 as amended).

1.04 REGULATORY REQUIREMENTS

A. Construction Safety Standards:

1. Comply with sanitation and potable water requirements of EPA Construction General Permit (CGP).

B. Laws, Regulations, and Permits:

1. Perform construction operations to comply, and ensure subcontractors comply, with:

a. Applicable Federal, Tribal, State, and local laws, orders, regulations, and Water Quality Standards concerning control and abatement of water pollution; and terms and conditions of applicable permits issued by permit issuing authority.

b. If conflict occurs between Federal, Tribal, State, and local laws, regulations, and requirements, the most stringent shall apply.

C. Contractor Violations:

1. If noncompliance should occur, verbally report noncompliance within 2 hours to CO. Submit specific written information within 24 hours including corrective actions.

2. Violation of applicable Federal, Tribal, State, or local laws, orders, regulations, or Water Quality Standards may result in CO stopping site activity until compliance is ensured.

3. Contractor shall not be entitled to extension of time, claim for damage, or additional compensation by reason of such a work stoppage.

4. Corrective measures required to bring activities into compliance shall be at Contractor's expense.

1.05 REQUIRED PERMITS

A. Wastewater Discharge Permit:

1. Permit:

   a. Prior to discharging other wastewater or other pollutants, provide draft permit(s) to discharge pollutants as required under Section 402 of Clean Water Act (Public Law 92-500 as amended) and New Mexico Environment Department Surface Water Quality Bureau.

   b. Submit permit applications to Government, for review and Government's submittal.

   c. Government will support Contractor in obtaining permits, as deemed appropriate by Government.

2. Terms and Conditions: Comply with terms and conditions as stated in permits.

3. Monitoring and Treatment:

   a. Provide monitoring and water treatment to achieve compliance with permit conditions.

   b. Provide recordkeeping required of permittee, as stated in Section 402 permit.

4. Sampling: Include sampling in monitoring required of Contractor to meet Section 402 requirements, as well as required laboratory tests to determine effluent characteristics.

5. Monitoring Results:

   a. Provide monitoring results to appropriate agency as required by permit.

   b. Send copies of information transmitted to appropriate agency to COR.

B. Stormwater Discharge Permit Associated with a Construction Site (EPA CGP):

1. Stormwater Pollution Prevention Plan (SWPPP): Comply with terms and conditions to obtain and maintain CGP.

2. Notice of Intent (NOI):

   a. Submit a NOI to obtain coverage under CGP.

   b. Submit NOI via the EPA's eNOI system (go to www.epa.gov/npdes/electronic-notice-intent-enoi)

      1) To obtain coverage under construction general permit to control stormwater and certain regulated non-stormwater discharges associated with construction activity including construction dewatering.

2) Outlined in Part 1 of Final 2022 CGP required under provisions of Clean Water Act, 33 U.S.C. § 1251.

3) Amended by Water Quality Act of 1987 (P.L. 100-4).

3. Monitoring and Water Treatment:

a. Provide monitoring and water treatment to achieve compliance with applicable Water Quality Standards and CGP.

b. Provide recordkeeping required by CGP associated with construction activity.

C. Stormwater Discharge Permit Associated with Industrial Activity:

1. Stormwater Discharge Permit:

a. If construction activities will entail use of a mobile CLSM plant, concrete plant, or nonmetallic borrow areas, a stormwater discharge permit associated with industrial activity may be required.

2. Terms and Conditions:

a. Comply with terms and conditions to obtain and maintain industrial stormwater discharge permit, including preparation of a Pollution Prevention Plan.

3. Monitoring and Water Treatment:

a. Provide monitoring and water treatment, if necessary, to achieve compliance with applicable Water Quality Standards.

b. Provide recordkeeping required by stormwater discharge permit associated with industrial activity.

D. 404 Permit and 401 Water Quality Certification:

1. Government will apply for a permit to discharge dredged or fill material into waters of the United States (including wetlands) as required under section 404 of Clean Water Act (Public Law 92-500 as amended).

2. If a 404 permit is required, Government will obtain a CWA 401 water quality certification from regulatory authority(s) having jurisdiction: NMED, NNEPA, and/or USEPA.

3. Contractor shall:

a. Coordinate with Government and meet requirements of 404 permit and water quality certifications.

b. Provide documentation to support permit application, monitoring, and close out.

1.06 CONTRACTOR RESPONSIBILITIES

A. For contractor obtained permits:

1. Obtain permits in Government and Contractor's name.
2. Any permits obtained by Government are exceptions to clause at FAR 52.236-7, Permits and Responsibilities, which requires the Contractor to obtain necessary licenses and permits.

B. Monitoring:

1. Conduct monitoring to meet requirements of permits which may include:
   a. Sampling.
   b. Site inspections.
   c. Required laboratory tests to determine effluent characteristics.

C. Reporting:

   a. Provide monitoring results to appropriate agency as required by permit.
   b. Send copies of information transmitted to appropriate agency to COR.

D. Recordkeeping: Retain records and data for life of project or as required by permits, whichever is longer.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 POLLUTION CONTROLS

A. Control pollutants by use of sediment and erosion controls, wastewater and stormwater management controls, construction site management practices, and other controls including Federal, Tribal, State, and local control requirements.

B. Sediment and Erosion Controls:

1. Establish methods for controlling sediment and erosion including vegetative practices, structural control, silt fences, straw dikes, wattles, sediment controls, and operator controls as appropriate.
2. Institute stormwater management measures as required, including velocity dissipators, and solid waste controls which address controls for building materials and offsite tracking of sediment.

C. Wastewater and Stormwater Management Controls:

1. Pollution prevention measures:

    a. Use methods of dewatering, unwatering, excavating, or stockpiling earth and rock materials which include prevention measures to control silting and loss of soil due to wind and water erosion, and which will intercept and settle runoff of sediment-ladened waters.

    b. Prevent wastewater from general construction activities such as drainwater collection, aggregate processing, , and drilling, grouting, or other construction operations, from entering flowing or dry watercourses without the use of approved turbidity control methods.

    c. Divert stormwater runoff from upslope areas away from disturbed areas.

2. Turbidity Prevention Measures:

    a. Use methods for prevention of excess turbidity which include, but are not restricted to, intercepting ditches, settling ponds, gravel filter entrapment dikes, flocculating processes, recirculation, combinations thereof, or other approved methods that are not harmful to aquatic life.

    b. Wastewaters discharged into surface waters shall meet conditions of permits.

    c. Do not operate mechanized equipment in waterbodies without having first obtained a Section 404 permit and 401 water quality certifications, and then only as necessary to construct crossings or perform required construction.

D. Construction Site Management:

1. Contractor construction operations:

    a. Perform construction activities by methods that will prevent entrance, or accidental spillage, of solid matter, contaminants, debris, or other pollutants or wastes into streams, flowing or dry watercourses, lakes, wetlands, reservoirs, or underground water sources:

        1) Pollutants and wastes include but are not restricted to refuse, garbage, cement, sanitary waste, industrial waste, hazardous materials, radioactive substances, oil and other petroleum products, aggregate processing tailings, mineral salts, and thermal pollution.

2. Stockpiled or deposited materials:

    a. Do not stockpile or deposit excavated materials or other construction materials, near or on, stream banks, lake shorelines, or other watercourse perimeters where they can be washed away by high water or storm runoff or can encroach upon watercourse.

3. Petroleum product storage tanks management:

    a. Place oil or other petroleum product storage tanks at least 100 feet from streams, flowing or dry watercourses, lakes, wetlands, reservoirs, and

other water sources. Do not place petroleum product storage tanks within a watercourse's 100-year floodplain.

b. Do not use underground storage tanks.

c. Construct storage area dikes at least 12 inches high or graded and sloped to permit safe containment of leaks and spills equal to storage tank capacity located in the area plus sufficient freeboard to contain 25-year rainstorm:

1) Line diked areas with an impermeable barrier at least 50 mils thick.

d. Areas for refueling operations: Lined with impermeable barrier at least 10 mils thick covered with 2 to 4 inches of soil.

3.02 CROSS DRAINAGE

A. Flows from natural drainage channels intercepted by work shall be directed through construction.

B. Provide and maintain temporary construction required to control, bypass, or otherwise cause flows to be harmless to work and public or private property.

C. Perform additional ditching and grading for drainage as directed by OGR.

D. Remove temporary construction and restore site to original condition, as approved by COR, when temporary construction is no longer needed and prior to acceptance of work.

E. Return drainage to original grade and slope once pipe is in place and backfilled.

F. Damage to work under these specifications and to public or private property shall be restored to original condition, as approved by COR, at Contractor's expense.

END OF SECTION

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SECTION 01 57 40
PESTICIDES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in Price Schedules for other items of work.
2. Contractor is responsible for damages resulting from use of pesticides under these specifications in accordance with the clause at FAR 52.236-7, Permits and Responsibilities.

1.02 DEFINITIONS

A. Pesticides: Includes herbicides, insecticides, fungicides, rodenticides, piscicides, avicides, surface disinfectants, animal repellants, and insect repellants.

1.03 REFERENCE STANDARDS

A. Code of Federal Regulations (CFR)

1.	40 CFR Subchapter E	Pesticide Programs
2.	40 CFR 156.10	Labeling Requirements for Pesticides and Devices

1.04 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 57 40-1, Qualifications

1. Certifications
2. Resubmit within 10 days of personnel change.

C. RSN 01 57 40-2, Pesticide Use Plan:

1. Not required for insect repellant to be applied directly to clothing, or for small quantities of aerosol insecticide, such as fly and spider sprays, to be applied within or directly to offices or shop buildings.
2. For each pesticide:
   a. Provide a SDS.
   b. Complete label as defined by 40 CFR 156.10, containing the following:
      1) Name, brand, or trademark product sold under.

2) Name and address of the producer or registrant.

3) Net contents.

4) Producing establishment's number.

5) Ingredient statement.

6) Warning or precautionary statements.

7) Directions for use.

8) Use classification.

1.05 QUALIFICATIONS

A. Certified to apply appropriate weed and vegetation control measures in the project area. Possess, at a minimum, a commercial applicator's license for the State of New Mexico to dispense herbicides in industrial-right-of-ways, rangeland settings, and in wetland areas for weed control work involving herbicide application.

1.06 DELIVERY, STORAGE, AND HANDLING

A. Comply with labeling and Safety Data Sheet (SDS) requirements when dealing with pesticides.

B. Keep records of pesticide types and amounts purchased, delivered, stored, mixed, and actually used, and disposal means of excess. Make records available for review by COR upon request.

PART 2 PRODUCTS

2.01 PESTICIDES

A. Apply herbicide after gravel surfacing has been placed.

B. Allowed pesticides:

1. Appropriate for site-specific conditions where applied
2. Refer to pesticides in Section 51 00 80 – BLM Herbicides
3. Not prohibited by State, Navajo Nation, or local agencies.

PART 3 EXECUTION

3.01 APPLICATION

A. Apply pesticides only as allowed by labeling.

B. In accordance with and only after approval of Pesticide Use Plan.

END OF SECTION

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SECTION 01 57 50
TREE AND PLANT PROTECTION

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in Price Schedules for other items of work, except as specified.
2. Costs for repair or treatment of injured vegetation and replacement of trees or shrubs are Contractor's responsibility.

1.02 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 57 50-1, Protection Plan:

1. Description of protective barriers or other methods used to protect vegetation from damage or injury caused by construction operations.

PART 2 PRODUCTS

2.01 REPLACEMENT TREES AND SHRUBS

A. Species: Same as removed tree or shrub or other species approved by COR.

B. Size: Same size as removed tree or shrub, or maximum practicable size that can be planted and sustained in the particular environment as approved by COR.

PART 3 EXECUTION

3.01 PRESERVATION AND PROTECTION

A. Preserve natural landscape and preserve and protect existing vegetation not required or otherwise authorized to be removed:

1. Submit requests to remove vegetation not specifically required to be removed to COR.

B. Conduct operations to prevent unnecessary destruction, scarring, or defacing of natural surroundings in vicinity of work.

C. Move crews and equipment within rights-of-way and over routes provided for access to the work in a manner to prevent damage to grazing land, crops, or property.

D. Protect vegetation from damage or injury caused by construction operations, personnel, or equipment by use of protective barriers or other methods approved by COR.

E. Minimize, to greatest extent practicable, clearings and cuts through vegetation. Irregularly shape authorized clearings and cuts to soften undesirable aesthetic impacts.

F. Do not use trees for anchorages except in emergency cases or as approved by COR:

1. For such use, wrap trunk with a sufficient thickness of approved protective material before rope, cable or wire is placed.

2. Submit requests to use trees for anchorage, except for emergencies. Include description of protective material.

G. Use safety ropes where tree climbing is necessary; do not use climbing spurs.

3.02 REPAIR, TREATMENT, OR REPLACEMENT

A. Contractor is responsible for injuries to vegetation caused by Contractor operations, personnel, or equipment. Repair, treat or replace injured vegetation as directed by COR.

B. Employ the services of an experienced arborist or licensed tree surgeon to direct repair, treatment, and replacement of injured vegetation. Submit qualifications of experienced arborist or licensed tree surgeon to COR prior to employment.

C. Repair or treat injured vegetation without delay and as recommended by and under direction of an experienced arborist or licensed tree surgeon.

D. Removed and dispose of trees or shrubs not required or otherwise authorized to be removed that, in the opinion of COR, are injured beyond saving.

E. Replace removed tree or shrub with tree or shrub approved by COR.

END OF SECTION

SECTION 01 57 60
PROTECTED SPECIES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work.

1.02 PROJECT CONDITIONS

A. Certain native plant and animal species in the State of New Mexico are protected under Federal, State and Tribal laws. Government has ascertained that protected species and/or their habitats may exist in the areas to be disturbed by construction activities. These species include, but may not be limited to, the following:

1. Migratory Birds.
2. Burrowing Owl.
3. Ferruginous Hawk.
4. Prairie Dog.
5. Kit Fox.
6. Naturita Milkvetch.
7. Mesa Verde Cactus.
8. Raptors.

B. Federal legislation provides for protection of Threatened and Endangered Species, which might otherwise be lost due to alteration of terrain as a result of any Federal construction projects.

C. Project is designed to comply with Environment Commitments of Navajo-Gallup Water Supply Project Final Environmental Impact Statement, Realignment of the Northern Portion of the San Juan Lateral Environmental Assessment, and Final Biological Opinion for the Navajo-Gallup Water Supply Project reissued by U.S. Fish and Wildlife Service on September 20, 2022.

D. Insert this section in subcontracts.

E. In accordance with Federal, State, and Tribal Law, Government may arrange for removal of protected species.

1. Cooperate with those performing such removal.
2. If these species are not removed, cooperate with and abide by protection plans developed by appropriate Federal, State, and Tribal entities to avoid damage to or disturbance of protected species.

F. Persons who, without permission contribute to the take of threatened and endangered species are subject to arrest and penalty of United States, Navajo Nation, and New Mexico laws.

1. Take: to harass, harm, pursue, hunt, shoot, wound, kill, trap, capture or collect, or to attempt to engage in any such conduct.
2. Harm: includes significant habitat modification or degradation that results in death or injury to listed species by significantly impairing essential behavioral patterns, including breeding, feeding, or sheltering.
3. Harass: intentional or negligent actions that create the likelihood of injury to listed species to such an extent as to significantly disrupt normal behavior patterns which include, but are not limited to, breeding, feeding or sheltering.
4. Incidental take: take that is incidental to, and not the purpose of, the carrying out of an otherwise lawful activity.
5. Offenders will be subject to United States, Navajo Nation, and New Mexico environmental protection laws and civil penalties.

1.03 SPECIAL REQUIREMENTS

A. Migratory Birds

1. Nesting may occur in area between March 15th and August 15th:
2. Government will survey for nesting birds in vegetated areas to be disturbed. Evaluation shall be performed no more than 5 days prior to area disturbance.
3. Notify COR 10 days before disturbing an area.
4. Do not disturb a nesting pair of birds with eggs.
5. Avoid birds and modify construction activities as directed by COR.

B. Raptors:

1. Raptor nesting may occur in the area from January 15th to July 15th:
    a. Minimize disturbance to raptors. Government will conduct raptor surveys prior to construction that is to begin between January 15th and July 15th. Evaluation shall be performed no more than 5 days prior to area disturbance. The survey will include all potential nesting habitat within specified buffer distances (0.25-mile to 0.625-mile, depending on the species) of the construction right- of-way.
2. Notify COR 10 days prior to construction to allow for raptor nest surveys.
3. Do not disturb an active raptor nest:
    a. Government will notify Contractor of nest locations.
    b. No construction will be allowed within the above months within 0.25-mile to 0.625-mile of active nests, depending on the species.

4. Raptor Safe Utility Poles:

   a. All above ground utility poles shall conform to a design standard(s) that comply with the Raptor Electrocution Prevention Regulations RCS-43-08 published September 10, 2008. This is to avoid unintentional electrocution of raptors that may perch on the utility pole.

1.04 REPLACEMENT PLANTS

A. Protected plant species shall not be removed or damaged by Contractor.

1. If inadvertent effects occur to protected species during construction:

   a. Notify COR:

      1) Verbal: Within 2 hours.

      2) Written: Within 24 hours.

   b. Cease work in affected work area until consultation commitments are completed.

2. Government, in consultation with Bureau of Indian Affairs (BIA) and State of New Mexico, will manage required removal/replacement planting in accordance with consultation commitments.

3. Replacement numbers/ratio in accordance with consultation commitments.

1.05 DISCOVERY OF RESOURCES

A. When Contractor or parties operating or associated with Contractor, in performance of this contract discover threatened and endangered species on ROW:

1. Immediately cease work within 50 feet of discovery location.

   a. Place temporary safety fence in accordance with 01 56 32 –Temporary Safety fence to prevent disturbance until biologist can investigate to avoid inadvertent disturbance by construction activities.

2. Verbally notify CO and OGR within 2 hours, giving location and nature of findings.

   a. Follow with written confirmation to CO within 12 hours.

3. Verbally notify Reclamation staff biologist within 2 hours.

   a. Name of assigned staff biologist and telephone number will be provided at preconstruction meeting.

B. Do not disturb or damage threatened and endangered species discovered during construction activities. Provide cooperation and assistance to preserve findings for removal or other disposition by CO.

C. Government, in consultation with appropriate regulatory agencies upon discovery of threatened and endangered species, will manage required removal/replacement planting in accordance with consultation commitments.

1. When discovery mitigation has been completed, CO will notify Contractor that work can resume in discovery area.

2. Do not resume work in area of discovery until receipt of written notice to proceed from CO.

D. Where appropriate by reason of discovery, CO may order delays in time of performance or changes in work or both. When such delays or changes are ordered, an equitable adjustment will be made in the contract in accordance with applicable clauses of the contract.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

SECTION 01 57 90
PRESERVATION OF HISTORICAL AND ARCHEOLOGICAL DATA

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work, except as provided for an equitable adjustment.

1.02 DEFINITIONS

A. Cultural Resources: Includes prehistoric, historic, architectural, and traditional cultural properties. These include, but are not limited to, human skeletal remains, archaeological artifacts, records, and material remains related to such property.

B. Cultural Items: Native American cultural items (i.e., funerary objects, sacred objects, objects of cultural patrimony, or human remains) for which protection is prescribed under the Native American Graves Protection and Repatriation Act (NAGPRA) - Public Law 101-601; 104 Stat. 3042, Section 3(d); and 43 CFR Part 10.4.

C. Human Remains: Physical remains of the body of a person.

D. Funerary Objects: Native American items that, as part of the death rite or ceremony of a culture, are reasonably believed to have been placed intentionally at the time of death or later with or near individual human remains.

E. Native American: Of, or relating to, a tribe, people, or culture that is indigenous to the United States.

F. Sacred Objects: Native American items that are specific ceremonial objects needed by traditional Native American religious leaders for the practice of traditional Native American religions by their present-day adherents. These items are specifically limited to objects that were devoted to a traditional Native American religious ceremony or ritual and which have religious significance or function in continued observance or renewal of such ceremony.

G. Sacred Sites (Traditional Cultural Properties): Native American traditional ceremonial locations identified for Native American practice of traditional Native American religious activities.

H. Objects of Cultural Patrimony: Native American items having ongoing historical, traditional, or cultural importance central to the Indian tribe itself, rather than property owned by an individual tribal member. These objects are of such central importance that they may not be alienated, appropriated, or conveyed by any individual tribal member.

1.03 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 57 90-1, Alternate Use Area:

1. When using areas other than those approved are to be used, submit map showing location of unapproved use areas, for approval and obtain cultural and environmental approvals in accordance with Sections 01 31 19 – Project Management and Coordination and 01 57 90 – Preservation of Historical and Archeological Data.

1.04 PROJECT CONDITIONS

A. Project site has been surveyed and cleared of cultural resources for construction within Rights-of-Way (ROW) by agencies having jurisdiction (e.g., Navajo Nation Heritage and Historic Preservation Department (NNHHPD), State Historic Preservation Office (SHPO) and/or Bureau of Indian Affairs (BIA), etc.).

B. Onsite Contractor personnel shall undergo cultural resource sensitivity awareness training to inform onsite personnel of cultural resource requirements and expectations during pre-construction meeting before initial groundbreaking takes place:

1. Government will provide training:

a. Training will take approximately 1 hour.

b. Schedule for training will be mutually agreed upon.

c. CD and/or PowerPoint of Cultural Sensitivity Training will be provided to Contractor.

2. Contractor shall:

a. Require onsite personnel to complete required cultural sensitivity awareness training before being allowed to work in project area.

b. Require onsite personnel added after completion of required cultural sensitivity awareness training to view Cultural Sensitivity Training CD/or PowerPoint provided to Contractor at Cultural Sensitivity Training before being allowed to work in project area.

c. Maintain training records.

C. Federal legislation provides for protection, preservation, and collection of scientific, prehistorical, historical, and archeological data, including relics and specimens, which might otherwise be lost due to alteration of terrain as a result of any Federal construction project.

D. Persons who, without permission, injure, destroy, excavate, appropriate, or remove historical or prehistorical artifacts, objects of antiquity, or archeological resources on private, State and Tribal lands of the United States are subject to arrest and penalty of

United States, Navajo Nation, and New Mexico Cultural Resource Protection laws. Offenders will be subject to Navajo Nation and New Mexico Cultural Resource Protection laws and civil penalties.

E. Comply with Federal, State and Tribal laws when operating on Federal, State, and Indian lands.

F. Attend weekly coordination meetings. Refer to Section 01 31 19 – Project Management and Coordination.

G. Discovery of Resources:

1. When Contractor or parties operating or associated with Contractor, in performance of this contract discover cultural resources on ROW:

   a. Immediately cease work within 50 feet of discovery location:

      1) Place temporary safety fence in accordance with 01 56 32 – Temporary Safety Fence to prevent disturbance until archaeologist can investigate to avoid inadvertent disturbance by construction activities.

   b. Verbally notify CO and OGR within 2 hours, giving location and nature of findings.

   c. Verbally notify Reclamation staff archaeologist within 2 hours. Name of assigned staff archaeologist and telephone number will be provided at pre-construction meeting.

   d. Follow with written confirmation to CO within 12 hours.

2. Do not disturb or damage cultural resources uncovered during construction activities. Provide cooperation and assistance to preserve findings for removal or other disposition by CO.

3. Upon completion of concurred upon discovery treatment by Cultural Resources Contractor, a 48-hour consultation period will be conducted:

   a. When discovery mitigation has been completed CO will be notified that work can resume as recommended by Cultural Resources Contractor and OGR will notify Contractor work can resume in discovery area.

4. Do not resume work in area of discovery until receipt of written notice to proceed from CO.

H. Where appropriate by reason of discovery, CO may order delays in time of performance or changes in work, or both. When such delays or changes are ordered, an equitable adjustment will be made in the contract in accordance with applicable clauses of the contract.

I. Mitigate cultural resources as directed by Government on lands outside of Permanent or Construction ROW, including private lands:

1. Specifically comply with cultural resource stipulations as shown on drawings.

J. Insert this section in subcontracts which involve performance of work on jobsite terrain.

K. Obtain Government clearance before disturbing lands outside of ROW in accordance with Section 01 14 10 – Use of Site.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

3.01 CULTURAL MONITORING AREA

A. 48-hour notice must be provided to OGR, Government Archeologist and Cultural Resources Contractor prior to approaching or entering a cultural monitoring area for construction activities.

B. Install temporary fencing of the ROW extents at identified required monitoring areas and/or in constricted width ROW segments before site disturbance including surveying activities begin in these areas. Refer to Section 01 56 32 – Temporary Safety Fence.

C. An archeological monitor is required on all new ground disturbing construction activities within identified required monitoring areas.

END OF SECTION

SECTION 01 60 00
PRODUCT REQUIREMENTS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. When Price Schedules include a separate item for furnishing a material, include cost of furnishing, hauling, storing, and handling in prices offered in Price Schedules for item.
2. When Price Schedules do not include a separate item for furnishing a material, include cost of furnishing, hauling, storing, and handling in prices offered in Price Schedules for items of work requiring the material.

1.02 DEFINITIONS

A. Essential Characteristics: As used in these specifications, the term "essential characteristics" is synonymous with the term "salient characteristics."

B. Salient Characteristics: Those qualities of an item that are essential to ensure that intended use of item can be satisfactorily realized.

1.03 REFERENCE STANDARDS

A. American Society of Mechanical Engineers (ASME)

1.	ASME B1.1-2024	Unified Inch Screw Threads, (UN and UNR Thread Form)
2.	ASME B1.20.1-2013(R2018)	Pipe Threads, General Purpose, Inch

B. Bureau of Reclamation (Reclamation)

1.	RSHS	Reclamation Safety and Health Standards, including revisions, available at https://www.usbr.gov/safety/rshs/index.html

1.04 DELIVERY, STORAGE, AND HANDLING

A. Transport and handle manufactured products in accordance with manufacturer's instructions.

B. Store and protect manufactured products in accordance with manufacturer's instructions and RSHS. Obtain instructions from manufacturer before delivery of materials to jobsite. Maintain a copy of instructions at jobsite.

C. Protect materials from adverse effects of moisture, sunlight, ultraviolet light or weather during storage at jobsite.

D. Remove and replace damaged items with new items.

E. Store curing compounds, sealants, adhesives, paints, coatings, sealers, joint compounds, grouts, and similar products at temperature and environmental conditions recommended by manufacturer.

1.05 MAINTENANCE

A. Extra Materials:

1. Furnish additional maintenance materials specified as "extra materials" in specifications. Provide maintenance material identical to installed material and provide from same manufacturer's production lot as installed material.
2. Package extra materials for storage and label with complete product identification on packaging.
3. Deliver extra materials to Government at storage location within 150 miles of project in accordance with written direction from Government. Place in storage as directed by COR.

PART 2 PRODUCTS

2.01 MATERIALS

A. Provide materials required for completion of work.

B. Provide type and quality described in these specifications. Make diligent effort to procure specified materials from all available sources.

C. Furnish new materials conforming to referenced standards unless otherwise specified.

D. For materials not covered by these or referenced specifications, furnish materials of standard commercial quality.

E. If materials to be used deviate from or are not covered by recognized specifications and standards, submit, for approval, justification for and exact nature of the deviation, and complete specifications for materials proposed for use.

F. Make parts accurately to standard gauge where possible:

1. Use unified screw threads conforming to ASME B1.1 or ASME B1.20.1 for threads, including but not limited to those of bolts, nuts, screws, taps, pipes, and pipefittings.

2. For internal connections only, Contractor may deviate from ASME standards, provided a complete set of taps and dies are furnished as required to facilitate repair or replacement.

G. Permanently mark fasteners with a symbol identifying manufacturer and with symbol(s) indicating grade, class, type, and other identifying marks in accordance with reference or applicable standard.

2.02 SUBSTITUTIONS

A. If materials required by these specifications become unavailable because of Government priorities or other causes, substitute materials may be used.

B. Obtain written approval to use substitute materials from CO. State in request for approval the amount of adjustment, if any, to be made in favor of Government.

C. Government's determination as to whether substitution will be permitted and as to what substitute materials may be used, shall be final and conclusive.

D. If approved substitute materials are of less value to Government or involve less cost to Contractor than specified material, a contract adjustment will be made in favor of Government. Where amount involved or importance of substitution warrants, a deductive modification to contract will be issued.

E. No payments in excess of prices offered in Price Schedule will be made because of substitution of one material for another or because of use of one alternate material in place of another.

2.03 WORKMANSHIP

A. Accurately manufacture and fabricate materials in accordance with best modern practice and requirements of these specifications, notwithstanding minor errors or omissions therein.

B. Use liberal factors of safety and adequate shock-absorbing features in designs, especially for parts subjected to variable stress or shock, including alternating or vibrating stress or shock.

C. Include provisions which prevent components from loosening for shock-absorbing features and parts subject to vibration.

2.04 GOVERNMENT CONTRACT SOURCE QUALITY ASSURANCE

A. Materials will be subject to inspection in accordance with clause at FAR 52.246-12, Inspection of Construction at following locations, as determined by CO:

1. Place of production or manufacture.

2. Shipping point.

3. Jobsite.

B. To allow sufficient time to provide for inspection, submit at time of issuance, copies of purchase orders, including drawings and other pertinent information, covering material on which inspection will be made as advised by CO, or submit other evidence if such purchase orders are issued verbally or by letter.

C. Inspection of materials at locations specified above or waiving of inspection shall not be construed as being conclusive as to whether materials and equipment conform to contract requirements under the clause at FAR 52.246-12, Inspection of Construction nor shall Contractor be relieved thereby of responsibility for furnishing materials meeting requirements of these specifications.

D. Acceptance of materials will be made only at jobsite.

PART 3 EXECUTION

3.01 INSTALLATION

A. Install products in accordance with manufacturer's recommendations unless otherwise specified.

3.02 GOVERNMENT CONTRACT FIELD QUALITY ASSURANCE

A. Final inspection and acceptance of materials will be made only at jobsite after installation and testing.

END OF SECTION

SECTION 01 61 10
BUY AMERICAN

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work.

1.02 DEFINITIONS

A. Buy American Material: In accordance with FAR 52.225-11 Buy American-Construction materials.

1.03 SUBMITTAL

A. Include applicable certification for material in each RSN listed in Table 01 33 00A – List of Submittals in accordance with FAR 52.225-11 Buy American-Construction Materials under Trade Agreements.

B. Submit the following in accordance with Section 01 33 00 - Submittals.

C. RSN 01 61 10-1, Buy American Certificates

1. Prior to project closeout provide applicable Buy American Certificate for materials used on the project.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 CERTIFICATION

A. Deliver applicate certificate in accordance with 52.225-11 Buy American-Construction Materials under Trade Agreements to OGR with material delivered to the site.

B. Reclamation will not accept material without proper certification.

END OF SECTION

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SECTION 01 71 20
SURVEYING

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work.

1.02 DEFINITIONS

A. GNSS: Global Navigation Satellite Systems.

B. GPS: Global Positioning System.

C. TPS: Terrestrial Positioning Systems such as total stations and automatic levels.

1.03 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 71 20-1, Surveying Plan:

1. Describe work layout and survey methods.
2. Surveying schedule.
3. Example of field records format on electronic data collection devices.

C. RSN 01 71 20-2, Resume(s):

1. Responsible Professional: Proof of Registration as a licensed surveyor or engineer in New Mexico.
2. Construction surveyor(s).

D. RSN 01 71 20-3, Accuracy Check Results:

1. Accuracy check of Government-established primary control.

1.04 PRIMARY CONTROL

A. Government will establish primary control, in grid (state) coordinates, to be used for establishing work lines and grades.

B. Primary control will consist of benchmarks and horizontal control points in work vicinity.

C. Government will provide complete listing and identification of primary control within 15 days after issuance of Notice to Proceed.

D. Check and verify primary control and resolve discrepancies with Government before beginning work.

E. Preserve and maintain primary control points until otherwise authorized. Government may reestablish damaged or destroyed primary control points and back charge reestablishment cost to Contractor.

1.05 QUALIFICATIONS

A. Responsible Professional:

1. Registered Land Surveyor or Professional Engineer.
2. Registered in New Mexico.
3. When GPS/GNSS is utilized, shall have working knowledge of geodesy, GNSS/GPS capabilities.

1.06 COORDINATION MEETING

A. Contractor and Construction Surveyor shall meet with COR, OGR and Government surveyor 14 days before submittal of Surveying Plan to review:

1. Project coordinates and benchmarks.
2. Survey submittals.

PART 2 PRODUCTS

2.01 SURVEYING MATERIALS AND EQUIPMENT

A. Provide surveying instruments and materials required for surveying work, including, but not limited to, stakes, spikes, steel pins, templates, platforms, and tools.

1. Except as required to be incorporated in work or left in place, surveying materials remain property of Contractor.
2. Type and brand of surveying equipment including serial numbers (GNSS Systems, Total Stations, and Data Collectors) to be used onsite.

PART 3 EXECUTION

3.01 GENERAL

A. Construction surveyor shall be available within 48-hours.

1. Contracted surveyor will use FCCO Survey Group code list for entire duration of the project. Surveyor Chief of FCCO office will provide code list to contracted surveyor.

B. Cultural Resources:

1. Obtain COR approval before entering work site with a vehicle.
2. If vehicular travel through an archaeological site is required, an archaeologist shall flag a route that avoids features within the site.
3. Refer to Section 01 57 90 – Preservation of Historical and Archeological Data if cultural resources are encountered.

3.02 LAYOUT OF WORK SURVEYS

A. Establish lines and grades for work layout from Government-established primary control points.

B. Establish measurements required for work execution to specified tolerances.

C. Provide stakes, markers, and other survey controls necessary to control, check, and guide construction. Place and mark controls so OGR can monitor progress without the use of survey equipment.

1. ROW staking:
   a. Labels: Front and back with stationing and offset.
   b. Flagging: Orange and pink.
   c. Turn inward towards centerline.
2. Centerline staking:
   a. Labels: Front and back with stationing.
   b. Flagging: Orange.
3. Install staking a minimum of 6-inches in ground or substantially secure to prevent movement from weather, livestock, and similar.

D. Stations and distances shown on drawings are based on ground coordinates.

E. Northings and eastings shown on drawings are based on grid (state plane) coordinates.

F. Perform GPS/GNSS work using North American Datum 1983 (NAD 83) State Plane New Mexico West FIPS 3003.

G. Perform work using North American Vertical Datum of 1988 (NAVD 88) in vertical units of U.S. Survey Feet.

H. Georeferencing shall employ Geoid 09 (GEOID09).

I. All ground-to-grid conversions, where necessary, shall be performed utilizing site ground-to-grid conversation factor as shown on drawings. Independent site conversion factors will be provided and also appear on Construction Plan and Profile drawings, as needed.

J. Collect survey data in grid coordinate format to remain in and be delivered in New Mexico West 3003 State Plane grid coordinates (NAD 83, NAVD 88).

3.03 QUANTITY SURVEYS

A. Perform surveys and computations to determine quantities of work performed or placed during each progress payment period.

B. Perform surveys necessary for Government to determine final quantities of work in place. Final payment quantities will be based on Government's original terrain data and submitted survey notes and computations.

C. Perform quantity surveys in presence of authorized Government representative, unless specifically waived. Notify Government at least 24 hours before performing a quantity survey.

3.04 SURVEY REQUIREMENTS

A. Alignment and ROW Staking: Each 50 feet on tangent, each 25 feet on curves and each corner.

B. Slope Staking: Each 50-feet on tangent and each 25-feet on curves, stake every 10-feet in elevation on slopes.

C. Structures: Stake out of structures and checkouts before and during construction.

D. Roads and Service Yards: Blue tops each 50-feet on tangent and each 25-feet on curves. Blue tops on 25-feet grid for service yards.

E. Quantity surveys: Original, final, and intermediate Digital Terrain Models (DTMs) for structure sites and other locations as required for quantity surveys. Volumetric quantities shall be determined by preliminary (before) and final (as-constructed) DTM.

F. Set additional stakes as required by COR.

G. As-builts: As required for structures and other features of work.

H. Set additional stakes as required by OGR.

3.05 ACCURACY

A. Degree of Accuracy:

1. Alignment of Tangents and Curves: Within 0.1 foot.
2. Structure Points: Set within 0.01 foot, except where installation or operation considerations require tighter tolerances.
3. Vertical Control Surveys: Close within 0.05 feet times the square root of circuit length in miles.

B. Include check measurement on known survey point(s) at the beginning and ending of each survey.

3.06 FIELD RECORDS

A. Record field notes, computations, and other surveying data on electronic data collection devices. Field records shall be complete and accurate record of survey.

1. Include:

a. Names of all survey crew members.

b. Type (GNSS Systems, Total Stations, and Data Collectors) and brand of surveying equipment including serial numbers.

c. Directional survey report and instrumentation report, or equal, made during drilling operations.

d. “X”, “y”, and “z” coordinates of final pipe location. Spacing between coordinate points shall be 30-foot maximum.

B. Data and notes may be collected on electronic data collection devices:

1. Electronic files of notes: In approved format.

2. File structure: Day/Month/Year/Method/Crew Chief Initials.

a. Example: 02062025 RTK FML for survey performed February 6, 2025, via Real Time Kinematic method by Crew Chief named First Middle Last.

3. Download raw data files from survey equipment at least once per week.

4. Include electronic files and paper copies of notes in submittals.

END OF SECTION

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SECTION 01 74 00
CLEANING AND WASTE MANAGEMENT

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in Price Schedule for other items of work, except as specified.
2. Cost of environmental site assessments are Contractor's responsibility.

1.02 DEFINITIONS

A. Hazardous waste: Defined as hazardous by 40 CFR 261.3; or by other Federal, Tribal, State, or local laws or regulations.

1.03 REFERENCE STANDARDS

A. Bureau of Reclamation (Reclamation)

1.	RSHS	Reclamation Safety and Health Standards, including revisions, available at https://www.usbr.gov/safety/rshs/index.html

B. Code of Federal Regulations (CFR)

1.	40 CFR 261.3	Definition of Hazardous Waste
2.	49 CFR 171-179	Transportation - Hazardous Materials Regulations

1.04 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 74 00-1, Waste Production and Disposal Plan:

1. For each type of waste, list estimated quantity and planned disposal location.

C. RSN 01 74 00-2, Waste Production and Disposal Records:

1. For each type of waste, list quantity and disposal location.
2. Certification waste was properly disposed.
3. Recycled waste materials verification.
4. Hazardous wastes manifests.

D. RSN 01 74 00-3, Environmental Consultant Resume:

1. Describe experience on similar project.

E. RSN 01 74 00-4, Environmental Site Assessment.

1.05 QUALIFICATIONS

A. Environmental consultant: Minimum 2 years' experience in conducting environmental site assessments for similar construction.

1.06 REGULATORY REQUIREMENTS

A. Comply with Federal, Tribal, State, and local laws and regulations.

B. Comply with RSHS.

C. Conform to most stringent requirement in cases of conflict between specifications, regulatory requirements, and RSHS.

1.07 PROJECT CONDITIONS

A. Report waste materials discovered at jobsite to COR.

1. Cease work in areas where waste may be hazardous until waste materials are investigated by Government.
2. Have Environmental Consultant investigate waste materials and make recommendations for continuing work. Government will review and approve of investigation and recommendations.
3. If waste is hazardous, CO may order delays in time of performance or changes in work, or both.
4. If such delays or changes are ordered, an equitable adjustment will be made in the contract in accordance with applicable clauses of the contract.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 PROGRESS CLEANING

A. Keep work and storage areas free from accumulations of waste materials and rubbish.

3.02 FINAL CLEANUP

A. Remove temporary facilities, buildings, rubbish, unused materials, and other similar materials which are not part of permanent work.

3.03 DISPOSAL

A. Nonhazardous waste materials:

1. Dispose by removal from jobsite. Reuse or recycle waste materials whenever possible.
2. Dispose of nonhazardous waste materials that are not reused or recycled at appropriately permitted disposal facilities.

B. Hazardous Waste Disposal:

1. Dispose by removal from jobsite.
2. Recycle hazardous waste whenever possible.
3. Dispose of hazardous waste materials that are not recycled at appropriately permitted treatment or disposal facilities.
4. Transport hazardous waste in accordance with 49 CFR 171-179.

C. Certify that wastes are disposed of in accordance with Federal, Tribal, State, and local regulations.

3.04 SITE ASSESSMENT

A. Upon completion of work, perform site assessment at following areas for work done under these specifications:

1. Hazardous waste accumulation areas.
2. Petroleum dispensing and storage areas where aggregate storage of petroleum at jobsite was over 110 gallons.
3. Hazardous material storage areas.

B. Employ qualified environmental consultant to perform assessments.

C. Demonstrate and document by appropriate analytical sampling that site contamination is less than State action cleanup levels. Submit written report with sampling locations, findings, manifests, photos, and other pertinent information.

3.05 RECORDS

A. Keep records of types and amounts of waste materials produced.

B. Keep records of waste material disposal.

END OF SECTION

SECTION 01 78 30
PROJECT RECORD DOCUMENTS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. As-Built Drawings:

1. Payment: Lump-sum price stated in Price Schedules.

B. Final Drawings:

1. Payment: Lump-sum price stated in Price Schedules.

1.02 DEFINITIONS

A. As-Built Drawings (Also known as “Red-line” Drawings):

1. Set of full size, hard-copy drawings, that contain both Government informational drawings supplied with the Specifications and Contractor produced drawings that have been submitted for approval.
2. Update progress as-built drawings to document work or services performed for progress payments in accordance with the clauses at FAR 52.232-27, Prompt Payment for Construction Contracts. The COR will retain from progress payments if project drawings are not updated.
3. At a minimum, meet monthly with OGR to review and approve progress made to full-sized record drawings prior to the submittal for progress payment. Meeting location to be approved by OGR.
4. Verify with the OGR that mark-ups to the drawings are current with the progress of the work prior to submitting invoice for progress payment.

B. Final Drawings:

1. Complete electronic set of drawings produced by Contractor updated to reflect all changes made in the field through installation of each item requiring drawings.

1.03 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 78 30-1, Electronic Media Release Form, Upon Request.

1. Refer to Article 1.04 Electronic Drawings below.

C. RSN 01 78 30-2, Progress As-Built Drawings:

1. Copies of as-built drawings maintained on site. Colored copies are not required.

2. Submit to OGR monthly for review and approval prior to the submittal for progress payment.

    a. Certify as being up to date and accurate as shown in the description of work.

D. RSN 01 78 30-3, Final As-Built Drawings:

1. Certified marked sets. As-built drawings that have been certified by Contractor as being up to date and accurate in the description of work.

2. Submit to COR for approval prior to Government possession.

3. For Information Only (FIO) Drawings: Revision date and details.

E. RSN 01 78 30-4, Final Drawings:

1. Submit to COR for Government approval no later than 14 calendar days following approval of RSN 01 78 30-3, Final As-Built Drawings.

2. Updated with all changes as marked on as-built drawings to reflect approval comments.

1.04 ELECTRONIC DRAWINGS

A. AutoCAD Files, will be distributed upon request, after:

1. Notice to Proceed.

2. Contractor signs form in Section 51 00 50 – Electronic Media Release.

1.05 AS-BUILT DRAWINGS

A. Maintain 2 sets of full-size prints of contract drawings marked to show accurate and complete records of as-built conditions. Keep drawings at jobsite and mark as work progresses:

1. Mark and dimension to show variations between actual construction and that indicated or specified in contract documents:

    a. Include buried or concealed construction and utilities.

    b. Include existing items, topographic features, and utility lines revealed during construction which differ from those shown on contract drawings.

2. Mark to define construction actually provided where choice of materials or methods is permitted in specifications, or where variations in scope or character of work from that of the original contract are authorized.

B. Use standard drafting practice to represent changes and include supplementary notes, legends, and details necessary to clearly portray progress as-built construction.

C. Mark drawings in the following colors:

1. Red: Additions to original drawings.

2. Green: Deletions to original drawings.

3. Blue: Notations necessary for explanation of as-built markings.

D. Allow Government to review drawings during weekly construction meetings and be available at all other times.

E. Upon completion of work and prior to training and/or Government possession, submit to COR for review and acceptance:

1. One set will be made available to the Government:

a. For maintenance of changes to Government provided information drawings.

b. For verification of Final Drawings.

2. Sign marked prints as certified correct:

a. Sign and date each drawing as certified correct. Do not include or certify Information Only drawings.

b. If no revisions were necessary to illustrate as-built conditions, mark drawing with "No Changes."

c. Provide as-built drawings as outlined in Section 01 33 00 – Submittals and as required in other sections.

1.06 FINAL DRAWINGS

A. Update all drawings and diagrams either generated by Contractor or impacted by Contractor-performed work with all modifications made during design and construction process, including all modifications made during testing. Include all as-built drawing modifications.

B. Drawings shall represent final installed condition of pipelines at project completion.

C. Drawings supplied as Final Drawings shall meet requirements of Final Drawings included in Section 01 33 00 – Submittals.

D. Computer drafted drawings provided in PDF format, compatible with AutoCAD, 2018 or newer.

E. Upon completion, sign prints as certified correct:

1. Sign and date each drawing as certified correct.

2. If no revisions were necessary, mark drawing with "No Changes."

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 CONSTRUCTION DOCUMENTATION

A. Provide and maintain at least 2 copies of as-built drawings and manufacturer's installation instructions at jobsite upon equipment shipment:

1. Drawings shall be in as-built condition at time of equipment shipment and shall reflect Government approval comments.

B. Mark drawings with changes and revisions made during installation and checkout of equipment following as-built marking procedures:

1. Mark and dimension to show variations between actual construction and that indicated or specified in contract documents.
2. Mark to define construction actually provided:
    a. Where choice of materials or methods is permitted in specifications.
    b. Where variations in scope or character of methods is permitted in specifications.
    c. Where variations in scope or character of work from that of original contract are authorized.

C. Upon completion of work:

1. Sign marked prints as certified correct.
2. Submit to COR for approval within 14 days following substantial project completion:
    a. One set to be retained by COR.
    b. One set to be used in preparation of Final Drawings for each unit:
        1) Due no later than 14 calendar days following approval of Final As-Built Drawings.

END OF SECTION

SECTION 03 11 10
CONCRETE FORMING

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for items requiring forming.

1. Includes furnishing and placing reinforcement for thrust blocks, collars, welded rectangular barriers, guard posts and other applicable items.

1.02 REFERENCE STANDARDS

A. American Concrete Institute (ACI)

1.	ACI 318-19	Building Code Requirements for Structural Concrete and Commentary

B. APA - The Engineered Wood Association (APA)

1.	APA PS 1-09	Structural Plywood

C. Bureau of Reclamation (Reclamation)

1.	RSHS	Reclamation Safety and Health Standards, including revisions posted at: https://www.usbr.gov/safety/rshs/

D. Western Wood Products Association (WWPA)

1.	WWPA WLGR-11	Western Lumber Grading Rules

1.03 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 - Submittals.

B. RSN 03 11 10-1, Formwork Design and Drawings:

1. Include design and drawings for:
   a. all elements requiring formwork
2. Designed and stamped by registered professional engineer.
3. Design Calculations, stamped by a registered professional engineer.
4. Include product data for any alternative formwork materials.

1.04 DESIGN

A. Design formwork in accordance with RSHS.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 FORM MATERIALS

A. Sheathing or Lining: Conform to Table 03 11 10A - Form Sheathing or Lining Materials, or provide other materials which will produce equivalent results.

Table 03 11 10A - Form Sheathing or Lining Materials

Finish	Wood sheathing or lining	Steel sheathing or lining
F1	Any grade common board or plywood	Steel sheathing permitted Steel lining permitted
F2	No. 2 common or better, shiplap, or plywood	Steel sheathing permitted Steel lining permitted if approved
F3	Plane surfaces: Plywood. Warped surfaces: Plywood, or lumber which is free from knots and other imperfections and which can be cut and bent accurately without splinters or splitting. Severe curvature: May be lined with continuously supported flexible material such as Masonite or thin plywood. Use of material is subject to COR approval.	Steel sheathing permitted Steel lining not permitted.
Textured Surface	Use form liner specified for textured surface	Steel sheathing not permitted Steel lining not permitted

B. Steel sheathing is steel sheets not supported by wood backing. Steel lining is thin sheets supported by wood backing.

C. Wood Sheathing or Lining: Softwood or plywood of such kind and quality and treated or coated so that deterioration or discoloration of formed concrete surfaces due to chemical action, contamination, or uneven absorption of water from concrete is prevented.

D. Plywood: APA PS 1, Exterior, Grade B-B Concrete Form, Class I, mill oiled and edge sealed.

E. Softwood Lumber:

1. Meet requirements of WWPA WLGR for dressed lumber or worked lumber of specified grade.
2. Use common boards surfaced on both edges (S2E) in accordance with WWPA WLGR.
3. Use 6- or 8-inch wide lumber for shiplap forms.
4. Use same lumber width in forms for F2 finishes.

F. Aluminum sheathing or formwork is not permitted to be in contact with fresh concrete.

3.02 INSTALLATION

A. Construct forms to confine and shape concrete to required lines so that completed work meets specified structural deviations, surface tolerances, and finish requirements.

B. Construct forms with sufficient strength to withstand pressure from placing and vibrating concrete. Maintain in proper position.

C. Adjust formwork design and concrete placing rate to compensate for hydraulic pressures exerted on forms by concrete with high fluidity.

D. Where form vibrators are to be used, construct forms with sufficient rigidity to effectively transmit energy from form vibrators to concrete without damaging formwork or altering form position.

E. Seal surfaces and joints of forms to prevent absorption of water into forms or loss of mortar from concrete.

F. Place chamfer strips in corners of forms and at tops of wall placements to bevel edges of permanently exposed concrete surfaces. Do not bevel interior angles of intersecting concrete surfaces and edges of construction joints except as indicated on drawings.

G. Install sufficient plumb and string lines to monitor formwork positions before concrete placement. Monitor plumb and string lines during concrete placement and correct deficiencies in formwork.

H. F2 Finish:

1. Use 1 type of form sheathing or lining material for exposed F2 surfaces.
2. Construct forms to produce a uniform and consistent texture and pattern on face of concrete. Metal patches on forms are not permitted.
3. Place form sheathing or lining so that horizontal form marks are continuous across entire surface.

4. For forms constructed of plywood form lining or shiplap panels, make vertical form marks continuous for entire height of surface.

5. For forms constructed of shiplap which is not paneled, cut boards square, stagger vertical joints in boards, and place vertical joints at studs.

I. F3 Finish:

1. Construct forms to produce a uniform and consistent texture and pattern on face of concrete. Metal patches on forms are not permitted.

2. Align sheathing or lining horizontally and vertically and place to minimize joint marks on surfaces.

3. Place form sheathing or lining so that horizontal form marks are continuous across entire surface.

4. Make vertical form marks continuous for entire height of surface.

5. Fill and smooth finish voids at joints in plywood form lining or sheathing.

6. Do not construct forms continuously from lift to lift. Remove forms after concrete in a lift has hardened and reset forms for next lift.

7. Reset forms to overlap hardened concrete in previous lift by 1 inch, maximum.

8. Tighten forms snugly against hardened concrete so that forms will not spread and cause offsets or loss of mortar at construction joints when concrete placement is resumed. Provide additional bolts or form ties required to hold reset forms tight against hardened concrete.

J. Form Ties and Anchors:

1. Embed ties for holding forms.

2. Terminate ties not less than 2 diameters or twice minimum dimension of tie, whichever is greater, from formed surface of concrete, except where F1 finish is permitted.

3. Install ties so ends or end fasteners can be removed without causing spalling at face of concrete.

4. Provide form anchors as required to ensure that concrete surfaces will meet specified tolerances. Replace form anchors embedded in concrete which are loosened before placement of adjoining concrete with other supports firmly embedded in hardened concrete.

K. Cleaning and Oiling Forms:

1. Clean form surfaces of encrustations of mortar, grout, or other foreign material.

2. Coat form surfaces with a form oil which will prevent sticking and will not soften or stain concrete surfaces or cause concrete surface to become chalky or dust producing.

3.03 REMOVAL

A. Remove forms within 24-hours after concrete has gained sufficient strength to prevent damage by form removal:

1. Non-supporting forms such as sides of beams, walls, columns, and similar items: Remove after cumulatively curing at not less than 50 degrees F for 24-hours from time of concrete placement when:

   a. Concrete is sufficiently hard so as not to sustain damage by form removal operations.

   b. Curing protection is maintained.

2. Forms for elevated structural slabs or beams:

   a. Remove in accordance with ACI 318, Chapter 6.

   b. Do not remove forms until concrete has reached compressive strength of at least 80 percent of specified 28-day compressive strength, as determined by test cylinders.

3. Forms on upper sloping faces of concrete, such as forms on waterside of warped transitions: Remove as soon as concrete has attained sufficient stiffness to prevent sagging.

4. Wood forms for wall openings: Loosen as soon as loosening can be accomplished without damage to concrete to prevent excessive stress in concrete from swelling of forms. Construct forms for openings to facilitate loosening.

B. Do not remove forms until concrete strength is such that form removal will not result in perceptible cracking, spalling, or breaking of edges or surfaces, or other damage to concrete.

C. Remove forms in a manner which prevents damage to concrete.

D. Repair damaged concrete in accordance with Section 03 30 00 – Cast-in-Place Concrete.

E. Begin required repair and curing immediately after form removal.

END OF SECTION

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SECTION 03 20 00
CONCRETE REINFORCEMENT

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for items requiring reinforcement.

1. Includes furnishing and placing reinforcement for thrust blocks, collars, welded rectangular barriers, guard posts and other applicable items.

1.02 REFERENCE STANDARDS

A. American Concrete Institute (ACI)

1.	ACI 315R-18	Details and Detailing of Concrete Reinforcement (Part of SP-66-04 ACI Detailing Manual)

B. ASTM International (ASTM)

1.	ASTM A615/A615M-25	Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement
2.	ASTM A996/A996M-24	Rail-Steel and Axle-Steel Deformed Bars for Concrete Reinforcement

1.03 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals.

B. RSN 03 20 00-1, Reinforcement Diagrams and Lists:

1. Bar-placing diagrams, bar lists, and bar-bending diagrams required for reinforcement fabrication and placement:

a. Prepare bar-placing diagrams, bar lists, and bar-bending diagrams in accordance with ACI 315 and drawings.

1.04 DELIVERY, STORAGE, AND HANDLING

A. Store reinforcement of different sizes and shapes in separate piles or racks raised above the ground to avoid rusting.

B. Protect from contaminants such as grease, oil, and dirt.

C. Furnish identification after bundles are broken and tags removed.

PART 2 PRODUCTS

2.01 STEEL REINFORCING

A. Reinforcing Bars:

1. ASTM A615, Grade 60; or ASTM A996, Type A, Grade 60.
2. Deformed steel bar.

B. Dowel Bar System:

1. Smooth Dowel Bars: ASTM A615, Grade 50, plain round.
2. Speed Dowel System, as manufactured by Greenstreak, Inc. www.greenstreak.com or equal having the following essential characteristics:
   a. Manufactured to accept smooth bars.
   b. Two component system, including a closed end sleeve to allow for axial movement and a reusable attachment base.
   c. Compressibility of the system shall exceed 4,500 lbs/in^2.

2.02 ACCESSORIES

A. Chairs, hangers, spacers, and other supports:

1. Material: Concrete, metal or other approved material:
   a. Compressive strength of precast concrete blocks: Equal to or greater than surrounding concrete.
2. Exposed portions of supports on concrete surfaces designated to receive F2 or F3 finish:
   a. Galvanized, plastic coated, or corrosion-resistant material.
   b. Concrete supports not allowed.

PART 3 EXECUTION

3.01 INSTALLATION

A. Clean reinforcement surfaces of heavy, flaky rust; loose mill scale; dirt; grease; and other foreign substances before placement.

B. Field bending not allowed unless approved by COR:

1. Do not use heat to bend.

C. Do not use torch to cut.

D. Accurately place reinforcement:

1. Place reinforcement as shown on standard drawing 40-D-60003, unless otherwise shown on reinforcement design drawings.
2. Unless otherwise prescribed, placement dimensions shall be to the centerline of the bars.
3. Place reinforcement with a clear distance of 1-inch, minimum, between reinforcement and anchor bolts, form ties, or other embedded metalwork unless otherwise shown on drawings.

E. Splices:

1. Locate where shown on drawings, unless otherwise approved by COR.
2. Reinforcement may be spliced at additional locations for convenience of Contractor, subject to approval of COR.

F. Do not weld or tack weld reinforcing bars.

G. Secure reinforcement in place so that it will not be displaced during concrete placement.

H. Do not disturb reinforcement in concrete that has been placed.

I. Tolerances:

1. Maintain concrete cover over reinforcement within 1/2-inch of specified cover where specified cover is greater than 2 1/2-inches.
2. Maintain concrete cover over reinforcement within 1/4-inch of specified cover where specified cover is 2 1/2-inches or less.
3. Maintain spacing of reinforcing bars within 1-inch of required spacing.

J. Embedded Dowels:

1. Place dowels at locations shown on drawings.
2. Position dowels parallel to each other and to concrete surface as shown on drawings.
3. Maintain dowels accurately in position during concrete placement.

END OF SECTION

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SECTION 03 20 60
CONCRETE ANCHORS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include cost of concrete anchors in applicable prices offered in Price Schedules for items of work requiring concrete anchors.

1.02 REFERENCE STANDARDS

A. American Concrete Institute (ACI)

1.	ACI 355.2-22	Qualification of Post-Installed Mechanical Anchors in Concrete and Commentary
2.	ACI 355.4-19(21)	Qualification of Post-Installed Adhesive Anchors in Concrete and Commentary

B. American Institute of Steel Construction (AISC)

1.	AISC 303-22	Code of Standard Practice for Steel Buildings and Bridges
2.	AISC 360-16	Structural Steel Buildings
3.	AISC RCSC-20	Structural Joints Using High-Strength Bolts available at https://www.boltcouncil.org

C. American Society of Mechanical Engineers (ASME)

1.	ASME B1.1-2019	Unified Inch Screw Threads (UN, UNR, and UNJ Thread Forms)
2.	ASME B18.2.1-2012(R2021)	Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series)

D. ASTM International (ASTM)

1.	ASTM A108-24	Steel Bar, Carbon and Alloy, Cold-Finished
2.	ASTM A123/A123M-24	Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
3.	ASTM A153/A153M-23	Zinc Coating (Hot-Dip) on Iron and Steel Hardware
4.	ASTM A194/A194M-24	Carbon Steel, Alloy Steel, and Stainless-Steel Nuts for Bolts for High Pressure or High Temperature Service, or Both.

5.	ASTM A320/A320M-24a	Alloy-Steel and Stainless-Steel Bolting for Low-Temperature Service
6.	ASTM F436/F436M-24	Hardened Steel Washers Inch and Metric Dimensions
7.	ASTM F844-19(2024)	Washers, Steel, Plane (Flat), Unhardened for General Use
8.	ASTM F1554-20	Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength

E. Commercial Item Description (CID)

1.	CID A-A-1923A	Shield, Expansion (Lag, Machine and Externally Threaded Wedge Bolt Anchors)

F. International Code Council (ICC)

1.	IBC-21	International Building Code

G. International Code Council Evaluation Service (ICC ES)

1.	ICC ES AC308-19 (R2021)	Acceptance Criteria for Post-Installed Adhesive Anchors in Concrete Elements

1.03 SUBMITTALS

A. In accordance with Section 01 33 00 – Submittals.

B. RSN 03 20 60-1, Instructions:

1. Manufacturer's storage, handling, hole drilling, installation, pretensioning, and epoxy grouting instructions for post-installed anchors.

C. RSN 03 20 60-2, Test Data and Results:

1. Date and weather conditions.
2. Test location.
3. Deformations and loading.
4. Result of test, pass or fail.

PART 2 PRODUCTS

2.01 GENERAL

A. Unless noted on drawings, anchor rod holes in equipment support frames shall not exceed bolt diameters by more than 25 percent, up to a limiting maximum over-sizing of 1/4 inch or as indicated in AISC 360:

1. Unless otherwise specified on manufacturer’s equipment drawings or specification drawings, minimum anchor bolt diameter shall be 1/2-inch.

B. Provide tapered washers where mating surface is not square with nut.

C. Expansion (torque-controlled and displacement controlled) or adhesive post-installed anchors set in holes drilled in the concrete after concrete is placed will not be permitted in substitution for cast-in headed anchor rods except where approved prior to concrete placement.

D. Adhesive Anchors: Drilled type, installed with a chemical adhesive system:

1. Seismic qualified per IBC and ICC ES AC308 for seismic design categories A through F and cracked concrete conditions.

2. Adhesive: HIT-HY 200 epoxy adhesive manufactured by HILTI, 5825 South, 129th East Avenue, Suite F, Tulsa, OK 74134; or equal, having the following essential characteristics:

   a. Injectable, two-component hybrid adhesive, Type A or Type R, V3.

   b. Dual packaging system designed for automatic mixing during injection.

   c. Weathering resistant and works in submerged conditions.

   d. Suitable for use in diamond cored or pneumatic drilled holes.

   e. Suitable for cold weather installation as low as 14 degrees F and for in-service temperature range of -40 degrees F to 248 degrees F.

3. Threaded carbon steel anchor rod: HAS Anchor rod manufactured by HILTI, P.O. Box 21148, Tulsa, OK 74121; or equal, having the following essential characteristics:

   a. Rod material:

      1) ASTM F1554, Grade 36, Hot-dip galvanized. Typical unless noted otherwise on drawings

      2) ASTM F1554, Grade 55 Hot-dip galvanized.Where noted on drawings

   b. Thread: ASME B1.1, continuously threaded (all-thread).

   c. Fabricate with 45-degree chisel point on one end to facilitate insertion into adhesive-filled hole.

   d. Standard Steel Nuts: ASTM A563, Grade A or ASTM A194, Grade 2H. Hot-dip galvanized.

   e. Standard Steel Washers: ASTM F436, Type 1. Hot-dip galvanized.

4. Threaded stainless-steel anchor rod: HAS-R 304 or 316 SS Anchor Rod manufactured by HILTI, 5825 South, 129th East Avenue, Suite F, Tulsa, OK 74134; or equal, having the following essential characteristics:

   a. Rod material: Stainless-steel, ASTM F593, Type 304 or 316.

   b. Thread: ASME B1.1, UNC course thread series, continuously threaded (all-thread).

c. Fabricate with 45-degree chisel point on one end to facilitate insertion into adhesive-filled hole.

d. Nuts: ASTM F594, unless otherwise specified.

e. Stainless-Steel Washers: ASTM A240.

5. Rod length: Provide a rod with 3 1/2-inch minimum embedment, as shown on drawings, or as recommended by manufacturer of equipment being installed; whichever is greater.

E. Expansion Anchors:

1. Seismic qualified per ICC IBC and ICC AC 193 for seismic design category A through F and cracked concrete conditions.

2. Steel expansion anchors: Kwik Bolt TZ2 torque controlled expansion anchor body, wedge, nut, and washer as manufactured by HILTI, P.O. Box 21148, Tulsa, OK 74121; or equal, having the following essential characteristics:

a. Externally threaded wedge bolt anchor.

b. Meets the requirements of a CID A-A-1923A, Type 4 anchor.

c. Anchor Body and Wedge: Carbon steel, zinc plated in accordance with ASTM F1941.

d. Nuts and Washers: As recommended by anchor manufacturer, or as specified.

3. Stainless-steel expansion anchors: Kwik Bolt TZ2 SS 304 or 316 torque-controlled expansion anchor body, wedge, nut, and washer as manufactured by HILTI, 5825 South, 129th East Avenue, Suite F, Tulsa, OK 74134; or equal, having the following essential characteristics:

a. Externally threaded wedge bolt anchor.

b. Meets the requirements of a CID A-A-1923A, Type 4 anchor.

c. Anchor Body and Wedge: Stainless-steel, Type 304 or 316.

d. Nuts and Washers: As recommended by anchor manufacturer, or as specified.

4. Bolt Length: Provide a bolt with 3 1/2-inch minimum embedment, as shown on drawings, or as recommended by manufacturer of equipment being installed; whichever is greater.

F. Cast-in Headed Anchor Rods:

1. Steel anchor rods:

a. Rod Material: ASTM F1554:

1) Unless otherwise shown on drawings, Grade A36.

2) Hot-dip galvanize in accordance with ASTM A123.

b. Standard Steel Nuts: ASTM A563, Grade A, Heavy Hex:

1) Hot-dip galvanize in accordance with ASTM A153.

2) After galvanizing, remove excess spelter by centrifugal spinning.

c. Standard Steel Washers: ASTM F436, Type 1:

1) Hot-dip galvanize in accordance with ASTM A153.

2) After galvanizing, remove excess spelter by centrifugal spinning.

2. Stainless-steel anchor rods:

a. Rod Material: ASTM A320:

1) Unless otherwise shown on drawings, Grade B8.

b. Nuts: ASTM A194, Grade 8, Heavy Hex, unless otherwise specified.

c. Stainless-steel washers to match rod material.

PART 3 EXECUTION

3.01 GENERAL

A. Fieldwork, including cutting and threading is not permitted on galvanized items.

B. Dissimilar metals: Protected from galvanic corrosion by means of pressure tapes, coatings, or isolators.

C. Grouting of anchor bolts with epoxy grouts, where specified.

D. For anchor rods, a minimum of three threads shall be exposed past the nut.

3.02 PREPARATION

A. Allow concrete to attain design strength as determined by Government before anchor installation.

B. Position reinforcement and cast-in anchors to avoid interference. No portions of stainless-steel anchors shall be in contact with the reinforcement.

C. Locate reinforcement to ensure it is not contacted during hole drilling.

1. Follow manufacturer's recommendations when embedded steel or reinforcement is encountered during drilling for anchors.

D. Cast-in Headed Anchor Rods:

1. Placed accurately and held in correct position while the concrete is placed to prevent movement.

2. Formed in concrete and metalwork grouted in place in accordance with Section 03 62 20 – Nonshrink Grout.

a. Thoroughly clean surfaces of metalwork in contact with concrete.

3. After anchor rods have been embedded, protect threads by grease and nuts run on.

E. Post-Installed Adhesive Anchors:

1. Use of adhesive or capsule anchors: Subject to the following conditions:

a. Approval for specific application and from supplier of equipment to be anchored, if applicable:

1) Overhead applications (such as pipe supports): Not allowed.

b. Anchor diameter and grade of steel: Per contract documents or equipment supplier specifications:

1) Anchor: Threaded or deformed full length of embedment and free of rust, scale, grease, and oils.

c. Adhesive capsules of different diameters may be used to obtain proper volume for embedment, but no more than two capsules per anchor may be used.

d. When installing different diameter capsules in same hole, install larger diameter first.

e. Any extension or protrusion of capsule from hole is prohibited.

2. Two-part injection adhesives are to be supplied in individually sealed, equal volume cartridges.

3. Epoxy components: Thoroughly mixed when dispensed through static spiral mixing/dispensing nozzle:

a. Bulk hand mixed and measured adhesives are not allowed.

F. Expansion Anchors:

1. Use of expansion or wedge type anchors: Subject to the following conditions. Expansion or wedge anchors may not be substituted for cast-in or adhesive set anchors:

a. Approval for specific application and from supplier of equipment to be anchored, if applicable:

1) Overhead applications (such as pipe supports): Not allowed.

G. Contractor shall supply current evaluation report from International Conference of Building Officials for brand of expansion or adhesive anchor systems to be used.

H. Galvanizing Repair:

1. Repair damaged galvanized surfaces of cast-in headed anchor rods in accordance with Section 09 26 20 – Coatings.

3.03 INSTALLATION

A. In accordance with manufacturer's instructions.

B. Locate anchors to ensure that anchors or drilling does not hit reinforcing bars.

1. If embedded steel or reinforcement is encountered during drilling, follow manufacturer's recommendations.

C. If drilling water is used, clean exposed concrete surfaces immediately after drilling to prevent discoloration.

D. Cast-in Headed Anchor Rods:

1. Installation of anchor rods shall be in accordance with AISC 303, Section 7.5.
2. Anchor rod nuts should be "drawn down tight" or "snug-tightened" in accordance with AISC RCSC as columns and bases are erected.

E. Post-Installed Adhesive Anchors:

1. Follow installation recommendations by anchor system manufacturer carefully, including maximum hole diameter.
2. When drilling water is used, clean surfaces of concrete to remain exposed immediately to prevent discoloration.
3. Holes shall have rough surfaces, such as can be achieved using rotary percussion drill.
4. Following drilling, clean holes with water to remove cuttings, followed by compressed air to ensure holes are dry.
5. Anchors shall be left undisturbed and unloaded for full adhesive curing period.
6. Anchors shall be installed in concrete having a minimum age of 21 days at time of anchor installation.
7. Concrete temperature required to be compatible with curing requirements of adhesives per adhesive manufacturer:
    a. Anchors: Not be placed in concrete below 25 degrees F.

F. Expansion Anchors:

1. Follow installation recommendations by anchor system manufacturer carefully, including maximum hole diameter.
2. When drilling water is used, clean surfaces of concrete to remain exposed immediately to prevent discoloration.
3. Following drilling, clean holes with water to remove cuttings, followed by compressed air to ensure holes are dry.

3.04 CONTRACTOR FIELD INSPECTION

A. Periodic Inspection:

1. In accordance with ACI 355.2 for post-installed anchors and ACI 355.4 for post-installed adhesive anchors.
2. Inspect initial installations of each type and size of post-installed anchor:
    a. Subsequent installations of the same anchor type and size by the same personnel do not require inspection.
    b. Changes to the anchor type, size or installation personnel require inspection.
3. Inspect:
    a. Drill bits.
    b. Drilling method and hole cleaning meet manufacturer's instructions.
    c. Anchor edge distance and spacing.
    d. Hole diameter and depth.
    e. Anchor type, material, diameter, and length.
    f. Adhesive identification and expiration date.

3.05 CONTRACTOR FIELD QUALITY TESTING

A. Pull-out test.

1. Test one of every 10 anchors installed of the same size and type. The COR will select anchor bars to be tested.
    a. If less than 10 anchors are installed of a certain size and type, test one anchor.
2. Test anchors after epoxy grout has achieved 4,000 psi compressive strength.
3. Notify COR in writing at least 7 calendar days in advance of performing tests.
4. Application of load shall not damage bar. Replace damaged bars at Contractor's expense.
5. Testing apparatus: In accordance with ASTM D4435.
6. Test load:
    a. 60 percent of yield strength of bar.
    b. Apply load smoothly and rapidly.
    c. Maintain load for at least 10 minutes.
7. Load measurement system:
    a. Calibrated by an independent testing laboratory within past 6 months.
    b. Accuracy: Plus or minus 5 percent of actual load.

8. Record data for each test:
    a. Record load, deformation, and time at three times during the test.
    b. Record at approximately 1 minute, 5 minutes, and 10 minutes of the test.
9. Acceptance criteria:
    a. Tested bar holds at least 60 percent of yield strength of bar during test period.
    b. Failure:
        1) Application of test load is not possible.
        2) Total bar movement between 1 minute and 10 minutes of test exceeds 0.08 inches.
10. Replace anchor bar that fails test with a new anchor bar and test two additional anchor adjacent bars.

END OF SECTION

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SECTION 03 30 00
CAST-IN-PLACE CONCRETE

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in the Price Schedules for Steel Manifold Piping as applicable:

1. Yard Pipe Thrust Blocks.
2. Welded Rectangular Barriers.

B. Cost: Include cost of concrete for line pipe in prices offered in Price Schedules for Line Pipe, as applicable:

1. Includes:
   a. Thrust Blocks and Collars.
   b. Guard Posts.
   c. Welded Rectangular Barriers.

C. Cost: Include cost of other concrete items not listed in prices offered in Price Schedules other items of work requiring concrete.

1.02 ACRONYMS

A. NRMCA: National Ready Mixed Concrete Association.

1.03 DEFINITIONS

A. Supplementary Cementitious Materials (SCM): Cementitious materials other than Portland cement.

1.04 REFERENCE STANDARDS

A. American Concrete Institute (ACI)

1.	ACI 117-10(2015)	Tolerances for Concrete Construction and Materials
2.	ACI 201.2R-16	Guide to Durable Concrete
3.	ACI 301-20	Structural Concrete
4.	ACI 304R-00(2009)	Measuring, Mixing, Transporting, and Placing Concrete
5.	ACI 305.1-14	Hot Weather Concreting

	6.	ACI 306.1-90(2002)	Cold Weather Concreting
B.	ASTM International (ASTM)		
	1.	ASTM C31/C31M-25b	Making and Curing Concrete Test Specimens in the Field
	2.	ASTM C33/C33M-24a	Concrete Aggregates
	3.	ASTM C39/C39M-24	Compressive Strength of Cylindrical Concrete Specimens
	4.	ASTM C42/C42M-20	Obtaining and Testing Drilled Cores and Sawed Beams of Concrete
	5.	ASTM C94/C94M-25	Ready-Mixed Concrete
	6.	ASTM C114-24	Chemical Analysis of Hydraulic Cement
	7.	ASTM C117-23	Materials Finer than 75-µm (No. 200) Sieve in Mineral Aggregates by Washing
	8.	ASTM C136-25	Sieve Analysis of Fine and Coarse Aggregates
	9.	ASTM C138/C138M-24a	Density (Unit Weight), Yield, and Air Content (Gravimetric) of Concrete
	10.	ASTM C143/C143M-20	Slump of Hydraulic-Cement Concrete
	11.	ASTM C150/C150M-24	Portland Cement
	12.	ASTM C157/C157M-24e1	Length Change of Hardened Hydraulic-Cement Mortar and Concrete
	13.	ASTM C171-20	Sheet Materials for Curing Concrete
	14.	ASTM C231/C231M-25	Air Content of Freshly Mixed Concrete by the Pressure Method
	15.	ASTM C260/C260M-24	Air-Entraining Admixtures for Concrete
	16.	ASTM C295/C295M-19	Petrographic Examination of Aggregates for Concrete
	17.	ASTM C309-25	Liquid Membrane-Forming Compounds for Curing Concrete
	18.	ASTM C494/C494M-24	Chemical Admixtures for Concrete
	19.	ASTM C566-25	Total Evaporable Moisture Content of Aggregate by Drying
	20.	ASTM C595/C595M-25	Blended Hydraulic Cements
	21.	ASTM C618-25a	Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete
	22.	ASTM C920-18(2024)	Elastomeric Joint Sealants

23.	ASTM C989/C989M-25	Slag Cement for Use in Concrete and Mortars
24.	ASTM C1064/C1064M-23	Temperature of Freshly Mixed Hydraulic-Cement Concrete
25.	ASTM C1260-23	Potential Alkali Reactivity of Aggregates (Mortar-Bar Method)
26.	ASTM C1293-23ae1	Determination of Length of Change of Concrete Due to Alkali-Silica Reaction
27.	ASTM C1315-25	Liquid Membrane-Forming Compounds Having Special Properties for Curing and Sealing Concrete
28.	ASTM C1567-25	Determining the Potential Alkali-Silica Reactivity of Combination of Cementitious Materials and Aggregate (Accelerated Mortar-Bar Method)
29.	ASTM C1602/C1602M-22	Mixing Water Used in the Production of Hydraulic Cement Concrete
30.	ASTM D1751-23	Preformed Expansion Joint Filler for Concrete Paving and Structural Construction (Nonextruding and Resilient Bituminous Types)
31.	ASTM D1752-18(2023)	Preformed Sponge Rubber Cork and Recycled PVC Expansion Joint Fillers for Concrete Paving and Structural Construction

C. U.S. Army Corps of Engineers (USACE)

1.	USACE CRD-C 662-10	Determining the Potential Alkali-Silica Reactivity of Combinations of Cementitious Materials, Lithium Nitrate Admixture and Aggregate (Accelerated Mortar-Bar Method)

D. International Concrete Repair Institute (ICRI)

1.	ICRI 310.2-13	Selecting and Specifying Concrete Surface Preparation for Sealers, Coatings, and Polymer Overlays (formerly No. 03732)

E. Bureau of Reclamation (USBR)

1.	USBR M-47	Repair of and Maintenance of Concrete, August 2015 (Part 2 of "Guide to Concrete Repair, Second Edition" available at: www.usbr.gov/tsc/techreferences/mands/ma

		nds-pdfs/Guide2ConcreteRepair2015_Final.pdf
2.	USBR Concrete Manual	Concrete Manual, Eighth Edition, Revised Reprint, 1981

F. National Sanitation Foundation International (NSF)

1.	NSF/ANSI 61-13	Drinking Water System Components - Health Effects

1.05 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals.

B. RSN 03 30 00-1, Approval Data:

1. Mix Design: For each concrete mix:
   a. Mixture proportion.
   b. Material sources:
      1) Name and manufacturer of each cementitious material.
         a) Cementitious Materials Certification and Test Reports:
            i. Less than 3-months old.
      2) Name of aggregate source(s).
      3) Product name and manufacturer of admixtures to be used in mix. Certify that admixtures:
         a) Contain no purposefully added chlorides.
         b) Chloride ion limits in accordance with ACI 201.2R.
      4) Government reserves the right to require submission of samples of concrete materials for testing before or during use in concrete.
   c. Physical properties:
      1) Compressive strength:
         a) Test data: ACI 301, paragraph 4.2.3.4.
         b) Field test data: Performed within past 24-months.
         c) Trial mixtures:
            i. Incorporate admixtures that will be used in production mixes into trial mixes.
            ii. Results from trial batches made within past 6-months.
            iii. Trial mix test results, three 6-inch diameter cylinders each at 7- and 28-days.

iv. Average compressive strength of trial batch cylinders at specified design age.

d. Resubmit mix design for change in material source or type.

2. Name and manufacturer of curing compounds, and joint filler.
   a. Include application instructions for curing compound.
3. Aggregate and Water Certifications and test reports:
   a. Sealed by Professional Engineer.
   b. Less than 12-months old.
   c. Certifications and test reports:
      1) Aggregate producer for:
         a) ASTM C33 physical properties
         b) ASR testing reports for each aggregate source.
      2) Mixing water: ASTM C1602.
   d. Submittal of certifications and test reports shall not relieve Contractor of responsibility for furnishing materials meeting specified requirements.
4. Name and manufacturer of any proposed finishing aid.
   a. Include data sheet and manufacturer's application instructions.

C. RSN 03 30 00-2, Concrete Placement Schedule:
   1. Complete, detailed concrete placement schedule showing Contractor's plan for individual placements including placement of reinforcement and embedded items.
   2. Detail as necessary to show location, sequence, and date of concrete placements scheduled for each item of concrete work.
   3. Show submittal schedule for placement and reinforcement drawings.

D. RSN 03 30 00-3, Certifications:
   1. NRMCA Certification of Production Facilities. NRMCA certification shall be current and include automatic digital recording of cementitious materials, aggregate, water, and chemical admixtures. NRMCA certification shall be maintained throughout duration of project.
   2. ACI Aggregate Testing Technician certification(s).
   3. ACI Concrete Field-Testing Technician certification(s).
   4. ACI Concrete Strength Testing Technician certification(s).

E. RSN 03 30 00-4, Test Reports.
   1. Aggregate test results as required by Table 03 30 00B – Contractor Batch Plant Quality Testing.

2. Concrete test reports as required by Table 03 30 00E – Contractor Field Quality Testing.

1.06 QUALIFICATIONS

A. Ready mix plant: Currently certified by NRMCA. NRMCA certification shall include automatic digital recording of cementitious materials, aggregate, water, and chemical admixtures. NRMCA certification shall be maintained throughout duration of project.

B. ACI Aggregate Testing Technician: Currently certified ACI Aggregate Testing Technician - Level 1.

C. Flatwork foreman, minimum 5-years' experience with similar type placements.

D. ACI Concrete Field-Testing Technician: Currently certified ACI Concrete Field-Testing Technician - Grade I.

E. ACI Strength Testing Technician: Currently certified ACI Strength Testing Technician.

1.07 DELIVERY, STORAGE, AND HANDLING

A. Furnish legible and digitized batch ticket with each batch of concrete in accordance with ASTM C94. Deliver ticket to COR at jobsite during batch delivery.

B. Sponge rubber joint filler storage:

1. Store in protected area at temperature of 70 degrees F (21 degrees C) or less.
2. Do not expose to direct sun.

1.08 PRE-PLACING MEETING

A. At least 14-days before first concrete placement, meet with COR for pre-placement meeting.

B. Coordinate time and place of meeting with COR at least 14-days before meeting.

C. Contractor attendees shall be at least the following:

1. Contractor onsite supervisor.
2. Concrete forming, placing, and finishing onsite supervisor(s); Contractor or subcontractor(s) employee as applicable.
3. Technical specialist from ready-mix supplier.
4. Concrete pump subcontractor onsite supervisor, if applicable.
5. Testing agency onsite supervisor(s).

PART 2 PRODUCTS

2.01 CEMENTITIOUS MATERIALS

A. Portland cement:

1. ASTM C150, Type V:

a. Meet equivalent alkalies ($Na_2O + 0.658K_2O$) of 0.60% maximum:

1) Low-alkali limitation for portland cement may be waived when tests of concrete aggregate source show that low-alkali cement is not required for ASR mitigation. See Concrete Aggregate Materials article (2.03).

b. Meet early stiffening requirements of ASTM C150 - Table 4.

B. Blended hydraulic cement:

1. ASTM C595, Type IL (MS), IS (MS) or IP (MS).

2. Meet equivalent alkalies (Na2O + 0.658K2O) of 0.60% maximum. Meet equivalent alkali's requirement of ASTM C595, Table 2, Option G or Table 3.

C. SCM:

1. Pozzolan:

a. ASTM C618, Class F or N:

1) Except:

a) Sulfur trioxide, maximum: 4.0 percent.

b) Loss on ignition, maximum: 2.5 percent (Class F) or 3.0 percent (Class N).

2) In addition:

a) Calcium oxide, maximum: 8.0 percent.

b) Pozzolan "R" factor less than 2.5. Pozzolan with this "R" factor shall not decrease sulfate resistance of concrete.

i. R: (C-5)/F

ii. C: Calcium oxide content of pozzolan in percent determined in accordance with ASTM C114.

iii. F: Ferric oxide content of pozzolan in percent determined in accordance with ASTM C114.

2. Slag Cement:

a. ASTM C989, Grade 100 or 120.

2.02 WATER

A. ASTM C1602, including optional requirements of Table 2.

2.03 AGGREGATE MATERIALS

A. Fine aggregate: ASTM C33:

1. Percent material passing No. 200 sieve: Less than 3 percent.

B. Coarse aggregate: ASTM C33, Class 4S, Size No. specified in Table 03 30 00A – Concrete Mixes.

C. Alkali Silica Reaction (ASR):

1. Test fine and coarse aggregates in accordance with ASTM C1260 for potential deleterious ASR:
    a. For ASTM C1260, and other tests when required, continue readings for 28-days after the zero readings.
    b. Acceptance criteria specified below are based on 28-day readings after the zero readings.
    c. Expansion is no greater than 0.10 percent: Aggregates are acceptable.
    d. Expansion is greater than 0.10 percent:
        1) Test aggregates according to ASTM C1567 using components (e.g. coarse aggregate, fine aggregate, cementitious materials, and ASR inhibiting admixtures) in proportions proposed for mixture design. Continue readings for 28 days:
            a) For mixes using lithium admixtures use test procedure USACE CRD-C 662.
            b) Expansion of proposed mixture design test specimens, tested in accordance with ASTM C1567 does not exceed 0.10 percent:
                i. Aggregates are acceptable.
            c) Expansion of proposed mixture design test specimens is greater than 0.10 percent:
                i. Aggregates are not acceptable unless adjustments to mixture design can reduce expansion to less than 0.10 percent or testing by ASTM C1293 indicates aggregates will not experience deleterious expansion.
        2) Use tested materials. Materials may be rejected if they do not match tested materials.
2. ASTM C1293 test results may be substituted for ASTM C1260 test results:

a. Average ASTM C1293 concrete prism expansion less than 0.04 percent at one year: Aggregates acceptable.

D. Appropriate for use in accordance with ASTM C295.

2.04 ADMIXTURES

A. Air-entraining admixture: ASTM C260.

B. Chemical admixtures:

1. Do not use chemical admixtures which contain more than 0.1 percent chloride, by weight.
2. Admixtures shall be compatible with each other.
3. Allowable chemical admixtures:
   a. ASTM C494, Type A, D, F, G, or S.
   b. ASTM C494, Type C and E, provided they do not contain chlorides.

C. Specialized chemical admixtures:

1. When batch plant has not previously used a specialized chemical admixture, admixture manufacturer shall provide on-site representative to assist with mix design and to train batch plant personnel in dispensing and mixing operations.
2. Do not use specialized chemical admixtures which contain more than 0.1 percent chloride, by weight.
3. Alkali Silica Reaction (ASR) Inhibiting Admixture:
   a. Lithium Nitrate Admixture for ASR mitigation of reactive aggregates having the following characteristics:
      1) Meets NSF/ANSI 61.
      2) Nominal 30 percent aqueous solution of Lithium Nitrate:
         a) Density: 10 pounds/gallon (1.2 kg/L).
         b) Approximate chemical constituents (percent by mass):
            i. LiNo3 (Lithium Nitrate): 30 plus or minus 0.5.
            ii. SO4-2 (Sulfate Ion), maximum: 0.1.
            iii. Cl- (Chloride Ion), maximum: 0.2.
            iv. NA+ (Sodium Ion), maximum: 0.1.
            v. K+ (Potassium Ion), maximum: 0.1.
   b. Coordinate with manufacturer regarding Lithium Nitrate dosage.
   c. Do not use Lithium Nitrate Admixture for concrete in continuous or nearly continuous contact with water.
4. Extended set control admixture:

a. MasterSet Delvo hydration controlling admixture manufactured by Master Builders Solutions. https://www.master-builders-solutions.com/en-us; or equal, with the following essential characteristics:

1) Meets ASTM C494, Type B.

2) Retards setting.

3) Does not reduce concrete strength.

b. Use within manufacturer's time limits.

c. Include admixture on batch ticket.

d. Admixture quantity required to stabilize concrete shall be pre-determined using jobsite materials. Initial concrete setting time shall be monitored and adjusted during project by qualified concrete technician.

2.05 CURING MATERIALS

A. Water: ASTM C1602, including optional requirements of Table 2.

B. Curing compound: Per Table 03 30 00F and 03 30 00G:

1. Capable of meeting moisture retention at manufacturer's application rate.

2. Meet Federal, State, Tribal and local regulations for VOCs.

C. Sheet materials:

1. Polyethylene film: ASTM C171, white opaque.

2. White burlap-polyethylene sheeting: ASTM C171.

2.06 ACCESSORIES

A. Sponge rubber joint filler:

1. ASTM D1752, Type I, except:

a. Test specimen compression load: 50 to 150 pounds per square inch.

2. Joint filler adhesive: Non-bituminous adhesive recommended by filler manufacturer.

B. Bituminous joint filler: ASTM D1751.

C. Elastomeric sealant: ASTM C920, polyurethane, Use M, Grade NS, Class 25.

D. Evaporation control:

1. MasterKure ER 50 manufactured by Master Builders Solutions. https://www.master-builders-solutions.com/en-us; or equal having the following essential characteristics:

a. Monomolecular film forming compound applied to exposed concrete slab surfaces for temporary protection from rapid moisture loss.

b. For application after finishing and prior to applying curing compound.

c. For use when the evaporation rate is high.

2. Do not use as finishing aid.

2.07 MIX

A. Contractor shall design and adjust concrete mix.

1. Government reserves the right to adjust mix proportions when need for adjustment is indicated by results of materials testing.

a. When required, adjustment of mix proportions by Government will be in accordance with USBR Concrete Manual.

B. Cementitious materials options:

1. Specified Portland cement plus specified pozzolan by percent weight specified in Table 03 30 00A – Concrete Mixes.

2. Specified Portland cement plus specified slag cement by percent weight specified in Table 03 03 00A – Concrete Mixes.

3. Blended Hydraulic: ASTM C595 provided, specified Portland cement with percent of specified pozzolan or specified slag cement specified in Table 03 30 00A – Concrete Mixes.

C. Design concrete mixes in accordance with Table 03 30 00A – Concrete Mixes. General concrete mix shall be used for concrete unless otherwise specified.

1. Net water-cementitious materials ratio (w/c) is maximum, by weight. Cementitious material weight is cement plus SCM.

2. Slump at point of placement: In accordance with ASTM C143.

3. Air Entrainment: Percent air by volume of concrete as discharged at point of placement, in accordance with ASTM C231.

Table 03 30 00A – Concrete Mixes

Mix No	Feature	f'c (lb/in^2)	Max w/c*	NMSA**	Percent SCM*** A: Class F Pozzolan B: Slag Cement	Slump (inches)	Air Content (percent)	Notes
1	General Concrete and other not specified	4500 at 28-days	0.45	57, 67	A: 20 plus or minus 5	2 to 4	4.5 to 7.5	1

2	High Slump Structural Concrete	4500 at 28-days	0.45	57, 67	A: 20 plus or minus 5	4 to 7	4.5 to 7.5	1, 2
3	Interior Floor Slab	4500 at 28-days	0.45	57, 67	A: 20 plus or minus 5	2 to 4	N/A	1, 2
4	Pipe Encasement / Thrust Blocks / Collars	4500 at 28-days	0.45	57, 67	A: 20 plus or minus 5	2 to 4	4.5 to 7.5	1,2
5	Low Water Crossing	4500 at 28-days	0.45	57, 67	A: 20 plus or minus 5	2 to 4	4.5 to 7.5	1,2

* Maximum water/cementitious materials ratio.
**Nominal Maximum Size Aggregate.
*** SCM as percent of total cementitious material, by weight.

NOTES:

1. Ternary blended cementitious materials which meet the specifications may be submitted for approval.
2. Concrete with ASTM C494 Type F high-range water-reducing admixtures, or Type G high-range water-reducing and retarding admixtures:
   - i Admixture shall be incorporated into trial batch or historical data.
   - ii. Use slump appropriate for placing conditions.
   - iii. On-Site additions of high-range water reducers to be done by concrete supplier's QC personnel only.

D. Submit design mixes for each type and strength of concrete substantiated by either laboratory trial batch or field performance methods as specified in ACI 301. For trial batch method, mix shall be proportioned and stamped by a professional engineer.

E. Concrete trial mixes:

1. Average compressive strength of trial batch cylinders at design age: Design strength plus 1,200 pounds per square inch for concrete between 3,000 and 5,000 psi.
2. Admixtures to be used in mix shall be incorporated into mix design submitted for approval.
3. Air content: Within 1 percent of top of specified range.
4. Slump: Within 1-inch of top of specified range.

2.08 BATCHING, MIXING, AND TRANSPORTING

A. Batch plant: NRMCA certified with automatic digital recording of cementitious materials, aggregate, water, and chemical admixtures.

B. Manufacture and deliver in accordance with ASTM C94 and ACI C304R:

1. Prepare batch ticket in accordance with ASTM C94 for every batch of concrete.

C. Cold weather: When air temperature has fallen to or is expected to fall below 40 degrees F, prepare ingredients and mix in accordance with ACI 306.1:

1. Do not use frozen materials or materials containing ice or snow.
2. Uniformly heat water and aggregates before mixing to obtain concrete mixture temperature of not less than 50 degrees.

D. Hot Weather: When precautions are necessary, prepare ingredients and mix in accordance with ACI 305.1:

1. Cool ingredients before mixing to maintain specified maximum concrete temperature at time of placement.
2. Mixing water may be chilled or chopped ice may be used to control temperature, provided water equivalent of ice is calculated to total amount of mixing water. Ice replacing batch water shall be melted prior to discharge.
3. Using liquid nitrogen to cool concrete is Contractor's option.

E. Prevent appreciable segregation of ingredients.

F. Place concrete within 90-minutes from introduction of cement to water or aggregates:

1. For placing times exceeding 90-minutes, extended set control admixtures may be used when approved by COR.

2.09 CONCRETE TEMPERATURE

A. Concrete temperature at placing:

1. 50 to 85 degrees F (10 to 29 degrees C).

2.10 CONTRACTOR SOURCE QUALITY TESTING

A. Perform sampling, testing, and reporting as required in Table 03 30 00B – Batch Plant Testing:

1. Independent testing agency shall meet requirements specified in Section 01 46 20 – Testing Agency Services.
2. Personnel conducting tests: Qualified as ACI Aggregate Testing Technician, Level 1; or equal.

B. Perform tests at least as often as frequencies specified in Table 03 30 00B – Batch Plant Testing.

C. Notify COR immediately of test results showing failure of materials to meet specifications. Provide passing test to COR within 24-hours. Submit reports of test results as specified.

2.11 BATCH PLANT TESTING

Table 03 30 00B – Batch Plant Testing

TESTS OF	TEST STANDARD	STANDARD TITLE	REQUIREMENT	TESTING FREQUENCY
Aggregate Gradation	ASTM C136	Sieve Analysis of Fine and Coarse Aggregates	Fine and Coarse Aggregate meets sizing requirements per ASTM C33.	At beginning of placing each mix. At change in mix design. At least every 500 yd^3 of placing a mix.
Aggregate Fines content	ASTM C117	Materials Finer than 75-µm (No. 200) Sieve in Mineral Aggregates by Washing	Fine aggregate meet specified allowable fines content (material passing No. 200 sieve)	At beginning of placing each mix. At change in mix design. At least every 500 yd^3 of placing a mix.
Aggregate moisture content	ASTM C566	Total Evaporable Moisture Content of Aggregate by Drying	Verify that moisture meter at batch plant is accurate with the material batched.	At beginning of placing each mix. At change in mix design. At least every 500 yd^3 of placing a mix.

PART 3 EXECUTION

3.01 PREPARATION

A. Remove standing water, mud, and debris from foundation surfaces to be covered by concrete.

B. Prepare rock surfaces free from oil, objectionable coatings, and loose, semidetached, and unsound fragments. Immediately before placement of concrete, wash rock surfaces with air-water jet and dry to uniform surface-dry condition.

C. Prepare earth foundations free from frost or ice.

D. Thoroughly moisten surfaces of absorptive foundations to be covered with concrete so that moisture will not be drawn from fresh concrete. Keep subgrade moisture uniform without puddles or dry areas.

E. Clean, roughen, and surface dry surfaces of construction joints to be covered with fresh concrete:

1. Remove laitance, loose or defective concrete, coatings, and sand, curing compound, and other foreign material.

2. Sandblast, steel shotblast, or high-pressure water jet surfaces, or use other method approved by COR to create a surface equivalent to or larger than CSP 5 in accordance with ICRI 310.2.

3. Wash surface thoroughly, and surface dry immediately before placement of adjoining concrete.

3.02 PLACING

A. Notify COR at least 24-hours before placing concrete.

B. Provide placement checkout cards in a watertight container. COR will approve placement card format:

1. Sign each line item indicating work is completed in accordance with specifications.

2. Obtain COR signature after each line item.

3. Do not place concrete without required signatures.

4. Use of placement checkout cards shall be required by COR.

5. Placement card shall be stored in a watertight container located at the placement area.

C. Do not place concrete without approval of COR.

D. Place concrete in presence of COR.

E. Placing concrete shall be performed under direct supervision of a qualified flatwork foreman.

F. Allow at least 7-days between adjacent placements, or as approved by COR.

G. Do not use aluminum pipes and chutes for placing or pumping concrete.

H. Adding water to concrete batch at site will be allowed only once and only when approved by COR:

1. Add additional water before concrete is discharged.

2. Do not exceed specified water to cement ratio.

3. After water is added, concrete shall be mixed for at least 30 revolutions of mixer drum at mixing speed.

4. Record added water on the batch ticket to nearest gallon.

I. Adding air entraining admixtures to concrete batch at site will be allowed only once when approved by COR:

1. After air entraining admixture is added, concrete shall be mixed for minimum of 30 revolutions of mixer drum at mixing speed.

2. Take slump and air content after air-entraining admixture addition and additional revolutions.

3. Record added air entraining admixture on batch ticket to nearest ounce.

J. Adding high range water reducer to concrete batch at site shall only be performed by concrete supplier's technical specialist.

1. After high range water reducer admixture is added, concrete shall be mixed for minimum of 30 revolutions of mixer drum at mixing speed.

2. Take slump and air content after high range water reducer addition and additional revolutions.

3. Record added high range water reducer admixture on batch ticket to nearest ounce.

K. Furnish legible and digitized batch ticket with each batch of concrete in accordance with ASTM C94. Ticket shall include additions made to the load after batching (water, air adjustment, admixtures). Deliver ticket to COR at jobsite during batch delivery.

L. Do not use concrete which has become so stiff that concrete cannot be properly placed.

M. Place formed concrete in continuous, approximately horizontal layers. Do not exceed 20-inches in depth of layers.

N. Vibrate concrete until concrete has been consolidated to maximum practical density, is free from pockets of coarse aggregate, and closes snugly against surfaces of forms and embedded materials.

O. Hot Weather:

1. Place concrete in accordance with ACI 305.1.

2. Protect reinforcing steel so that steel temperature does not exceed ambient air temperature immediately before placing concrete.

3. Fog spray forms, reinforcing steel, and subgrade just before placing concrete. Keep subgrade moisture uniform without puddles or dry areas.

P. Cold weather:

1. Place concrete in accordance with ACI 306.1.

2. Do not place concrete on frozen subgrade or subgrade containing frozen materials.

3.03 FINISHING

A. Notify COR before finishing concrete.

B. Finish concrete in presence of Government inspector unless inspection is waived in each specific case.

C. Perform concrete finishing under the direct supervision of a qualified flatwork foreman.

D. Finish surfaces as specified in Table 03 30 00F – Formed Surfaces and Table 03 30 00G – Unformed Surfaces.

E. Where finishes are not specified or shown on drawings for a particular surface, finish concrete as specified for similar work.

F. Formed surfaces:

1. Finish class is designated by symbols F1, F2.
2. Finish F1:
    a. Applies to formed surfaces to be covered by fill material, grout, or concrete, and construction joint surfaces as specified in Table 03 30 00F – Formed Surfaces.
    b. Protect form tie rod ends on surfaces in contact with fill material from moisture where they will be below water table or waterline:
        1) Recess tie rod ends and fill recess with dry pack or other material approved by COR.
    c. Cut off flush with formed surface form tie rod ends on surfaces in contact with concrete or fill material and above maximum water table or waterline elevation.
3. Finish F2:
    a. Applies to exposed formed surfaces not permanently concealed by fill material, grout, or concrete, and not required to receive finish F4, and to contraction joint surfaces and expansion joint surfaces as specified in Table 03 30 00F – Formed Surfaces:
        1) Recess tie rod ends and fill recess with dry pack or other material approved by COR.

G. Unformed surfaces:

1. Do not use dry portland cement or additional water during finishing.
2. Do not use “jitterbugs” or other tools to force coarse aggregate away from surface.
3. Finish class is designated by symbols U1, U2, or broom finish.
4. Finish U1 (Screeded Finish):
    a. Applies to unformed surfaces to be covered by fill material, grout, or concrete as specified in Table 03 30 00G – Unformed Surfaces.
    b. Use as first stage of finish U2.
    c. After concrete is placed and consolidated, strike off and level concrete to produce even uniform surface.
5. Finish U2 (Floated Finish):

a. Applies to unformed surfaces not permanently concealed by fill material, grout, or concrete, as specified in Table 03 30 00G – Unformed Surfaces.

b. Begin floating as soon as screeded surface has sufficiently stiffened and bleed water sheen has disappeared.

c. Use hand- or power-driven equipment.

d. Finish surface with minimum floating necessary to produce surface that is free of screed marks and is uniform in texture.

6. Finish U3 (troweled finish):

a. Applies to unformed surfaces where appearance and porosity is considered by Government to be of special importance as specified in Table 03 30 00G – Unformed Surfaces.

b. Begin steel troweling after bleed water has disappeared and floated surface has sufficiently hardened to prevent an excess of fine material from being drawn to surface.

c. Trowel with firm pressure to flatten sandy texture of floated surface.

d. Trowel to a dense uniform surface free from blemishes and trowel marks. Do not excessively trowel surface.

7. Broomed finish:

a. Apply broom finish immediately after concrete receives U2 finish.

b. Edge transverse joints before brooming.

c. Produce a scored surface by brooming with fiber-bristle brush in direction transverse to that of traffic with adjacent strokes slightly overlapping.

d. Finished surface shall have uniform appearance and be free of abrupt corrugation exceeding 1/8-inch in depth.

e. Brooming shall eliminate flat surface left by the surface face of the edger.

8. Slope interior surfaces for drainage where shown on drawings or as directed by COR. Slope surfaces exposed to weather for drainage as directed by COR.

9. Slope narrow surfaces, such as tops of walls and curbs, approximately 3/8-inch per foot of width, unless use of other slopes or level surface is indicated on drawings or is directed by COR.

10. Slope broader surfaces; such as walks, platform, and decks; approximately 1/4-inch per foot unless use of other slopes or level surfaces is indicated on drawings or is directed by COR

3.04 JOINTS AND EDGES

A. Construction joints (CJ):

1. Construction joints are joints which are purposely placed in concrete to facilitate construction, reduce initial shrinkage stresses and cracks, allow time for

installation of embedded metalwork, or allow for subsequent placing of other concrete.

2. Bond is required at construction joints regardless of whether or not reinforcement is continuous across joint.

3. Locate construction joints where shown on drawings. Relocation, addition, or elimination of construction joints will be subject to approval by COR.

4. Clean, roughen, and surface dry surfaces of construction joints to be covered with fresh concrete. See Preparation article.

5. Do not use a mortar layer on construction joints.

B. Control joints (Ct.J):

1. Control joints are joints placed in concrete to provide for control of initial shrinkage stresses and cracks of monolithic units.

2. Construct control joints so no bond exists between concrete surfaces forming the joint:

a. Construct control joints by placing concrete on one side of joint and allowing it to set before concrete is placed on other side of joint.

b. Coat surface of concrete first placed at control joint with curing compound that prevents bond before placing concrete on other side of joint.

3. Reinforcement is continuous across tooled or saw cut control joints.

C. Partial Contraction Joints, (Partial Cr. J):

1. Construct partial contraction joints so no bond exists between concrete surfaces forming the joint:

a. Construct partial contraction joints by placing concrete on one side of joint and allowing it to set before concrete is placed on other side of joint.

b. Coat surface of concrete first placed at partial contraction joint with curing compound that prevents bond before placing concrete on other side of joint.

2. At partial contraction joints, discontinue every other reinforcement bar perpendicular to the joint; i.e. 1/2 of perpendicular reinforcement shall cross the joint. End discontinued bars 2-inches from face of joint.

D. Contraction joints (Cr.J):

1. Contraction joints are joints placed in concrete to provide for volumetric shrinkage of a monolithic unit or movement between monolithic units.

2. Construct contraction joints so no bond exists between concrete surfaces forming the joint:

a. Construct contraction joints by placing concrete on one side of joint and allowing it to set before concrete is placed on other side of joint.

b. Coat surface of concrete first placed at contraction joint with curing compound that prevents bond before placing concrete on other side of joint.

3. Except as provided for dowels, reinforcement is not continuous across a contraction joint.

E. Expansion joints (EJ):

1. Cut sponge rubber joint filler to size and shape of joint surface to receive filler.

2. Adhere filler to concrete in accordance with adhesive manufacturer's recommendations.

3. Butt sections of filler with tight-fitting butt joints to prevent mortar from seeping through joint.

F. Expansion joints (EJ):

1. Form joint in concrete.

2. Fill joint with bituminous joint filler.

3. Butt sections of filler with tight-fitting butt joints.

G. Edges:

1. Permanently exposed concrete, except slabs and top edges of curbs: Chamfer edges with 45-degree bevel 3/4-inch by 3/4-inch; unless otherwise shown on drawings.

2. Exposed edges of slabs and top edges of curbs: Tool to radius of 1/4-inch.

H. Preformed joints consisting of plastic or metal strips not allowed.

3.05 STRUCTURAL DEVIATIONS AND SURFACE TOLERANCES

A. Structural deviations are defined as allowable variations from specified lines, grades, and dimensions.

B. Surface tolerances are defined as maximum allowable magnitude of surface irregularities.

C. Specified structural deviations and surface tolerances are consistent with modern construction practice and governed by effects that permissible variations may have upon a structure. COR reserves right to diminish specified structural deviations and surface tolerances where such variations impair structural action, operational function, or architectural appearance of structure or portion of structure.

D. Construct concrete within stated variations even though more than one may be specified:

1. Specified variation for one element of structure will not apply when it will permit another element of same structure to exceed its allowable variation.

2. Where variations are not specified or shown on drawings for structure, variations shall be those specified for similar work. As exception to clause at FAR 52.236-

21 "Specifications and Drawings for Construction", specific tolerances shown on drawings in connection with dimension governs.

E. Structural Deviations:

1. Check variations from specified lines, grades, and dimensions in hardened concrete to determine that structures are within tolerances specified in Table 03 30 00C – Deviations from Specified Lines, Grades, and Dimensions.

2. Variation is distance between actual position of structure or element of structure and specified position in plan for structure or element:

   a. Plus, or minus variations indicate a permitted actual position up or down and in or out from specified position in plan.

   b. Variations not designated as plus or minus indicate maximum deviation permitted between designated successive points on completed element of construction.

3. Specified position in plan is defined as lines, grades, and dimensions described in this section, shown on drawings, or prescribed by COR.

Table 03 30 00C – Deviations from Specified Lines, Grades, and Dimensions

A. MANIFOLD PIPING STRUCTURES			
1. Footings:			
	(a)	Variation in length and width dimensions from those specified.....	minus 1/2-inch plus 2-inches
	(b)	Horizontal misplacement or eccentricity:	2 percent of footing width in direction of misplacement, but not more than 2-inches
	(c)	Reduction in thickness from that specified.....	5 percent of specified thickness, not to exceed 1-inch

Table 03 30 00C – Deviations from Specified Lines, Grades, and Dimensions

2.	Variation from plumb or specified batter for lines and surfaces of walls:	
	(a) When overall height of line or surface is:	
	Less than 10-feet	plus, or minus 1/4-inch
	10-feet or more	plus, or minus 3/8-inch
	(b) For any 2 successive intermediate points on the line or surface separated by:	
	10- to 20-feet, inclusive	1/4-inch
	More than 20-feet	3/8-inch
3.	Variation from level or specified grades for slabs:	
	(a) When overall length of line or surface is:	
	Less than 10-feet	plus, or minus 1/4-inch
	10- to 20-feet, inclusive	plus, or minus 3/8-inch
	More than 20-feet	plus, or minus 3/4-inch
	(b) For any 2 successive intermediate points on the line or surface separated by:	
	10- to 20-feet, inclusive	1/4-inch
	More than 20-feet	3/8-inch
4.	Variation in cross-sectional dimensions from those specified and in thicknesses of slabs and walls from those specified	minus 1/4-inch plus 1/2-inch
5.	Variation from specified elevation for top of concrete for foundations	plus, or minus 1/2-inch
6.	Variation from specified grade or alignment for trenches:	
	(a) When overall length is:	
	Less than 10-feet	plus, or minus 1/4-inch
	10- to 20-feet, inclusive	plus, or minus 3/8-inch
	More than 20-feet	plus, or minus 3/4-inch
	(b) For any 2 intermediate points separated by:	
	10- to 20-feet, inclusive	1/4-inch
	More than 20-feet	3/8-inch

Table 03 30 00C – Deviations from Specified Lines, Grades, and Dimensions

B. STRUCTURES AND OTHER BUILDINGS			
1.	Footings:		
	(a)	Variation in length and width dimensions from those specified.....	minus 1/2-inch plus 2-inches
	(b)	Horizontal misplacement or eccentricity:	2 percent of footing width in direction of misplacement, but not more than 2-inches
	(c)	Reduction in thickness	5 percent of specified thickness
2.	Variation of horizontal dimensions at floor and roof levels from specified position in plan:		
	(a)	Overall building dimensions	plus, or minus 1/2-inch per 100foot length with maximum for entire length of plus or minus1-inch
	(b)	Overall bay dimensions limited by (c)below, but not to exceed	plus, or minus 3/8-inch
	(c)	Intermediate dimensions for column, wall, and partition locations:	
		For dimensions less than 10-feet.....	plus, or minus 1/4-inch
		For dimensions equal to or greater than 10-feet but less than 20-feet	plus, or minus 3/8-inch
		For dimensions 20-feet or more.....	plus, or minus 1/2-inch

Table 03 30 00C – Deviations from Specified Lines, Grades, and Dimensions

3.	Variation of vertical dimensions from specified position in plan:		
	(a)	Overall building dimensions	plus, or minus 1/2-inch
	(b)	Overall story height limited by (c) below, but not to exceed	plus, or minus 3/8-inch
	(c)	Intermediate dimensions:	
		Less than 10-feet	plus, or minus 1/4-inch
		10- to 20-feet, inclusive	plus, or minus 3/8-inch
		20-feet or more.....	plus, or minus 1/2-inch
4.	Variation from plumb or specified batter for lines and surfaces of columns, piers, walls, and for arrises:		
	(a)	When overall height of line or surface is:	
		Less than 10-feet	plus, or minus 1/4-inch
		10- to 20-feet, inclusive	plus, or minus 3/8-inch
		More than 20-feet, but less than 40-feet	plus, or minus 1/2 -inch
		40-feet or more	plus, or minus 1-inch
	(b)	For any 2 successive intermediate points on the line or surface separated by:	
		10- to 20-feet, inclusive	1/4-inch
		More than 20-feet	3/8-inch

Table 03 30 00C – Deviations from Specified Lines, Grades, and Dimensions

5.	Variation from plumb for lines and surfaces of corner columns, control joint grooves, and other conspicuous lines:	
	(a) When overall height of line or surface is:	
	Less than 10-feet	plus, or minus 1/4-inch
	10- to 20-feet, inclusive	plus, or minus 3/8-inch
	More than 20-feet	plus, or minus 1/2-inch
	(b) For any 2 successive intermediate points on the line or surface separated by:	
	10- to 20-feet, inclusive	1/4-inch
	More than 20-feet	3/8-inch
6.	Variation from level or established grades for floors, roof decks, ceilings, beam soffits, and arrises:	
	(a) When overall length of line or surface is:	
	Less than 10-feet	plus, or minus 1/4-inch
	10- to 20-feet, inclusive	plus, or minus 3/8-inch
	More than 20-feet, but less than 40-feet	plus, or minus 1/2-inch
	40-feet or more	plus, or minus 3/4-inch
	(b) For any 2 successive intermediate points on the line or surface separated by:	
	10- to 20-feet, inclusive	1/4-inch
	More than 20-feet	3/8-inch

Table 03 30 00C – Deviations from Specified Lines, Grades, and Dimensions

7.	Variation from level or specified grades for exposed lintels, sills, parapets, horizontal grooves, and other conspicuous lines:	
	(a) When overall length of line or surface is:	
	Less than 10-feet	plus, or minus 1/4-inch
	10- to 20-feet, inclusive	plus, or minus 3/8-inch
	More than 20-feet	plus, or minus 1/2-inch
	(b) For any 2 successive intermediate points on the line or surface separated by:	
	10- to 20-feet, inclusive	1/4-inch
	More than 20-feet	3/8-inch
8.	Variation in location from specified position in plan of sleeves and wall openings	plus, or minus 1/2-inch
9.	Variation in sizes from those specified for sleeves, floor openings, and wall openings, except wall openings for swinging doors and louvers without tubular frames	plus, or minus 1/4-inch
10.	Variation in sizes from those specified for wall openings for swinging doors and louvers without tubular frames	minus 0-inch plus 1/4-inch
11.	Variation in cross-sectional dimensions from those specified for columns and beams and in thicknesses from those specified for slabs and walls	minus 1/4-inch plus 1/2-inch
12.	Variation in rise and tread of steps from that specified:	
	(a) Flight of stairs:	
	Rise	plus, or minus 1/8-inch
	Tread	plus, or minus 1/4-inch
	(b) Consecutive steps:	
	Rise	1/16-inch
	Tread	1/8-inch

Table 03 30 00C – Deviations from Specified Lines, Grades, and Dimensions

C. PIPELINE STRUCTURES			
1.	Footings:		
	(a)	Variation of length and width dimensions from those specified.....	minus 1/2-inch plus 2-inches
	(b)	Misplacement or eccentricity:.....	2 percent of footing width in direction of misplacement, but not more than 2-inches
	(c)	Reduction in thickness from that specified.....	5 percent of specified thickness not to exceed 1-inch
2.	Culverts:		
	(a)	Departure from established alignment.....	plus, or minus 2-inches
	(b)	Departure from established profile grade	plus, or minus 2-inches
	(c)	Variation from specified thickness	minus 2.5 percent of specified thickness or minus 1/4-inch, whichever is greater; or plus, 5 percent of specified thickness or plus 1/2-inch, whichever is greater
	(d)	Variation from specified inside dimensions	plus, or minus 0.5 percent of inside dimensions

Table 03 30 00C – Deviations from Specified Lines, Grades, and Dimensions

3.	Checks, overchutes, drops, turnouts, inlets, chutes, and similar structures:		
	(a)	Departure from established alignment	plus, or minus 1-inch
	(b)	Departure from established grade	plus, or minus 1-inch
	(c)	Variation from plumb or specified batter for lines and surfaces of columns, piers, walls, and for arrises:	
		(1) When overall length of line or surface is:	
		Less than 10-feet	Exposed: plus, or minus 3/8-inch Buried: plus, or minus 3/4-inch
		10-feet or more	Exposed: plus, or minus 1/2-inch Buried: plus, or minus 1-inch
		(2) For any 2 successive intermediate points on the line or surface separated by:	
		10- to 20-feet, inclusive	Exposed: 3/8-inch Buried: 3/4-inch
		More than 20-feet	Exposed: 1/2-inch Buried: 1-inch

Table 03 30 00C – Deviations from Specified Lines, Grades, and Dimensions

4.	Variations from level or specified grades for slabs, beams, and horizontal grooves:	
	(a) When overall length of line or surface is:	
	Less than 10-feet	Exposed: plus, or minus 3/8-inch Buried: plus, or minus 3/4-inch
	10-feet or more	Exposed: plus, or minus 1/2-inch Buried: plus, or minus 1-inch
	(b) For any 2 successive intermediate points on the line or surface separated by:	
	10- to 20-feet, inclusive	Exposed: 3/8-inch Buried: 3/4-inch
	More than 20-feet	Exposed: 1/2-inch Buried: 1-inch
5.	Variation in cross-sectional dimensions of columns, piers, slabs, walls, beams, and similar parts of structure(s) in 3 above from those specified	minus 1/4-inch plus 1/2-inch
6.	Variation in sizes and locations from those specified for slab and wall openings	plus, or minus 1/2-inch
7.	**Variation from plumb or level for sills and sidewalls for radial gates and similar watertight joints	Not greater than a rate of 1/8-inch in 10-feet
	_____	
**	Dimensions between sidewalls for radial gates: No more than shown on drawings at sills and not less than shown on drawings at top of walls.	
8.	Variation from plumb of pipe erected vertically in any length of 10-feet	plus, or minus 1/2-inch

F. Surface irregularities:

1. Bulges, depressions, and offsets are defined as surface irregularities or roughness.
2. Surface irregularities are classified as “abrupt” or “gradual” and allowable tolerances are specified in Table 03 30 00D – Surface Tolerances:

a. Surface tolerance designated by capital "T" followed by number 1 through 5.

b. Surface tolerance designations are separate from surface finishes and structural deviations.

3. Abrupt surface irregularities:

a. Abrupt surface irregularities are defined as offsets such as those caused by misplaced or loose forms in which maximum dimension of irregularity perpendicular to surface is greater than maximum dimension of irregularity in plane of surface.

b. Abrupt surface irregularities include isolated surface irregularities which exceed specified gradual irregularities.

4. Gradual surface irregularities:

a. Gradual surface irregularities are defined as bulges and depressions resulting in gradual changes on surface.

b. Gradual surface irregularities are further defined as isolated undulations on surface. Maximum dimension of undulation perpendicular to surface is small relative to maximum dimension of undulation in plane of surface.

5. Check magnitude of surface irregularities of formwork and finished surfaces to ensure that surfaces are within specified tolerances.

G. Surface tolerances:

Table 03 30 00D – Surface Tolerances

Concrete surface	Maximum allowable surface irregularity tolerance	
	Abrupt	Gradual
T1	1-inch	1/4-inch/inch
T2	1/2-inch	1/8-inch/inch
T3	1/4-inch	1/16-inch/inch
T4	1/8-inch	1/32-inch/inch
T5	1/32-inch	1/120-inch/inch

H. Repair of hardened concrete not within specified tolerances:

1. Repair hardened concrete which is not within specified tolerances to bring it within those tolerances.

2. Perform repair after consultation with Government inspector regarding method of repair. Notify COR as to time when repair will be performed.

3. Repair concrete which will be exposed to view in manner which will result in concrete surface with uniform appearance:

a. When grinding surfaces exposed to view, limit depth of grinding such that no aggregate particles are exposed more than 1/16-inch in cross section at finished surface.

b. Where grinding has caused or will cause exposure of aggregate particles greater than 1/16-inch in cross section at finished surface, repair concrete by excavating and replacing concrete.

I. Field verification of surface tolerances:

1. Determine compliance of surface with specified surface tolerances.

2. Evaluate surface roughness:

a. Measure roughness height or depth and check for compliance with values specified in Table 03 30 00D – Surface Tolerances and Table 03 30 00C – Deviations from Specified Lines, Grades, and Dimensions.

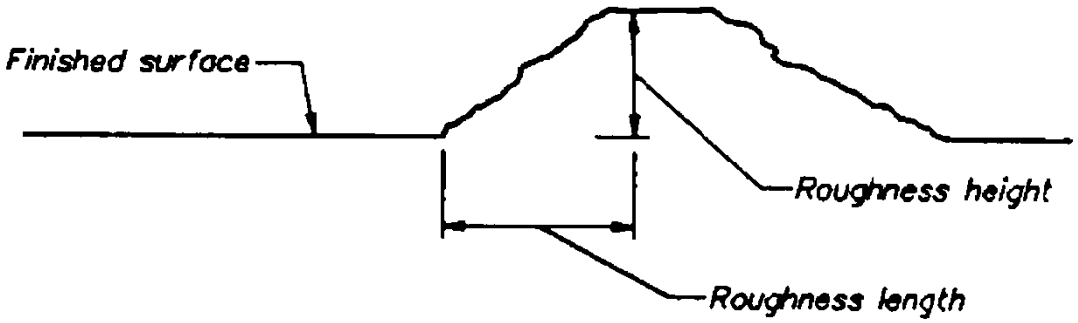
b. When measured height or depth of roughness is less than value in abrupt tolerance specification and height or depth of roughness does not cause structure to exceed applicable value specified in Table 03 30 00C – Deviations from Specified Lines, Grades, and Dimensions, surface roughness is acceptable.

c. When roughness height or depth exceeds abrupt tolerance specification, determine roughness slope for comparison to gradual tolerance specification:

1) Measure roughness length and determine roughness slope by dividing roughness height or depth by roughness length (see Figure 1).

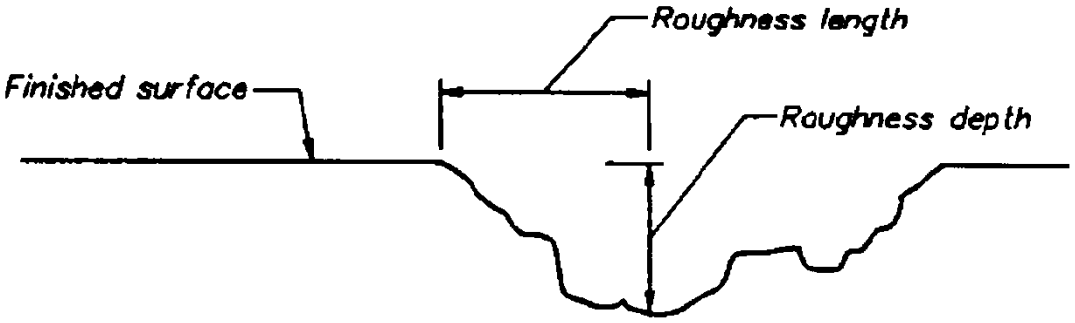
2) When roughness slope is greater than slope specified by gradual tolerance specification, surface roughness is unacceptable.

3) When roughness slope is less than gradual slope specified and gradual roughness does not cause structure to exceed allowable structural deviations, surface roughness is acceptable.



$$\text{Roughness slope ratio} = \frac{\text{Roughness height}}{\text{Roughness length}}$$

CASE 1 = Offset on the Surface



$$\text{Roughness slope ratio} = \frac{\text{Roughness depth}}{\text{Roughness length}}$$

CASE 2 = Offset into the Surface

FIGURE 1

J. Measuring surface roughness:

1. Examples below illustrate how to make necessary surface measurements for typical roughness.

2. Case 1 - Roughness Protruding above Surface:

   a. Measure roughness protruding above surface with straightedge that is at least 20 times longer than roughness height being measured.

   b. Position straightedge with one end resting on top of roughness, as shown in Figure 2 (case 1).

   c. Determine roughness height by measuring maximum gap that occurs normal to straight edge.

   d. Note position on the straightedge from which normal distance is measured.

   e. To determine roughness length, measure distance along straightedge from point where the height was measured to point of contact between straightedge and top of roughness.

   f. Roughness slope is defined as ratio of roughness height to roughness length.

   g. As roughness is seldom symmetric, moving position of straightedge about roughness may be necessary to locate point where maximum height and slope exists.

3. Case 2 - Roughness Extending below the Surface:

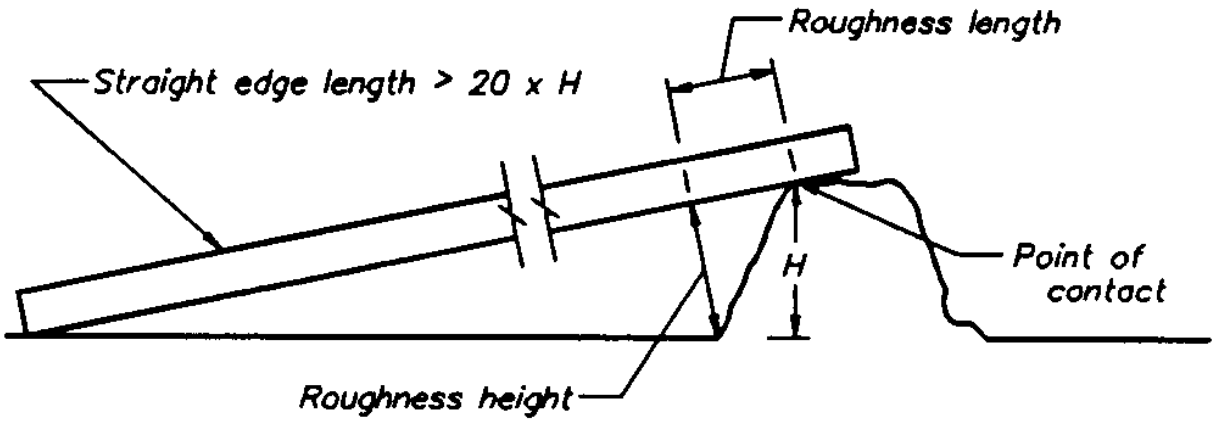
   a. A roughness occurring as an indentation to surface is measured by placing straightedge across indentation, as shown in Figure 2 (case 2).

   b. Measure maximum gap between straightedge and surface and note location of measurement on straightedge.

   c. From point of depth measurement, measure along straightedge in both directions to point of contact with surface.

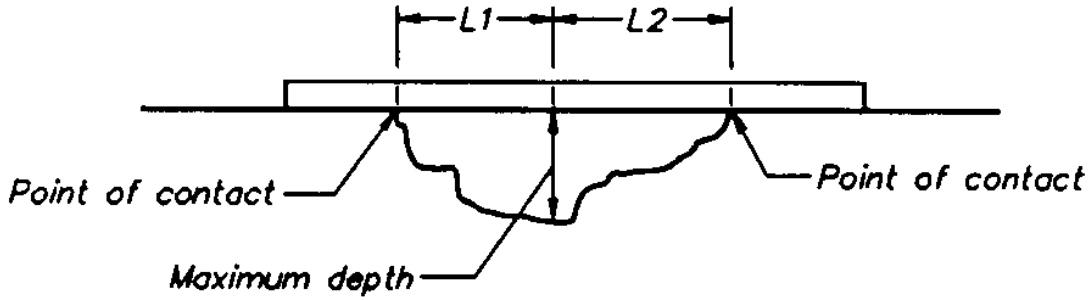
   d. Use shortest length measured as roughness length.

   e. Divide roughness depth by roughness length to determine roughness slope.



CASE 1

$$\text{Roughness slope} = \frac{\text{Roughness depth}}{\text{The shortest distance L1 or L2}}$$



CASE 2

FIGURE 2

K. Prevention of repeated failure to meet tolerances:

1. When concrete placements result in hardened concrete which does not meet specified tolerances, submit to COR an outline of preventive actions such as modifications to forms, modified procedure for setting screeds, and different finishing techniques to be implemented to avoid repeated failures. Submit when requested by COR.

2. Government reserves the right to delay concrete placements until preventive actions which have been approved by COR are implemented.

3.06 CURING

A. Water Curing:

1. Keep concrete surface wet for 14-days, minimum, from time concrete has attained enough set to prevent detrimental effects to surface.

2. Cure methods:

a. Water-saturated material.

b. System of perforated pipes, mechanical sprinklers, or porous hose.

c. Other methods which will keep surfaces wet.

d. Subject to approval by COR.

B. Curing with Curing Compound:

1. Apply to concrete surface to provide water-retaining film. Reapply as necessary to maintain continuous, water-retaining film on surface for 28-days.

2. Thoroughly mix compound and spray apply in one coat to provide continuous, uniform film over surface.

3. Do not exceed coverage rate recommended by curing compound manufacturer. Decrease coverage rate on rough surfaces as necessary to obtain required continuous film.

4. Ensure ample coverage on edges, corners, and rough surfaces.

5. Use spray equipment recommended by curing compound manufacturer.

C. Sheet Material Curing:

1. Thoroughly moisten concrete surface by lightly spraying with water as soon as concrete has hardened sufficiently to prevent damage.

2. Completely cover concrete surface with sheet material to provide airtight, water-retaining film over entire surface.

3. Lap edges of sheet material to seal adjacent sheets.

4. Place tightly against concrete surface at extreme edge of curing area.

5. Secure sheet material to withstand wind and prevent circulation of air inside sheet material.

6. Keep surface covered for 14-days, minimum.

3.07 CONTRACTOR FIELD QUALITY TESTING

A. Perform sampling, testing, and reporting as required in Table 03 30 00E – Field Testing.

1. Independent testing agency shall meet requirements specified in Section 01 46 20 – Testing Agency Services.

2. Personnel conducting plastic concrete field tests: Qualified as ACI Concrete Field-Testing Technician, Grade 1; or equal.

3. Personnel conducting concrete specimen tests: Qualified as ACI Concrete Strength Testing Technician; or equal.

B. Perform tests at least as often as frequencies specified in Table 03 30 00E – Field Testing.

C. Notify COR immediately of test results showing failure of materials to meet specifications. Notify COR within 2-hours of test results showing materials meet specifications. Submit reports of test results as specified.

3.08 FIELD QUALITY ASSURANCE

A. In addition to specified Contractor Quality Field Testing, Government may also perform tests listed in Table 03 30 00E – Field Testing.

Table 03 30 00E – Field Testing

TESTS OF	TEST STANDARD	STANDARD TITLE	TESTING FREQUENCY
Fresh Concrete Properties - tests performed at site	ASTM C143	Slump of Hydraulic-Cement Concrete	1 set of tests per load for first 2 loads. When tested concrete meets specifications, 1 set of tests each day of placement for each mixture for first 50 or less cubic yards, and 1 set of tests for each additional 100 cubic yards of concrete. Minimum of 1 set of tests per hour during placements. When concrete does not meet specifications, test each load until 2 consecutive loads meet specifications, then resume testing frequency specified above.
	ASTM C231	Air Content of Freshly Mixed Concrete by the Pressure Method (alternative to ASTM C138 gravimetric method)	
	ASTM C1064	Temperature of Freshly Mixed Hydraulic-Cement Concrete	
	ASTM C138	Density (Unit Weight), Yield, and Air Content (Gravimetric) of Concrete	

Table 03 30 00E – Field Testing

TESTS OF	TEST STANDARD	STANDARD TITLE	TESTING FREQUENCY
Compressive Strength	ASTM C31 ASTM C39	Making and Curing Concrete Test Specimens in the Field Compressive Strength of Cylindrical Concrete Specimens	1 set of samples (6- by 12-inches for each day of placement for each mixture for first 50 or less cubic yards, and 1 set of samples for each additional 100 cubic yards of concrete. Minimum of 5 samples for strength testing shall be made each time strength samples are collected. 2 additional field cured test cylinders during placement in adverse (hot or cold) weather. Cure samples on jobsite under same conditions as concrete cylinders represent for minimum 7 days, then transfer to testing laboratory until testing at strength design days. Test 2 cylinders each at 7-days age and 2 cylinders at strength design age. Maintain last cylinder for testing in event strength design age test results fall below required strength.
Concrete Cores	ASTM C42	Obtaining and Testing Drilled Cores and Sawed Beams of Concrete	At discretion of Government when cylinder strengths fail to meet minimum requirements. Contractor shall obtain core specimens in accordance with ASTM C42 at locations directed by COR, at no additional cost to Government. Contractor shall repair core holes in accordance with USBR M-47 as directed by COR

B. Acceptance criteria:

1. Cylinder compressive strength:

   a. In accordance with ASTM C94, except as follows:

      1) 90 percent of test cylinders exceed specified compressive strength at design age.

      2) Average compressive strength of six consecutive test cylinders exceeds specified compressive strength at design age.

      3) No individual strength test falls below specified compressive strength by more than 500 pounds per square inch.

2. Drilled concrete cores:

   a. Concrete in placement represented by core tests will be considered structurally adequate when average compressive strength of three cores is equal to at least 85 percent of specified compressive strength and no single core has compressive strength of less than 75 percent specified compressive strength.

3.09 PROTECTION

A. Protect concrete from damage until final acceptance by Government:

1. Do not load, remove forms or shoring, or backfill against concrete until concrete has gained sufficient strength to safely support its weight and imposed loads.

2. Protect fresh concrete against erosion from rain, hail, sleet, or snow; contamination from foreign materials; and damage from foot traffic until the concrete has hardened.

3. Protect concrete from heavy foot traffic and other construction activities by covering with plywood or other suitable material. Remove and dispose of temporary covering when no longer required.

B. Protect concrete when freezing temperatures are imminent:

1. Maintain concrete at a temperature of 50 degrees F (10 degrees C) or greater for 72-hours, minimum, after placement. Vent heater and prevent concrete from drying where artificial heat is employed.

2. Protect concrete from freezing during water curing. After discontinuance of water curing, maintain at a temperature of 50 degrees F (10 degrees C) or greater for next 72-hours.

3. Discontinue protection against cold weather such that the drop in temperature of the concrete will be gradual and will not exceed 5 degrees F per hour and 40 degrees F in 24-hours for thin sections and 5 degrees F per hour and 20 degrees F in 24-hours for massive sections greater than 36-inches.

3.10 REPAIR

A. Repair concrete in accordance with USBR M-47.

B. Use repair or replacement method directed by COR.

3.11 FINISH, SURFACE TOLERANCES, AND CURING SCHEDULES

Table 03 30 00F – Formed Surfaces

Surface	Finish	Maximum Allowable Tolerances	Acceptable Curing Methods
Surfaces upon or against which fill material will be placed	F1	T1	Water, White ASTM C309 Class A or B curing compound for horizontal surfaces, and White ASTM C309, Class B curing compound for vertical or sloped surfaces.
Surfaces not permanently concealed by fill material or concrete where appearance is not critical	F2	T2 and T3	Water, White ASTM C309 Class A or B curing compound for horizontal surfaces, and White ASTM C309, Class B curing compound for vertical or sloped surfaces.
Interior walls and ceilings of vaults, sumps, pullboxes, and entry boxes	F2	T4	White ASTM C309, Class B curing compound
Indoor and outdoor curbs	F2	T4 T5	Polyethylene film or White Burlap-polyethylene, ASTM C309 Class A or B curing compound
Pedestals and equipment foundations	F2	T5	ASTM C1315 Class A Clear curing compounds, water-emulsified resin-base if outdoors, Polyethylene film
Construction joints and surfaces to be covered by grout	F1	T 4	Water White, ASTM C309 Class A curing compound
Expansion joints	F2	T 4	Water White ASTM C309 Class A or B curing compound for horizontal surfaces, and White ASTM C309, Class B curing compound for vertical or sloped surfaces
Contraction joints	F2	T 4	White ASTM C309 Class A curing compound

Table 03 30 00G - Unformed Surfaces

Surface	Finish	Maximum Allowable Tolerances	Acceptable Curing Methods
Surfaces to be covered by fill material or concrete	U1	T1	White ASTM C309 Class A or B curing compound for horizontal surfaces, and White ASTM C309, Class B curing compound for vertical or sloped surfaces
Outdoor entrance slabs, and walks and stoops	U2	T3	ASTM C1315 Class A Clear curing compounds, water-emulsified resin-base, Polyethylene film or White Burlap-polyethylene sheet
Outdoor curbs and gutters	U2	T3	White ASTM C309 Class A or B curing compound Polyethylene film or White Burlap-polyethylene sheet
Outdoor decks and roofs	U2	T5	White ASTM C309 Class A or B curing compound Polyethylene film or White Burlap-polyethylene sheet
Outdoor equipment slabs and foundations	U2	T3	White ASTM C309 Class A or B curing compound, Polyethylene film or White Burlap-polyethylene sheet
Tops of walls not prominently exposed to public view	U2	T4	White ASTM C309 Class A or B curing compound Polyethylene film or White Burlap-polyethylene sheet
Tops of walls prominently exposed to public view	U2	T5	ASTM C1315 Class A Clear curing compounds, water-emulsified resin-base
Floors of vaults, sumps, pullboxes, and entry boxes	U2	T5	White wax-base or white water-emulsified resin-base curing compound
Subfloor surfaces not otherwise listed	U2	T4	Water, Polyethylene film or White Burlap-polyethylene sheet

END OF SECTION

SECTION 03 48 00
PRECAST CONCRETE MANHOLES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for applicable Line Pipe.

1.02 REFERENCE STANDARDS

A. American Association of State Highway and Transportation Officials (AASHTO)

1.	AASHTO HB-17	Highway Bridges, Seventeenth Edition

B. ASTM International (ASTM)

1.	ASTM C33/C33M-24a	Concrete Aggregates
2.	ASTM C443/C443M-21	Joints for Concrete Pipe and Manholes, Using Rubber Gaskets
3.	ASTM C478/C478M-22	Circular Precast Reinforced Concrete Manhole Sections
4.	ASTM C857-19	Minimum Structural Design Loading for Underground Precast Concrete Utility Structures
5.	ASTM C858-19	Underground Precast Concrete Utility Structures
6.	ASTM C877-25	External Sealing Bands for Concrete Pipe, Manholes, and Precast Box Sections
7.	ASTM C920-18(2024)	Elastomeric Joint Sealants
8.	ASTM C923/C923M-20(2025)	Resilient Connectors Between Reinforced Concrete Manhole Structures, Pipes, and Laterals
9.	ASTM C990/C990M-25	Joints for Concrete Pipe, Manholes, and Precast Box Sections Using Preformed Flexible Joint Sealants
10.	ASTM C1478/C1478M-20(2025)	Storm Drain Resilient Connectors Between Reinforced Concrete Storm Sewer Structures, Pipes, and Laterals
11.	ASTM D751-19	Test Methods for Coated Fabrics
12.	ASTM D2136-19e1	Test Method for Coated Fabrics – Low-Temperature Bend Test

C. Code of Federal Regulations (CFR)

1.	29 CFR 1910.27	Fixed Ladders

D. National Precast Concrete Association (NPCA)

1.	NPCA QC Manual	Quality Control Manual for Precast Concrete Plants

1.03 REFERENCE SPECIFICATION SECTIONS

A. 05 50 00 – Metal Fabrications.

1.04 SUBMITTALS

A. Submit the following in accordance with 01 33 00 – Submittals.

B. RSN 03 48 00-1, Certificates:

1. Quality control procedures in accordance with NPCA QC Manual or verification of current NPCA Plant Certification.

C. RSN 03 48 00-2, Shop Drawings:

1. Precast Structure:
    a. Indicate design assumptions.
    b. Show conformance to project drawings and requirements.
    c. Details of steel reinforcement size and placement.
    d. Details of joint sealant placement.
    e. Details of pipe entry connectors.
    f. Installation and construction information.
    g. Signed and sealed by engineer registered in one of 50 States of the United States or District of Columbia.

D. RSN 03 48 00-3, Product Data:

1. Delivery, storage, and handling instructions.
2. Anchorage, Lifting Inserts and Devices:
    a. Product data sheets and installation instructions
    b. Indicate unit dimensions and safe working loads.
3. Accessory Items:
    a. Including, but not limited to sealants, gaskets, pipe entry connectors, steps, racks, and other items installed before or after delivery.
    b. Installation instructions and relevant product data upon request.
4. Test Reports:

a. Upon request, precast concrete producer shall supply copies of material certifications and/or laboratory test reports, including mill tests and all other test data, for portland cement, blended cement, pozzolans, ground granulated blast-furnace slag, silica fume, aggregate, admixtures, and curing compound proposed for use on this project.

b. Upon request, precast concrete producer shall submit copies of test reports showing that mix has been successfully tested to produce concrete with properties specified and will be suitable for project conditions. Such tests may include compressive strength, flexural strength, plastic or hardened air content, freeze-thaw durability, abrasion and absorption. Special tests for precast concrete items shall be clearly detailed in the specifications.

c. Upon request, precast concrete producer will supply copies of in-plant QA/QC inspection reports.

1.05 QUALIFICATIONS

A. NPCA Plant Certification: Prior to and during production of products.

B. Producer Qualifications:

1. Minimum 5-years business experience producing similar products.
2. Maintain permanent quality control department or retain independent testing agency on continuing basis. Agency shall issue a report, signed by a licensed PE, detailing ability of precast concrete producer to produce quality units meeting requirements.

1.06 DELIVERY, STORAGE AND HANDLING

1. Per producers' recommendations.

PART 2 PRODUCTS

2.01 SYSTEM DESCRIPTION

A. Precast concrete manholes are located at air valves and blow offs along the line pipe. See line pipe drawings for locations and details. Conceptual details for precast concrete manhole are shown on 1695-D-62553.

2.02 DESIGN REQUIREMENTS:

A. Standard Precast Concrete Unit:

1. Structure and covers and/or grating: Withstand HS 20-44 as defined in AASHTO HB and ASTM C857.
2. Consider stresses induced during handling, shipping, and installation
3. Indicate design loads on shop drawings
4. Provide block-outs and pipe entry connectors as required.

B. Non-Standard Precast Concrete Unit Design:

1. Design calculations and drawings of non-standard precast units: Prepared and signed by licensed professional engineer registered in State of New Mexico and submitted for approval prior to fabrication.
2. Calculations shall include the analysis of units for lifting stresses and sizing of lifting devices.
3. Provide block-outs and pipe entry connectors as required.

C. Joints and Sealants:

1. Meet specified design requirements.
2. Show on shop drawings.

2.03 MATERIALS

A. Concrete and Reinforcement: ASTM C858.

B. Inserts and Embedded Metal:

1. Refer to Section 05 50 00 – Metal Fabrications, unless noted elsewhere.
2. Proprietary Items: In accordance with manufacturers published literature.

C. Joint Sealants and Joint Gaskets:

1. Rubber gaskets for circular concrete: ASTM C443.
2. External sealing bands for noncircular pipe: ASTM C877.
3. Preformed flexible joint sealants: ASTM C990.
4. Elastomeric joint sealants: ASTM C920.

D. Pipe Entry Connectors:

1. ASTM C923.
2. ASTM C1478.

E. Nonshrink grout in accordance with Section 03 62 20 – Nonshrink Grout for Equipment and Metalwork.

F. Cementitious Grout: Suitable mix design for intended use, consisting of portland cement, sand, and water.

2.04 PRECAST CONCRETE MANHOLE

A. Precast Reinforced Concrete Manhole Sections:

1. Riser section(s) and slab top lid as shown on drawings:
   a. Size: As indicated on drawings.

2. Concrete manhole conforming to ASTM C478, and design requirements indicated on drawings:
    a. Bell and spigot joints.
    b. No elliptical reinforcement allowed.

B. Joints and Joint Sealant: ASTM C990.

C. Steps:
1. In accordance with 29 CFR 1910.27.
2. Spacing: 16-inches on centers.
3. Manhole steps shall be made of 1/2-inch steel rod encapsulated with copolymer polypropylene or approved equal and shall conform to ASTM C478.

D. Steel Manhole Lid: Fabricate in accordance with Section 05 50 00 – Metal Fabrications.

2.05 COARSE AGGREGATE BASE

A. ASTM C33, coarse aggregate, Size No. 57.

2.06 GUARD POSTS

A. Fabricate in accordance with Section 05 50 00 – Metal Fabrications.

2.07 BEDDING

A. ASTM C33, coarse aggregate, Size No. 57.

2.08 GOVERNMENT INSPECTION

A. Government may inspect plant during production. Notify COR 14 days prior production.

PART 3 EXECUTION

3.01 INSTALLATION

A. Site Access:
1. Contractor shall be responsible for providing adequate access to site to facilitate hauling, storage, and proper handling of precast concrete units.

B. Installation:
1. Precast concrete units shall be installed to lines and grades shown in contract documents or otherwise specified.
2. Precast concrete units shall be lifted by suitable lifting devices at points provided by precast concrete producer.

3. Precast concrete units shall be installed in accordance with applicable industry standards. Upon request, precast concrete producer shall provide installation instructions.
4. Field modifications to product shall relieve precast producer of liability, including if such modifications result in failure of precast concrete unit.
5. Place coarse aggregate base below precast concrete manholes as shown on drawings and in accordance with Section 31 23 02 – Compacting Earth Materials.
6. Erect precast concrete structures per Manufacturer's recommendations and as located on drawings.
7. Clean precast concrete manhole section pipe ends with wire brush prior to ends being joined. Place bell end down.
8. Install plumb in accordance with Manufacturer's recommendations.
9. Seal joints between manhole sections.
10. Place and compact backfill materials uniformly about manhole structure to prevent unequal loading and displacement of structure in accordance with Section 31 23 02 – Compacting Earth Materials.
11. Install guard posts in accordance with Section 05 50 00 – Metal Fabrications.
12. Paint exposed interior and exterior surfaces of steel manhole covers in accordance with Section 09 96 20 – Coatings.

C. Watertightness:

1. Where watertightness is a necessary performance characteristic of precast concrete units end use, watertight joints, pipe-entry connectors, and inserts should be used to ensure integrity of entire system.

END OF SECTION

SECTION 03 62 20
NONSHRINK GROUT FOR EQUIPMENT AND METALWORK

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for items of work requiring nonshrink grout.

1.02 REFERENCE STANDARDS

A. ASTM International (ASTM)

1.	ASTM C1107/C1107M-20	Packaged Dry, Hydraulic-Cement Grout (Nonshrink)

1.03 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals.

B. RSN 03 62 20-1, Packaged Nonshrink Grout Data:

1. Manufacturer's product data.
2. Manufacturer's certification that product meets specifications.
3. Manufacturer's instructions:
    a. Surface preparation, mixing, installation, finishing, curing, and protection recommendations:
        1) Include specific recommendations based on anticipated jobsite temperatures and environmental conditions.
        2) Include manufacturer's recommended methods for float finishing grout.
        3) Include manufacturer's recommended minimum temperature and protection period for protecting grout from freezing and damage after placement.

PART 2 PRODUCTS

2.01 PACKAGED NONSHRINK GROUT

A. Only allowable use for nonshrink grout is pipe penetrations and electrical channel bases. All other references in spec to grout shall be epoxy grout in accordance with Section 03 63 00 – Epoxy Grout.

B. Pre-mixed commercial grout mixture:

1. ASTM C1107.

C. Water:

1. Clean.
2. Free of detrimental quantities of silt, organic matters, salts, or other impurities.
3. As recommended by grout manufacturer.

D. Mix: In accordance with manufacturer's instructions to consistency recommended by manufacturer for application.

PART 3 EXECUTION

3.01 PREPARATION

A. Clean and prepare base concrete surface to which nonshrink grout will be bonded in accordance with manufacturer's instructions or as directed by COR:

1. Roughen surface.
2. Remove dirt, laitance, loose or defective concrete, curing compound, coatings, and other foreign material.
3. Wash with water.

B. Lubricate concrete surface by washing with water immediately before placing nonshrink grout when delay occurs between washing and placing nonshrink grout.

3.02 PLACING

A. Place nonshrink grout in accordance with manufacturer's instructions to completely fill space to be grouted.

B. Use forms, where required, to confine nonshrink grout.

3.03 CURING

A. Cure exposed surface of packaged nonshrink grout by method recommended by manufacturer.

B. Do not apply loads sooner than 72-hours after placement and only after nonshrink grout has attained compressive strength of at least 3,000 lb/in^2.

END OF CONTENTS

SECTION 03 63 00
EPOXY GROUT

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for items requiring epoxy grout.

1.02 REFERENCE STANDARDS

A. ASTM International (ASTM)

1.	ASTM C579-23	Compressive Strength of Chemical-Resistant Mortars, Grouts, Monolithic Surfacing and Polymer Concretes
2.	ASTM C580-18(2023)	Flexural Strength and Modulus of Elasticity of Chemical-Resistant Mortars, Grouts, Monolithic Surfacing's, and Polymer Concretes
3.	ASTM D638-22	Tensile Properties of Plastics
4.	ASTM D696-24	Coefficient of Linear Thermal Expansion of Plastic Between -30 degree C and 30 degree C with a Vitreous Silica Dilatometer

1.03 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals.

B. RSN 03 63 00-1, Approval Data:

1. Grout manufacturer's product data.
2. Grout manufacturer's environmental, product storage, preparation, mixing, installation and curing instructions.

1.04 DELIVERY, STORAGE, AND HANDLING

A. Deliver epoxy materials to jobsite in manufacturer's original unopened packaging with labels and seals intact.

B. Store epoxy materials in protected area in accordance with manufacturer's instructions.

PART 2 PRODUCTS

2.01 EPOXY GROUT

A. Three part flowable epoxy grout; resin, hardener, and aggregate filler.

1. CHOCKFAST RED manufactured by ITW Polymer Technologies, 130 Commerce Drive, Montgomery PA; or equal, having following essential characteristics:
    a. ASTM C579 Compressive Strength: Minimum of 15,000 psi.
    b. ASTM C579 Compressive Modulus: 2,000,000 psi.
    c. ASTM D696 Coefficient of Linear Thermal Expansion:
        1) 11.2×10^{6} degrees F.
    d. ASTM C580 Flexural Strength: Minimum 4,000 psi.
    e. ASTM D638 Tensile Strength: Minimum 1,800 psi.
    f. Service temperature: Up to 140 degrees F.
    g. Pot life: Approximately 3-hours at 70 degrees F.
    h. Shelf life: 2-years in dry storage.

B. Aggregate filler as recommended by manufacturer.

PART 3 EXECUTION

3.01 INSTALLATION

A. Mix and install grout in accordance with manufacturer's instructions to fill space to be grouted:

1. Concrete shall be a minimum of 28-days old.
2. Clean and roughen concrete surfaces.
3. Remove dirt, dust, oil, grease, debris, paint, curing compounds, sealers, and unsound concrete.
4. Mechanically prepare concrete surfaces in accordance with manufacturer's instructions to give surface profile of a minimum of 1/8-inch and expose coarse aggregate of concrete.
5. Concrete shall have an open surface texture.

B. Placement:

1. Place grout at air temperature recommended by manufacturer.
2. Bring surfaces and materials to be epoxied to 70 degrees F minimum before grouting.

3. Do not place grout over frozen concrete.
4. Hold grout in place with forms. Prepare forms to prevent sticking of grout.
5. Minimum 3/4-inch by 3/4-inch, 45 degree chamfer.
6. Mix grout components in accordance with manufacturer's instructions and technical data sheet.
7. Place grout in accordance with manufacturer's instructions.
8. Finish surface of grout in accordance with manufacturer's instructions.
9. Smooth sharp edges.

C. Protection:

1. Protect placed grout from damage during construction.
2. Maintain grout at 50 degrees F or greater for 72-hours, minimum, after placement unless otherwise recommended by manufacturer.
    a. After protection period, discontinue protection against cold weather such that the drop in temperature of the grout will be gradual and will not exceed 5 degree F per hour and 40 degree F in 24-hours.
3. The more stringent of grout manufacturer's recommendation or section shall apply.
4. Initial Loading:
    a. Do not apply loads epoxy grout sooner than 72-hours after placement:
        1) Apply load after epoxy grout has attained a compressive strength of 3,000 psi, minimum.
    b. Time required for epoxy grout used to attain this strength will be determined by OGR.
    c. Take care when applying loads on hardened epoxy grout.
    d. Contractor shall be responsible for damage resulting from impact loads when positioning equipment or metalwork.

D. Protect from movement until grout has fully cured.

END OF SECTION

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SECTION 05 50 00
METAL FABRICATIONS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include cost of the following in prices offered in Price Schedules for applicable items of work:

1. Guard posts.
2. Welded rectangular barriers.
3. Manhole lids.
4. Electrical equipment supports.
5. Removal and disposal of guard post disturbed by construction activities.
6. Fasteners for complete installation of metalwork.

1.02 DEFINITIONS

A. Miscellaneous metalwork: Where either shown on drawings or specified elsewhere in this section or these specifications means metal fabrications as used in this section.

1.03 REFERENCE STANDARDS

A. American Institute of Steel Construction (AISC)

1.	AISC 360-16	Specification for Structural Steel Buildings

B. American Society of Mechanical Engineers (ASME)

1.	ASME B18.2.1-2012(R2021)	Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series)

C. ASTM International (ASTM)

1.	ASTM A36/A36M-19	Carbon Structural Steel
2.	ASTM A53/A53M-24	Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
3.	ASTM A123/A123M-24	Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
4.	ASTM A153/A153M-23	Zinc Coating (Hot-Dip) on Iron and Steel Hardware
5.	ASTM A307-21	Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength

6.	ASTM A385/A385M-22	Providing High-Quality Zinc Coatings (Hot-Dip)
7.	ASTM A563-24	Carbon and Alloy Steel Nuts
8.	ASTM A615/A615M-26	Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement
9.	ASTM F844-19(2024)	Washers, Steel, Plain (Flat), Unhardened for General Use

D. American Welding Society, Inc. (AWS)

1.	AWS D1.1/D1.1M-25	Structural Welding Code – Steel

E. International Code Council (ICC)

1.	ICC ES AC 308	Acceptance Criteria for Post-Installed Adhesive Anchors in Concrete, November 2009
2.	ICC IBC – 2021	International Building Code

F. Society of Protective Coatings (SSPC)/NACE International (NACE)

1.	SSPC-SP 6/NACE 3-2006	Joint Surface Preparation Standard - NACE No. 3/ SSPC-SP 6 – Commercial Blast Cleaning

1.04 QUALIFICATIONS

A. Qualify welders in accordance with AWS D1.1 using procedures, materials, and equipment of the type required for the work.

1.05 DELIVERY, STORAGE, AND HANDLING

A. Protect from corrosion, deformation, and other types of damage.

B. Store items in an enclosed area free from contact with soil and weather.

C. Remove and replace damaged items with new items.

PART 2 PRODUCTS

2.01 MATERIALS

A. Arc-Welding Electrodes:

1. Use filler metal and shielding gases suitable for base materials, positions, and other conditions.
2. Filler metal and required shielding gases or fluxes: AWS D1.1.

3. Use filler metal with a minimum tensile strength of 70,000 pounds per square inch (psi) for steel.

B. Bolts, Nuts, And Washers:

1. Bolts:

a. ASTM A307.

b. Length of bolt threads: ASME B18.2.1.

c. Thread class: Two free-fit, American National coarse-thread series.

2. Nuts:

a. Steel: ASTM A563.

3. Washers:

a. Steel: ASTM F844, unhardened for general use.

C. Structural Steel:

1. Bars and Plates: ASTM A36.

2.02 ELECTRICAL EQUIPMENT SUPPORTS

A. Metal Framing:

1. Combination channels as manufactured by Unistrut Corporation, 35660 Clinton Street, Wayne, MI 48184; Cooper B-Line, 509 West Monroe Street, Highland, IL 62249; or equal, having the following essential characteristics and accessories:

a. Combination channels: Cooper B-Line B12A, Unistrut P5501; or equal, having the following essential characteristics:

1) No. 12 gauge steel.

2) Channel: 1 5/8-inch wide by 4 7/8-inch deep with 7/8-inch opening.

3) Hot-dipped galvanized.

b. Connection plates: Cooper B-Line B532, Unistrut P1726; or equal, having the following essential characteristics.

1) Thickness: 1/4-inch.

2) Hole spacing (from end): 13/16-inch.

3) Hole spacing (on center): 1 7/8-inch.

4) Width: 1 5/8-inch.

5) Hole diameter: 9/16-inch.

6) Hot-dipped galvanized.

c. Bolts, nuts, and washers:

1) Bolts: ASTM A307.

2) Nuts: Spring nuts, suitable for channel strut attachments.

3) Washers: ASTM F844.

4) Hot-dipped galvanized.

B. Fabricated Plates:

1. ASTM A36 steel, galvanized.

2.03 GUARD POSTS AND WELDED RECTANGULAR BARRIERS

A. Steel Pipe: ASTM A53, Type E or S, Grade B:

1. Unless otherwise shown on drawings, provide standard-weight, black, steel pipe.
2. Government inspection at mill and hydrostatic tests will not be required.

B. Concrete:

1. In accordance with the applicable requirements of Section 03 30 00 – Cast-In-Place Concrete.
2. 3/4-inch maximum size aggregate.
3. Minimum compressive strength of 4,500 pounds per square inch at 28-days.

C. Reinforcing Bars:

1. ASTM A615, Grade 60.
2. Deformed steel bar.

D. Painting:

1. In accordance with Section 09 96 20 – Coatings.

2.04 ANCHORS

A. In accordance with 03 20 60 – Concrete Anchors.

2.01 GROUT

A. Nonshrink Grout: In accordance with Section 03 62 20 – Nonshrink Grout for Equipment and Metalwork.

2.02 FABRICATION

A. Fabricate metalwork in accordance with AISC 360, and these specifications:

1. Perform welding and related work in accordance with AWS D1.1.

B. If straightening is necessary, use methods that will not injure metal.

C. Provide vent holes in pipe as required for galvanizing.

D. After shop work completion and before galvanizing, clean material of foreign substances.

E. Galvanizing:

1. Galvanize items of metalwork as specified or shown on drawings. Use hot-dip galvanizing in accordance with ASTM A123 and ASTM A385.

2. Galvanize bolts, nuts, washers, and locknuts in accordance with ASTM A153. Remove excess spelter by centrifugal spinning.

3. Fabricator's Galvanizing Repair:

a. Re-dip material with damaged galvanizing unless damage is local and can be repaired by 2 component epoxy zinc primer.

b. If galvanized coating becomes damaged after being dipped twice, material will be rejected.

c. Where local repair is authorized: Repair damage to galvanizing in accordance with Section 09 96 20 – Coatings.

PART 3 EXECUTION

3.01 PREPARATION

A. Where locations and dimensions of miscellaneous metalwork shown on drawings are dependent upon existing equipment and/or equipment furnished, confirm locations and dimensions prior to fabrication of miscellaneous metalwork.

3.02 REMOVAL AND DISPOSAL

A. Remove and dispose of existing guard post as identified on Contract Drawing 1695-D-62537.

B. Dispose of removed materials in accordance with Section 01 74 00 – Cleaning and Waste Management.

3.03 INSTALLATION

A. Embedded Metalwork:

1. Locate metalwork to be embedded in concrete. Hold metalwork in correct position and alignment and protect metalwork from damage and displacement during placing and setting of concrete.

2. Unless otherwise specified, use only metal braces, supports, and other items to position and align embedded metalwork, which will be embedded in concrete. Do not use wooden braces, supports, or other items to position and align embedded metalwork if they will also be embedded in concrete.

3. Clean surfaces of metalwork to be in contact with or embedded in concrete grouting mortar in accordance with SSPC-SP 6/NACE 3.

B. Guard Posts and Welded Rectangular Barriers:

1. Install posts plumb in concrete footings as shown on drawings.

2. Fill posts with concrete and trowel top of post round for water drainage. Posts may be capped in lieu of filling posts with concrete.

C. Installer's Galvanizing Repair:

1. Repair damage to galvanizing in accordance with Section 09 96 20 – Coatings.

D. Holes in Metalwork:

1. Drill, or drill and tap as required, holes in metalwork required for installation.

E. Grout:

1. Where shown on drawings, fill spaces under metalwork completely as required with nonshrink grout in accordance with Section 03 62 20 – Nonshrink Grout for Equipment and Metalwork.

2. Clean surfaces of metalwork to be in contact with or embedded in concrete or nonshrink grout in accordance with SSPC-SP6/ NACE 3.

F. Painting:

1. Paint surfaces of miscellaneous metalwork exposed after installation; except galvanized steel and cast iron; in accordance with Section 09 96 20 – Coatings.

3.04 ELECTRICAL GROUNDING

A. Ground metalwork in accordance with Section 26 05 26 – Grounding and Bonding.

END OF SECTION

SECTION 09 96 20
COATINGS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work.

1.02 REFERENCE STANDARDS

A. ASTM International (ASTM)

1.	ASTM A380-17	Cleaning, Descaling, and Passivation of Stainless Steel Parts, Equipment, and Systems
2.	ASTM A780-20	Repair of Damaged and Uncoated Areas of Hot Dip Galvanized Coatings
3.	ASTM C920-18	Elastomeric Joint Sealants
4.	ASTM D870-17	Testing Water Resistance of Coatings Using Water Immersion
5.	ASTM D2794-93(2019)	Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact)
6.	ASTM D3276-15e1	Guide for Painting Inspectors
7.	ASTM D4060-19	Abrasion Resistance of Organic Coatings by the Taber Abraser
8.	ASTM D4285-83(2018)	Indicating Oil or Water in Compressed Air
9.	ASTM D4541-17	Pull-Off Strength of Coatings Using Portable Adhesion Testers
10.	ASTM D4587-11(2019)	Conducting Tests on Paint and Related Coatings and Materials Using a Fluorescent UV-Condensation Light- and Water-Exposure Apparatus
11.	ASTM D5894-21	Cyclic Salt Fog/UV Exposure of painted Metal, (Alternating Exposures in a Fog/Dry Cabinet and a UV/Condensation Cabinet)
12.	ASTM G8-96(2019)	Cathodic Disbonding of Pipeline Coatings

B. American Water Works Associations (AWWA)

1.	AWWA C 116-16	Protective Fusion-Bonded Coatings for the Interior and Exterior Surfaces of Ductile-Iron and Gray-Iron Fittings
2.	AWWA C 213-15	Fusion-Bonded Epoxy Coatings and Linings For Steel Water Pipe and Fittings
3.	AWWA C 222-25	Polyurethane Coatings and Linings for Steel Water Pipe and Fittings
4.	AWWA C 550-24	Protective Interior Coatings for Valves and Hydrants

C. NSF International (NSF)

1.	NSF 61-2016	Drinking Water System Components

D. The Society for Protective Coatings (SSPC)/NACE International (NACE)

1.	SSPC-AB1-19	Mineral and Slag Abrasives
2.	SSPC-AB2-16	Cleanliness of Recycled Ferrous Metallic Abrasives
3.	SSPC-AB3-17	Newly Manufactured or Re-Manufactured Steel Abrasives
4.	SSPC-Guide 15-13	Field Methods for Retrieval and Analysis of Soluble Salts on Substrates and Other Nonporous Substrates
5.	SSPC-PA1-16	Shop, Field, and Maintenance Painting of Steel
6.	SSPC-PA2-22	Measurement of Dry Paint Thickness with Magnetic Gages
7.	SSPC-QP1-15	Evaluating Painting Contractors (Field Application to Complex Industrial Structures)
8.	SSPC-QP3-10/AISC 420	Evaluating the Qualification of Shop Painting Contractors
9.	SSPC-SP1-16	Solvent Cleaning
10.	SSPC-SP2-04	Hand Tool Cleaning
11.	SSPC-SP3-04	Power Tool Cleaning
12.	SSPC-SP5/NACE 1-07	White Metal Blast Cleaning
13.	SSPC-SP10/NACE 2-15	Near-White Blast Cleaning
14.	SSPC-SP11-20	Power Tool Cleaning to Bare Metal, Level 3 minimum requirement

15.	SSPC-VIS1-12	Guide and Reference Photographs for Steel Surfaces Prepared by Abrasive Blast Cleaning
16.	SSPC-VIS3-07	Visual Standard for Power- and Hand-Tool Cleaned Steel
17.	SSPC Paint Manual	Volume 1, Good Painting Practices, Fourth Edition 2002
18.	NACE SP 0188-06	Discontinuity (Holiday) Testing of New Protective Coatings on Conductive Substrates
19.	NACE SP 0274-11	High Voltage Electrical Inspection of Pipeline Coatings
20.	NACE SP 0287-16	Field Measurement of Surface Profile of Abrasive Blast Cleaned Steel Surfaces Using a Replica Tape
21.		
22.	NACE RP 0288-88	Recommended Practices for Inspection of Linings on Steel and Concrete.
23.	NACE SP 0178-07	Detail, Finish, and Design Considerations for Immersion Service.

1.03 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 - Submittals.

B. RSN 09 96 20-1, Approval Data:

1. For each coating material selected under Coating Tabulations 2 through Tabulation 5:

   a. Manufacturer's product data sheet, application sheet, and Safety Data Sheet (SDS).

   b. Identification of "Items to be coated" including sub-letter and sub-number listed in Coating Tabulations.

   c. For Coating Tabulations 2 and 5: For use in potable water complete both tables in full for item(s) requiring National Sanitation Foundation (NSF) 61 or Underwriter Laboratory (UL)certification.

Table 1: Coating Material Selection and Item Details

Coating Tabulation Number and Letter	Item to be coated	Item Size Range*	Coating Product Name	Selected Color

*Include size ranges in case of multiple items. Gallons for tanks, pipe diameter, pump size (diameter), etc.

Table 2: NSF/UL Certification Requirements and Limitations

Coating Product Name	Equipment Type**	Certified Structure Size Range	Certified Color(s)	Certified DFT Range	Reference*** (Website or Documents)

**State what equipment the NSF/UL requirements are for i.e. pumps, valves, pipes, tanks, etc.

***Provide website or documentation if website is not available

2. For each caulk and filler:
   a. Manufacturer's product data, application, and SDS sheets.
   b. Identification of areas of use including sub-letter and sub-number listed in Coating Tabulations.
3. For each substituted coating material "equal" to specified product in Coating Tabulation:
   a. List of not less than 5 similar successful material applications:
      1) Project name, location, type of structure, and application dates.
      2) Owner's name, address, and telephone number.
   b. Certification that material meets or exceeds specified requirements:
      1) Manufacturer's certification including technical representative's signature and date of signature.
      2) Independent laboratory's certified test reports.
4. Products under Tabulations No. 01, 05, and 06 do not require submission for coating approval data.

C. RSN 09 96 20-2, Final Approval Data:
   1. For each coating material:
      a. Purchase orders including order number, date, manufacturer's designated product name, batch number(s), and quantity purchased.

b. Supplier's name, address, and phone number.

D. RSN 09 96 20-3, Paint Chip Samples:

1. Color chip samples approximately 4 by 6-inch.
2. Label each sample to include manufacturer's designated product name, color, and gloss.
3. Supply paint chip samples to the COR for selection and approval.

E. RSN 09 96 20-4, Qualifications:

1. Coating Contractor:
   a. Provide current AMPP-QP1 certification for field coating and AMPP-QP3 certification for shop coating to be requested as needed and at the discretion of the COR for Tabulations 4 and 5.
   b. List not less than 5 comparable completed jobs for each coating type including owner's name, address, and telephone numbers completed within the last five years.
2. Coating Applicator(s):
   a. List of not less than 5 comparable jobs for each coating type including owner's name, address, and telephone number.
   b. Provide one of the options below indicating personnel are qualified with relevant knowledge of surface preparation, coating materials, mixing, thinning and application practices:
      1) Statement from contractor.
      2) Statement from coating or equipment supplier
3. Coating Contractor Inspector(s):
   a. Provide one of the following to document qualifications to inspect specified coating materials:
      1) Current certification for NACE Certified Coatings Inspector (CIP) Level 3 or Association for Materials Protection and Performance (AMPP) Senior Certified Coatings Inspector.
      2) Letter from contractor listing the personnel performing inspections and current certifications for each.
         a) Inspector: certified at a minimum as NACE CIP Level 1 or AMPP Basic Coating Inspector.
         b) Supervisor: review and certify inspection results as a NACE CIP Level 3 of AMPP Senior Certified Coatings Inspectors.

F. RSN 09 96 20-5, Quality Control Plan:

1. Quality Control Manager:
    a. Name and title.
2. Quality Control Procedure:
    a. Quality control plan.
        1) Factory, shop, or field testing procedures shall be in accordance with these specifications, product manufacturer's written instructions, and NACE RP 0288, and ASTM D3276.
    b. Procedure for addressing identified non-conformities.

G. RSN 09 96 20-6, Contractor Quality Testing Report:
1. Submit for coating Tabulations 3 through 5
2. Make daily reports for review by the COR, including:
    a. Date of work.
    b. Description of areas or items and work performed.
    c. Datasets clearly labeled with corresponding Tabulation number, item number, coating system, and coat.
    d. Environmental conditions.
        1) Measurements to be collected near area of work:
            a) Air temperature
            b) Surface temperature
            c) Dew point
            d) Relative humidity
    e. Results from the following quality control tests:
        1) Soluble salt.
        2) Surface profile.
        3) Visual comparison of prepared surfaces.
        4) Assessment of dust contaminants.
            a) Provide photo of test results or confirmation of test completion.
        5) Compressed air quality.
        6) Coating color for NSF 61 coatings
        7) Dry film thickness (DFT).
            a) Provide spot measurements, as defined per SSPC PA 2, for each coat of each selected coating system.

b) Each dataset shall be clearly labeled with corresponding Tabulation number, item number, coating system, and coat.

c) Provide DFT data organized sequentially by Tabulation number.

8) Discontinuity (Holiday) testing.

9) Coating adhesion testing (when directed by the COR).

1.04 DELIVERY, STORAGE, HANDLING

A. Deliver materials to jobsite in original, undamaged, unopened containers labeled with manufacturer's name, designated product name, batch number, date of manufacture, and any special instructions.

B. Deliver materials in containers not larger than 5 gallons as packaged by manufacturer unless suitable equipment is provided at jobsite to handle and thoroughly mix materials in larger containers.

C. Store coating materials and thinners in accordance with the manufacturer's recommendations, and as a minimum, store off the ground, under cover, and at 50- to 95-degrees Fahrenheit (F).

D. Do not use coating material which has exceeded manufacturer's specified storage stability period (shelf life).

1.05 AMBIENT CONDITIONS

A. Maintain environmental conditions to meet coating manufacturer's requirements during surface preparation, coating application, and curing period. Provide heating or cooling, and dehumidification as needed. Complete curing before placing the coating systems into service.

B. Coatings shall be applied when the substrate surface temperature is 5 degrees F or more above the dew point temperature and for at least 8 hours after application.

1.06 PROJECT CONDITIONS

A. Galvanic isolation of dissimilar metals shall be required. Coatings in this specification section shall not serve as the primary galvanic isolation.

B. Contractor shall comply with applicable federal, state, and local air pollution and environmental control regulations for surface preparations, blast cleaning, disposal of spent materials and debris, and coatings application.

PART 2 PRODUCTS

2.01 EQUIPMENT

A. Surface preparation and spray application equipment:

1. Sized in accordance with coating manufacturer's recommendations.
2. Provide:
    a. Pressure gauges and pressure regulators.
    b. Air supply lines free from oil and moisture.

2.02 ABRASIVES

A. Mineral and slag abrasives: Meets SSPC-AB1, type I (natural minerals) and type II (slags), class A.

B. SSPC-AB2 for recycled cleanliness.

C. SSPC-AB3 Class I (steel) or II (iron) for angular shaped grit. Shot abrasives are not acceptable.

D. Commercially available products or processes to render the hazardous waste stream non-hazardous and not exceeding toxicity threshold limits.

1. Additive stabilizers for leachable toxic metal wastes are allowed, except elemental iron is not permitted.

2.03 COATINGS

A. Specified in Coating Tabulations:

1. Apply only one coating system per option in Coating Tabulations.
2. Apply compatible products of same manufacturer for coating system components.

B. Volatile Organic Compounds (VOC): Do not exceed maximum permitted by Federal, State, and local air pollution control regulations as supplied in container or by addition of thinner material.

C. Factory color or tint only. Do not color or tint at jobsite.

D. Use manufacturer's recommended thinners for coating materials.

E. Do not use accelerator products unless approved by COR.

2.04 FILLERS AND CAULKS

A. Caulks:

1. Coating manufacturer's standard or compatible caulk material.

2. Polyurethane, silicone, or silicone acrylic caulk material: Meets or exceeds ASTM C920 type S or M, grade NS, class 25, suitable for water immersion service.

B. Fillers:

1. Apply compatible products of same manufacturer as coating system components.

2. Approved fillers:

a. Carboguard 501, as manufactured by Carboline, www.carboline.com.

b. Devmat 142HB, as manufactured by International Paint, www.international-pc.com.

c. PPG Amercoat 114 A, as manufactured by PPG, www.ppgpmc.com.

d. Steel Seam FT910, as manufactured by Sherwin Williams, www.sherwin-williams.com.

e. Surfacing Epoxy Series 215 as manufactured by Tnemec, www.tnemec.com.

3. Provide one of the above fillers, or equal with following essential characteristics:

a. Coating manufacturer's standard filler material.

b. Approved use by manufacturer.

2.05 MISCELLANEOUS MATERIALS

A. Deionized Water.

B. Chlor-Rid as manufactured by Chlor-Rid, www.Chlor-rid.com,

C. HoldTight 102 as manufactured by Holdtight, Corrosion Innovations , www.holdtight.com,

D. Aquron 7000, as manufactured by Aquron, www.aquron.com

E. Provide one of the above salt removers, or equal with following essential characteristics:

1. Commercially available additive for soluble salt removal.

2. Water soluble.

3. Biodegradable.

4. Zero volatile organic compounds.

PART 3 EXECUTION

3.01 SHOP PAINTING

A. Shop coated and lined equipment shall be primed and finished in the shop and touched up in the field with identical coating materials after installation and in accordance with the coatings manufacturer's standard surface preparation and application instructions.

3.02 FIELD PAINTING

A. Follow specified information in the Coating Tabulations. When application not specified in Coating Tabulations, follow manufacturer's standard surface preparations and application instructions for field applied coatings.

3.03 COATING OF METAL WORK EMBEDDED IN CONCRETE

A. For atmospheric service exposure with item partially embedded in concrete:

1. Coat exposed surfaces completely.
2. Extend prime coat on steel six inches minimum into concrete embedment measured from face of concrete and along surface of embedded item.

B. For burial or immersion service exposure on either side of concrete faces:

1. Extend primer coat on steel continuously through solid concrete structure.
2. Examples include buried pipe which will be embedded in concrete through anchor blocks, walls, and related concrete structures.

C. Caulk the concrete/steel interface to seal gaps at exposed faces in accordance with manufacturer's instructions with maximum thickness of 1/2-inch.

3.04 GALVANIC AND DIELECTRIC ISOLATION

A. Galvanic isolation of dissimilar metals shall be required.

B. Dielectrically isolated ductile iron pipe, valves, and fittings from other metals.

C. Coatings shall not serve as the primary galvanic isolation.

3.05 PROTECTION OF ADJACENT SURFACES

A. Remove, mask, or otherwise protect items or surfaces not coated from contamination and damage during cleaning and coating operations.

B. Examples include protecting mating and machined surfaces, mechanical and electrical equipment (open or enclosed), instruction and similar plates, and wet and newly coated surfaces.

3.06 SURFACE PREPARATION

A. Remove or repair surface irregularities and impart profile as specified:

1. Welds:

a. Welds: Smooth and continuous.

b. Weld spatter, protrusions, laminations, slag, buckshot, and slivers: Remove and grind smooth in accordance with NACE SP 0178.

c. Undercuts and porosity: Fill with weld material and grind smooth.in accordance with NACE-SP 0178.

d. Projections, sharp edges, high points, fillets, and corners: Grind smooth to a radius of at least 1/16-inch.

2. Grind full length of edges and corners smooth to a radius of at least 1/16-inch.

B. Surface preparation method:

1. See Coating Tabulations.

2. Compare prepared steel surfaces to following visual reference photographs for allowable visible contaminants and stains:

a. SSPC-VIS1 for abrasive blast cleaning.

b. SSPC-VIS3 for power and hand tool cleaning.

C. Surface profile:

1. Prepare as specified in Coating Tabulations.

2. Where not specified, prepare in accordance with manufacturer's instructions for metallic substrates or existing coating surfaces and service environment.

3. Where manufacturer's instructions do not specify a surface profile, prepare blasted surfaces to following profile:

a. Atmospheric Service Environments: 1-mil or greater angular profile and less than specified millage of first applied coat.

b. Burial and Immersion Service Environments: Angular profile 2-mil or greater and less than specified millage of first applied coat.

4. Inspect surface profile in accordance with NACE SP 0287:

a. Use replica tape suitable for surface profile depth range.

b. Perform at least two tests per 1,000-square feet.

D. Assessment of dust contaminants:

1. Perform clear tape test method in accordance with ISO 8502-3.

2. Perform three tests per shift.

3. Acceptance Criteria: Degree of cleanliness shall meet Class 3.

E. Soluble salts:

1. Measure concentration of chlorides, nitrates and sulfates in accordance with SSPC Guide 15.
2. Perform at least two tests per 1,000-square feet of surface area or per item. Include corrosion pit areas.
3. Acceptance criteria: Do not exceed 3-micrograms per square centimeter for chlorides and 5-micrograms per square centimeter for nitrates and sulfates.
4. Re-clean or perform additional surface preparation of contaminated surfaces before coating application.

F. Heat cast iron components to between 140- and 180-degrees Fahrenheit to evaporate moisture for 2-hours minimum.

3.07 COMPRESSED AIR EQUIPMENT

A. Keep air compressor and spray application equipment lines free of oil and moisture during work.

B. Compressed air quality:

1. Perform tests in accordance with ASTM D4285.
2. Perform test at the beginning of every shift.

3.08 FILLERS AND CAULK APPLICATION

A. Caulk areas including crevices, gaps, plate seams, skip welds, and conduit penetrations.

B. Fill areas including plate seams, and gaps at bolts and rivets.

C. Apply in a uniform texture, neatness, and color matched appearance.

D. Apply in accordance with manufacturer’s instructions and achieve tight bond to substrate or previous coats.

3.09 COATING APPLICATION

A. Coating shall be done in a competent manner to produce an even film of uniform thickness. The finished surfaces shall be free from runs, drops, ridges, waves, laps, sags, curtains, brush marks, and variations in color, texture, and finish.

B. Shop-coated and shop-lined equipment shall be primed and finished in the shop and touched up in the field with identical coating materials after installation and in accordance with the coating manufacturer’s application instructions.

C. Follow specified systems in the Coating Tabulations and coating manufacturer’s application instructions for applied coatings. In event of a conflict, the more stringent requirement shall apply.

D. Stripe coat:

1. Brush-apply stripe coat to each coat applied.
2. Apply stripe coat prior to spray application of the primer, when practical.
3. Apply to edges, corners, interior angles, pits, plate seams, crevices, rivets, nuts, threads, bolts, washers, welds, and similar features.

E. Apply number of coats and coating thickness specified in Coating Tabulations.

1. Alternate coat colors to differentiate between coats.
2. Apply in accordance with manufacturer's instructions.
3. Apply in accordance with SSPC PA1.

F. Prevent overspray or dry spray. When such conditions occur, clean surfaces prior to application of subsequent coats.

G. Re-clean contaminated surfaces between coats.

H. Dry Film Thickness (DFT):

1. After an individual coat has cured, measure DFT in accordance with SSPC PA2.
    a. Acceptance Criteria:
        1) Each area measurement must fall within specified DFT range.
            a) One area is the average of 5 spot measurements.
        2) Each spot measurement must fall within 80-120 percent of the specified DFT range.
            a) One spot measurement is the average of three gauge measurements within 1.5-inch diameter area.

I. Discontinuity (Holiday) Testing:

1. Inspect in accordance with NACE SP 0188:
    a. Use maximum test voltage for DFT as recommended by coating manufacturer to prevent coating damage.
2. Inspect pipeline coatings in accordance with NACE SP 0274:
    a. Use maximum test voltage for any DFT as recommended by coating manufacturer to prevent coating damage.

J. Coating Adhesion:

1. Test in accordance with ASTM D4541 at the direction of the COR.

3.10 CONTRACTOR QUALITY TESTING

A. Inspection Equipment:

1. Calibrate inspection devices and equipment in accordance with the manufacturer's instructions.
2. Verify calibration prior to use daily.

B. Qualified personnel shall conduct tests and sign reports.

1. Perform tests in presence of the COR:
2. Provide rigging for COR to observe inspections.

C. Make daily reports for review by the COR, including:

1. Date of work.
2. Description of areas or items and work performed.
3. Datasets clearly labeled with corresponding Tabulation number, item number, coating system, and coat.
4. Environmental conditions.
    a. Measurements to be collected near area of work:
        1) Air temperature.
        2) Surface temperature.
        3) Dew point.
        4) Relative humidity.
5. Results from the following quality control tests:
    a. Soluble salt.
    b. Surface profile.
    c. Visual comparison of prepared surfaces.
    d. Assessment of dust contaminants.
        1) Provide photo of test results or confirmation of test completion.
    e. Compressed air quality.
    f. Dry film thickness (DFT).
        1) Provide spot measurements, as defined per SSPC PA 2, for each coat of each selected coating system.
        2) Each dataset shall be clearly labeled with corresponding Tabulation number, item number, coating system, and coat.
        3) Provide DFT data organized sequentially by Tabulation number .
    g. Discontinuity (Holiday) testing.
    h. Coating adhesion testing (when directed by the COR).

D. Final inspection will take place after repairs, but before placing into service.

3.11 REPAIR AND REMEDIAL WORK

A. Repair defective or damaged areas of coatings in accordance with coating manufacturer's recommendations.

3.12 COATING TABULATIONS

Tabulation No. 01		
Items to be coated - Commercial or manufactured items with factory or shop applied coating system using coating manufacturer's standard coating material: a. Valve actuators and motors b. Electrical cabinets, enclosures, and junction boxes c. Gutters d. Steel ornamental fence		
Coating materials	Number and thickness of coats	Surface preparation method
Factory or Shop applied permanent coating system: 1. Factory or shop standard surface preparation and permanent coating system. 2. Items subject to sunlight shall be topcoated with ultraviolet (UV) stable materials. 3. Permanent coating system shall have 10-year corrosion-free protection without significant defects stored in atmospheric exposure. 4. Color and gloss: Manufacturer's standard color unless specific color is otherwise specified in color schedule. 5. Unless otherwise specified, unexposed surfaces that require coatings, such as interior of cabinets, enclosures, and equipment, shall be given manufacturer's standard permanent coated finish.		
Field Repair: Repair damaged areas of coated surfaces with compatible materials to equal thickness and color match of undamaged areas, unless otherwise tabulated herein.		

Tab AE-05.DOC

Tabulation No. 02		
Items to be coated - Commercial or manufactured items with factory or shop applied NSF 61 approved coating system using coating manufacturer's standard coating material: a. Gate valves b. Globe style silent check valves c. Electro-magnetic flowmeter d. Flow control valves e. Ball valves f. Combination air valve assemblies g. Grooved-end flexible couplings h. Orifice plate for flow control valve		
Coating materials	Number and thickness of coats	Surface preparation method

Tabulation No. 02
Factory or Shop applied permanent coating system: 1. Factory or shop standard surface preparation and permanent coating system. 2. Surfaces designed to be in contact with water shall be coated with NSF 61 approved materials designed for potable water. 3. Items subject to sunlight shall be topcoated with ultraviolet (UV) stable materials. 4. Permanent coating system shall have 10-year corrosion-free protection without significant defects stored in atmospheric exposure. 5. Color and gloss: Manufacturer's standard color unless specific color is otherwise specified in color schedule. 6. If orifice plate is stainless steel do not coat and use galvanic isolation kit
Field Repair: Repair damaged areas of coated surfaces with compatible materials to equal thickness and color match of undamaged areas, unless otherwise tabulated herein.

Tab AE-05.DOC

Tabulation No. 03
Items to be coated: Exterior Atmospheric Exposure a. Doors and door frames for Turnout Buildings
Notes: 1. Repair damaged areas with same shop applied primer and finish coat materials or compatible materials. 2. Stainless steel, aluminum, brass, and non-metallic parts need not be painted.
Coating Option 1: Manufacturer’s standard surface preparation and applied permanent coating system that meets the following: 1. 10-year corrosion-free warranty without significant defects in sunlight-exposed atmospheric exposure.
Coating Option 2: Baked-on, coil, extrusion, or fusion bonded that consists of 1 or more coats of UV stable system.

Tabulation No. 03		
Coating materials – Option 3 Waterborne acrylic	Number and thickness of coats	Surface preparation method
Prime or spot repair Carboline: Carbocrylic 3358/ Carbocrylic 3359 International Paint: Devflex 4020PF DTM/ Devflex 4216HP Sherwin Williams: Pro Industrial Procryl Universal Acrylic Primer series B66-W00310/ Pro Industrial Procryl DTM Acrylic Color and gloss: See Table 09 96 20A – Color Schedule	1 or more prime coats, apply at 2 to 3 mils DFT per coat, plus stripe coats. 2 or more finish coats, apply at 2 to 3 mils DFT per coat. Total system excluding stripe coats: 6-mil DFT, minimum 9-mil DFT, maximum	Undamaged coated surfaces to receive additional coats: SSPC-SP 1 Uncoated surfaces: SSPC-SP 6 Damaged coated surfaces: SSPC-SP 6 and SSPC-SP 11 where abrasive blasting is impractical Aluminum, brass, bronze, and copper surfaces: lightly abrade by SSPC-SP 2 or SSPC-SP 3 Galvanizing: SSPC-SP 7, SSPC-SP 3, or manufacturer's chemical treatment
PERFORMANCE REQUIREMENTS:		
Testing Protocol	Acceptance Criteria	
Abrasion resistance, ASTM D4060, CS-17 wheel, 1,000 cycles, 1-kg load:	250 milligram loss or less	
Direct impact, ASTM D2794:	40 inch-pounds for alkyd/silicone alkyd 140 inch-pounds for waterborne acrylics	
Pulloff adhesion, ASTM D4541, annex A2, type II tester:	250 psi or greater for alkyd/silicone alkyd 500 psi or greater for waterborne acrylics	
Cyclic testing salt fog/UV, ASTM D5894:	1/4 inch or less undercutting at 3000 hrs	
QUV accelerated weathering test, ASTM D4587:	Passes 3,000 hour test with no blisters evident on either scribed or unscribed sides, or color difference, ASTM D2244	

Tab AE-02.DOCX

Tabulation No. 04
Items to be coated - Shop or field applied exterior coatings for non-UV atmospheric and burial exposure: a. Exterior surface of 2-, 3-, 6-, 8-, and 12-inch steel pipe and fittings b. Exterior surface of pipe reducers c. Flange pipe supports d. Touch up and spot repairs e. Interior and exterior surfaces of manhole covers

Notes:

1. Do not paint aluminum-brass, bronze, and stainless-steel metalwork.
2. Steel pipe shall have a minimum holdback of 3-inches from welding areas.
3. Coat holdbacks and complete touchup repairs with same material or compatible material approved by coating manufacturer.
4. Complete spot repairs using same or compatible material in accordance with manufacturer recommendation of existing coating on pipe.
5. Do not coat seals, mating surfaces, and machined surfaces where paint could interfere with operation or fit. Coat exposed machined surfaces after installation.
6. Coated ferrous surfaces permanently exposed to direct sunlight shall require a compatible UV resistant protective topcoat.
7. Coat miscellaneous metalwork not otherwise tabulated.
8. Coat all associated fasteners to include nuts, bolts, washers, and exposed anchors.
9. Topcoat exterior surface of manhole cover exposed to sunlight with manufacturer compatible UV resistant polyurethane or polysiloxane.
10. Do not coat 2-inch pipe is stainless steel or non-ferrous and galvanically isolate from steel pipe using isolation kit.

COATING OPTIONS

OPTION 1 - Polyurethane exterior in accordance with AWWA C222.

OPTION 2 - Fusion bonded epoxy in accordance with either AWWA C213 interior and exterior of steel pipe and/or fittings or AWWA C116 interior and exterior of ductile iron or grey-iron pipe and/or fittings.

OPTION 3 - Coating materials Epoxy	Number and thickness of coats	Surface preparation method
Carboline: Carboguard 690 Sherwin Williams: Duraplate 235 Tnemec: Epoxoline Series N69 or V69 PPG: Amercoat 235 International: Bar Rust 235 Color and gloss: see Color Schedule	3 or more base coats apply at 6 to 8 mils DFT per coat, plus stripe coats. Total base coat system excluding stripe coats: 18-mil DFT, minimum 24-mil DFT, maximum	SSPC-SP1 followed by: For steel: SSPC-SP10/NACE 2 For ductile iron: NAPF 500-03-04 For cast-iron: NAPF 500-03-05 2.5-4.0 mil profile Follow manufacturer's surface preparation and application instructions to apply subsequent coats.
OPTION 4 – 100% Solids Epoxy	Number and thickness of coats	Surface preparation method

Carboline Plasite 4500 Tnemec: Series 431 Duromar: HPL-2510 PPG Amercoat 113 International Paint: Enviroline 376F-30LT Sherwin Williams: Dura-Plate UHS	Follow coating manufacturer's instructions for number of coats and thickness per coat. Total coating system excluding stripe coats: 20-mil DFT, minimum 40-mil DFT, maximum	SSPC-SP1 followed by: For steel: SSPC-SP10/NACE 2 For ductile iron: NAPF 500-03-04 For cast-iron: NAPF 500-03-05 2.5-4.0 mil profile Follow manufacturer's surface preparation and application instructions to apply subsequent coats.

PERFORMANCE REQUIREMENTS	
Testing Protocol	Acceptance Criteria
Fresh/deionized water immersion test, ASTM D870:	passes, 2 years test with aerated water held at ambient temperature with no blisters evident on either scribed or unscribed sides.
Dilute Harrison immersion test, Modified ASTM D870:	passes, 2 years test with aerated water held at ambient temperature with no blisters evident on either scribed or unscribed sides.
Abrasion resistance, ASTM D4060, CS-17 wheel, 1,000 cycles, 1-kg load:	100 milligram or less
Direct impact, ASTM D2794:	30 inch-pounds
Pulloff adhesion, ASTM D4541, annex A2, type II tester:	1000 psi or greater
Cyclic testing salt fog/UV, ASTM D5894:	1/4 inch or less undercutting at 3000 hrs
Cathodic disbondment, ASTM G8:	1/4 inch or less disbondment at 120-day test

Tab BE-04.DOCX

Tabulation No. 05
Items to be coated: For potable water service, NSF 61/600 approved coating:

Tabulation No. 05

a. Interior surface of 2-, 3-, 6-, 8-, and 12-inch steel pipe and fittings
b. Interior surface of pipe reducers

Notes:

1. Materials listed below require certified NSF/ANSI 61 approval for contact with potable water. Apply products in accordance with manufacturer's requirements and NSF/ANSI 61 requirements to meet NSF/ANSI 61 certification for specific items as applied.
2. Steel pipe shall have a minimum holdback of 3-inches from welding areas.
3. Coat holdbacks and complete touchup repairs with same material or compatible material approved by coating manufacturer.
4. Do not coat stainless steel, bronze, or non-metallic surfaces.
5. Do not coat mating surfaces, machine surfaces, seals, and surfaces where the coating could interfere with proper operation or fit.
6. Coat exposed machined surfaces after assembly.
7. Coat all fasteners to include nuts, bolts, washers, and exposed anchors.
8. Remove amine blush prior to service.
9. Do not coat 2-inch pipe is stainless steel or non-ferrous and galvanically isolate from steel pipe using isolation kit.

OPTION 1 – Fusion Bonded Epoxy in accordance with NSF/ANSI 61 and either AWWA C213 interior and exterior of steel pipe and/or fittings or AWWA C116 interior and exterior of ductile iron or grey-iron pipe and/or fittings

When applicable in accordance with AWWA C550 Protective Interior Coatings for Valves and Hydrants

OPTION 2 – Liquid Applied Epoxy in accordance with NSF/ANSI 61 and either AWWA C210 Liquid-Epoxy Coatings and Linings for Steel Water Pipe and Fittings.

When applicable in accordance with AWWA C550 Protective Interior Coatings for Valves and Hydrants

Tab IE-03.DOC

Tabulation No. 06
Items not to be coated and to receive a uniform passivation layer - Damaged Stainless Steel: a. Stainless steel surfaces not otherwise tabulated herein.
Note: 1. Do not coat stainless steel items listed above. Stainless steel surfaces to receive uniform passivation. 2. Do not damage attached parts, adjacent parts, or materials by field cleaning and passivation methods of stainless steel.
Clean, restore, and inspect uniform passivation layer on damaged, contaminated, or depassivated stainless steel surfaces in accordance with ASTM A380.

Tab AE-14.DOCX

Tabulation No. 07		
Items to be coated for UV-Atmospheric Exposure: a. Exterior surface of guard posts b. Exterior surfaces of welded rectangular barriers		
Notes: 1. Do not coat stainless steel, bronze, or non-metallic surfaces. 2. Do not coat mating surfaces, machine surfaces, seals, and surfaces that would be functionally impaired by the coating. 3. Coat exposed machined surfaces after assembly. 4. Coat all associated fasteners to include nuts, bolts, washers, and anchors after installation.		
Coating materials – Option 1 Zinc Primer/Epoxy/ Polyurethane	Number and thickness of coats	Surface preparation method
For ferrous surfaces - Carboline: Carbozinc 859/Carboguard 890/Carbothane 134 HB, HG, or VOC PPG: Amercoat 68HS or Amercoat 68HS VOC/Amerlock 400 or Amerlock 2/Amercoat 450H or Amershield VOC Sherwin Williams: Zinc Clad III HS or	1 or more prime coats apply at 3 to 5 mils DFT, per coat, plus stripe coats. 1 or more intermediate coats apply at 6 to 8 mils DFT, per coat, plus stripe coats. 1 or more compatible manufacturer's finish coats apply at 3 to 5 mils DFT, per coat. Total system excluding stripe coats:	SSPC-SP1 followed by: For steel: SSPC-SP10/NACE 2 or SSPC-SP11 where abrasive blasting is impractical For ductile iron: NAPF 500-03-04 For cast-iron: NAPF 500-03-05 2.5-4.0 mil surface profile

Tabulation No. 07		
Zinc Clad III HS 100/ Duraplate 235 or Macropoxy 646/ Hi solids polyurethane or Hi solids polyurethane 250 Tnemec: Tneme-zinc 90-97/ Epoxoline Series N69 or Epoxoline Series V69/ Endura-shield 175 Color and gloss: As selected by OGR (see RSN 09 96 20-3)	12-mil DFT, minimum 18-mil DFT, maximum	Follow manufacturer's surface preparation and application instructions to apply subsequent coats.
Coating materials – Option 2 Zinc Primer/Epoxy/ Polysiloxane	Number and thickness of coats	Surface preparation method
Carboline: Carbozinc 859/ Carboguard 690/ Carboxane 2000 PPG: Amercoat 68HS or Amercoat 68HS VOC/ Amerlock 2 or 400/ PSX 700 Sherwin Williams: Zinc Clad III HS or 4100/ Duraplate 235/ Sherloxane 800 Color and gloss: see color schedule	1 or more prime coats, apply at 3 to 5 mils DFT per coat, plus stripe coats. 1 or more intermediate coats, apply at 5 to 7 mils DFT per coat. 1 or more finish coats, apply at 3 to 7 mils DFT per coat. Total system excluding stripe coats: 11-mil DFT, minimum 19-mil DFT, maximum	SSPC-SP1 followed by: For steel: SSPC-SP10/NACE 2 or SSPC-SP11 where abrasive blasting is impractical For ductile iron: NAPF 500-03-04 For cast-iron: NAPF 500-03-05 2.5-4.0 mil surface profile Follow manufacturer's surface preparation and application instructions to apply subsequent coats.
PERFORMANCE REQUIREMENTS:		
Testing Protocol	Acceptance Criteria	
Abrasion resistance, ASTM D4060, CS-17 wheel, 1,000 cycles, 1-kg load:	Epoxy/Polyurethane: 100 milligram loss or less Epoxy/Polysiloxane: 120 milligram loss or less	

Tabulation No. 07	
Direct impact, ASTM D2794	Epoxy/Polyurethane: 30 inch-pounds Epoxy/Polysiloxane: 40 inch-pounds
Pulloff adhesion, ASTM D4541, annex A2, Type II tester:	Epoxy polyurethane: 800 psi or greater Epoxy/Polysiloxane: 1000 psi or greater
Cyclic testing salt fog/UV, ASTM D5894:	1/4-inch or less undercutting at 3,000-hours
QUV accelerated weathering test, ASTM D4587:	Passes 3,000-hour test with no blisters evident on either scribed or unscribed sides, or color difference ASTM D2244.

Tabulation No. 08
Repair of Damaged Galvanizing: a. Combination channels, connection plates, and associated electrical mounting equipment b. Cathodic Protection test station pipe
Note: Repair damaged galvanized surfaces in accordance with ASTM A780. Repair materials containing cadmium and lead are not permitted.

Tab AE-11.DOCX

3.13 COLOR SCHEDULE

A. Colors and glosses of finished coats:

1. To be selected and approved by COR.

END OF SECTION

SECTION 10 14 00
SIGNAGE

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for Turnout Buildings.

1.02 REFERENCE STANDARDS

A. National Fire Protection Association (NFPA)

1.	NFPA 704 2022	Standard System for the Identification of the Hazards of Materials for Emergency Response

B. International Code Council (ICC)

1.	IFC 2021	International Fire Code
2.	IBC 2021	International Building Code
3.	ICC/ANSI A117.1-17	Accessible and Usable Buildings and Facilities

1.03 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals.

B. RSN 10 14 00-1, Shop Drawings and Approval Data:

1. Manufacturer's specifications and technical data indicating compliance with this section.
2. Submit list of required signs.
3. Complete set of manufacturer's standard color and product samples for selection of signs, lettering, character, and background colors, material, and style. Color copies of these items are not acceptable and will be returned not approved.
4. Submit shop drawings for each type of sign showing dimensions of plaque and characters, and text, symbols, and braille to be included on each sign. Shop drawings should also include proposed locations for each sign on a floor or site plan and typical horizontal and vertical locations of signage.

PART 2 PRODUCTS

2.01 GENERAL

A. "No Smoking" signs in accordance with IFC Sections 5003.7.

B. Hazard Identification Signs in accordance with IFC Sections 5003.5 and 5003.6 and NFPA 704.

C. Label piping within the plant with fluid type (air, water, etc.) and flow direction signage/stickers.

D. Provide metal nameplates (valve tags) for valves as detailed in Section 35 22 14 – Valves and Equipment.

2.02 EXTERIOR ALUMINUM SIGNS

A. Coated aluminum exterior signs manufactured by Signs.com, 1550 South Gladiola Street, Salt Lake City UT, 84104; www.signs.com, or equal, having the following essential characteristics:

1. One-sixteenth-inch-thick coated aluminum material; corrosion-resistant, abrasion-resistant, rust-resistant, heavy-duty, rigid, weather-resistant.
2. Two panels, approximately 22"x30", each panel.
3. UV-resistant graphics in blue and red on white reflective background.
4. NTUA logo prominently at upper portion of each panel.
5. Characters: Sans serif. Bilingual signage with English text on one panel and Navajo text on a companion panel:
   a. English: COMMUNITY DRINKING WATER
   b. Navajo: Nihidine'é Bá Tó 'Adoodíłígíí
6. Pre-drilled holes.
7. Rounded corners.
8. Braille: No.
9. Mounting: On perimeter site fence adjacent to gate where visible from roadway, as recommended by manufacturer for conditions.

10. Basis of Design example:



2.03 INTERIOR SIGNS

A. Coated aluminum interior signs manufactured by Signs.com, 1550 South Gladiola Street, Salt Lake City UT, 84104; www.signs.com, or equal, having the following essential characteristics:

1. One-sixteenth-inch-thick coated aluminum material; corrosion-resistant, abrasion-resistant, rust-resistant, heavy-duty, rigid.
2. Non-reflective.
3. Pre-drilled holes.
4. Rounded corners.
5. Characters: Upper case, sans serif.
6. Color: To be determined by Government from manufacturer's standard color chart.
7. Braille: No.
8. Mounting: On interior walls or other features as required, as recommended by manufacturer for surface conditions.

B. Acrylic interior signs manufactured by Office Sign Company, LLC, 310 NP Avenue N., Fargo ND 58102; www.officesigncompany.com or equal, having the following essential characteristics:

1. In accordance with IBC Section 1013.4.
2. One-eighth-inch-thick acrylic material.
3. Self-extinguishing.
4. Rounded corners.
5. Characters: Upper case, raised (1/32-inch minimum depth), sans serif.
6. Color: Non-glare, white characters on black background.
7. Braille: Yes.
8. Mounting: On interior walls, as recommended by manufacturer for surface conditions.

PART 3 EXECUTION

3.01 INSTALLATION

A. In accordance with manufacturer's instructions.

B. Locations: In accordance with reference standards. Height: Unless noted otherwise, signage shall be 48-inch minimum and 60-inch maximum above the adjacent floor, measured from the baseline of the characters.

C. Distance from door: Signs shall have a minimum 18- by 18-inches space on wall, centered on the sign, beyond the arc of door swing between closed position and 45-degree open position.

D. Protect exposed surfaces until final acceptance of the work in a manner sufficient to prevent damage or discoloration. Work damaged or discolored prior to final acceptance shall be replaced without additional cost to Government.

E. Final cleaning of surfaces shall be carefully done in accordance with the manufacturer's instructions.

END OF SECTION

SECTION 10 14 26
UTILITY MARKERS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for applicable Line Pipe.

1.02 REFERENCE STANDARDS

A. Navajo Tribal Utility Authority NTUA

1. NTUA Color Codes NTUA Blue

1.03 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals.

B. RSN 10 14 26-1, Utility Marker Plan:

1. Show proposed locations of utility markers.

PART 2 PRODUCTS

2.01 UTILITY MARKERS

A. CRM3007, Utility Marker, manufactured by Carsonite, www.carsonite.com; or equal, with the following essential characteristics:

1. Catalog # RPH3-66-U Hybrid 3-Rail Marker Post.
    a. Durable.
    b. UV resistant.
    c. T-cross section with support rubs along each edge.
    d. Resistant to impact, ozone and hydrocarbons.
    e. Service Temperature Range: Minus 40 degrees F to 140 degrees F.
    f. Width: 4-inches.
    g. Length: 66-inches minimum.
    h. Anchor Barb.
2. Color: NTUA Blue.
3. Label: NTUA Potable Water Pipeline, phone number and logo as shown in Section 51 00 40 – NTUA Markers.

PART 3 EXECUTION

3.01 INSTALLATION

A. Use manual driving tool drive per manufacturer's recommendations.

B. Install in locations listed below unless otherwise determined by the COR, provide offsets as needed to ensure marker posts are located outside of access roads and drive lanes.

1. Fabricated horizontal pipe bends.
2. Buried gas line crossings.

END OF SECTION

SECTION 10 44 20
FIRE EXTINGUISHERS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for Turnout Buildings.

1.02 REFERENCE STANDARDS

A. National Electrical Manufacturers Association (NEMA)

1.	NEMA Z535.2-2023	Environmental and Facility Safety Signs

B. National Fire Protection Association (NFPA)

1.	NFPA 10-2022	Portable Fire Extinguishers

C. National Fire Protection Association (NFPA)

1.	UL 299-2022	Dry Chemical Fire Extinguishers
2.	UL 711-2018	Rating and Fire Testing of Fire Extinguishers

1.03 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals.

B. RSN 10 44 20-1, Approval Data:

1. Manufacturer's technical data and installation instructions for all portable fire extinguishers and cabinets.

PART 2 PRODUCTS

2.01 FIRE EXTINGUISHERS

A. Hand-Held Portable Dry Chemical Fire Extinguishers:

1. UL 299 Dry Chemical Fire Extinguisher.
2. Ten-pound agent capacity.
3. Minimum UL rating 10A:120B:C, multi-purpose pressurized dry chemical, manufacturer's rated temperature range above -27 degrees F and below 120 degrees F.
4. Red glossy coated steel shell.

5. Minimum 12-year hydrostatic test requirement.

6. Waterproof pressure gauge, wall mounted, hand-operated, minimum 15-foot stream.

7. Provide all bolts, screws, expansion anchors, and mounting brackets required for installation of hand-held portable fire extinguishers. Mounting hardware and fasteners shall be corrosion resistant stainless steel.

B. Outdoor Cabinet for Dry Chemical Extinguishers:

1. FB Series Fiberglass Fire Extinguisher Cabinet by J.L. Johnson Industries, Incorporated, or equal, with the following characteristics:

a. Fiberglass tub and door construction.

b. UV resistant.

c. Full-length stainless steel hinge.

d. Rubber-gasketed door.

e. Red finish.

f. Safety glass or acrylic view window.

g. Coordinate size of outdoor fire extinguisher cabinets required with extinguisher selected for complete system integration.

C. Labeling:

1. Provide signs to indicate location of extinguishers.

2. NEMA Z535.2

2.02 COATINGS

A. Manufacturer's standard finish for fire extinguishers in accordance with Section 09 96 20 – Coatings.

PART 3 EXECUTION

3.01 INSTALLATION

A. Hand-Held Portable Fire Extinguishers: Install one extinguishers in each Turnout Building.

END OF SECTION

SECTION 13 34 00

PREFABRICATED ENGINEERED STRUCTURES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Turnout Buildings:

1. Payment: Lump-sum prices offered in the Price Schedules.

   a. Includes cost of foundation design, structural excavation and backfill, concrete stoops, precast concrete building, building HVAC, building electrical, building lighting and other building accessories.

B. Cost:

1. Include cost of security and surveillance system in price offered in the Price Schedule for Security and Surveillance System.

2. In cost of communication systems in price offered in the Price Schedule for Telephone System.

1.02 REFERENCE STANDARDS

A. American Concrete Institute (ACI)

1.	ACI 318-19	Building Code Requirements for Structural Concrete
2.	ACI 360R-10	Guide to Design of Slabs-on-Grade

B. American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE)

1.	ASHRAE 62.1-22	Ventilation and Acceptable Indoor Air Quality
2.	ASHRAE 90.1-22	Energy Standard for Sites and Buildings Except Low-Rise Residential Buildings
3.	ASHRAE 183-34	Peak Heating and Cooling Load Calculations in Buildings Except Low-Rise Residential Buildings

C. Americans with Disabilities Act (ADA)

1. ADA Standards for Accessible Design-2010, available from https://www.ada.gov/law-and-regs/design-standards/2010-stds

D. Architectural Barriers Act (ABA)

1. ABA Accessibility Standards (ABAAS)-2015, available from https://www.access-board.gov/aba

E. Institute of Electrical and Electronics Engineers (IEEE)

1.	IEEE C2-23	National Electrical Safety Code

F. International Code Council (ICC)

1.	IBC-21	International Building Code
2.	IFC-21	International Fire Code
3.	IECC-21	International Energy Conservation Code
4.	IMC-21	International Mechanical Code

G. InterNational Electrical Testing Association

1.	ANSI/NETA ATS-25	Standard for Acceptance Testing Specifications for Electrical Power Equipment

H. National Fire Protection Association (NFPA)

1.	NFPA 70-23	National Electrical Code (NEC)
2.	NFPA 780-23	Installation of Lightning Protection Systems

I. National Precast Concrete Association (NPCA)

1.	NPCA Quality Control Manual-25	NPCA Quality Control Manual for Precast Concrete Plants, 18th Edition, available at https://precast.org/wp-content/uploads/QCM-001_18th_Edition.pdf

J. Precast/Prestressed Concrete Institute (PCI)

1.	PCI MNL-120-25	PCI Design Handbook, 9th Edition

K. Sheet Metal and Air Conditioning Contractors (SMACNA):

1.	SMACNA 006-2021	HVAC Duct Construction Standard-Metal and Flexible

L. Steel Door Institute (SDI)

1.	ANSI/SDI A250.8-23	Standard Steel Doors and Frames (SDI-100), available at https://steeldoor.org/wp-content/uploads/2020/04/A250_8-1.pdf

M. Underwriters Laboratories (UL)

1.	UL-467	Grounding and Bonding Equipment
2.	UL-752	Testing and Certification for Bullet-Resistant Materials

N. U.S. Public Law (PL)

1.	PL 90-480	Architectural Barriers Act (ABA) of 1968
2.	PL 101-336	Americans with Disabilities Act (ADA) of 1990

1.03 SUBMITTALS

A. In accordance with Sections 01 33 00 – Submittals and 01 33 26 – Electrical Drawings and Data.

B. RSN 13 34 00-1, Approval Drawings and Data:

1. Manufacturer's product data listing specified essential characteristics and proof of manufacturer's National Precast Concrete Association (NPCA) certification.
2. Manufacturer's product data for HVAC, plumbing, electrical, and miscellaneous installed fixtures demonstrating compliance with these specifications.
3. Manufacturer's bill of materials (BOM) including Contractor-supplied components.
4. Shop Drawings:
    a. Building and floor panel layout, dimensions, and details, including wall anchoring details.
        1) If used, cast-in-place concrete foundation layout, dimensions, and details, including wall anchoring details.
    b. Lighting and receptacle layouts, include switches and lighting control component locations.
    c. Layout for HVAC equipment.
    d. Building penetration locations for pipe, conduit or other equipment.
        1) Identify any penetration areas to be filled with cast-in-place concrete, including dimensions and reinforcement details.
    e. Power panel schedules and electrical wiring and connection diagrams. Include equipment electrical ratings for power panels, and electrical protection components.
    f. Indicate design loads and other information pertinent to the structural design as required by IBC Section 1603 Construction Documents.
5. Engineering calculations indicating that structural design complies with specified design requirements. Signed and sealed by a professional engineer licensed in New Mexico.
6. Manufacturer's delivery, storage, handling, and installation instructions.
7. Manufacturer's standard warranty.

C. RSN 13 34 00-2, Sample and Color Sample Kit:

1. Complete set of manufacturer's standard color samples, finish samples, and texture samples for selection of color, finish, and texture. Color copies of these items are not acceptable and will be returned not approved.

D. RSN 13 34 00-3, Geotechnical Report:

1. In accordance with IBC Section 1803.6.
2. Include site specific recommendations and drawings for foundations, subsurface excavation depths, fill placement, and maximum permanent and temporary slopes.
3. Signed and sealed by a professional civil engineer registered in the State of New Mexico.

E. RSN 13 34 00-4, Factory Test Reports:

1. Certified electrical factory test reports after testing equipment at the factory.
2. Equipment shall not be shipped, released, or transported without written approval of Factory Test Reports.
3. Provide Factory Test Reports for:
   a. All power distribution equipment.
   b. All internal wiring.
   c. Control systems.

F. RSN 13 34 00-5, Field Quality Test Reports:

1. Certified electrical test reports after field quality testing.
   a. All power distribution equipment.
   b. Lighting
   c. Internal wiring.
   d. Receptacles.
   e. Control Systems.

G. RSN 13 34 00-6, Final Data:

1. Operation and maintenance manuals for electrical and mechanical equipment in Part 2 Products.

1.04 SYSTEM DESCRIPTION

A. Design Loading:

1. Design prefabricated engineered structure in accordance with IBC for required loads, or combination of loads thereof.
   a. Dead Loads: Dead loads shall be the weight of prefabricated engineered structure and as determined by system manufacturer.

b. Live Loads: As shown on drawings or as determined by system manufacturer, whichever is more stringent.

c. Wind Loads: As determined by system manufacturer using structural design data shown on drawings.

d. Snow Loads: As determined by system manufacturer using structural design data shown on drawings.

e. Seismic Loads: As determined by system manufacturer using structural design data shown on drawings.

f. Collateral Loads: Support collateral loads imposed by doors, cable trays, pipe supports, cabinets and other equipment and accessories supported by building structure.

2. See drawings for design criteria and load designations.

a. It is the responsibility of the system manufacturer to determine any design load or parameter not explicitly listed in the specifications or drawings.

B. Building System Design

1. Design, fabricate, furnish, and field assemble precast concrete building with precast concrete floor panels.

a. In lieu of precast concrete floor panels, the precast concrete building can be assembled on a cast-in-place foundation.

2. Concrete design in accordance with ACI 318, ACI 360 and PCI MLN-120, as applicable.

3. Building dimensions: As shown on drawings.

4. Gable roof pitch: Minimum 1/2-inch: 12-inch.

5. Access, doors, clearances, and hardware designed to meet ADA/ABA requirements.

C. Coordinate with the following specifications to accommodate installation of the Control System, Telephone System, Security and Surveillance System equipment within the building.

1. Section 25 00 01 – System Control and Monitoring.

2. Section 27 30 01 – Telephone System.

3. Section 28 10 01 – Security and Surveillance System.

D. Include turnout pipe corrosion collar in floor panel, or cast-in-place foundation, design. See drawing 1695-D-62568 for details.

E. Prefabricated building manufacturer shall be a plant-certified member of the National Precast Concrete Association (NPCA), the Precast/Prestressed Concrete Institute (PCI), or equal.

F. Prefabricated building manufacturer shall have a minimum of 5 years' experience manufacturing and setting precast concrete.

1.05 CONTRACTOR RESPONSIBILITY

A. Project Conditions.

1. Refer to Section 53 10 00 – Geologic Investigations and Division 31 Sections for additional information and requirements.

2. Turnout site are likely to have expansive soils.

B. The Contractor is responsible for the Foundation Design of the Turnout Buildings, which at a minimum shall include the following requirements and those indicated in the Division 31 Sections.

1. Field Investigations:

a. The field investigation shall be directed by a registered professional civil engineer.

b. A minimum of one boring (drill hole) to a depth of at least 20 feet below the lowest elevation point of the foundation, shall be advanced at each site using a drilling method determined by the design engineer. Include geologic logs of borings.

c. For each boring, standard penetration test (SPT) blow count (N) data and soil/bedrock samples shall be collected at intervals no less frequent than every 5 feet and bulk samples shall be collected as determined by the design engineer.

d. Additional geotechnical investigations as deemed necessary by the design engineer in accordance with IBC Section 1803 Geotechnical Investigations.

2. Geotechnical laboratory testing shall be directed by a registered professional civil engineer.

a. Swell/consolidation testing shall be performed on potentially expansive and collapsible soils and expansive bedrock.

b. Include soil classification and Standard Proctor testing at each site.

c. Additional laboratory testing as deemed necessary by the design engineer in accordance with IBC Section 1803 Geotechnical Investigations.

3. Frost Protection: In accordance with IBC Section 1809.5 Frost Protection.

4. Foundation Maximum Settlement, Differential Settlement, or Heave: 1 inch.

5. Prepare and submit to Government a Geotechnical Engineering Report as indicated in RSN 13 34 00-3.

PART 2 PRODUCTS

2.01 PREFABRICATED BUILDING

A. Precast concrete building (roof, walls and floor panels).

B. Modular Connections, LLC. (Utility Buildings, 1090 Industrial Blvd, Bessemer, AL 35022, https://modularconnections.com/utility-buildings); or equal, having the following essential characteristics:

1. Dimensions as shown on drawings.
2. Other vendors:
    a. Easi-Set Buildings, https://easiset.com/products/easiset-buildings.
    b. CXT Structures, https://lbfoster.com/infrastructure/precast-concrete-buildings/multipurpose-buildings.
    c. UBC Precast, https://ubcprecast.com/CustomUnits.html.
3. Materials:
    a. Precast Concrete: Concrete, concrete reinforcement, concrete materials, and hot and cold weather concreting shall conform to the NPCA Quality Control Manual.
        1) Concrete in contact with soil shall be sulfate resistant per IBC 21.
    b. Concrete Finish:
        1) Building Interior: Smooth form finish on interior concrete surfaces unless exterior finish is produced using a form liner, then smooth hand-troweled finish.
        2) Building Exterior: Form liner, exposed aggregate, or other simulated finish on exterior wall and roof surfaces as selected by COR from manufacturer's standard, exterior finishes.
        3) Doors as indicated in drawings.
    c. Minimum thermal envelope requirements. Provide R-15 to 24-inches below grade; minimum values of R-49 at roof and R-11.4 (continuous) at walls per ASHRAE 90.1 by providing one of the following:
        1) Continuous rigid insulation with composite liner system.
        2) Furred or framed wall with cavity and continuous insulation.
        3) Insulated precast sandwich panel system.
    d. Provide durable, moisture resistant and cleanable finished interior wall and ceiling surfaces over insulation system with structural backing and localized backing for equipment maintaining exterior dimensions as shown in drawings. Provide one of the following acceptable systems:

1) Minimum 0.03-inch thick white FRP, HDPE, or rigid vinyl liner panel over minimum ½-inch plywood composite system.

2) Engineered stud system behind liner panel.

e. Provide support for wall and ceiling mounted equipment as specified, with:

1) Minimum point load of 200-pounds.

2) Embedded plates, inserts or continuous backing.

3) Corrosion resistant unistrut or reinforced zones, where necessary.

f. Fully adhered, fleece-backed, white roof membrane, minimum 50 mil thick, over smooth roof panel.

g. Provide floor with broom finish.

4. Bullet resistance: In accordance with UL 752 Test Method Level 5.

5. Building wall thickness, including concrete wall, insulation system and interior finishes, should not exceed 10-inches.

2.02 CAST IN CONCRETE

A. Optional cast-in-place concrete building foundation for precast concrete building:

1. Concrete shall be in accordance with Section 03 30 00 – Cast-In-Place Concrete.

2.03 ACCESSORIES AND FIXTURES

A. Doors and Frames:

1. Comply with ANSI/SDI A250.8 (SDI-100) and as specified herein.

2. Galvanized doors and frames shall meet ANSI/SDI A250.8 Level 2. Manufacturers: Republic, Steelcraft, Ceco, Black Mountain, Pioneer, Curries, Mesker, MPI, DCI Hollow Metal on Demand; or equal

B. Prefinished metal gutter and downspout.

C. Splash Blocks:

1. As shown on drawings.

2. Precast concrete.

3. 12-inches wide by 36-inches long by 3-inches high.

D. Fasteners, anchors, and brackets shall be of structural quality and galvanized or stainless steel.

E. Security: Coordinate building and door design with the requirements in Section 28 10 01 – Security and Surveillance System.

F. Heating, Ventilating, and Air Conditioning:

1. Design HVAC systems to maintain indoor building temperatures listed in this specification based on heating and cooling loads calculated in compliance with ASHRAE Standard 183.

2. Protect equipment from environmental conditions, manage heat loads from equipment and meet occupancy requirements.

3. Design and select HVAC equipment based on:

a. Weather data conditions found in the ASHRAE Fundamentals Handbook for the geographically closest weather station:

1) Use 99.6 percent heating dry bulb and the 0.4 percent cooling dry bulb for the basis of the analysis.

4. Indoor building temperatures:

a. Minimum winter temperature: 40 deg F or higher if required for equipment protection.

b. Maximum summer temperature: 102 deg F or lower if required for equipment protection.

5. Ventilation rates per ASHRAE 62.1.

6. Meet the criteria of the IBC, IMC, IECC, ASHRAE, IFC and NFPA.

7. Equipment Requirements:

a. Energy Star rated. If Energy Star products not available, equipment to be per ASHRAE 90.1.

b. Refrigerants per ASHRAE 15 and the IMC.

c. Duct per SMACNA 006.

d. Heating shall be electric. Fossil fuels not available at the site.

8. Evaporative cooling is not permitted.

G. Electrical:

1. Provide electrical system including conductors, cables, raceways, grounding, and bonding in accordance with NFPA 70 and IEEE C2. Include HVAC, fire, lighting, and receptacle components supplied with the building for fully functional systems.

2. Power Panels:

a. 120/240V Single-Phase Distribution Panelboard:

1) Provide panels shown on the Single-Line Diagram, Drawing 1695-D-62578.

a) For each building panel, provide a UL 1449–listed Type 2 SPD, integrated into the panelboard, sized and rated for the system voltage, configuration, and available fault current.

SPD must also be compliant with applicable NFPA 70 (NEC) codes.

2) Update panel design and circuit breakers as necessary to accommodate building loads.

b. Connect building loads and components to the panelboards supplied with the building.

c. Supply power to all equipment located in the building.

3. Cable and Raceway:

a. Provide cable and raceway as necessary to connect all components provided with the building. This includes wiring all HVAC, Control System, Telephone System, Security System, instrumentation, valve, fire, lighting, and receptacle components supplied with the building for fully functional systems.

b. Supply power to all equipment provided with the building from pre-installed panelboard in accordance with drawings and specifications.

c. Cable and raceway shall be provided in accordance with Section 26 05 10 – Conductors and Cables and Section 26 05 33 – Raceway and Boxes.

4. Grounding:

a. In accordance with Section 26 05 20 – Grounding and Bonding.

b. Provide grounding loop around the interior of the building, installed near the ceiling. Ground loop should meet the minimum conductor size required for ground risers in Section 26 05 20 – Grounding and Bonding and shall connect to manufacturer supplied ground pads located on the exterior of the building.

c. Provide a minimum of two ground pads on the exterior of the building near the base of the building. Ground pads should include ground lugs to connect exterior grounding system ground risers.

5. Lighting:

a. Interior Lighting:

1) Interior lighting for each room in the building shall be operated by a switch located next to the door entering the room.

2) Provide the following minimum average maintained area lighting levels: 40 foot-candles measured 30 inches above finished floor.

3) Provide Uniformity Ratio (max/min) of no more than 3:1 in each area.

4) UL listed

5) LED

b. Provide illuminated exit signs with integral emergency lighting fixtures in each room.

1) UL listed

2) LED

3) Battery backup with 90 minutes of battery backup power

c. Exterior lighting:

1) Provide photocell controlled exterior lighting above each exterior door

2) UL listed

3) LED

4) 4000k color temperature

6. Power Receptacles:

a. UL listed

b. Interior:

1) Provide 20A, 120 VAC, duplex power receptacles on each of the interior walls and next to each door.

2) Interior receptacles shall be located at least 16 inches above finished floor.

c. Exterior:

1) Provide 20A, 120 VAC, GFCI, duplex receptacles next to each exterior door:

a) Exterior receptacles shall be weatherproof with weatherproof covers.

b) Exterior receptacles shall be located at least 30 inches above exterior final grade.

H. Lightning Protection: In accordance with NFPA 780. Lightning protection system shall include air terminals, roof conductors, down conductors, and ground connections. Include all necessary accessories and hardware required to complete and install a robust and properly functioning lightning protection system.

I. Fire Extinguishers: Provide 1 type ABC fire extinguisher in each room.

J. Provide a UL-listed smoke alarm in each room.

2.04 FABRICATION

A. Fabricate in accordance with NPCA Quality Control Manual.

2.05 CONTRACTOR SOURCE QUALITY TESTING

A. Inspection and Witnessing:

1. Government may:

a. Inspect equipment and witness factory tests.

b. Perform additional functional tests with assistance from factory personnel.

2. Notify COR at least 21 days in advance of factory testing. Government will inform Contractor if the Government will perform a factory inspection and witness the factory testing.

B. Minimum Factory Testing:

1. Inspect structure and accessory components supplied with the structure.

2. Verify completed structure meets specification requirements.

3. Verify that the dimension shown on the structure's drawings are accurate.

4. Energize building and verify operation of the electrical distribution system:

PART 3 EXECUTION

3.01 PREPARATION

A. In accordance with applicable requirements of the codes and standards listed herein as well as manufacturer's requirements.

3.02 EARTHWORK

A. Excavation, backfill, and compaction in accordance with Division 31 specifications.

3.03 BUILDING ERECTION

A. Lift units with suitable listing hardware at points provided by manufacturer. Temporary bracing, if necessary, shall comply with manufacturer's recommendations.

B. Seal wall-to-floor joints and adjacent wall panel joints.

C. Install foundation and erect buildings with flow, orientation, and location as shown in site plan drawings (1695-D-62528 to 1695-D-62542).

3.04 HEATING, VENTILATING, AND AIR CONDITIONING INSTALLATION

A. Install units level and plumb, maintaining manufacturer's recommended clearances and tolerances.

B. Install wall sleeves in finished wall assembly; seal and weatherproof.

C. Install and anchor wall sleeves to withstand, without damage to equipment and structure, seismic forces required by building code.

3.05 CONTRACTOR FIELD QUALITY TESTING

A. Contractor shall perform all electrical testing in accordance with NETA ATS acceptance testing requirements.

B. Inform COR in writing of scheduled activities a minimum of 14 days prior to performing activity to allow Government opportunity to witness.

C. Government may:

1. Witness inspection, testing, or commissioning activities.
2. Perform additional equipment testing on site with cooperation of Contractor to verify operability of equipment.

D. Contractor to supply all testing equipment.

E. Inspection and test documentation including:

1. Name and signature of person performing the test.
2. Date on which the test was performed.
3. Identification of component, equipment, or system tested.
4. Identification of tests performed.
5. Summary of results of the test or inspection activity, including positive identification for each activity of whether the system or component passed or failed.
6. Data or readings taken during the test to verify component, equipment, or system acceptance.

F. Inspect structure and accessory components:

1. Verify completed structure meets specification requirements.
2. Check building against submitted drawing package verifying layout and dimensions shown on supplied drawings.
3. Energize building and verify operation of the electrical distribution system:
    a. Perform cable testing in accordance with Section 26 05 10 – Conductors and Cables.
    b. Verify operation of each light fixture and power receptacle, including verifying the phase, neutral, and grounding connections in each power receptacle is correct.
4. Verify operation of all electrically powered equipment.

G. Generate the following documentation:

1. Test Plan:
    a. Specific detailed test plan of the testing activities planned for the site. Includes:
        1) Schedule and sequencing of activities and specific tests to be performed on individual components and the system.
    b. Photographs of constructed elements, prior to testing.
2. General procedures for planned testing activities including how flow and velocity will be measured, any initial water surface measurement equipment, steps for adjustments as necessary.
3. Test Procedures.
    a. Detailed procedures for planned testing activities.
    b. In accordance with specifications and manufacturer's recommendations.
    c. If references are used in testing procedures, include a copy of references used in testing procedures.
4. If pressurization of system is required in testing procedures or specifications, describe how system shall be pressurized, how pressurized water or air will be discharged, and where water or air source will be obtained.
    a. Use potable water to test potable water systems.
5. Test Reports:
    a. Typed, 8 1/2- by 11-inch sheets.
    b. Certified, signed, dated, and attested to having passed requirements and tests.
    c. Provide data summary and necessary graphs and plots of tests performed.
    d. Available for COR review upon request.
    e. Use in preparing final drawings.
6. As-Built Drawings: In accordance with Sections 01 78 30 – Project Record Documents, and 01 33 26 – Electrical Drawings and Data.
7. Maintain a list of discrepancies found during inspection and testing activities and how corrected.

END OF SECTION

SECTION 25 00 01
SYSTEM CONTROL AND MONITORING

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. System Control and Monitoring:

1. Payment: Lump sum prices offered in the Price Schedules.

1.02 DEFINITIONS AND ACRONYMS

A. FIT: Flow Indicating Transmitter.

B. GFE: Government Furnished Equipment.

C. HART: Highway Addressable Remote Transducer.

D. I/O: Input/Output.

E. LCD: Liquid Crystal Display.

F. LED: Light-Emitting Diode.

G. NTUA: Navajo Tribal Utility Authority.

H. NGWSP: Navajo-Gallup Water Supply Project.

I. OIT: Operator Interface Terminal.

J. P&ID: Piping and Instrumentation Diagram.

K. PLC: Programmable Logic Controller.

L. PoE: Power-over-Ethernet.

M. RTD: Resistance Temperature Detector.

N. SCADA: Supervisory Control and Data Acquisition.

O. SJLWTP: San Juan Lateral Water Treatment Plant.

P. UPS: Uninterruptible Power Supply.

Q. VPN: Virtual Private Network.

1.02 REFERENCE STANDARDS

A. Electrotechnical Commission (IEC)

1.	IEC 61131-3-2025	Programmable Controllers – Part 3: Programming Languages

B. Institute of Electrical and Electronics Engineers (IEEE)

1.	IEEE 802.3-2022	Local and Metropolitan Area Networks - Specific Requirements Part 3:Carrier Sense Multiple Access with Collision Detection Access Method and Physical Layer Specifications
2.	IEEE 1815-2015	Electric Power Systems Communications – Distributed Network Protocol (DNP3)

C. International Society of Automation (ISA)

1.	ISA 101.01-2015	Human Machine Interfaces for Process Automation Systems
2.	ISA 62443-1-1(99.01.01)-2007	Security for Industrial Automation and Control Systems Part 1-1: Terminology, Concepts, and Models
3.	ISA 62443-2-1(99.02.01)-2009	Security for Industrial Automation and Control Systems Part 2-1: Establishing an Industrial Automation and Control Systems Security Program

D. National Electrical Manufacturers Association (NEMA)

1.	NEMA 250 - 2014	Enclosures for Electrical Equipment (1,000-Volts Maximum)
2.	NEMA BI 50009-2024	Low Voltage Cartridge Fuses
3.	NEMA IA 10030-2024	Industrial Control and Systems: Enclosures
4.	NEMA IA 10039-2025	Industrial Control and Systems: Control

E. National Fire Protection Association, Inc. (NFPA)

1.	NFPA 70-2023	National Electrical Code (NEC)

F. National Institute of Standards and Technology (NIST)

1.	NIST SP 800-82 Rev2	Industrial Control Systems (ICS) Security

G. Underwriters Laboratories, Inc. (UL)

1.	UL 508-1999	Industrial Control Equipment, Seventeenth Edition

1.03 SUBMITTALS

A. Submit the following in accordance with Sections 01 33 00 – Submittals and 01 33 26 – Electrical Drawings and Data.

B. Drawings and data in sufficient detail to show full compliance with this specification and applicable codes.

C. RSN 25 00 01-1, Qualifications – Contractor, Designer, Programmer, and Installer work summary, include:

1. Description of projects and experience intended to be used to meet experience requirements listed in Article 1.04 Qualifications.
2. Include name, date, and location of projects.

D. RSN 25 00 01-2, System Description Documentation:

1. Functional System Description:
    a. Describe and document the proposed Control System, its features and limitations, compliance with the requirements, operation, software algorithms, display screens, and software configuration.
2. System overview:
    a. Overview of hardware, software, control functions, interfaces, and general data flow associated with entire system.
    b. Include software block diagram of entire system.

E. RSN 25 00 01-3, Approval Drawings and Data:

1. Drawings:
    a. Instrumentation Loop Diagrams, Schematic Diagrams, and Wiring Diagrams:
        1) Show all wiring and cable connections for supplied components, including all connections made inside cabinets, panels, or other enclosures.
        2) Include external connections showing all connections to Control System components, including components required under other Specifications. Identify the remote end of the cable connections on the drawings including identifying the name and tag of the connected device as well as the terminals used for the connection.
    b. Networking Diagram.
        1) Show all communications links, including, but not limited to servers, network switches, media converters, and communication links between the PLC, OIT, and other field devices.

c. Provide in accordance with the requirements for Computer drafted drawings in Section 01 33 00 – Submittals.

2. Bill of Materials.

3. Manufacturer's product data and catalog sheets on proposed system components.

a. Identification of model or catalog number proposed, equipment capabilities, and electrical characteristics.

4. Uninterruptible power supply calculations to meet requirements found in Article 2.13 E.

5. Spare part list.

F. RSN 25 00 01-4, Layout Drawings:

1. Dimensioned drawings showing relative positions of components. Includes component identification and layout information.

2. Enclosure and Control Panel Layout Drawings. Exterior and interior arrangements.

3. Include nameplate information.

4. Provide in accordance with the requirements for Computer drafted drawings in Section 01 33 00 – Submittals.

G. RSN 25 00 01-5, Software Development Documentation:

1. Include the following for the Programmable Controller:

a. I/O List, including at a minimum the following information:

1) Point designation.

2) Point description.

3) Card location.

4) Type and rating information.

5) Termination data.

b. Data points list for soft data points internal to programmable controller.

1) Includes information on logical and calculated datapoints, as well as datapoints that are communicated between components.

2) Point designation.

3) Point description.

4) Type and rating information.

5) Address.

c. Communication link information.

d. Graphic display drawings (proposed LCD display and OIT touch panel) for each screen available on the LCD or OIT displays.

e. Logic diagrams: Functional and binary logic drawings for the entire Control System.

f. Manufacturers data on all software used and any software licenses provided for the project. Includes all software applications used on or for the OIT, PLC, etc.

H. RSN 25 00 01-6, Final Drawings, Documentation, and Operations and Maintenance Manuals:

1. Final drawings:

a. Includes final versions of all previously submitted diagrams and drawings with as-built changes incorporated into them.

b. Layouts, schematic diagrams, and wiring diagrams including “as-built” changes.

2. Final version of approved software documentation including “as-built” changes.

3. Operations and Maintenance Manuals:

a. Include operation of following system features and hardware items:

1) Screen displays and control operations.

2) General description of all control actions.

3) Limit and trend settings.

4) Manual status entry.

5) Report generation.

6) Control access methods.

4. Final I/O point list.

5. Final uninterruptible power supply calculations to meet requirement found in Section 2.13 E.

6. Final spare parts list.

7. Handwritten updates or any updates that do not meet the requirements for computer drafted drawings in Section 01 33 00 – Submittals are not acceptable. Drawing modifications must be made in the program originally used to generate the drawing and modifications made by a pdf editing program, like Adobe Acrobat or Bluebeam, are not acceptable.

1.04 QUALIFICATIONS

A. Contractor:

1. Minimum 7-years’ experience with similar projects including systems with programmable controllers.

2. Completed at least 5 projects of similar size, type, application, and complexity. At least three projects must have included using proposed Schnieder Electric programmable controller product line, software, and graphics package

B. Designers and Programmers:

1. Minimum 5-years' experience programming systems of similar size and complexity containing programmable controllers.

2. Completed at least 2 projects of similar size and complexity using proposed Schnieder Electric programmable controller product line, software, and graphics package

C. Installer:

1. Minimum 5-years' experience installing control panels and control equipment.

2. Minimum 5-years' experience performing any testing or commissioning activity installer is expected to perform.

1.05 DESIGN REQUIREMENTS

A. General:

1. Provide a Control System for each of the following facilities:

a. Sanostee Turnout.

b. Newcomb Turnout.

c. Naschitti Turnout.

d. BBN9 Turnout.

e. Coyote Canyon Turnout.

f. Rock Springs Turnout.

g. Window Rock Turnout.

2. Control System shall be the primary control, automation, and monitoring system for instruments and equipment located in each Turnout Building and on the Turnout Building site:

a. Provide the Control System with associated equipment, software, terminations, and wiring to implement control and monitoring functions described in these Specifications.

b. Design Control System with functional bi-directional data links to other systems, components, PLCs, PACs, and networks as required.

c. Control System shall provide support for the following minimum communication protocols:

1) DNP 3.0 in accordance with IEEE 1815.

2) Modbus Serial (RTU and ASCII).

3) Modbus Network (TCP).

4) HART Communications Protocol.

d. Fully integrated system, including the following minimum features:

1) Processing units, I/O components, network and communication components, power supplies, and other miscellaneous control components.

2) Graphic displays viewable from OIT.

3) At minimum, I/O identified in Table 25 00 01A – Typical Turnout Building I/O and Data Point List, and the P&ID shown in drawings 1695-D-62569 through 1695-D-62577.

e. Provide Control System programming, control logic, and macros:

1) Includes development of logic diagrams, flow charts, and OIT graphical displays.

2) Provided documentation and descriptions listed within software, including descriptions for each rung, block, or line of ladder diagrams, function block diagrams, and structured text logic.

f. Provide interface design:

1) Include development of schematics, datalinks, and other applicable items, required for interfaces between components.

3. Provide Programmable Controller with required I/O and local control at the Turnout Building:

a. Provide media converters, network switches, and fiber optic patch panels as required to complete installation.

b. I/O components, communication components, backplanes, and other components utilized in the Control System shall be of the same manufacturer. Minimize number of different types and models utilized.

c. Failure of remote I/O hardware or network communication link must be individually alarmed at the OIT in the Turnout Building.

4. Software packages used in the installation and for development of system graphics, control logic, shall be up-to-date and compatible with the most recent version of major operating systems, including Microsoft Windows. Provide software not mentioned specifically, but essential for system to meet functional requirements.

5. Control System shall interface to a long-haul fiber optic network at each site, which is provided either by NTUA (utility) or Government (underground).

6. Provide additional equipment not specified in this section but necessary to make system complete and operational at no cost to Government.

B. Control System Function:

1. Utilize the long-haul optical fiber transmission lines to communicate telemetry, monitoring, and control data from the Turnout Building Control System to the SJLWTP SCADA system.
2. Monitoring and Alarming:
    a. Monitor the instruments and equipment in the Turnout Building and record and display local telemetry and monitoring information.
    b. Display information from devices or systems on graphic displays. Display information in units of process measurement where applicable.
    c. Monitor and provide alarming for instruments and devices shown on the supplied P&IDs 1695-D-62569 through 1695-D-62577, the I/O table of Article 3.01 of this Specification, and where the Specifications reference interfaces with PLCs, facility control, or SCADA systems.
    d. Status alarms for Control System equipment malfunctions.
    e. Where Flow Indicating Transmitter (FIT) devices are shown on P&IDs, display the following minimum information on graphic display:
        1) Flow direction.
        2) Flow rate.
        3) Total flow.
3. Control:
    a. Where solenoid operated flow control valve is shown on P&ID, display the status of the flow control valve on graphic display.
        1) Flow Control Valve Operating.
        2) Flow Control Valve Closed.

C. Remote Interface:

1. Provide remote interface from the Control System at each site to the remote San Juan Lateral SCADA System (the San Juan Lateral SCADA System is by Others). The Control System shall transmit and receive monitoring and control data to and from the San Juan Lateral SCADA System. Provide system programming, logic development and register address assignment as necessary to complete the remote communication link. Interface will be over the long-haul optical fiber transmission network connection.
    a. Contractor shall coordinate with BOR for setup of interface signals between the facilities and San Juan Lateral SCADA system.
2. Long Haul Optical Fiber Transmission Lines:
    a. For the Sanostee, Newcomb, Naschitti, BBN9, and Coyote Canyon Turnout Buildings, an NTUA (utility) provided long-haul optical fiber transmission system shall be the long-haul transmission lines that are used

for the long-haul optical transmission network connection to the SJLWTP SCADA System at each site.

b. For the Rock Springs and Window Rock Turnout Buildings, a Government provided underground long-haul transmission lines that are used for the long-haul optical transmission network connection to the SJLWTP SCADA System at each site.

c. The Government will supply a Network Interface Panel for each site, that will be used to interface to the long-haul optical fiber transmission lines, as shown on drawing 1695-D-62588.

1) Install the Government supplied Network Interface Panel for each site. Provide power and communication connections to the Network Interface Panel.

2) The Network Interface Panel shall be utilized by the Control System to connect the respective long-haul optical fiber transmission line network via the Government furnished Network Interface Panel network switch and fiber bypass unit.

3) Network Gateway switch in Network Interface Panel will act as the firewall between the Control System and the long-haul communication network.

d. Primary communication system utilized for transmission of the following signals:

1) SCADA/Control Data: Modbus TCP communication protocol.

3. Remote Interface Communication Link:

a. Design and provide communication link between the Network Interface Panel and Contractor supplied Control System equipment.

b. Contractor is responsible for programming, setup, and testing required for Contractor supplied equipment, including the Contractor supplied PLC to support the communication link. The SJLWTP Contractor will program the Network Interface Panel communication equipment.

c. Control data leaving and entering the Control System site shall be secured utilizing IPSec VPN tunneling. Control data includes data passed through Programmable Controllers. The IPSec VPN shall be implemented by the network interface switch located in the Network Interface Panel.

D. Graphic Displays

1. Show monitoring feedback and status of components and system processes tied into control and monitoring system.

2. View graphic displays from OIT.

E. Provide Door/Gate/Hatch Intrusion Monitoring:

1. Utilize balanced magnetic contacts to provide door, gate, control panel, network interface panel, and security and communication network rack intrusion monitoring.
2. Provides specific indication of which entry point opened.
3. Minimum Locations:
    a. Doors or entry points for entering the building. This includes personnel doors, equipment doors, and roll-up doors.
    b. Equipment control panels and network racks.
    c. Gates allowing entry to the site or through fences.
4. Alerts: Provide hardwired (not network or communication based) alarms to Control System for system alarm or unauthorized entry.

F. Power:

1. Provide power distribution to Control System components, including protection, to support full operation of system. This distribution system shall include analog loop power supplies and contact interrogation voltages for dry contact inputs.
2. Provide a UPS with battery backup for programmable controller, network, and fiber optic equipment located inside Control System Enclosure, Network Interface Panel, and the Security and Communication Rack. Components located on same site shall share a common UPS system.
    a. Provide UPS power distribution to the Network Interface Panel.
    b. Solenoid valves shall be powered by UPS with battery backup to allow for automatic operations and valve control when site power is lost.

G. Software:

1. Software, including graphics and control logic implementations shall be configurable by Government to allow for future modifications to Control System:
    a. Graphic display configuration software shall include libraries of standard displays, nameplates, and components utilized by Government for future Control System modifications
    b. Modules used in control logic implementation shall be viewable and configurable by Government, including internal implementation of those modules.
2. Password protect access to PLC program loading. Provide the following access levels in the software:
    a. Status access:
        1) View only access to facility displays and trends.
        2) No access to process control or alarm acknowledgment.
    b. Operator access:

1) Access to facility displays, trends, process control, and alarm acknowledgement.

c. Administrator:

1) Access to facility displays, trends, process control and alarm acknowledgement.

2) Access to view and modify Control software.

1.06 DELIVERY, STORAGE, AND HANDLING

A. Handle and store hardware in accordance with manufacturer's instructions. Include copy of manufacturer's instructions with equipment at time of shipment.

B. Coordinate with Government regarding GFE delivery schedule.

1.07 EXTRA MATERIALS

A. Furnish a set of special tools and accessories required for installation, normal operation, and maintenance for each facility. Include:

1. Contractor and manufacturer recommended tools and accessories.
2. Tools and accessories recommended for field maintenance and those designed to aid in the diagnosis, maintenance, repair, and replacement of supplied equipment.

B. Furnish spare parts. Furnish a set of the following components:

1. I/O Cards: Minimum of 10 percent spare of each type used, no less than one (1) of each size and type used.
2. Network and Communication Cards or Components: Minimum of one (1) spare of each type of control system component.
3. Programmable Controllers: One (1) fully redundant spare set of automation controllers.
4. Back planes: Minimum of one (1) spare of each type of control system component.
5. Communications Converters: Minimum of one (1) spare of each type of communications converter used.
6. Miscellaneous Components: Minimum of one (1) spare of each type of control system component.
7. OIT: One spare of each type.
8. Contractor or manufacturer recommended spare parts.
9. Replace spare parts used during the installation, testing, and commissioning process to provide a complete set of spare parts.

C. Furnish two copies of system software. Include:

1. Software required to re-install software required for the system to function properly, for each facility.
2. Copies supplied shall contain the final version of furnished software.

PART 2 PRODUCTS

2.01 PROGRAMMABLE LOGIC CONTROLLER

A. Modicon M340 Programmable Controller including accessories shall meet the following requirements:

1. Chassis: DIN rail mountable.
2. 24 VDC Power Supply.
3. Modbus and Ethernet capable.
4. Provide with a minimum of 40 percent spare capacity, including memory utilization, processing capacity, and other functions.

2.02 OPERATOR INTERFACE TERMINAL (OIT)

A. Provide OIT to perform as primary operator interface to the Control System. Harmony ST6 Touch Screen Panel as manufactured by Schnieder Electric, or similar.

1. Mount on Control System Enclosure.
2. Color TFT LCD. 15-inch touch screen display, minimum.
3. Modbus TCP/IP, RS485, RS232.
4. EcoStruxure Operator Terminal Expert Software.

2.03 INPUT/OUTPUT (I/O)

A. Provide I/O as required to meet specified functionality.

1. Provide I/O cards intended for use with the Modicon M340 product line, manufactured by Schneider Electric http://www.schneider-electric.com.
2. Provide 10 percent spare installed I/O of each size and type at each I/O location.
3. Discrete and Analog Input and Output modules as required to implement monitoring and control functions described.
4. Provide capability to install 20 percent additional future I/O cards at each I/O location. Includes connection capability, space on backplanes and sufficient margin in power and operating capacity for a true ‘plug and play’ installation of additional I/O cards.

2.04 PROGRAMMING CONTROL LOGIC

A. Provide capability of programming control logic in Control System using the following programming languages and in accordance with IEC 61131-3:

1. Ladder Diagram.
2. Function Block Diagram.
3. Structured Text.

B. Implement majority of control logic in Control System using Ladder Diagram or Function Block Diagram, with only select areas implemented using other methods if necessary.

C. Emphasize readability and maintainability in control logic implementation.

D. Fully document control logic developed for Turnout Building.

2.05 GRAPHIC DISPLAYS

A. Provide graphic displays for each Control System in accordance with requirements and recommendations in ISA 101.01. ISA 101.01 recommendations and requirements in addition to requirements included in this Specification represent minimum level of functionality for graphic displays.

B. Control System graphic displays are subject to approval by COR.

C. Spare capacity for an increase of up to 20 percent in the number of graphic pages.

D. Designed to support user situational awareness, intuitive navigation, and intuitive system status and condition information.

E. Designed to support types of tasks and activities required of the operator. Control of system and modification of operational setpoints shall normally be accomplished through OIT.

F. Separate graphic display screens shall be designed and provided for screens with different functions and locations.

G. Use consistent design philosophy across graphic displays, however, the specific functions desired for the individual screens may require unique arrangements of graphics and information.

H. Control System graphic displays shall incorporate a hierarchy of displays that provide progressive exposure to detail. The following list provides a general guideline for the hierarchy of displays:

1. Highest level display shall show entire system within operator's control. It is intended to act as an overall indicator as to how the site is running by tracking key performance indicators.
2. Displays shall consist of detailed information on system operations.
3. Equipment operation if not included in system operation.

I. User interaction techniques:

1. Clear, with graphics that are consistent in the method that they present similar information.
2. Graphic functions: Standardized, intuitive, and straight forward with minimal user manipulation.
3. Error tolerance in user interaction for critical devices shall be included, with simple notification of error and effective methods for recovery.

J. Develop a color scheme for graphic displays for approval by COR:

1. Grey backgrounds for graphics, with minimal use of color in graphics.
2. Utilize color very specifically and consistently.
3. Color alone shall not be the sole differentiator of an important condition or status.

K. Include analog representation of important measurements, indicating their value relative to normal, abnormal, and alarm conditions.

L. Include visual indication of an Operator input field, or other operator-initiated control function, has been disabled.

M. Response time requirements for the graphic displays:

1. Graphic displays shall take no longer than 3-5 seconds to appear when called up.
2. Update live information on display every 2-seconds unless critical nature of process information displayed requires more frequent updating.

N. Provide feedback on the completion of an action or the failure to complete an action.

O. Component Depiction:

1. Utilize standard shapes for components. Shapes shall be simple and avoid unnecessary detail.
2. Show status of components clearly and consistently.
3. Include operating hours on operating and standby rotating equipment.
4. Provide faceplates to display detailed component information and user input fields.
5. Vessels and Tanks: Size shown shall be relative to process importance of component and relate to its physical size.

6. Process Lines: Limited number of line types and line thickness.
7. Process Controllers:
    a. Depicted as a physical entity on the graphics, with key values like set point, output, and control mode displayed.
    b. Provide faceplate type displays for process controllers with detailed information and user input fields, where required, for the process controller.

P. Text and Values:
1. Do not display leading zeros when depicting values unless the value is less than 1.
2. Differentiate live values from static text, for example by depicting live values with a color. Units of measure shall not be depicted as a live value.
3. Date and time display:
    a. Display date in form of "YYYY-MM-DD".
    b. Display time of day in form "hours:minutes:seconds". Update time display automatically once every second.

Q. Page title: Each format shall have appropriate page title.

R. Provide visual display of lockout/tagout application.

S. Provide faceplates for sequences, including automatic startup and shutdown.

T. Annunciate active trip conditions on the graphic.

U. Alarms:
1. Alarms and indications of abnormal situations shall be clear, prominent, and consistently distinguishable.
2. Provide the following minimum alarm displays:
    a. Alarm status.
    b. Alarm log.
3. Displays shall provide the Operator the ability to acknowledge alarms.
4. Place alarms out of service (access-controlled operation).
5. Modify alarm attributes (access-controlled operation).
6. Alarm indication in the Control System shall have audible alarms associated with them. Sound level of the alarm shall be enough for easy detection.

V. No animation except for specific alarm related graphics.

W. Alarm and Events:
1. Maintain a consistent approach to alarming across Turnout Buildings.

2. Allow for alarm monitoring for discrete and analog inputs/outputs and communication links.

3. Document the following for each alarm:

   a. Alarm type.

   b. Alarm setpoint.

4. Indicate acknowledged alarms, unacknowledged alarms, and unacknowledged alarms which have returned to normal.

5. Time stamped logs for alarm history, event history, diagnostic, and audit history for the alarm system.

2.06 CONTROL

A. Designed to aid operator over complete operating range of the facility while controlled equipment is in service. Available and configurable parameters useable to control equipment within manufacturer's limitations and enhance equipment life shall be incorporated into the system design.

B. Provide modifiable setpoints and control variables, including alarm setpoints, delays, and control setpoints that are modifiable by operator at OIT. Sufficient setpoints shall be included to allow operator adjustment of system through desired operating range. Values referred to as 'setpoints' in these Specifications shall be modifiable from OIT.

C. Provide control logic to mitigate bad quality or out of range signals. Logic shall drive devices to a safe state, substitute a value, or hold last value as appropriate for application.

D. Fault checking shall be included and cause appropriate response, change over, or alarm.

E. Upon initial system power up or system power restoration, systems shall be in manual with outputs in "off" or fail-safe mode.

F. Provide rapid shut down and emergency shutdown of equipment.

G. Make appropriate adjustments to Control Systems for areas where actual Contractor supplied equipment and installation varies from the information included in Contract Documents.

H. Flow Control Valve Control:

1. Provide Flow Control Valve control from Control System in accordance with Section 35 22 14 – Valves and Equipment.

   a. When the Control System issues a close command to the Flow Control Valve, the Flow Control Valve solenoid shall be energized to cause the valve to close.

b. When the Control System removes it command to close the Flow Control Valve, the Flow Control Valve solenoid will be deenergized to allow Flow Control Valve normal operations.

I. Monitoring and Alarming:

1. Where monitoring or alarming are required in these Specifications at a minimum provide the following functionality. This does not preclude additional functionality required by other requirements in these Specifications or required system functionality.

2. Monitoring

a. Display information received from end devices on graphic displays. Display information in units of process measurement where applicable.

b. Where Flow Indicator Transmitter (FIT) devices are shown on P&IDs, display the following minimum information on graphic displays:

1) Flow Direction.

2) Flow.

3) Total Flow.

3. Alarming:

a. Monitor and provide alarming for instruments and devices shown interfacing with the Control System Enclosure on the supplied P&ID drawings, Table 25 00 01A – Typical Turnout Building I/O and Data Point List, and where the Specifications reference interfaces with PLCs or Control System:

b. Provide status alarms for Control System equipment malfunctions.

J. Turnout Building Operation:

1. Operational logic should incorporate data-points and system inputs shown in Table 25 00 01A – Typical Turnout Building I/O and Data Point List and supplied Process and Instrumentation Diagrams.

2. Additional Facility Monitoring:

a. Monitor access to monitoring system in accordance with Section 28 10 01 – Security and Surveillance System.

b. Status alarms for UPS system(s).

c. Status alarms for Control System equipment malfunctions.

K. Remote System Interfaces:

1. Provide the following remote-control signals for interface with the future San Juan Lateral SCADA System as shown on Table 25 00 01B – SJLWTP SCADA System Remote Interface Data Point List. Provide system programming, logic

development and register address assignment to implement remote control interface to complete the remote communication link.

2. Interface will be via Modbus TCP communication over the long-haul optical fiber network connection:
    a. Control System Local/Remote Control.
        1) Local Control shall allow operator in Turnout Building to operate Control System from OIT.
        2) Remote Control shall allow the San Juan Lateral SCADA System to operate the equipment.
    b. Automatic Operation Control Signals:
        1) Flow Control Valve Close.
        2) Flow Control Valve Not Closed.
3. Contractor shall coordinate with Government and San Juan Lateral SCADA System installer for setup of interface signals between the two systems.

L. Safety Measures:
1. Failure of PLC, remote I/O hardware, or network communication link shall be individually alarmed at OITs.
2. Watchdog function to monitor:
    a. Internal processor clock failure.
    b. Processor memory failure.
    c. Loss of communication between processor and I/O modules.
    d. Processor ceases to execute logic program.
3. Initial safety conditions:
    a. Utilize program module to dictate output states in known and safe manner prior to running of control program.
    b. Utilize program each time PLC is re-initiated and control program activated.
4. Monitoring of internal faults and display:
    a. Internal PLC system status and faults shall be monitored and displayed:
        1) Monitored items shall include:
            a) Memory ok/loss of memory.
            b) Processor ok/processor fault.
            c) Scan time overrun.
5. Operator intervention:
    a. Logic system failure shall not preclude proper operator intervention:

1) Safety shutdown of equipment or system shall require manual operator intervention before equipment or system operation is reestablished.

2.07 LOCKOUT/TAGOUT

A. Lockout/Tagout functionality shall provide the operator with the following minimum control features:

1. Ability to lockout individual components.
2. Ability to lockout process systems.

B. Upon application of a lockout/tagout command system shall automatically stop, or remove from service, affected component or system. This operation shall be performed regardless of control mode that system is in:

C. Control System shall include logic to automatically react to the application of lockout/tagout on a component, or system, by making appropriate adjustments or shutting down impacted systems.

2.08 USER ACCESS CONTROL AND MONITORING SECURITY

A. Provide the following minimum access control capabilities, in accordance with recommendations in ISA 62443-1-1, ISA 62443-2-1 and NIST SP 800-82:

1. Configurable account password solution to include requirements for password length, frequency of change, minimum required password complexity, number of login attempts, inactive session logout, screen lock by application, and denial of repeated or recycled use of the same password. Adjustments to these requirements require explicit written approval of COR upon receipt of written justification.
2. Network Components: Provide individual accounts for each user with customizable access privileges for each account. Accounts shall be configurable into role-based functions allowing varying levels of interface with and modifications to the system components. Include account authentication by approved method.

B. Prior to testing of Control System and components, remove or disable software components that are not required to provide specified system functions and capabilities or required for operation or maintenance of the system.

C. Provide logging of all events, including access events for network components. Forward logs to central server for long term storage.

D. Provide operating logs for windows operating systems.

E. Provide approved anti-virus protection on components where applicable.

2.09 NETWORK

A. Provide high speed data network between automation controllers and processors:

1. Network shall be designed to be scalable and facilitate future expansion of the system.
2. Network design shall allow any component associated with a given system partition, including controllers, backplanes, communication cards, and I/O modules, to be removed without affecting communication with other partitions.
3. Network shall be designed to facilitate speed of communication between networked components and shall not introduce significant data latency.
4. Network shall be designed to have 50 percent spare base system capacity without exceeding performance limits.
5. Provide optical fiber cable, in accordance with Section 26 05 10 – Conductors and Cables, for connections between Network Interface Panel and long-haul fiber optical transmission system.

B. Provide hardware, software, and design to provide fully functional bi-directional data links to other systems, components, PLCs, PACs, and networks as required.

C. Control System shall provide support for the following minimum communication protocols:

1. DNP 3.0 in accordance with IEEE 1815.
2. Modbus Serial (RTU and ASCII).
3. Modbus Network (TCP).
4. HART Communications Protocol.

D. Provide capability for continual capture of NetFlow data between components for network and protocol analysis.

E. Control System Network Switch:

1. Provide Control System network switch.
2. Network Features: Layer 2 Ethernet switching.
3. All network switches shall provide the following minimum features:
    a. Supports linear, star, and ring topology.
    b. Loop Prevention.
    c. Resiliency.
    d. Segmentation.
    e. Prioritization.
    f. Time synchronization.

g. Security and Diagnostics.

h. Provide ports and connections, as required. Spare to include, at a minimum, the following: 20 percent spare ports of each type used at the switch.

4. Physical Features:

a. Copper ports: Eight 10/100BaseTX, RJ-45, minimum.

b. Fiber ports: Two 100BaseFX, single-mode 1310nm, ST connectors, minimum.

c. Status indicators on each port.

d. Power: Redundant inputs.

e. Enclosure: Metal case.

5. Network Features:

a. IEEE 802.3 compliance.

b. Full/Half Duplex operation on all ports.

c. 100-Mbps throughput, minimum.

6. Allows enabling/disabling ports.

2.10 CONTROL EQUIPMENT ENCLOSURE

A. Location:

1. As required for mounting system equipment and components.

2. A cabinet shall be located in each Turnout Building and, at a minimum, utilized for mounting the system OIT and related components.

3. Provide adequate room for at least 90 degree opening of cabinet door.

4. No less than 3- by 3-feet accessibility in front of cabinet.

B. Enclosures design shall allow free and easy access to internal components.

C. Type:

1. Wall mounted.

2. NEMA 250 Type 12 enclosure for indoor installations and NEMA 250 Type 4X enclosures for outdoor installations.

3. Size: As required to house components. All new enclosures shall include a minimum of 25 percent spare space.

4. Constructed of sheet metal no lighter than 14-gauge.

5. Seams continuously welded and ground smooth; no holes or knockouts.

6. Continuous hinge on door. Doors shall be lockable.

7. Finish: Manufacturer's standard coating.

8. Back panel.

D. Enclosures shall have louvers, or ventilation, as required to maintain temperature of equipment within manufacturer recommended range. Interior temperature shall be maintained at a minimum of at least 10 degrees F below maximum operating temperature of internal devices at maximum ambient temperature:

1. Fans or other forced ventilation devices shall be provided as redundant pairs and shall be provided with temperature alarms.

2. Louvers and openings shall be equipped with screens and changeable filter material to prevent the entrance of dust, insects, and rodents.

E. Provide thermostatically controlled cabinet space heaters to control condensation in cabinets not in a temperature-controlled environment, or subject to condensation.

F. Provide 120 VAC LED light fixture sufficient to illuminate panel interior for each new enclosure installed by Contractor.

2.11 NETWORK RACK

A. Belden XNM Series as manufactured by Belden Inc., 1N. Brentwood Blvd., St. Louis, MO 63105, or equal having the following essential characteristics:

1. 19-inch standard size.

2. 12U minimum.

3. Lockable.

4. Mesh cooling design.

5. Built-in utility outlets to plug into UPS.

6. Contractor to determine final size based on components to be installed for System Control and Monitoring equipment and Section 28 10 01 – Security and Surveillance System equipment.

2.12 INSTRUMENTS

A. RTD Type Ambient Temperature Sensors:

1. Dwyer Series O-4, or equal, having the following essential characteristics:

a. Platinum type.

b. Ambient temperature sensor.

c. NEMA 4X enclosure.

2. Provide RTD type ambient temperature sensors to monitor ambient temperature inside Turnout Building.

2.13 POWER

A. Provide power distribution system to supplied components, including protection, to support full operation of the system. This distribution system shall include analog loop power supplies and contact interrogation voltages for dry contact inputs where required.

B. Failure of power supply, or power feed, shall be alarmed in the system.

C. Provide fuses, circuit breakers, power supplies, transformers etc., required to supply and transform power feeds at the desired voltage. Provide DIN Rail disconnect for each 120-volt circuit entering enclosure.

D. DC Power Supply:

1. Type:
    a. Switch-mode power supply.
    b. DIN-Rail mountable.
2. Input Voltage: 100-240 VAC.
3. Output Voltage: 24-VDC plus-and-minus 1 percent.
4. Output Current: 5-Amps, minimum.
5. Connection Type: Screw clamp.
6. Operating Temperature Range: -20 to 60 degrees C.

E. Uninterruptable Power Supply (UPS):

1. Type: Tower.
2. Minimum status display and outputs to Control System:
    a. On line.
    b. On battery.
    c. Overload.
    d. Replace battery.
    e. UPS trouble.
3. Input voltage: 120-volts, 60-hertz.
4. Output voltage: 120-volts, 60-hertz.
5. Batteries:
    a. Leak-proof.
    b. Hot-swappable, user replaceable.
    c. Maintenance-free sealed Lead-Acid battery with suspended electrolyte.

6. Size to provide at least 200 percent of full load current for the following equipment inside the Control System cabinet, the Network Interface Panel, and the Communication and Security Rack for at least 30-minutes:
    a. Everything inside Control System Enclosure, Network Interface Panel, and Security and Communication Rack.
    b. Networking and Control System equipment outside of panels and rack.
    c. Fiber optic equipment.
    d. Security and Surveillance Equipment in accordance with Section 28 10 01 – Security and Surveillance System.
7. Receptacle strip(s):
    a. Provide adequate receptacles for each enclosure.
    b. Voltage: 120-volts, 60-hertz.
    c. Current: 20-amperes.
    d. UL listed.

F. Fuse Block and Fuses:
1. Type: Finger-Safe terminal block.
2. DIN-Rail mountable.
3. Fuse:
    a. Rated for size required.
    b. 250-volts, current limiting.

G. Ground Bus:
1. Material: Copper.
2. Size: Sufficient to attach equipment ground conductors.
3. Length: Sufficient to attach equipment ground conductors plus 20 percent spare.
4. Insulated mounting bracket.

2.14 INSULATED CONDUCTORS

A. Insulated conductors used to connect components external to enclosure are in accordance with Section 26 05 10 – Conductors and Cables.

2.15 SURGE SUPPRESSION

A. 4 to -20 mA Transient Protection Modules:
1. Triple stage hybrid protection.
2. Rated Voltage: 24-VDC

3. Rated Current: 10-Amps
4. Surge Current (8/20 microsecond wave form): 2.5kA, minimum.
5. Clamping Voltage: 30-VDC, maximum.
6. Response Time: Less than 5-nanoseconds.
7. Throughput Resistance: 12-Ohms, maximum.
8. Suppressor condition status indicator.
9. Automatically resets after each transient.

B. Line Voltage Surge Protection Device:
1. Rated Voltage: 120-volt, 60-hertz.
2. Rated Current: 15-Amps.
3. Single-phase service.
4. Surge Current (8/20 microsecond wave form): 7 kA, minimum.
5. Clamping Voltage: 270-volts, maximum.
6. Maximum Continuous Operating VAC (MCOV): 115 percent rated line voltage.
7. Response Time: Less than 5-nanoseconds.
8. Modes of Protection: L-N, L-G, N-G.
9. Suppressor condition status indicator
10. Automatically resets after each transient.

C. For automation equipment in outdoor enclosures, surge suppression is required for power supply inputs and wiring leaving enclosure, including digital I/O, analog I/O, and communication wiring.

2.16 COMPONENT LIFE CYCLE / SUPPORT

A. Components for Control System shall be chosen to maximize installation life-cycle and extend the time to replacement to the extent possible. Components shall be chosen with the following characteristics to help provide maximum life-cycle of the installation:
1. Components from established product lines and manufacturers.
2. Components from the most recent generation of equipment available from their manufacturer. Components where a later generation of equipment with a similar functionality is available shall not be acceptable except where specifically identified in this Section.

B. Software packages used in the installation and for development of system graphics and control logic, shall be up-to-date and compatible with the most recent version of major operating systems, including Microsoft Windows.

2.17 SOURCE QUALITY CONTROL

A. Government reserves the right to inspect equipment at supplier's manufacturing plant prior to final testing.

PART 3 EXECUTION

3.01 INSTALLATION

A. In accordance with manufacturer's recommendations, with suitable and secure fasteners.

B. Incorporate into the Control System, at a minimum, the instrumentation and control I/O points identified in Table 25 00 01A – Typical Turnout Building I/O and Data Point List and the P&IDs shown in drawings 1695-D-62569 through 1695-D-62577.

C. For connections to the programmable controllers which run outside the Turnout Building, surge suppression is required, including digital and analog I/O.

D. Install Network Interface Panel in Turnout Building.

1. Network Interface Panel is GFE to be provided to Contractor by Government and installed in Turnout Building by Contractor. Coordinate with COR to receive, handle, store, and install GFE.
2. Provide final power and communication connections to the Network Interface Panel to complete installation and integration.
3. Do not make any modifications or alterations to the Network Interface Panel. Minor modifications are allowed to route incoming communication and power connections and terminate final wiring and fiber connections.

E. The contractor shall furnish and install optical fiber in the Turnout Building site per Section 26 05 10 – Conductors and Cables.

F. Contractor shall complete fiber connection between long haul fiber communication system splice box to the Turnout Building Network Interface Panel. Utilize fiber splice handholes shown on supplied drawings to route fiber to Turnout Building.

G. Contractor is responsible for determining the final installed layout, wiring connections and schematic diagrams based on field installed equipment and design coordination meetings.

H. Make wire connections between the Control System equipment cabinet terminal blocks and the programmable controller's field wiring arms with NEC type SIS No. 18 AWG switchboard wire.

I. Make connections necessary to place the Control System into operation.

J. Terminal Blocks:

1. Arrange terminal blocks to segregate signal types and account for spare space.
2. Arrange terminal blocks to allow external cabling to be supported.
3. Mount terminal blocks at least 3-inches from panel edges and other devices. Mount adjacent rows of terminal blocks with 12-inches separation minimum. Access to front of terminal blocks shall be unobstructed.
4. Machine lettered terminal numbers on terminal block marking strips. Do not use hand lettering.
5. Identify vertical columns of terminal blocks with single letter such as "A", "B", or "C". Vertical columns may contain more than 12 termination points.
6. Terminals:
    a. Binding-head or washer-head screws with serrated or grooved contact surfaces.
    b. Spring-clamp type terminal blocks are not acceptable.
7. Voltage rating: 300-volts.
8. Current rating: 20-amps.
9. Analog inputs shall be disconnecting switch type terminal blocks.

K. Cable and Wiring:

1. Provide cable and wiring in accordance with Section 26 05 10 – Conductors and Cables and Section 26 05 02 – Basic Electrical Materials and Methods.
2. Wire Connections:
    a. Make connections at device terminals or terminal blocks. Maximum two wires per termination point.
    b. Install incoming 120-volt branch circuits on adjacent terminal block terminals.
    c. Install ring tongue connectors for terminations at terminal blocks and electrical devices. Use pin-type terminal connectors where use of ring tongues is not practical such as terminations at selector switches, pushbuttons, indicating lamps, and auxiliary relays.
    d. Do not terminate wire without terminal connector.
    e. Wire splices are not acceptable.

L. Line power for control and monitoring equipment, including Network Interface Panel, shall be supplied from Contractor provided UPS system.

M. Network cabling shall provide a designated and easily identified to-and-from location marker at each end of cable.

N. Make cable and conduit connections. Use watertight conduit hubs for conduit connections.

O. Make grounding and bonding connections in accordance with Section 26 05 20 – Grounding and Bonding.

P. Control Cable and Wire Identification: In accordance with Section 26 05 10 – Conductors and Cables.

Q. Provide software and associated programming/configuration required to meet performance requirements of Contract Documents:

1. At substantial completion of Project, turn current licenses for software over to Government in Bureau of Reclamation's name and install latest version, upgrade or service pack for all software.

2. Provide respective software supplier's Comprehensive Support Contract for software and network components covering full one-year warranty period following substantial completion. Provide software upgrades, service packs and tech support from software or network component suppliers at no cost.

R. Nameplates: Provide nameplates for Contractor supplied equipment in accordance with Section 26 05 02 – Basic Electrical Materials and Methods.

S. Installation shall look complete and professionally done, damage to enclosures or installation area shall be repaired and left in a finished condition.

T. I/O and Remote Interface Data Lists:

1. List is solely for information of Contractor. I/O list may change throughout Contractor system design period.

2. I/O Table and Data Point List are shown as applied to Sanostee Turnout Building, but are the same for every Turnout Building, only the location tag number will change. All I/O and Data shown here apply across all Turnout Buildings.

Table 25 00 01A – Typical Turnout Building I/O and Data Point List

Equipment	Description	Type	Signal
S1205-FIT-001	Flowmeter Flowrate	MB	RTU
S1205-FIT-001	Flowmeter Totalizer	MB	RTU
S1205-FIT-001	Flowmeter Direction	MB	RTU
S1205-EX-001	UPS Online	MB	RTU
S1205-EX-001	UPS On Battery	MB	RTU
S1205-EX-001	UPS Overload	MB	RTU
S1205-EX-001	UPS Replace Battery	MB	RTU
S1205-EX-001	UPS Trouble	MB	RTU
S1205-FCV-001	Flow Control Valve Close Command	DO	24 VDC

S1205-ZSC-001	Flow Control Valve Closed	DI	24 VDC
S1205-ZSC-001	Flow Control Valve Not Closed	DI	24 VDC
S1205-TE-001	Indoor Ambient Temperature	AI	4-20 mA
S1205-ZSC-002	Control System Enclosure Intrusion	DI	24 VDC
S1205-ZSC-003	Network Interface Panel	DI	24 VDC
S1205-ZSC-004	Comm. & Security Rack Intrusion	DI	24 VDC
S1205-ZSC-005	Site Entrance Gate Intrusion	DI	24 VDC
S1205-ZSC-006	Building Door 1 Entrance Intrusion	DI	24 VDC
S1205-ZSC-007	Building Door 2 Entrance Intrusion	DI	24 VDC
S1205-ZSC-008	Fiber Splice Vault Intrusion	DI	24 VDC

Table 25 00 01B – SJLWTP SCADA System Remote Interface Data Point List

Equipment	Description	Type	Signal
S1205-FIT-001	Flowmeter Flowrate	Output	MB TCP
S1205-FIT-001	Flowmeter Totalizer	Output	MB TCP
S1205-FIT-001	Flowmeter Direction	Output	MB TCP
S1205-EX-001	UPS Trouble	Output	MB TCP
S1205-FCV-001	Flow Control Valve Close Command	Input	MB TCP
S1205-ZSC-001	Flow Control Valve Closed	Output	MB TCP
S1205-ZSC-001	Flow Control Valve Not Closed	Output	MB TCP
S1205-TE-001	Indoor Ambient Temperature	Output	MB TCP
Control System	General Intrusion Alarm	Output	MB TCP
Control System	Control System Trouble	Output	MB TCP
Control System	In Remote	Output	MB TCP

3.02 CONTRACTOR FIELD QUALITY TESTING

A. Perform field wiring checkout and test in accordance with Section 26 05 90 – Wiring Checkout and Tests.

B. Perform system SPAT and Lateral Integration Testing in accordance with Section 25 08 10 – Automation System Testing and Acceptance.

END OF SECTION

SECTION 25 08 10
AUTOMATION SYSTEM TESTING AND ACCEPTANCE

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for System Control and Monitoring.

1.02 DEFINITIONS

A. I/O: Input/Output.

B. OIT: Operator Interface Terminal.

C. PLC: Programmable Logic Controller.

D. SCADA: Supervisory Control and Data Acquisition.

E. SJLWTP: San Juan Lateral Water Treatment Plant.

F. SPAT: System Performance Availability Test.

1.03 REFERENCE STANDARDS

A. International Society of Automation (ISA)

1.	ISA 62381-2011	Automation Systems in the Process Industry – Factory Acceptance Test (FAT), Site Acceptance Test (SAT), and Site Integration Test (SIT)
2.	ISA 62382-2012	Automation Systems in the Process Industry – Electrical and Instrumentation Loop Check

1.04 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals and Section 01 33 26 – Electrical Drawings and Data.

B. RSN 25 08 10-1, SPAT Procedure.

1. Detailed procedures for planned testing, including individual test and inspection activities that will be performed and a step-by-step procedure for performing the tests.

C. RSN 25 08 10-2, SPAT Start Date.

1. Provide start date of SPAT testing for each Turnout Site. Resubmit RSN if there are any changes to start date.

D. RSN 25 08 10-3, SPAT Test Report:

1. Certification of performance and results of the SPAT for each Turnout Building.

E. RSN 25 08 10-4, Lateral Integration Test Start Date.

1. Provide start date of lateral integration testing for each Turnout Site. Resubmit RSN if there are any changes to start date.

F. RSN 25 08 10-5, Lateral Integration Test Report

1. Documentation of all hardware and software modifications made to the Turnout Building Control System during the conduct of Lateral Integration Testing.

1.05 DESIGN REQUIREMENTS

A. Automation Testing and Acceptance shall be performed for each Turnout Building individually. This Specification describes testing and test documentation requirements for one Turnout Building but all provisions within shall apply to each Turnout Buildings individually.

B. Automation Testing and Acceptance Process Documentation:

1. General: SPAT documentation prepared by Contractor shall represent testing program in accordance with guidelines in ISA 62381, ISA 62382, and the requirements found in this Specification.

2. Inspection, testing, and certification documentation shall include the following:

a. Name and signature of person performing test.

b. For which Turnout Building test was performed.

c. Date at which test was performed.

d. Identification of component, equipment, or system tested.

e. Identification of tests performed.

f. Any data or readings taken during test to verify component, equipment, or system acceptance.

3. Each Turnout Building shall have an individual test report. Do not submit one test report that contains testing for more than one Turnout Building.

4. Each Turnout Building will have a separate test report. A singular test procedure may be submitted as long as differences between tests in the different Turnout Buildings are accounted for in the test procedure.

PART 2 PRODUCTS

Not Used.

PART 3 EXECUTION

3.01 SYSTEM PERFORMANCE AVAILABILITY TEST (SPAT)

A. Purpose of SPAT is to demonstrate functionality, performance, and stability of the Control System.

B. Government will witness test. Notify COR in writing at least 20-days prior to beginning of SPAT.

C. Perform site acceptance testing and site integration testing in accordance with ISA 62381.

D. Perform loop checks in accordance with ISA 62382. Perform verification for each loop and each input or output.

E. Demonstrate correct operation of components and systems including, but not limited to the following:

1. Control system.
2. Information technology and cyber security.
3. Communication links.
4. Instrumentation.

F. Control System:

1. Demonstrate accurate operation of graphic displays and operator interfaces to graphic displays and OIT. Operate field components and instruments for operating configurations and signals represented on OIT and graphic displays to verify accuracy of feedback signals.
2. Verify system alarms:
    a. Ensure any alarms operate as intended.
    b. Where simulation of feedback signal is required to verify an alarm, perform simulation to verify system interlock unless device is inaccessible.
    c. Provide feedback by operating or forcing feedback from device, if it does not damage device or present safety hazard.
3. Check alarms and annunciation points.
4. Adjust settings and graphic displays.
5. Flow Control Valve (FCV) Solenoid Pilot Valve Operation:

a. Ensure solenoid pilot valve repositions upon application and removal of control power from Turnout Building Control System.

b. Confirm that the FCV limit switch status indications transmitted to the Turnout Building Control System OIT are accurate. The OIT graphical display shall correctly reflect valve position from limit switch feedback.

G. Information Technology and Cyber Security:

1. Capture NetFlow data between components for network and protocol analysis at time of system testing.

2. Perform vulnerability scan and correct deficiencies.

3. Verify account-based access control.

4. Validate network architecture including port address lists and device address lists.

5. Validate and document communication restrictions, filtering, and monitoring rules through firewalls and other network components.

6. Perform additional Contractor- and manufacturer-recommended testing.

7. Government may perform additional cyber security related testing and verification of system with cooperation from Contractor.

H. Communication Links:

1. Verify cable is free of defects and meets design and manufacturer's guidelines.

2. Verify communication links between devices.

3. Demonstrate feedback signals are transmitting correctly and are operating correctly between facilities.

I. Instrumentation:

1. Inspect each gauge, sensor, transmitter, thermometer, and other instruments. Replace defaced or broken items.

2. Inspect instrument tubing.

3. Field verify calibration and performance of each instrument. Document results for each instrument on separate field calibration sheet. Include, at minimum:

a. Input, output, and error at 0, 25, 75, and 100 percent of span for analog instruments.

b. Switch setting, contact action, and deadband, if applicable, for discrete elements.

4. Flowmeter Simulated Tests

a. Apply simulated flow signal inputs. Simulated signals to cover minimum and maximum flows over the velocity range.

b. Check displays for correct readouts and check each I/O point for correct I/O signal.

c. Test each output connection for correct output conformance.

d. Verify all instrumentation and indication devices function properly. Test all functions of flowmeter transmitter for correct operation.

J. Required I/O and feedback data not visible on OIT shall be demonstrated by temporarily connecting a computer to the PLC to verify the data is available to be transmitted to the SJLWTP SCADA System.

3.02 CONTRACTOR FIELD QUALITY TESTING

A. Perform field wiring checkout and test in accordance with Section 26 05 90 – Wiring Checkout and Tests.

B. Wiring checkout and tests shall occur prior to SPAT activities.

3.03 LATERAL INTEGRATION TESTING

A. Test communication links between the Turnout Building Control System and the SJLWTP SCADA System.

B. Minimum duration of lateral integration testing per Turnout Building: 2 days.

C. Government will witness test. Notify COR in writing at least 40-days prior to beginning of Lateral Integration Testing.

D. Provide the following minimum technical personnel to support integration with the lateral communication system. Technical personnel shall be onsite throughout the entire duration of the lateral integration testing:

1. Two electricians
2. SCADA/Control System representative capable of performing onsite modifications to the Control System.

END OF SECTION

SECTION 26 05 02
BASIC ELECTRICAL MATERIALS AND METHODS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Complete Electrical System for Outdoor Equipment Only:

1. Payment: Lump sum price offered in Price Schedules includes cost of:
    a. Performing required design.
    b. Furnishing recently certified calibrated testing equipment.
    c. Performing testing.
    d. Transporting and storing equipment and materials.
    e. Modifying and relocating equipment.
    f. Placing, leveling, and grouting channel bases.
    g. Assembling, adjusting, and installing equipment.
    h. Painting equipment and materials.
    i. Brackets, fasteners, bolts, nuts, lock washers, and other accessories required for mounting or installing electrical equipment and materials.
    j. Drilling holes in steel structures (other than tubular structures) as required for mounting or installing electrical equipment and materials.
    k. Furnishing, handling, and storing spare parts for electrical equipment.
    l. Furnishing special tools and appliances for maintenance and adjustment of equipment.
    m. Making electrical connections; furnishing all miscellaneous materials, which are required for making the connections to electrical equipment. Contractor estimates the number of connections to be made to the equipment based on prior knowledge or experience with similar equipment. No additional compensation will be allowed Contractor if actual connections exceed number estimated by Contractor at time of bidding.
    n. Performing wiring checkout and tests.

1.02 REFERENCE STANDARDS

A. ASTM International (ASTM)

1. ASTM A123/A123M-17 Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

2.	ASTM A153/A153M-16a	Zinc Coatings (Hot-Dip) on Iron and Steel Hardware

B. Institute of Electrical and Electronics Engineers (IEEE)

1.	IEEE C2-2023	National Electric Safety Code (NESC)

C. National Electrical Installation Standards (NEIS)

1.	NECA 1-2015	Good Workmanship in Electrical Contracting

D. National Fire Protection Association (NFPA)

1.	NFPA 70-2023	National Electrical Code (NEC)

1.03 EXTRA MATERIALS

A. Provide to Government special tools and equipment necessary to properly install, adjust, test, and check operation of electrical equipment. Furnish all additional tools and equipment as necessary to properly install, adjust, and check operation of electrical equipment.

PART 2 PRODUCTS

2.01 MATERIALS AND EQUIPMENT

A. Conform to NFPA 70 and IEEE C2.

B. Furnish all electrical materials and equipment.

C. Provide galvanized steel or non-corrosive metal for mounting bolts, nuts, and washers for minor items of electrical equipment and lighter weight items. Do not use cadmium-plated mounting hardware.

D. Change designs as required where Contractor-furnished electrical equipment and materials differ in size, type, ratings, or other physical properties from designs in these specifications. CO will approve changes at Contractor's expense, unless Contractor can demonstrate changes are necessary regardless of manufacturer.

E. Provide special tools and appliances furnished by manufacturer for maintenance and adjustment of manufacturer's electrical equipment.

F. Use test equipment that has been calibrated within 365-days of its use with calibration sticker clearly visible.

G. Deliver all spare parts required by these specifications to Bureau of Reclamation only after completion of contract.

PART 3 EXECUTION

3.01 INSTALLATION

A. In the event a conflict is discovered by Contractor, it will be Contractor's responsibility to inform CO of the conflict:

1. In the event of conflicting requirements, establish precedence in the following order:

a. These specifications or as directed by CO.

b. Drawings included in these specifications.

c. Latest edition of "National Electric Code" of the National Fire Protection Association for electrical wiring apparatus and the adopted electrical code of the State in which construction is located. Cooperate with any agency designated by that State to inspect electrical installation for conformance with applicable State code.

2. Notwithstanding the order above, the most stringent requirement shall apply.

B. Do not install or ship electrical equipment until approval drawings, factory tests, and data for associated equipment have been approved by Government.

C. Electrical Installations, Assembly Operations, and Adjustments: Comply with NFPA 70, IEEE C2, and NECA 1.

D. Make electrical installations complete and ready for service.

E. Install electrical equipment in accordance with directions furnished by manufacturer's instruction books.

F. Make all electrical wire, cable, conduit, and grounding connections and furnish all miscellaneous materials which are required for making these connections to equipment.

G. Install items of electrical equipment required to be in a lineup, so appearance is uniform, including height, depth, color, and general construction.

H. Tighten nuts used in electrical equipment assembly with torque wrenches to torque values recommended by equipment manufacturers.

I. Drill holes in bolted steel structures and provide fastenings required for mounting or installing electrical equipment and materials.

J. Do not drill holes in tubular steel structures. Fasten to tubular steel structures by means of properly drilled and tapped pads or brackets welded to tubular members.

K. Installation of Electrical Equipment includes:

1. Leveling and grouting channel bases.

2. Drilling holes, furnishing hardware, and assembling components to each other.
3. Furnishing materials for and making all connections correctly in accordance with final wiring diagrams.
4. Tagging wires and cables at each end.
5. Correcting any errors made by Contractor in installation at no expense to Government.
6. Wiring and checkout of all equipment in accordance with Section 26 05 90 – Wiring Checkout and Tests.

L. Set electrical equipment installed on concrete foundations and grout in place to provide full and even bearing.

3.02 REPAIR

A. Repair or replace damaged devices and repair damaged painted surfaces of equipment to match original finish.

B. Correct by repair or replacement, at Contractor's expense, all damage to or failure of any part of the items of material and equipment which in opinion of COR was caused by faulty installation, faulty mechanical assembly, or mishandling.

3.03 FIELD QUALITY ASSURANCE

A. Government will inspect electrical equipment during construction. Acceptance of equipment will be made after equipment is operational.

END OF SECTION

SECTION 26 05 10
CONDUCTORS AND CABLES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for Complete Electrical System for Outdoor Equipment Only.

1.02 DEFINITIONS

A. Cable: Cable, cables, wire, or wires of one or more insulated conductors.

B. Power Cable: For power loads including site power, receptacle outlets; motors; alternating- and direct-current distribution circuits; heating, ventilating, and air-conditioning and lighting circuits; and cable that is used for controlling heating, ventilating, air-conditioning, and lighting equipment.

C. Building Premises Wire: For connection of separately-housed components such as wiring devices and luminaire circuits to branch circuit breakers and to one another.

D. Multiconductor Control Cable: For control, metering, indication, annunciation, and relaying circuits; and circuits not identified as power circuits.

E. Instrumentation Cable: Power limited cable for RTD, transducer output circuits, and can be used for serial communication (RS-232 & RS-485).

F. Telecommunications Cable: For supervisory computer control, data, voice communication, and logic level signals.

G. Enclosure Wiring: For wiring inside of enclosures or control boards.

1.03 REFERENCE STANDARDS

A. ASTM International (ASTM)

1.	ASTM B8-11-2017	Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft

B. Fiber Optic Association (FOA)

1.	FOA-4a	OTDR Testing of Fiber Optic Cable Plant

C. Institute of Electrical and Electronics Engineers (IEEE)

1.	IEEE C2-2017	National Electrical Safety Code (NESC)

D. Insulated Cable Engineer Association (ICEA)

1.	ICEA S-87-640-2016	Fiber Optic Outside Plant Communications Cable
2.	ICEA S-116-732-2019	Standard for Category 6 and 6A, 100 Ohm, Individually Unshielded Twisted Pairs, Indoor Cables (With or Without and Overall Shield) for Use in LAN Communication Wiring Systems

E. National Electrical Manufacturers Association (NEMA)

1.	NEMA WC 57-2014	Control, Thermocouple Extension, and Instrumentation Cables
2.	NEMA WC 70-2009	Power Cables Rated 2000V or Less for the Distribution of Electrical Energy

F. National Fire Protection Association (NFPA)

1.	NFPA 70-2023	National Electrical Code (NEC)

G. Telecommunications Industry Association (TIA)

1.	TIA-526-1992	Standard Test Procedures for Fiber Optic Systems
2.	TIA-568-C.1-2009	Commercial Building Telecommunications Cabling Standard
3.	TIA 568-C.2-2011	Balanced Twisted-Pair Telecommunications Cabling and Components Standards
4.	TIA 568-C.3-2008	Optical Fiber Cabling Components Standard
5.	TIA -598-2014	Optical Fiber Cable Color Coding
6.	TIA-1152-2009	Requirements for Field Test Instruments and Measurements for Balanced Twisted-Pair Cabling

H. Underwriters Laboratories, Inc. (UL)

1.	UL 508-1999	Industrial Control Equipment, Seventeenth Edition

1.04 SUBMITTALS

A. Submit the following in accordance with Sections 01 33 00 – Submittals and 01 33 26 – Electrical Drawings and Data.

B. RSN 26 05 10-1, Approval Data:

1. Product data sheets

C. RSN 26 05 10-2, Field Test Reports

D. RSN 26 05 10-3, As-Built Drawings

1. Include cable schedule

1.05 DESIGN REQUIREMENTS

A. Determine conductor sizes, except where shown or specified, in accordance with NFPA 70. In event of conflict between design requirements and NFPA 70, more stringent requirement applies.

B. Where supplied equipment differs from that specified, determine conductor sizes in accordance with NFPA 70.

PART 2 PRODUCTS

2.01 CABLE, GENERAL

A. Manufactured no more than 24-months prior to Notice to Proceed.

B. Round, except for 2-conductor cable with parallel conductors.

C. Conductors:

1. Copper in accordance with ASTM 8, class B or C.
2. Stranded conductors.

D. AWG or kcmil designation.

E. Coverings or insulation: Suitable for installation in vertical position without injury to covering or deformation of insulation when supported in accordance with NFPA 70.

2.02 PHASE IDENTIFICATION

A. Power Cable Phase Conductors:

1. 120/240, single-phase: Black.

B. Neutral (Grounded) Conductors: In accordance with NFPA 70.

C. Equipment Grounding Conductors:

1. Green insulation.

2.03 COLOR CODING

A. Control Cable:

1. Colored insulation or jacket compound. Do not apply color coatings to insulation or jacket surface.
2. In accordance with NEMA WC 57, Table E-2.

2.04 POWER CABLE AND MULTICONDUCTOR CONTROL CABLE

A. 600-volt rated insulation.

B. Power Cable Type: Single-conductor or multi-conductor

1. Suitable For installation in raceways

C. Control Cable Type: Shielded

1. Suitable For installation in raceways

D. Insulation Type:

1. Single-conductor cable: XHHW-2.
2. Multi-conductor cable: XHHW-2 for 14, 12, 10 AWG. TFFN or TFN for 18 and 16 AWG.

E. Multiconductor Overall Jacket:

1. Type: PVC (Polyvinyl chloride) or better, flame-retardant.
2. Jacket thickness:
   a. 16 AWG conductors: 45 mils for 12 conductor or less. 60 mils for greater than 12 conductors.
   b. 14 AWG conductors: 45 mils for 9 conductor or less. 60 mils for greater than 9 conductors.

F. Single-conductor power cable in accordance with NEMA WC 70. Multiconductor cable in accordance with NEMA WC 57. Cable UL listed and labeled.

G. Provide multi-conductor power cable with integral ground conductor. Size ground conductor in accordance with NEMA WC 70.

2.05 BUILDING PREMISES WIRE

A. Type: Single, solid conductor THWN-2 rated for 600 volts.

B. Size: 14 AWG for 15 ampere branch circuits, 12 AWG for 20 ampere branch circuits.

C. Suitable for installation in raceways.

D. Conform to NEMA WC 70. UL listed and labeled.

2.06 INSTRUMENTATION CABLE

A. Type: Twisted, single-pair or twisted triad shielded or multiple individual twisted shielded pairs or twisted triads with overall shield.

B. Suitable for installation in raceways and cable tray.

C. Size: 16 AWG or as shown.

D. Insulation: 300-volt, 105 degrees C, minimum.

E. Single-Pair or Triad Shield: Aluminum-Mylar tape providing 100 percent coverage with tinned, stranded copper drain wire.

F. Multiple Pair or Triad Shield:

1. Pair or triad shield: Aluminum-Mylar tape providing 100 percent coverage with tinned, stranded copper drain wire.
2. Cable shield: Aluminum-Mylar tape providing 100 percent coverage with tinned, stranded copper drain wire.
3. Each pair or triad sequentially numbered for identification.

G. Jacket: Black PVC with rip-cord.

2.07 TELECOMMUNICATIONS CABLE

A. Optical Fiber Cable:

1. Provide cable in accordance with ICEA S-87-640, as appropriate.
2. Cable performance shall not be impacted by immersion in water.
3. UL listed, where available.
4. Color Coding: Per TIA-598.
5. Single Mode Optical Fiber Cable:
   a. Major cable components:
      1) Protect each fiber in a buffer tube of gel free, standard loose-tube type construction.
      2) 2.5 mm buffer tubes, color coded.
      3) Central dielectric strength member: Glass reinforced plastic (GRP) rod filler.
      4) Cable core: Symmetrically stranded buffer tubes around dielectric central member.
      5) Water swellable yarn or water blocking protection.
   b. Optical fibers:

1) 100 percent synthetic silica.

2) Cladding diameter: 125-microns.

3) Core/clad concentricity error: < 0.5 microns, < 0.2 microns typical.

4) Cladding non-circularity: < 1.0.

5) Primary buffer coating: 250 microns.

6) Attenuation of 1310/1383/1550 nanometer wavelength light: < 0.5/0.5/0.4 dB per kilometer.

7) Attenuation: Directly proportional to cable length for an arbitrary cable length section after steady-state conditions are reached.

8) Bend Insensitive Single-Mode Fiber.

9) Color coded.

10) Optical fibers: 12 minimum.

c. Cable Jacket:

1) Type: PVC (Polyvinyl chloride) or CPE (Chlorinated Polyethylene) or better, UV resistant, flame-retardant.

2) Ripcord under sheath.

d. Strength members: With expansion and contraction characteristics similar to glass fibers.

e. Cable Bend Radius:

1) Loaded: 7.7-inch.

2) Unloaded: 5.1-inch.

f. Cable tensile strength:

1) Long term (installed): 135 pounds per foot.

2) Short term (during installation): 600 pounds per foot.

g. Operating temperature range: -40 to 150 degrees F.

h. Suitable for installation in innerduct.

B. Optical Fiber Patch Cable:

1. Type fiber: Single-mode.

2. Length: As required.

3. Connectors: As required to connect to fiber optic equipment.

4. Loss per connector: < 0.5 decibels.

C. CAT 6 Cable:

1. Provided per ICEA S-116-732 and TIA-568-C.2.

2. Four twisted pair, minimum size 24 AWG conductors.
3. Provide communications rated cabling for installation purpose installed (plenum, riser, general purpose, etc.) in accordance with NFPA 70.
4. UL listed and labeled for wet locations.
5. Cable shall be imprinted with manufacturers name or identifier, flammability rating, gauge of conductor, transmission performance rating and length marking at regular intervals.

2.08 ENCLOSURE WIRING

A. Wiring inside control panels not intended for use extending outside of enclosure. Enclosure wiring for instrumentation or telecommunications shall maintain characteristics of those cable types.

B. Minimum 250-volt rated insulation.

C. Type:
1. NEC type SIS.
2. VW-1 Rated.
3. UL Listed.

D. Minimum Conductor Size:
1. 14 AWG.
2. Conductor sized according to application requirements, and in accordance with requirements and recommendation in NEC.

E. Insulation Type:
1. XLPE, or equal.

F. Insulation Color:
1. White or Grey, unless special purpose application.

G. Flame Test Rating:
1. UL 1581 VW-1.

2.09 ACCESSORIES

A. Cable Tags:
1. Type: Heat-shrink type, machine printable sleeve.
2. Color: White with black text.
3. Text size: As large as practical but at least 7 point (0.06-inch).

4. Be 1 inch in length, minimum, and have diameter suitable for installation on the intended conductor.

B. Multiconductor Cable Tags:

1. Type: Cross-laminated polyethylene film or Nomex. Be resistant to water, oil, solvent, abrasion, and tearing.
2. Rectangular shape and have minimum dimensions of 2- by 0.5-inches.
3. Color: White with black text.
4. Text size: As large as practical but at least 16 point (0.16-inch).
5. Attach to cable with standard nylon self-locking tie wraps or cable ties.

C. Terminal Connectors for Control and Instrumentation Cable:

1. Heavy-duty, insulated, pressure-crimp-type with ring tongues. 600-volt, pressure-crimp pin type may be used where use of ring tongue connectors is not practical.
2. Tin-plated copper.
3. Serrated inner barrel.
4. UL listed.
5. Compatible with conductor size and type of cable for which used.

D. Telecommunications Cable Faceplates and Connectors:

1. Connectorize fiber cables and connect to patch panels using ST type connectors. Make connections to components as required be component terminal.
2. Provide patch cords to connect dielectric fiber cable to fiber optic equipment.
3. Terminate unshielded twisted pair (UTP) cables in accordance with TIA-568-C.1 and TIA-568-C.2 with T-568A color configuration.
4. Use registered jack (RJ) 45 modular connector.
5. Provide each location with 2 port RJ-45 faceplates unless otherwise noted on drawings.
6. Faceplate color: White.

E. Fiber Optic Cable Patch Panels:

1. Type: Rack mount or wall mount.
2. Metal housing with splice tray holder, splice tray, adaptor plates, and radius limiters.
3. Splice Trays:
   a. Same manufacturer as housing.
   b. Fusion splice organizer to retain each splice and its shrink tubing.

c. Manufactured for loose tube type fiber optic cable.

4. Adaptor Plates:

a. Same manufacturer as enclosure.

b. Mount at minimum 6 fibers.

c. Use ST Connectors.

F. Optical Fiber Splice Closure:

1. Re-enterable splice enclosure:

a. Factory installed sealing materials.

b. Re-enterable and reusable.

c. Telcordia GR-771 compliant.

d. Flame retardant UV stabilized material.

e. International Electrotechnical Commission (IEC) Ingress Protection Rating (IP), IP-68 Rating compliant.

2. 12 fiber splice, minimum.

3. Grommet sealing system to support flat or round cable profiles.

PART 3 EXECUTION

3.01 INSTALLATION

A. Install in accordance with NFPA 70, IEEE C2, TIA 568-C.2 and manufacturer's requirements and recommendations.

B. When using cable ties, do not over tighten, to ensure tie does not cut conductor insulation or cable outer jacket. Do not use cable ties to support cables.

C. Use building premises wire for lighting circuits and 120-volt receptacles.

D. Maintain manufacturer recommended minimum bending radius for cables.

E. Cable Splicing:

1. Fiber Optic Cable:

a. Do not splice fiber optic cables outside of patch panels.

b. Fiber optic splicing:

1) Method: Arc-fusion.

2) Splices per set of fusion tips: 50.

a) Splice loss: 0.1 decibels, average bi-directional.

2. All other cable types: Do not splice cables.

F. Cable Installed in Conduit:

1. Do not pull cable into conduit until they have been cleaned and are free from obstructions and sharp corners.
2. Draw a clean, dry, tight-fitting rag through conduit immediately before installing cable.
3. Install cable using gradual and uniform pulling stresses. Do not exceed allowable pulling tensions and sidewall pressures as recommended by cable manufacturer.
4. Install cable in such a manner as to prevent cuts or abrasions in insulation or protective covering and kinks in cable.
5. When a lubricant is needed as an aid to pulling, use only soapstone or other suitable lubricant that is not injurious to cable insulation.
6. When mechanical cable pulling equipment is used, use a tension meter release mechanism to ensure pulling tension remains below tensile strength of the cable.
7. Remove and replace cables damaged during installation.
8. All conductors, in a single conduit, shall be pulled in at the same time.

G. Cable in Boxes and Enclosures:

1. Provide slack in cable in all boxes, and enclosures. Do not pull cable tight against any edge or component in enclosures and boxes.
2. If cable is terminated in box or enclosure include enough spare length in terminated conductor to reach any other terminal in enclosure. In multi-panel control cabinets conductor should be able to reach any other terminal within that panel of enclosure.

H. Fiber optic cables:

1. Termination:
   a. Terminate fiber optic cable installed between enclosures at patch panels.
   b. Coil 10-feet of fiber cable inside each patch panel.
   c. Leave spare fibers in each patch panel.
   d. Retain fibers at each cable end equal in length to longest single fiber.
2. Provide fiber optic coils inside pull boxes, junction boxes, and other electrical and control enclosures, and where recommended by manufacturer.
3. Clamp cable entry with fiber type cable clamp.

I. CAT 6 Cable:

1. Install in accordance with TIA-568-C.2 and NFPA 70 as applicable.

2. Install per manufacturer's instructions and do not exceed manufacturers' cable pull tensions for copper cables.
3. Install back boxes for RJ-45 connectors.
4. Do not install power and communication cable in same conduit.
5. No cable shall contain unterminated elements.
6. Do not untwist Category 6 cables more than 1/2-inch from point of termination to maintain cable geometry.
7. Terminate networking cables in T568A configuration per TIA-568-C.2.

J. Excavation and backfill for buried cable: In accordance with Section 31 23 10 – Earthwork for Site and Structures.

K. Grounding.

1. Provide one nonactive conductor grounded at both ends in each multiconductor control cable.
2. Ground spare conductors by wire jumpers from equipment terminal block points to equipment ground bus.

L. Wiring methods:

1. Terminate control cable with terminal connectors.
2. At termination point of multiconductor cable, form conductors into neat packs and tie conductors with self-locking cable ties.
3. Clamp cable entering equipment with commercial type cable clamps.
4. Leave sufficient length cable ends to make connections conveniently to equipment, fixtures, and devices.
5. Retain spare single conductors at each end of a multiconductor cable in a length equal to longest single conductor of multiconductor cable.
6. Terminate spare conductors of multi-conductor cable at terminal blocks. Ground spare conductors by using wire jumpers between terminations and equipment ground bus.

3.02 CABLE AND CONDUCTOR IDENTIFICATION

A. Power Cable:

1. Identify power conductors that are service conductors and panelboard feeders with colored insulation or colored phase tape in accordance with this section. An additional conductor marker is not required.
2. Provide a single-line marker on the load end of feeder and branch-circuit conductors that originate at panelboards. The marker to include designation of source equipment and circuit number.

B. Conductor Markers:

1. Text to be machine or computer generated. Handwritten text is not acceptable.
2. Provide individual conductors of multiconductor control cable with conductor marker. Print conductor designation, as shown on approved schematic diagram, on first line. Print conductor destination and device terminal number on second line.
3. A single line conductor marker will be acceptable provided the conductor designation is enclosed in parenthesis followed by the conductor destination.
4. Do not heat shrink marker onto conductor. Allow marker to rotate about the conductor.

C. Multi-Conductor Cable Tags:

1. Tag multiconductor cables at each end.
2. Text to be machine or computer generated. Handwritten text is not acceptable.
3. Mark tags with cable designations as shown on approved wiring and interconnection diagrams. Cable designation to be a single-line of text that includes cable number, originating equipment designation, and destination equipment designation. Develop cable designations when not shown. Examples of cable designations are:
   a. 1DSA-IM1 - First cable from electrical panel DSA to In-line motor No. 1.
   b. 3DSA-CP1 - Third cable from electrical panel DSA to control panel 1.
4. Attach tags to cable with self-locking tie wraps or cable ties.

3.03 CONTRACTOR FIELD QUALITY TESTING

A. Perform testing after installation, but before connection to equipment.

B. Test cables in large groups to minimize number of testing sessions.

C. Notify COR, in writing, of date, time, and cables to be tested at least 5 working days before testing. Testing dates to be mutually agreeable between Contractor and COR.

Perform following tests:

1. Continuity test: Test continuity of each conductor prior to performing insulation resistance test.
2. Insulation resistance tests for power, control, and instrumentation cables:
   a. Measure insulation resistance between each conductor and station ground with all other conductors in same cable or conduit grounded.
   b. Test voltage: 1,000 volts direct current, minimum.
   c. Minimum acceptable insulation resistance: R in MΩ (rated voltage in kV + 1) x 1,000/length in feet.

3. Insulation resistance test for telemetry cable:
   a. Test cable before making ground connections,
   b. Test voltage: 1,000 volts direct current, minimum.
   c. Measure insulation resistance between each conductor and remaining conductors, and between conductor and shield.
4. Record following test data on test report for each conductor tested:
   a. Date of test.
   b. Name or names of Contractor's personnel who performed test.
   c. Name of Government inspector who witnessed test.
   d. Conductor identification.
   e. Megohms between conductor and ground.
   f. Total length of conductor.

D. Fiber Optic Cable:
1. Test each fiber using an Optical Time Domain Reflectometer (OTDR):
   a. Purpose and intent of testing:
      1) Determine if installed cable is free from defects.
      2) Compare measured attenuation with cable manufacturer's specified attenuation of 0.4 decibels per kilometer for 1,300 nanometer light.
      3) Compare measured bi-directional splice loss with specified splice loss of 0.1 decibels per splice.
      4) Compare measured connector loss with specified connector loss of 0.5 decibels.
      5) Verify cable length.
   b. OTDR requirements:
      1) Fiber type: Single-Mode.
      2) Dynamic Range: 25 decibels, minimum.
      3) Resolution: Selectable 0.1- to 10-meters.
      4) Pulse width: Selectable. 1 for all tests.
      5) Reflectance accuracy: Plus or minus 2 decibels.
      6) Loss accuracy (Linearity): 0.02 decibel per decibel.
   c. Obtain an OTDR waveform for each fiber to:
      1) Determine actual cable length.
      2) Determine attenuation and splice loss.

3) Identify and locate splices and step discontinuities/possible fiber breaks.

d. Prepare test reports:

1) Generate PDF copy and paper copy of each trace made during testing for inclusion in report:

2) OTDR waveform for each fiber showing “A” and “B” range marks:

a) Do not move or change location of either mark.

3) Include Analysis Summary Results Table for each fiber:

a) Indicate numerical values associated with each event.

b) Do not use question marks, dashes, or blanks.

4) Include the following information for each trace:

a) Trace name.

b) Operator’s name.

c) Date and time.

d) Fiber type being tested.

e) Bundle color.

f) Fiber color.

g) Fiber number.

h) Launch reel length.

i) OTDR resolution.

j) Pulse width.

k) Wavelength.

l) Index of refraction.

m) Averaging number.

n) Receiver bandwidth.

o) Sampling point count.

p) End to end loss, decibels.

q) Reflectance, decibels.

r) Fiber Break, decibels.

s) Backscatter.

5) Traces to be viewable and reproducible on PC using Windows 7 or later.

E. CAT 6 Cable: Perform the following tests:

1. Test equipment shall conform to TIA-1152.
2. Inspect network cabling terminations for T568A color configuration.
3. Perform testing for each outlet.
4. Performance test:
    a. Perform Category 6 link tests in accordance with TIA-568-C.1 and TIA-568-C.2.
    b. Include the following in Tests shall include wire map, length, insertion loss, NEXT, PSNEXT, ELFEXT, PSELFEXT, return loss, propagation delay, and delay skew.

F. If conductor fails continuity or insulation resistance testing, remove, and replace cable and conductor.

END OF SECTION

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SECTION 26 05 20
GROUNDING AND BONDING

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in price offered in Price Schedules for Complete Electrical System for Outdoor Equipment Only.

1.02 REFERENCE STANDARDS

A. ASTM International (ASTM)

1.	ASTM B3-13(2018)	Soft or Annealed Copper Wire
2.	ASTM B8-11(2017)	Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft
3.	ASTM B228-11a(2016)	Concentric-Lay-Stranded Copper-Clad Steel Conductors

B. Institute of Electrical and Electronic Engineers (IEEE)

1.	IEEE 81-2017	Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System
2.	IEEE 142-2010	Grounding of Industrial and Commercial Power Systems
3.	IEEE 837-2014	Qualifying Permanent Connections Used in Substation Grounding
4.	IEEE C2-2017	National Electrical Safety Code (NESC)

C. National Fire Protection Association (NFPA)

1.	NFPA 70-2023	National Electrical Code (NEC)

D. Underwriters Laboratories (UL)

1.	UL 467-2013	Grounding and Bonding Equipment

1.03 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals and Section 01 33 26 – Electrical Drawings and Data.

B. RSN 26 05 20-1, Approval Drawings and Data:

1. Manufacturer’s catalog data.

C. RSN 26 05 20-2, As-built Drawings:

1. Specification drawings indicating as-built changes made to grounding systems.

D. RSN 26 05 20-3, Test Reports.

PART 2 PRODUCTS

2.01 MATERIALS

A. Ground Cable:

1. Annealed bare-copper cable:
   a. ASTM B8, Class B.
   b. Solid wires used in forming copper cable: ASTM B3.
2. Copper-clad steel cable: ASTM B228.
3. Concentric stranded.
4. Sizes:
   a. As indicated on drawings.
   b. Minimum sizes:
      1) Ground risers and electrode conductors: 4/0 AWG.
      2) Equipment grounding conductors:
         a) Major equipment: Equal in size to electrode conductors.
         b) Minor equipment: 4 AWG.
      3) Electrode interconnecting conductors: Equal in size to electrode conductors.

B. Ground Rods:

1. Type: Solid, copper-clad steel:
   a. Bond 0.010-inch thick layer of copper inseparably to steel core.
2. Length: 10-feet.
3. Diameter: 3/4-inch, nominal.
4. In accordance with UL 467.

2.02 ACCESSORIES

A. Cable Connectors:

1. Welded or bolted solderless type.
2. Current-carrying capacity: Equal to cable with which they are used.

3. Connectors for ground cables, including fittings, lugs, bolts, nuts, and washers: Copper alloy containing not more than 4 percent zinc.
4. Bolted solderless type connectors: IEEE 837.
5. Ground connectors that will be direct buried or embedded in concrete: Listed for direct burial use in accordance with requirements of UL 467.

B. Ground Plates:

1. High-strength, high-conductivity, cast-copper alloy.
2. Four-hole type, except where indicated on drawings.
3. Suitable for concrete construction.

C. Flexible Copper Braid:

1. Flat, extra flexible, tinned, pure copper braid.
2. Un-plated, seamless, pure copper ferrules formed into a rectangular shape on each end.
3. Outdoor ampacity: 400 amperes, minimum.
4. Overall length: Suitable for application.

2.03 WELDING PROCESS

A. Use Cadweld, Thermoweld, or equivalent exothermic process.

B. Molds and Weld Metal:

1. Use fresh stock from same manufacturer.
2. Weld metal and starting material: No significant quantities of hazardous ingredients.

PART 3 EXECUTION

3.01 INSTALLATION

A. Install in accordance with NFPA 70, IEEE C2, and this Section.

B. Make grounding connections in accordance with standard drawings 40-D-4334, 40-D-4335, and 40-D-4753.

C. Do not connect equipment grounding conductors directly to steel reinforcing bars.

D. Ground or Bond as applicable:

1. Metal enclosures for electrical equipment.
2. Shielding and armor on insulated conductors.

3. Steel structural columns.

4. Building fence.

5. Metal structures located within 200-feet of power or distribution line centerlines.

E. Grounding Electrodes at Turnout Buildings:

1. Ground ring:

a. Attach to ground rods to serve as grounding electrode for Turnout Buildings.

b. Encircle each building completely.

c. Bury in direct contact with earth.

d. Depth: Not less than 2 1/2-feet.

e. Conductor: At least 20-feet of 4/0 AWG stranded bare copper.

F. Service Yard Fence:

1. Ground service yard fence, gate posts, and gate frames.

2. Ground fence fabric on a location where fabric is not continuous.

G. Ground Electrode Interconnections:

1. Install minimum of two physically separate grounding electrode tie conductors as interconnections between ground electrodes of the Turnout Buildings ground system and the site grounding electrode system.

2. Installation depth: At least 2 1/2-feet.

3. Separation: At least 10-feet.

H. Ground Rods:

1. Drive ground rods vertically until tops of rods are within 2-inches above installation location of ground cables to which rods will be connected.

2. If solid rock is encountered within 3-feet of subgrade, grout ground rod into 1 1/4-inch vertical hole such that top of rod is within 2-inches above installation location of ground cables to which rod will be connected.

3. If solid rock is encountered deeper than 3-feet below subgrade, drive ground rod to refusal at approximately 45-degree angle, then bend ground rod horizontally such that top of rod is within 2-inches above installation location of ground cables to which rod will be connected.

I. Connections:

1. Make all ground connections between equipment, miscellaneous metalwork, and ground plates whether or not such grounding connections are shown.

2. Number of grounding connections for equipment may be one, two, or more in some cases.

3. Remove paint, enamel, scale, oil, grease, or other foreign nonconductive material from point of contact on metal surfaces before making ground connections.

4. Repair damaged or removed paint or galvanizing material on metal finishes.

5. Make ground connections that are direct buried or embedded in concrete using an exothermic process and in accordance with manufacturer's instructions:

   a. In accordance with manufacturer's instructions.

   b. Inspect in accordance with pocket handbook A7D Installers and Inspectors Guide for CADWELD® Electrical Connections written by ERICO® Products, Inc., Cleveland, Ohio.

   c. Remove and replace rejected connections.

J. Excavate, place, and compact backfill in accordance with Sections 31 23 10 – Earthwork for Site and Structures and 31 23 02 – Compacting Earth Materials, respectively.

3.02 CONTRACTOR FIELD QUALITY TESTING

A. Contractor shall perform all electrical testing in accordance with NETA ATS acceptance testing requirements.

B. Test Resistance of Grounding System:

1. Notify COR in writing at least 3 working days in advance of time of test in order that test may be witnessed by COR.

2. Perform test no sooner than 30-days after completion of structural concrete work.

3. Measured resistance: Less than 5 ohm.

4. Use fall of potential method in accordance with section 8.2.1.5 of IEEE 81.

5. Testing device:

   a. Earth resistance tester:

      1) Heavy-duty, low-resistance type with direct-reading and direct-current ohmmeter as described in bulletin No. 25 and any one of bulletins Nos. 25J, 25J-2, and 25T, all by James G. Biddle Company; 3 point electronic Vibroground® by Associated Research, Incorporated; or equal.

      2) Capable of measuring ground resistance to an accuracy of 0.5 ohms or better.

6. Test performance:

   a. Initial distance between remote current electrode (C) and grounding system under test (X): At least six times longest diagonal dimension of grounding system.

b. Distance between electrode C and electrode X if grounding electrode system consists of one or two ground rods: At least 100-feet or as directed by COR.

c. Beginning at electrode X, drive potential-reference electrode (P) into ground at a number of test points on a straight line between electrode X and electrode C.

d. Distance between test points: Approximately 5 percent of actual distance between electrodes X and C.

e. Continuously log apparent resistance readings for each test point and plot curve of resistance versus distance to ensure that electrode C location is giving acceptable results as indicated in IEEE 81.

f. Measure and record test points until reaching electrode C or as directed by COR. If plot of resistance -vs- distance curve indicates remote current electrode is in area of influence of grounding system, increase distance between electrode C and grounding system, and measure and record new set of test data.

g. Allow 40 calendar days after receipt of grounding system resistance test reports for Government to determine location and size of any additional grounding is required.

h. Retest grounding system resistance in accordance with these paragraphs if additional grounding is added.

END OF SECTION

SECTION 26 05 33
RACEWAYS AND BOXES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Electrical Conduit:

1. Measurement: Length of installed conduit.
2. Payment: Linear foot prices offered in Price Schedules for applicable conduit:
   a. Includes cost of work and materials for conduit.

B. Fabricated Sheet Steel Boxes, Wiring Troughs, and Precast Handholes:

1. Payment: Lump sum prices offered in Price Schedules:
   a. Includes cost of drilling and punching holes.

1.02 REFERENCE STANDARDS

A. American Association of State Highway and Transportation Officials (AASHTO)

1. AASHTO HB-17-02 Highway Bridges, 17th Edition

B. American National Standards Institute (ANSI)

1.	ANSI C80.1-2020	Electric Rigid Steel Conduit (ERSC)
2.	ANSI/SCTE 77-2017	Specification for Underground Enclosure Integrity

C. ASTM International (ASTM)

1.	ASTM C857-19	Minimum Structural Design Loading for Underground Precast Concrete Utility Structures.

D. Institute of Electrical and Electronics Engineers (IEEE)

1.	IEEE C2-2017	National Electrical Safety Code (NESC)

E. National Electrical Manufacturers Association (NEMA)

1.	NEMA 250-2014	Enclosures for Electrical Equipment (1,000 Volts Maximum)

2.	NEMA FB 1-2014	Fittings, Cast Metal Boxes, and Conduit Bodies for Conduit, Electrical Metallic Tubing, and Cable
3.	NEMA RN-1-2018	Polyvinyl Chloride (PVC) Externally Coated Galvanized Rigid Steel Conduit and Intermediate Metal Conduit
4.	NEMA TC 2-2020	Electrical Polyvinyl Chloride (PVC) Tubing and Conduit
5.	NEMA TC 3-2016	Polyvinyl Chloride (PVC) Fittings for Use with Rigid PVC Conduit and Tubing

F. National Fire Protection Association, Inc. (NFPA)

1.	NFPA 70-2023	National Electrical Code (NEC)

G. Underwriters Laboratories, Inc. (UL)

1.	UL 360-13	Liquid-tight Flexible Steel Conduit
2.	UL 6-2007	Electrical Rigid Metal Conduit – Steel
3.	UL 651-11	Schedule 40 and 80 Rigid PVC Conduit

1.03 DESIGN REQUIREMENTS

A. Determine/verify size of junction and pull boxes, fabricated sheet steel boxes, and precast handholes in accordance with NPFA 70.

B. Determine/verify size of wiring trough in accordance with NFPA 70.

C. Provide underground communication pull boxes with a minimum horizontal clearance from Turnout Building footprint of at least 2-feet from the nearest edge of the pull box.

1.04 SUBMITTALS

A. Submit the following in accordance with Sections 01 33 00 – Submittals and 01 33 26 – Electrical Drawings and Data.

B. RSN 26 05 33-1, Approval Drawings and Data:

1. Manufacturer’s drawings and catalog data.

C. RSN 26 05 33-2, As-built Drawings:

1. Marked prints of specifications drawings indicating “As-built” changes made to electrical raceways during construction.

PART 2 PRODUCTS

2.01 RACEWAYS

A. Rigid Metal Conduit (RMC):

1. ANSI C80.1.

B. Rigid Polyvinyl Chloride Conduit (PVC):

1. Schedule 80 PVC conduit. Conduit and elbows furnished with integral bell end.
2. Length: Manufacturer's standard 10- or 20-foot lengths.
3. Temperature Rating: Equal to or greater than temperature rating of enclosed conductors.
4. Conduit: Conform to NEMA TC 2, NFPA 70, and UL 651. Fittings and conduit bodies conform to NEMA TC 3, NFPA 70, and be UL listed.

C. Liquidtight Flexible Metal Conduit (LFMC):

1. Interlocked steel construction with PVC jacket.
2. Ultra-violet resistant.
3. Color: Black.
4. Conform to NFPA 70 and UL 360.
5. Fittings conform to NEMA FB 1.

D. Wiring Trough:

1. Material: 16-gauge or 14-gauge galvanized steel.
2. Enclosure:
    a. NEMA Type 3R in accordance with NEMA 250.
    b. Drip shield top and no knockouts.
3. Slip-on removable cover fastened with captive screws along bottom edge and provided with provisions for padlocking.
4. Embossed mounting holes on back of enclosure.
5. Finish: Manufacturer's standard grey coating inside and outside of enclosure.

2.02 OUTLET BODIES, DEVICE BOXES, JUNCTION AND PULL BOXES

A. Outlet Bodies and Fittings:

1. Material: Cast iron or malleable iron.
2. Covers: Cast or sheet metal.

B. Cast Devices Boxes: Type FD and GRF manufactured by Cooper Crouse-Hinds; or equal, having the following essential characteristics:

1. Rectangular or round, deep-wiring device outlet boxes with threaded hub connections for rigid steel conduit.
2. Material: Iron alloy.
3. Number of gangs: As required.
4. Provide plugs for unused openings.
5. Extension rings: Iron alloy with corrosion-resistant finish.

C. Boxes for Dry Wall Construction:

1. Standard galvanized sheet steel switch and outlet boxes.
2. Switch boxes:
   a. Depth: 2 1/8-inches, minimum.
   b. Number of gangs: As required.
   c. Furnish with raised device covers (plaster rings).

D. Fabricated Sheet Steel Boxes:

1. Used as junction and pull boxes.
2. Sheet steel thickness: No. 14 United States Standard gauge, minimum.
3. Boxes installed in pumping plant building to be NEMA Type 12. Boxes installed outdoors to be NEMA type 4X.
4. Hinged cover with oil-resistant gasket.
5. Finish: Manufacturer's standard grey coating.
6. Conform to NEMA 250.

2.03 PRECAST HANDHOLE

A. Type: Precast polymer concrete with open bottom and Tier 15 rating.

B. Cover:

1. Bolt down type with gasket and stainless-steel bolts.
2. Tier level rating embossed on the surface.
3. Electrical pull box: Embossed logo "ELECTRIC" on the surface.

C. Provide embedded pulling eyes with ratings suitable for installation. Locate pulling eyes on wall opposite each conduit entrance.

D. Conform to test provisions of ANSI/SCTE 77.

2.04 UNDERGROUND COMMUNICATIONS PULL BOX

A. Type: Precast concrete.

B. Loading: Design, design loads, concrete, and reinforcing steel materials, concrete mixes, and construction of components shall meet the requirements of ASTM C858. The components and the complete installation shall be designed for H-20 loading rating in accordance with AASHTO HB-17 and ASTM C857.

C. Concrete Strength: Minimum 5,000 psi compressive strength at 28 days.

D. Casting:

1. In accordance with ASTM A48, class 35.
2. Uniform quality, free of defects.

E. Cover:

1. Solid cover, bolt down type with gasket.
2. Emboss or engrave logo “COMMUNICATIONS” on the surface.
3. Load rating meets handhole loading requirements.
4. Rated for heavy traffic applications.

F. Knockouts: Provide knockouts for a minimum of two conduits on opposite ends of the handhole.

G. Provide pre-cast concrete 6-inch or 12-inch risers between top of handhole and finished grade.

H. Dimensions (inches): Contractor to size pull box.

I. Provide the following accessory components or features in each electrical handhole:

1. Drainage outlet located on the bottom of the handhole.
2. Pulling irons, irons shall be galvanized.
3. Stainless Steel hardware: nuts, bolts, washers, etc.

2.05 CONDUIT FITTINGS

A. Expansion-deflection Couplings:

1. Furnish for movement in any direction.
2. Watertight, raintight, concrete tight.
3. Deflection and expansion: 3/4-inch, minimum.

4. Angular misalignment: 30-degrees, minimum.

B. Watertight Conduit Hubs:

1. Myers hub manufactured by Cooper Industries, Crouse-Hinds Division, Syracuse, NY, catalog series ST-, STA- and SSTG-; catalog series HUB- and HUBG- manufactured by Appleton, Rosemont, IL; catalog series CHM- manufactured by O-Z Gedney, Rosemont, IL; or equal with the following essential characteristics:

a. Suitable for terminating rigid steel electrical conduit through the walls of enclosures.

b. Recessed O-ring for raintight connection.

c. In accordance with NEMA FB-1.

d. Galvanized malleable iron.

e. Insulated throat.

2.06 ACCESSORIES

A. Fittings required to complete electrical conduit systems. Includes, caps, connectors, couplings, unions, nipples, reducers, elbows, pipe plugs, bondnuts, and any other fittings.

B. Devices required to fasten, clamp, attach, and support conduit in place. Includes supports and clamps complete with bolts, washers, and nuts.

C. Locknuts and Bushings:

1. Locknuts: Steel. Die cast locknuts are not acceptable.

2. Bushings: Insulated and pressure cast or malleable iron. Grounding bushing to be lay-in saddle type.

D. Conduit Riser Grips:

1. Suitable to support vertical cable runs in rigid steel conduit.

2. Single weave closed or split mesh type.

3. Mesh material: Tin-coated bronze strands.

E. Conduit Tags: Round brass or stainless-steel.

F. Polyethylene Warning Tape:

1. Type: Heavy-duty polyethylene, detectable underground warning tape.

2. Width: 6-inches.

3. Color: Red.

4. Copy: Continuously imprinted with "CAUTION BURIED ELECTRIC LINE BELOW" or equivalent wording.

5. Suitable for direct burial.

G. Thread Lubricant: Lithium, graphite, or zinc-based material designed to inhibit corrosion and to lubricate metal-to-metal joints.

H. Electrical duct seal for sealing ends of raceways.

I. Protective Sealant: Water repellant, and resistant to peeling and cracking.

J. PVC Solvent Cement: In accordance with NEMA TC 2 and conduit manufacturer's recommendations.

K. Corrosion Protection Tape:

1. ScotchrapTM 51 All-Weather Corrosion Protection Tape manufactured by 3M, St. Paul, MN, and Plymouth Plywrap 11 Vinyl Pipe Wrap manufactured by Plymouth Ruber Company, Canton, MA; or equal, having the following essential characteristics:

a. Thickness: 20 mils minimum.

b. Material: Polyvinyl chloride (PVC).

PART 3 EXECUTION

3.01 INSTALLATION

A. Conduit Type: As listed below unless shown otherwise:

1. PVC Conduit:

a. For direct-buried conduit inside and outside the fenced Turnout yard.

b. For conduit installed beneath concrete slab floors.

2. Rigid Metal Conduit: Exposed conduit, unless shown otherwise. Vertical risers through concrete slab floor.

3. Liquidtight Flexible Metal Conduit: used for the last 6 feet to terminations on vibrating or moving equipment.

B. Install as shown and in accordance with NFPA 70 and IEEE C2 unless directed otherwise.

1. Allow a minimum of 3' of vertical separation between electrical conduit/fiber optic conduit and water pipe or as otherwise determined by the COR.

C. Determine routing of exposed, buried, and embedded conduit when not shown.

D. Determine exact location of embedded conduit stub-ups based on equipment being furnished.

E. Bends:

1. Make smooth, gradual bends to permit pulling insulated conductors without undue stress or damage to conductors or conduit.
2. Ensure bends are free from kinks, indentations, or flattened surfaces.
3. Make metal conduit bends onsite with radii in accordance with NFPA 70.
4. Bend conduit cold to prevent damage to protective coating.

F. Remove burrs and sharp corners at ends of metal conduit.

G. Coat male threads of rigid metal conduit joints with suitable graphite or zinc sealing material before making joints.

H. Tighten conduit joints securely to ensure electrical continuity and to prevent entrance of moisture or foreign material.

I. Install with necessary fittings and supports.

J. Transition to RMC prior to PVC conduit vertical transition. Conduit elbow, vertical segments, and above grade conduit shall be RMC.

K. Make transitions from PVC conduit to RMC with suitable fittings designed for application.

L. Tighten conduit to electrical equipment mounted on tubular metal structures securely and support rigidly in place by conduit clamps, hex head nuts, and threaded 3/8-inch stud fasteners driven by powder-actuated tools.

M. Install expansion couplings where exposed or embedded conduits cross expansion or contraction joints. Install expansion couplings and expansion-deflection couplings in accordance with manufacturer's instructions.

N. Exposed Conduit Runs:

1. Straight and parallel with each other and with centerlines of room or structure.
2. Support conduit rigidly in place and in accordance with NFPA 70. Do not weld conduit to conduit supports or structural steel.

O. Conduit Beneath Concrete Slab Floors:

1. Transition horizontal runs of PVC conduit to rigid metal conduit before making 90-degree bends using appropriate conduit fitting.
2. Make 90-degree bend with rigid metal conduit and extend vertically through concrete slab floor. Ensure conduit risers are vertically plumb.
3. Wrap rigid metal conduit with anti-corrosion tape from transition coupling to at least 6-inches above concrete slab floor.

4. Apply manufacturer's recommended pipe primer to rigid metal conduit before wrapping with corrosion tape.
5. Apply anti-corrosion tape in accordance with manufacturer's instructions and recommendations.
6. Make adequate provisions to protect anti-corrosion tape wrap from physical damage during backfill operations.

P. Conduit Embedded in Concrete:

1. Conduit to be rigid steel (RMC) type.
2. Conduit not to displace more than 3 percent of gross area of cross-section of slab (Calculate cross sectional area as 12-inches times slab thickness).
3. Conduit outside diameter to be no larger than 1/3 of overall thickness of slab in which they are embedded.
4. Conduit to be spaced at least 3 diameters or widths on center.
5. Conduit to occupy middle third of cross-sectional depth of slab in which they are embedded, where possible.
6. Protect ends of conduit with coupling and plug to prevent entrance of concrete, sand, or other foreign material.
7. Tie conduit from reinforcing bars or otherwise support to prevent sagging while concrete is being placed.
8. Conduit stub-ups:
    a. Minimum length above finished floor: 4-inches, not including coupling.
    b. Terminate conduit with coupling and plug until ready to connect remaining conduit sections. Replace plug with bushing or Chase-type nipple before installing insulated conductors.
    c. Approximately 1-foot of horizontal embedded conduit and entire vertical stub-up to be plastic-coated rigid steel conduit.
9. Seal conduit boxes with rubber gasketed blank cover. Clean concrete from inside of conduit boxes immediately after forms are removed.
10. Swab conduit within 24-hours after removal of forms with clean dry rags until conduit is thoroughly cleaned and dried.
11. Terminate ends of conduit that do not terminate at boxes with couplings and pipe plugs or insulating bushings and caps.

Q. Direct-Buried Conduit:

1. Schedule 80 unless shown otherwise.
2. Depth: 24-inches, minimum unless shown otherwise.
3. Install 2-inches of sand around conduit.

4. Install detectable warning tape 18-inches above buried conduit.
5. Backfill and compact trench in accordance with Section 31 23 22 – Pipe Trench Earthwork.

R. Liquidtight Flexible Metal Conduit:

1. Use flexible conduit for connections to equipment for no more than the last 6 feet that is subject to vibration, where flexibility is required, or where shown.
2. Maximum length of a flexible conduit section is not to exceed 24-inches.

S. Hand Holes and Underground Communication Pull Boxes:

1. Install in locations shown and in accordance with manufacturer's instructions and recommendations, and as described in these Specifications
2. Place a minimum of 6-inches of pea gravel or crushed rock in bottom of excavation before installing hand hole.
3. Install such that top of hand hole is at finished grade level.
4. Backfill and compact around hand hole in accordance with Section 31 23 10 – Earthwork for Site and Structures and as directed by COR.

T. Conduit Terminations:

1. Make conduit terminations to indoor boxes, cabinets, and panel boards in accordance with standard drawing 104-D-254.
2. Make conduit terminations to outdoor boxes with watertight conduit hubs. Use of locknuts, bushings, or bushed nipples at outdoor conduit terminations is not acceptable.
3. Install bushings or Chase-type nipples on ends of conduit to protect insulation of insulated conductors from abrasion.
4. Install locknuts and bond nuts to provide tight ground connections between conduit and boxes, panel boards, and cabinets.
5. Seal ends of conduit terminations with electrical duct seal to prevent air circulation and entrance of vermin through conduits into boxes, panel boards, or cabinets. Install duct seal to a minimum depth of 1-inch.

U. Conduit Tags:

1. Provide metal conduit tags for conduit terminations.
2. Stamp or engrave tag with conduit number as shown on "As-Built" cable and conduit schedule.
3. Permanently attach tag to conduit with wire and metal crimp-type connector.

END OF SECTION

SECTION 26 05 90
WIRING CHECKOUT AND TESTS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in the Price Schedule for Complete Electrical System for Outdoor Equipment Only.

1.02 DEFINITIONS

A. Qualified Person: In accordance with Article 100 of NFPA 70.

B. Cost:

1. Include in price offered in Price Schedule A for Complete Electrical System.

1.03 REFERENCE STANDARDS

A. National Fire Protection Association (NFPA)

1. NFPA 70-23 National Electrical Code (NEC)

1.04 QUALIFICATIONS

A. Testing Personnel:

1. Competent and qualified persons to perform wiring checkout, operational and functional testing and make revisions to drawings.
2. Have thorough knowledge of the electrical equipment, installation, and drawings.
3. Qualified to operate equipment being tested.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 GENERAL

1. NFPA 70-2023 National Electrical Code (NEC)

3.02 DEFINITIONS

A. Qualified Person: In accordance with Article 100 of NFPA 70.

3.03 PERFORMANCE REQUIREMNENTS

A. Tests:

1. Demonstrate proper interaction between equipment through cabling interconnections and verify these interconnections are accurately documented.
2. In addition to tests required by Section 26 05 10 – Conductors and Cables.

B. Furnish materials, including test instruments, required for checkout and testing.

C. Make wiring changes and drawing revisions to ensure there are no discrepancies between check prints, and internal equipment wiring and external cabling.

D. Replace defective equipment and make device setting adjustments as required for operation of electrical systems in accordance with this sectionspecifications and manufacturer's instructions.

E. If required during testing, remove and replace wire connections.

F. Make wiring changes, wire tag revisions, and other repairs or revisions during the testing session the discrepancy is discovered.

G. If required during testing, remove, and replace wire connections.

H. Make wiring changes, wire tag revisions, and other repairs or revisions during testing session discrepancy is discovered.

I. Perform wiring checkout, and operational and functional tests in presence of COR.

3.04 CONTRACTOR FIELD QUALITY TESTING

3.05 QUALIFICATIONS

A. Testing Personnel:

1. Competent and qualified person to perform wiring checkout, and operational and functional testing.
2. Have thorough knowledge of electrical equipment, installation, and drawings.
3. Qualified to operate equipment being tested.

PART 4 PRODUCTS

Not Used

PART 5 EXECUTION

5.01 FIELD QUALITY CONTROL

A. Notify COR in writing, 10 working 10-days in advance of each testing session.

B. Provide the following at time wiring checkouts and tests are performed:

1. Two complete sets of full-size (ANSI D-size) electrical schematic and wiring diagram check prints for equipment being tested:

a. One set to be used and maintained by COR.

b. One set to be maintained by Contractor and used to produce final drawings.

C. Perform wiring checkout and tests in accordance with the following.:

1. Wiring checkout:

a. Use electrical schematic and wiring diagram check prints to perform internal wiring inspection of factory wired equipment. A detailed wire-by-wire verification of pre-wired assemblies is not required to be performed in the field unless errors are noted during the equipment inspection.

b. Use electrical schematic and wiring diagram check prints to perform internal wiring check of equipment.

c. Use electrical schematic and wiring diagram check prints to perform wiring check of external cabling.

d. Perform a complete wire-by-wire, terminal-by-terminal (point-to-point) check of interconnected circuits.

e. Perform a complete wire-by-wire, contact-by-contact, terminal-by-terminal (point-to-point) check of circuit logic.

f. Verify agreement with approved wiring diagrams. Revise to reflect circuit logic corrections made during checkout.

g. Verify agreement with approved schematic diagrams. Revise to reflect circuit logic corrections made during checkout.

h. Verify contact configuration and contact numbering.

i. Verify continuity of wiring.

j. Verify proper wire tagging at ends of interconnecting conductors and cables.

k. Verify proper wire tagging at ends of conductors and cables.

l. Check conductors for insulation damage. Perform additional insulation resistance test of damaged conductors.

m. Field verify custom cabinet wiring.

2. Operational and functional testing:
   a. Perform after completion of wiring checkout of equipment being tested.
   b. If required, adjust, and calibrate protective devices in accordance with specifications, and manufacturer's instructions and recommendations.
   c. Check electrical systems and controls for proper sequence of operation, correct adjustments and settings, and agreement with schematic diagram.
3. See Section 01 80 15 Commissioning and Testing.

END OF SECTION

SECTION 26 27 10
METER SOCKET

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in price offered in Price Schedules for Complete Electrical System for Outdoor Equipment Only.

1.02 REFERENCE STANDARDS

A. American National Standard Institute (ANSI)

1.	ANSI C12.7-2014	Requirements for Watthour Meter Sockets

B. National Electrical Manufacturers' Association (NEMA)

1.	ANSI/NEMA EN 10250 2024	Enclosures for Electrical Equipment (1,000 V Maximum)

C. National Fire Protection Association (NFPA)

1.	NFPA 70-2023	National Electrical Code (NEC)

D. Underwriters Laboratories, Inc (UL)

1.	UL 489-2016	Safety Molded-Case Circuit Breakers, Molded-Case Switches, and Circuit-Breaker Enclosures

1.03 SUBMITTALS

A. In accordance with Sections 01 33 00 - Submittals and 01 33 26 - Electrical Drawings and Data.

B. RSN 26 27 10-1, Approval Drawings and Data:

1. Layout drawings.
2. Manufacturer's technical data.

1.04 DESIGN REQUIREMENTS

A. Meter-Main: Meet the requirements of local utility.

B. Main Circuit Breaker:

1. Mounted in meter-main enclosure.
2. Molded case.

3. Size: In accordance with NFPA 70.

PART 2 PRODUCTS

2.01 METER SOCKET

A. System: 1-phase, 3-wire.

B. Location: Contractor to coordinate with utility.

C. Voltage Rating: 250 V.

D. Nominal Voltage: 120/240 V.

E. Current Rating: 100A, minimum.

F. Enclosure: ANSI/NEMA EN 10250-2024, Type 4X.

G. Provided with instrument transformers, jaws, and sealing ring as required by local utility.

H. One self-contained socket and main circuit breaker.

I. Conform to the requirements of local utility, ANSI C12.7, and be UL (Underwriters Laboratories) listed.

2.02 MOLDED CASE CIRCUIT BREAKER

A. Type: Thermal-magnetic.

B. Voltage: 120/240 volts.

C. Amps/Poles: 100A, two-pole.

D. Automatic: Inverse-time trip characteristics on overload and instantaneous trip on short circuit with an interrupting rating of 10kA symmetrical RMS amperes, minimum.

E. Arc quenchers.

F. Quick-make and quick-break toggle mechanisms.

G. Operating Handle:

1. Trip-free.
2. Single operating handle. Handle ties on multiple-pole circuit breakers are not acceptable.
3. Meet OSHA Lock-out/Tag Out requirements. Padlockable in “OFF” position.

H. Conform to UL 489 and be UL listed.

2.03 SERVICE ENTRANCE

A. Underground.

B. Voltage Rating: 250 V.

C. Ratings: 100 Amperes.

2.04 ELECTRICAL EQUIPMENT

A. Furnish all cables, conduits, terminations, and accessories required for interconnecting the equipment included in this section.

2.05 FINISHES

A. Manufacturer's standard finish appropriate for installed location.

PART 3 EXECUTION

3.01 INSTALLATION

A. Install in accordance with manufacturer's instructions and recommendations.

B. Items anticipated to be furnished and installed by local utility:

1. Distribution power transformer
2. Cables between power transformer and meter-main
3. Revenue Meter

C. Make cable and conduit connections on load side of meter socket.

D. Coordinate with local utility to ensure utility requirements and standards are met.

E. Ground and bond in accordance with Section 26 05 20 – Grounding and Bonding and NFPA 70.

END OF SECTION

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SECTION 26 28 20
SERVICE DISCONNECT SWITCH

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in price offered in Price Schedules for Complete Electrical System for Outdoor Equipment Only.

1.02 REFERENCE STANDARDS

A. InterNational Electrical Testing Association (NETA)

1.	NETA ATS	Acceptance Testing Specifications for Electrical Power Equipment and Systems

B. National Electrical Manufacturers Association (NEMA)

1.	ANSI/NEMA EN 10250 2024	Enclosures for Electrical Equipment (1,000 Volts Maximum)
2.	NEMA FU 1-2012	Low Voltage Cartridge Fuses
3.	NEMA KS 1-2013	Heavy Duty Enclosed and Dead-Front Switches (600 Volts Maximum)

C. National Fire Protection Association (NFPA)

1.	NFPA 70-2023	National Electrical Code (NEC)

D. Underwriters Laboratory (UL)

1.	UL 98-2016	Enclosed and Dead-Front Switches

1.03 SUBMITTALS

A. In accordance with Sections 01 33 00 - Submittals and 01 33 26 - Electrical Drawings and Data.

B. RSN 26 28 20-1, Approval Data:

1. Manufacturer’s technical data.

C. RSN 26 28 20-2, Field Test Reports

1. Certification of performance and results of test.

PART 2 PRODUCTS

2.01 FUSED SERVICE DISCONNECT SWITCH

A. Type: Heavy-duty industrial, single-throw, fused and rated for service entrance duty.

B. Number of Poles: 2.

C. Enclosure: ANSI/NEMA EN 10250-2024 Type 4X in accordance with NEMA 250.

D. Switch Voltage Rating: 250 volts, single-phase, 60 Hz, 3-wire.

E. Switch Current Rating: 100 amperes, minimum.

F. Switch Mechanism/Handle:

1. Switch Operating Mechanism: Made from steel, with no plastic parts.
2. Switch Operating handle:
    a. Integral part of the switch enclosure. Use of cover as disconnect device is not acceptable.
    b. Made from steel, with no plastic parts other than handle grip.
    c. Have provisions for padlocking in the OFF position.

G. Switch Cover Interlock:

1. Prevent unintentional opening of switch cover when switch is ON.
2. Prevent closing switch with switch cover open.

H. Current Carrying Parts: Plated to resist corrosion.

I. Fuses:

1. 240-volt, 60 Hz, current-limiting type suitable for service entrance equipment and feeder protection.
2. Size: Contractor to determine.

J. Spring reinforced fuse clips.

K. Disconnect switch conform to NEMA KS 1 and UL listed.

L. Fuses conform to NEMA FU 1 and UL listed.

M. Engraved Nameplates:

1. Conform to Standard Drawing 40-D-6234 using Type A material and 1/2-inch lettering.
2. Engraving:

a. Sanostee: “Main Service Disconnect – S1205-DSC-001”

b. Newcomb: “Main Service Disconnect – S1305-DSC-001”.

c. Naschitti: “Main Service Disconnect – S1405-DSC-001”.

d. BBN9: “Main Service Disconnect – S1505-DSC-001”.

e. Rock Springs: “Main Service Disconnect – S1605-DSC-001”.

f. Coyote Canyon: “Main Service Disconnect – S1705-DSC-001”.

g. Window Rock: “Main Service Disconnect – S1805-DSC-001”.

PART 3 EXECUTION

3.01 FIELD TESTS

A. Test per NETA ATS

3.02 INSTALLATION

A. Install in accordance with manufacturer’s instructions and recommendations including but not limited to NEMA 250 and UL98.

B. Service disconnect switch enclosure is to be strut mounted on concrete pad. See Section 03 30 00 - Cast-in-Place Concrete for more details on the concrete pad.

C. Contractor to determine final location.

D. Ground and bond in accordance with NFPA 70 and Section 26 05 20 – Grounding and Bonding.

E. Make conduit and cable connections.

END OF SECTION

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SECTION 26 42 00
CATHODIC PROTECTION SYSTEMS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Corrosion System(s) for Turnout Yard Pipe:

1. Payment: Lump sum prices offered in the applicable Price Schedule.

1.02 REFERENCE STANDARDS

A. ASTM International (ASTM)

1.	ASTM B3-13(2024)	Soft or Annealed Copper Wire
2.	ASTM B843-18(2025)	Magnesium Alloy Anodes for Cathodic Protection
3.	ASTM C33/C33 M-24a	Concrete Aggregate

B. Bureau of Reclamation (USBR)

1.	USBR M-47	Guide to Concrete Repair, Second Edition available at: https://www.usbr.gov/tsc/techreferences/mands/mands-pdfs/Guide2ConcreteRepair2015_Final.pdf

C. Association for Materials Protection and Performance (AMPP, formerly NACE)

1.	NACE/ASTM G193-2022	Standard Terminology and Acronyms Relating to Corrosion
2.	NACE SP0169-2024	Control of External Corrosion on Underground or Submerged Metallic Piping Systems
3.	NACE TM0497-2022	Measurement Techniques Related to Criteria for Cathodic Protection on Underground or Submerged Metallic Piping Systems

1.03 DEFINITIONS

A. Bonded Dielectric Coating:

1. Protective barrier coating system with high electrical resistivity bonded directly to underlying structure, acting to physically and electrically isolate metallic structure from electrolyte.

B. Cathodic Protection:

1. Galvanic Anode Cathodic Protection:

a. Reduction of corrosion of metal in electrolyte by galvanically coupling to more anodic metal:

1) Also referred to as Sacrificial Protection.

2. Impressed Current Cathodic Protection:

a. Reduction of corrosion of metal in electrolyte by supplying DC current through external power source and employing impressed current anode.

C. Electrolyte:

1. Electrically conductive solution, such as soil or water:

a. Terms for specific conductive solutions may be substituted for "electrolyte" in these definitions:

1) Conductive solutions include:

a) Water.

b) Soil.

D. Reference Electrode:

1. Electrode whose open circuit potential is constant under similar conditions of measurement; used for measuring relative potentials of other electrodes (e.g., protected structures):

a. Sometimes referred to as reference half-cell.

b. Copper/Copper Sulfate Reference Electrode (CSE) is often used.

E. Structure-to-Electrolyte Potential, (structure-to-soil and structure-to-water potentials):

1. Potential of structure in electrolyte as compared to potential of reference electrode contacting same electrolyte.

2. Static Structure-to-Electrolyte Potential:

a. Structure-to-electrolyte potential determined without any external current:

1) (e.g., prior to energizing cathodic protection system, and with no galvanic couple, pH cell, interference currents, or like present) or,

2) After such current source has been disconnected for an extended time.

3) Also referred to as native structure-to-electrolyte potential.

3. Uncorrected Structure-to-Electrolyte Potential:

a. Structure-to-electrolyte potential determined with cathodic protection system energized and cathodic protection current flowing:

1) Also referred to as Protective Potential (ON Potential) and may contain significant IR drop error.

4. Polarized Structure-to-Electrolyte Potential:

a. Structure-to-electrolyte potential determined after cathodic protection system has been energized, but immediately after cathodic protection current is interrupted:

1) Also referred to as “instant off” structure-to-electrolyte potential.

5. Polarization:

a. Change from static or native potential as result of current flow across electrode/electrolyte interface:

1) Also considered difference between polarized and native potentials.

1.04 QUALIFICATIONS

A. Cathodic Protection Technologist:

1. Certified by Association for Materials Protection and Performance (AMPP).

1.05 SYSTEM DESCRIPTION

A. Provide Corrosion Monitoring Systems for the following features:

1. Metallic turnout pipe, fittings, and associated metalwork.

B. In accordance with Section 26 42 11 – Buried Galvanic Anode Cathodic Protection

1.06 QUALITY

A. Installation and testing conducted by or under direction of Cathodic Protection Technologist. Other qualified individuals may assist installation and testing personnel.

B. Manufacturer’s Standards: Comply with manufacturer’s recommendations and standards unless otherwise specified.

C. Materials, Installation, and Testing:

1. Consistent with sound corrosion engineering principles and practices in accordance with NACE TM0497, and NACE SP0169.

1.07 SYSTEM PERFORMANCE

A. Provide corrosion-engineering services to furnish, install, and test corrosion monitoring and/or cathodic protection systems:

1. Include electrical continuity bonds, test stations, isolation joint flange kits, galvanic anodes, and accessory equipment and features as directed in this specification, specifications listed in Article 1.05, or as otherwise indicated.

B. Corrosion Monitoring Systems shall have capacity to:

1. Provide minimum service-life of 20-years for all components.

C. Cathodic Protection Systems:

1. Provide minimum service-life of 20-years for all components and system as determined by final data.

2. Maintain on all locations of protected metallic structures, with reference to copper/copper sulfate electrode placed in contact with electrolyte close to structure:

a. Polarized structure-to-electrolyte potential of minus 0.850 volts or more negative.

b. Alternately, at discretion of Government, minus 0.100 volts of polarization if the minus 0.850 volt criterion cannot be obtained due to extenuating circumstances (e.g., on bare or poorly coated structures).

c. Polarized potential at any point on dielectrically coated structure not more negative than minus 1.100 volts.

d. Modified under the presence or possible presence of microbial induced corrosion (MIC), such as with sulfate-reducing bacteria, to the following:

1) minus 0.950 volts or more negative on structure.

1.08 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 26 42 00-1, Certification and Data:

1. Copy of Cathodic Protection Technologists' AMPP certification for personnel performing or directing installation and testing of corrosion monitoring system(s).

2. Manufacturer’s Data:

a. Test station.

b. Metallurgical bonding system.

c. Dielectric coating system for bonds.

d. Cables.

e. Cable identification labeling system.

f. Isolation Joint Flange Kit.

PART 2 PRODUCTS

2.01 CABLES

A. Conform to ASTM B3.

B. Manufactured not more than 24 months prior to notice-to-proceed.

C. Single-Conductor: Stranded, plain, annealed copper cable.

D. Cable Insulation:

1. Rated for 600-volts and direct immersion or burial.
2. High Molecular Weight Polyethylene (HMWPE) outer jacket with minimum thickness of 0.100-inch:
    a. Black, unless otherwise noted.

E. Suitable lengths to permit installation from terminus to terminus (e.g., structure-to-test station or structure-to-structure) free of splices and without stress, unless otherwise specified.

Table 26 42 00A – Cable Gauge

Gauge (AWG), or larger:	Use for	Additional Notes
No. 12	Structure cable	Black Insulation
No. 12	Galvanic Anodes	Black Insulation
No. 12	Electrical Continuity Joint Bonds (less than 12-inch dia. pipe)	Black Insulation
No. 6	Electrical Continuity Joint Bonds (12-inch dia. pipe)	Black Insulation

2.02 BONDS

A. Metallurgical Bonds:

1. Exothermic Metallurgical Bonds:
    a. Exothermic metallurgical bond system manufactured by ThermOweld, 6100 S. Yale Ave., STE 1110, Tulsa, OK 74136; or equal, having the following essential characteristics:
        1) Designed for:
            a) Cathodic protection or monitoring systems.
            b) Metallic substrate material.
            c) Bond orientation.
        2) Exothermic reaction produces molten copper, which produces permanent, high conductivity connection.

3) Uses special alloy to provide minimum heat effect on substrate material.

4) Current carrying capacity equal or better than that of conductor.

2.03 DIELECTRIC COATING FOR BONDS

A. Dielectric Material: Ci thermOcap® PC, manufactured ThermOweld, 6100 S. Yale Ave., STE 1110, Tulsa, OK 74136; or equal, having the following essential characteristics:

1. Specifically designed for cathodic protection systems.
2. Applied with primer coat, as needed.
3. Approved dielectric coating material.
4. Suitable for intended environment.
5. Cover exposed weld nugget, exposed cable, or structure metal.

B. Refer to Section 09 96 20 – Coatings.

2.04 TEST STATION ENCLOSURE

A. Refer to Drawing 1695-D-62591 for details.

B. Test Stations: TESTOX Series 100 or Series 700, manufactured by Gerome Electric Supply Co., 336 E. Main St., Uniontown, PA 15401; or equal, having the following essential characteristics:

1. Pipe-mounted, above-ground type.
2. Cast aluminum enclosure.
3. Specifically constructed for cathodic protection system installations and intended environment.
4. Protected as required for permanency.
5. Secured to pipe with setscrew or threaded.
6. Enough terminals, including associated hardware for number and size of cables and required accessories: Minimum 5 terminals.
7. Equip with 0.01-ohm calibrated shunt for each anode such that current output of each anode can be determined (anode test stations only).
8. Equip with variable resistors for each anode (if necessary) such that the current output of each anode can be adjusted (anode test stations only).

C. Test station cables in accordance with cable requirements of this Section.

2.05 TEST STATION PIPE

A. Steel Pipe:

1. In accordance with Section 05 50 00 – Metal Fabrications:
    a. Pipe size indicated on Drawing 1695-D-62591.

B. Painting:
1. Test station support post galvanized in accordance with Section 09 96 20 – Coatings.

2.06 ISOLATION JOINT FLANGE KIT (IJFK)

A. Isolation kit manufactured by GPT Industries, 4990 Iris Street, Wheat Ridge, CO 80033; or equal, with the following essential characteristics:
1. Consist of an isolating gasket, bolt sleeve, and one or more isolating washer.
2. Sized for turnout pipe diameter in accordance with Drawings 1695-D-62561 thru 1695-D-62567.

2.07 SAND BACKFILL

A. Sand Backfill: ASTM C33, fine aggregate.

2.08 WARNING TAPE

A. Polyethylene warning tape for anode cables:
1. Minimum 3-inches wide.
2. Red or yellow with black lettering.
3. Suitable for direct burial.
4. Print full length of tape with the following:

"Caution –Cathodic Protection Cable Buried Below"

PART 3 EXECUTION

3.01 GENERAL INSTALLATION

A. Installation and testing shall be performed or directed by Cathodic Protection Technologist certified by AMPP.

B. Installation details indicated on Drawing(s) 1695-D-62590 and 1695-D-62591.

C. Electrical isolation:
1. All structures to be cathodically protected shall be electrically isolated from other extraneous metal (casings, foreign structures, and rebar in concrete).

D. Dielectric materials, other than non-metallic carrier pipe, that might cause shielding of cathodic protection current (plastic debris) shall not be left in excavation with structure.

E. Cable:

1. Inspect for insulation defects prior to backfilling:
    a. Replace cable if insulation is damaged.
2. Install without kinks, stresses, and/or splices.
3. Buried applications:
    a. Horizontal segments:
        1) In accordance with Drawing 1695-D-62590.
        2) For all horizontal segments of cable (long segments of jumper bonds or cables) the following requirements must be met in accordance with drawing(s) and Section 31 23 02 – Compacting Earth Materials:
            a) Minimum burial depth: 30-inches.
            b) Surround cable with minimum 6-inches of sand backfill.
            c) Place warning tape approximately 12-inches above cable for entire length of cable segments.
    b. Run cable through rigid galvanized conduit, test station pipe, or similar protection once cable emerges from ground.
4. Connect to test station with crimped, ring-tongue connectors.
5. Identify origin of cables terminating in an enclosure:
    a. Cable identification as to the distinct originating structure or anode in accordance with Drawing 1695-D-62590.
    b. Printed letters on shrinkable label attached to cables clearly visible within enclosure.
    c. Encase printed labels in clear heat shrink tubing.

F. Structure Cables:

1. Cable information for each test station type shown on Drawing 1695-D-62591.
    a. Two cables per structure at each test station.
2. Connect cables and jumper bonds to structures by exothermic metallurgical bond as shown on Drawing 1695-D-62590:
    a. Make exothermic metallurgical bonds in accordance with bonding supply manufacturer's instructions.
    b. Bonds shall not damage linings inside pipes or fittings.
    c. Test metallurgical bond integrity by striking side of weld nugget with 16-ounce hammer, in presence of OGR.
    d. Coat bare copper, weld nugget, and ferrous materials at metallurgical bonds with approved dielectric metallurgical bond coating.

e. Allow dielectric material to cure before repair of damaged structure coating or lining:

1) Repair dielectric coatings/linings in accordance with 09 96 20 – Coatings

2) Repair mortar/concrete coating and lining in accordance with USBR M-47.

G. Electrical Continuity Joint Bonds:

1. Provide metallurgical bonds at mechanical type joints (non-welded joints) between ferrous parts to be included in cathodic protection system as indicated in this section or as necessary to ensure electrical continuity.

2. Minimum number and size of bond cables per bond joint for long metallic pipe sections is shown in material table on Drawing 1695-D-62590.

3. Minimum of two cables per bond joint for fittings on non-metallic pipe.

4. Bond cable installed with sufficient slack to prevent stress or tugging.

a. Allow for minimum 1/2-inch of joint movement.

5. Joint Bond Locations:

a. Between non-welded ferrous pipe sections and ferrous pipe and fittings.

b. Between star grips and fittings for non-metallic pipe.

c. Between pipe installed under this contract and pipe installed by others: specifically:

1) Sanostee Turnout:

a) Between Reach 4C and turnout pipe.

2) Newcomb Turnout:

a) Between Reach 6 and turnout pipe.

3) Naschitti Turnout:

a) Between Reach 8 and turnout pipe.

4) BBN9 Turnout:

a) Between Reach 9A and turnout pipe.

5) Coyote Canyon Turnout:

a) Between Reach 11 and turnout pipe.

6) Rock Springs Turnout:

a) Between Reach 12.1 and turnout pipe.

7) Window Rock Turnout:

a) Between Reach 12.2 and turnout pipe.

H. Test Stations:

1. Locations:

a. Upstream and downstream side of turnout buildings.

1) Secured against outside wall of turnout building.

b. Locations requiring galvanic anodes.

c. Locate the test stations to make them readily accessible.

d. Where test stations will not likely be damaged or interfere with personnel and/or equipment (e.g., adjacent to above ground structures). Exact location is subject to approval of OGR.

e. As close to each current carrying structure connection as possible.

2. Details for each type of test station are shown in Drawing 1695-D-62591.

3. Permanently, uniquely, and clearly identify each test station with station number. Hand lettering inside of test station cap with a permanent marker is acceptable.

4. Terminate cables for given location within same test station enclosure.

5. Contain two cables, at minimum:

a. Specific cable information for each test station type is shown on Drawing 1695-D-62591.

6. Protect test stations using turnout building outside wall in accordance with Drawing 1695-D-62591:

a. Secure test station support pipe to outside wall of turnout buildings with metal straps.

I. Isolation Joint Flange Kits (IJFK):

1. Locations:

a. Specifically located:

1) In accordance with Drawings 1695-D-62559 and 1695-D-62560.

2) At first flange and last flange inside turnout building.

a) To isolate turnout building valves and flow meters from pipe both upstream and downstream of turnout building.

2. Install isolation joint flange kits in accordance with Drawing 1695-D-62590:

a. Test each bolt and flange to ensure isolation is satisfactory.

3.02 INSTALLATION OF GALVANIC ANODE CATHODIC PROTECTION SYSTEMS

A. In accordance with Section 26 42 11 – Buried Galvanic Anode Cathodic Protection.

3.03 FIELD TESTING

A. Include energizing, adjusting, and testing cathodic protection systems.

B. Perform testing in presence of OGR.

C. Inform OGR of date, time, and tests to be performed at minimum 5 working days prior to testing.

D. Testing equipment:

1. Portable Voltmeter:

a. Minimum input impedance of 10-megohms and capable of measuring DC voltages between plus or minus 0.1-millivolt to ±100-volts.

b. Portable $Cu/CuSO_4$ reference electrode.

E. Verify each test station is properly labeled.

F. Testing shall include the following:

1. Record and report all readings with final test data submittal.
2. Native pipe-to-soil potentials at all test stations.
3. Verify electrical continuity.
4. Verify isolation at locations where IJFK's were installed.

G. Conduct testing in 2 testing cycles, minimum:

1. Check isolation flanges to ensure adequate electrical isolation using appropriate equipment
2. First Testing Cycle:

a. Prior to energizing system:

1) Test structure electrical continuity.

2) Test station integrity:

a) Determine voltage difference between cables within each test station enclosure.

3) Measure static structure-to-soil potentials of the buried metalwork:

a) At each individual test station location.

b) Place portable reference electrode as close to buried metalwork as possible for structure-to-soil potential measurements.

c) Ground shall be saturated with water to reduce reference cell contact resistance.

4) Measure static anode-to-soil potentials of galvanic anodes:

a) At each individual test station location.

b) Ground shall be saturated with water to reduce reference cell contact resistance.

b. After energizing system:

1) Determine protective (uncorrected) and polarized structure-to-soil potentials in accordance with capacity requirements of this Section:

a) At each individual test station location.

b) Place portable reference electrode as close to buried metalwork as possible for structure-to-soil potential measurements.

c) Saturate ground with water to reduce reference cell contact resistance.

2) Determine current output of each individual anode in test stations.

3. Second and Subsequent Testing Cycles:

a. Time between testing cycles shall be 30- to 60-days.

b. Cathodic protection system shall not be adjusted between testing cycles.

c. Repeat testing required after energizing system during first testing cycle.

d. If testing cycle data indicates that cathodic protection system requires adjustment to meet specifications requirements, adjust cathodic protection system and conduct subsequent testing cycles within 30- to 60-days window.

H. Troubleshoot and correct any problems determined by OGR.

I. Submit final data after last testing cycle when performance criteria have been met and no further adjustments are needed.

J. After reviewing submittal, Government may require additional testing and re-submittal to determine if corrosion monitoring system conforms to specification requirements.

END OF SECTION

SECTION 26 42 11
BURIED GALVANIC ANODE CATHODIC PROTECTION

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Buried Galvanic Anode Cathodic Protection for Turnout Yard Pipe:

1. Include in lump sum prices offered in Price Schedules for Corrosion Systems.

1.02 REFERENCE STANDARDS

A. In accordance with Section 26 42 00 – Cathodic Protection Systems.

1.03 DEFINITIONS

A. In accordance with Section 26 42 00 – Cathodic Protection Systems.

1.04 QUALIFICATIONS

A. In accordance with Section 26 42 00 – Cathodic Protection Systems.

1.05 SYSTEM DESCRIPTION

A. Provide Galvanic Anode Cathodic Protection Systems for the following features:

1. Metallic turnout pipe, fittings, and associated metalwork.

1.06 QUALITY

A. In accordance with Section 26 42 00 – Cathodic Protection Systems.

1.07 SYSTEM PERFORMANCE

A. In accordance with Section 26 42 00 – Cathodic Protection Systems.

1.08 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 26 42 11-1, Certification and Data:

1. Copy of Cathodic Protection Technologists' AMPP certification for personnel performing or directing installation and testing of cathodic protection systems (if different from personnel submitted under 26 42 00).
2. Manufacturer’s data if applicable, including catalog data sheets when available, including but not limited to:

a. Galvanic anodes.

b. Test station(s) (included with Specification section 26 42 00).

c. Cables (included with Specification section 26 42 00).

d. Cable identification labeling system (included with Specification section 26 42 00).

e. Shunts

f. Variable Resistors.

g. Cable trench backfill material (included with Specification section 26 42 00).

3. Preliminary drawings, including:

a. Proposed location of test stations.

b. Proposed location of anode beds.

C. RSN 26 42 11-2, Final Data:

1. Information and data including:

a. Include data for all continuity, isolation, and cathodic protection required for the project.

b. Test equipment and methods utilized.

c. Placement of reference electrode during each test, if required.

d. Tests results.

e. Photos.

2. As-built and Final Drawings:

a. Provide as-built and final installation drawings in accordance with Section 01 78 30 – Project Record Documents:

1) Final anode locations and connection details.

2) Locations of anodes and/or anode beds.

3) Electrical continuity bonds and isolation details.

4) Locations of test stations and cable runs:

a) Locations shall be given using unique identifier and GPS coordinates.

b) Location shall include type of test station in accordance with Drawing 1695-D-62591.

5) List of anode cable identifications in test stations.

PART 2 PRODUCTS

2.01 CABLES

A. In accordance with Section 26 42 00 – Cathodic Protection Systems.

2.02 BONDS

A. In accordance with Section 26 42 00 – Cathodic Protection Systems.

2.03 DIELECTRIC COATING FOR BONDS

A. In accordance with Section 26 42 00 – Cathodic Protection Systems.

2.04 ELECTRICAL CONDUIT

A. In accordance with Section 26 42 00 – Cathodic Protection Systems.

2.05 GALVANIC ANODES

A. 20 lb or larger High Potential Magnesium Anode manufactured by Mesa Products, 4445 S. 74th E Avenue, Tulsa, OK 74145; or equal, with the following essential characteristics:

1. Twenty (20) pounds magnesium anode material per anode, minimum.
2. High potential magnesium anode material specifically designed for cathodic protection systems and intended environment.
3. Anode material meeting or exceeding requirements of ASTM B843 and having minimum potential of -1.700 volts referenced to copper/copper sulfate reference electrode (CSE).
4. Contains mild steel core extending essentially entire length of anode. Center mild steel core within anode material and exposed on one end of anode for factory made anode-to-cable connection.

B. Factory anode-to-cable connection, exposed mild steel core, and exposed copper cable potted in epoxy.

C. Silver-solder connection between anode cable and mild steel core.

D. Anode Cable: In accordance with cable requirements above.

2.06 SAND BACKFILL

A. In accordance with Section 26 42 00 – Cathodic Protection Systems.

2.07 TEST STATION ENCLOSURE

A. In accordance with Section 26 42 00 – Cathodic Protection Systems.

B. Refer to Drawing 1695-D-62591 for details.

2.08 TEST STATION PIPE AND BOLLARDS

A. In accordance with Section 26 42 00 – Cathodic Protection Systems.

2.09 WARNING TAPE

A. In accordance with Section 26 42 00 – Cathodic Protection Systems.

PART 3 EXECUTION

3.01 INSTALLATION OF GALVANIC ANODE CATHODIC PROTECTION SYSTEMS

A. Installation details indicated on Drawing 1695-D-62591.

B. Installation and testing shall be performed or directed by a Cathodic Protection Technologist certified by AMPP.

C. Cable:

1. Inspect for insulation defects prior to backfilling:
    a. Replace anode if cable insulation is damaged.
2. Install without kinks, stresses, and/or splices.
3. Buried applications:
    a. Horizontal segments:
        1) In accordance with Drawing 1695-D-62590.
        2) For all horizontal segments of cable (long segments of jumper bonds or cables) the following requirements must be met in accordance with drawing(s) and Section 31 23 02 – Compacting Earth Materials:
            a) Minimum burial depth: 30-inches.
            b) Surround cable with minimum 6-inches of sand backfill.
            c) Place warning tape approximately 12-inches above cable for entire length of cable segments.
    b. Run cable through rigid galvanized conduit, test station pipe, or similar protection once cable emerges from ground.
4. Connect to test station terminals with crimped, ring-tongue connectors.
5. Identify origin of cables terminating in an enclosure:
    a. Cable identification as to the distinct anode in accordance with Drawing 1695-D-62590.

b. Printed letters on shrinkable label attached to cables clearly visible within enclosure.

c. Encase printed labels in clear heat shrink tubing.

D. Galvanic Anodes:

1. Install horizontally or vertically as follows and as directed by OGR

a. Location:

1) Metallic pipeline.

2) Minimum 20-feet from nearest part of structures to be protected.

3) Minimum 10-feet from one another if multiple anodes are required on the structure as indicated by test data.

4) Minimum one anode length from non-metallic piping.

5) Bond all metalwork within 100-feet.

6) Install a minimum of one anode on metallic fittings and structures as shown on table on Drawing 1695-D-62591.

b. Depth:

1) At or below invert of pipe,

2) Minimum of 6-feet below final grade.

2. Remove outer water-resistant covering on pre-packaged anodes before installation, taking care not to damage wettable covering containing backfill and magnesium.

3. At no time shall anodes be supported by their cables.

4. Do not wet pre-packaged anodes until in the ground, surrounded with compacted earth, and covered by at least 1-foot of compacted backfill.

5. Do not connect anodes to structures at test stations, energizing systems only at time of testing.

6. Connect individual galvanic anodes to structures through individual calibrated 0.01-ohm shunts within test station for current measurement.

7. Where multiple anodes are involved, a calibrated shunt shall be included for each individual anode.

8. Connect individual anodes to structure through variable resistors within test station to allow for equalizing current from individual anodes (if necessary).

E. Test Stations:

1. Locate at metallic structures where galvanic anodes are required.

2. Cables for test stations as shown on Drawing 1695-D-62591.

F. Placement of Anodes:

1. Provide adequate distribution of cathodic protection current, as indicated by polarized potential readings.

G. Dielectric materials that might cause shielding of cathodic protection current (e.g., plastic debris) shall not be left in excavation with structure.

3.02 FIELD TESTING

A. In accordance with Section 26 42 00 – Cathodic Protection Systems.

END OF SECTION

SECTION 27 30 01
TELEPHONE SYSTEM

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Telephone System:

1. Payment: Lump sum prices offered in Price Schedules.

1.02 REFERENCE STANDARDS

A. Institute of Electrical and Electronic Engineers (IEEE)

1.	IEEE 802.3-2015	Standard for Ethernet

B. National Electrical Manufacturer's Association (NEMA)

1.	NEMA 250-2020	Enclosures for Electrical Equipment (1,000-Volts Maximum)

C. National Fire Protection Association (NFPA)

1.	NFPA 70-2023	National Electrical Code (NEC)

D. Telecommunications Industry Association (TIA)

1.	TIA 222-I-2024	Structural Standard for Antenna Supporting Structures and Antennas and Small Wind Turbine Support Structures

1.03 SUBMITTALS

A. Submit the following in accordance with Sections 01 33 00 – Submittals and 01 33 26 – Electrical Drawings and Data.

B. RSN 27 30 01-1, Communication System Design:

1. Communication Diagram:

a. Provide location of the closest cellular towers to each Turnout Building and identify:

1) Cellular Provider

2) Frequency Band

3) Signal Strength

4) Tower Locations and coverage (superimposed on topographical map/photo).

2. Communication Design:
    a. Provide equipment configuration recommendations.
        1) Interior antenna type and quantity.
        2) Exterior antenna type (directional and omnidirectional):
            a) Provide direction of exterior antenna if directional type.
        3) Identification of manufacturer and component model numbers for signal booster and interior and exterior antennas.

C. RSN 27 30 01-2, Approval Data:
1. Bill of Materials.
2. Manufacturer's product data.

D. RSN 27 30 01-3, Test Procedure and Test Schedule:
1. Detailed test procedure, identifying:
    a. Equipment and systems being tested
    b. Methods of testing equipment and systems:
        1) Strategy
        2) Tools and test equipment to be used
        3) Metrics to be collected
        4) Number of equipment and system configurations to be tested
        5) Special requirements or procedures for testing, if any
    c. Test criteria:
        1) Test data to be collected
        2) Pass/Fail criteria
        3) Proposed test result logs.
2. Test schedule.

E. RSN 27 30 01-4, Operations and Maintenance Manual:
1. Operations and Maintenance Manuals.
2. Spare parts list.

F. RSN 27 30 01-5, Field Operational Checkout Test Report:
1. Certification of performance and results of test.

1.04 DELIVERY, STORAGE, AND HANDLING

A. Handle and store telephone equipment in accordance with manufacturer's instructions. Include copy of these instructions with equipment at time of shipment.

1.05 PROJECT CONDITIONS

A. Ambient Temperature Range: -40 to 40 degrees C.

1.06 PERFORMANCE REQUIRMENTS

A. Provide a Cellular Signal Booster System at each of the following locations:

1. Sanostee Turnout.
2. Newcomb Turnout.
3. Naschitti Turnout.
4. BBN9 Turnout.
5. Coyote Canyon Turnout.
6. Rock Springs Turnout.
7. Window Rock Turnout.

B. Cellular Phone Signal Booster System:

1. Provide cellular phone signal booster system at each Turnout Building. Installation of signal booster system includes installation of signal booster, interior and external antennas, cables, etc., to complete a working system.
2. Signal booster system shall be of "Consumer" type as described in FCC-12-31 and 47 CFR § 20.21 and be covered under the Blanket Authorization Under Provider Licenses.
3. Internal antennas shall provide cellular coverage for the entire interior area of the Turnout Building.
4. External antenna shall:
    a. Be mountable on roof of Turnout Building.
    b. Include guy assemblies as needed per manufacturer's recommendations and TIA-222-H.
5. Functional testing of Contractor installed signal booster system hardware.

1.07 EXTRA MATERIALS

A. Any additional equipment not specified in these specifications but necessary to make system complete and operational shall be provided by Contractor.

PART 2 PRODUCTS

2.01 CELLULAR PHONE SIGNAL BOOSTER

A. Cel-Fi Go G32 as manufactured by Nextivity Inc., San Diego, California or equivalent.

B. Physical Features:

1. Supports major carriers (AT&T, Verizon, T-Mobile).
2. Max signal gain: 100 dB.
3. 3G/4G/5G multi-carrier support.
4. Carrier switching through phone app.
5. Environmental:
    a. Temperature: 0 to 65 degrees C.
    b. Relative Humidity: 0 to 95 percent, noncondensing.
    c. Indoor/outdoor NEMA 4 Rating.

2.02 EXTERNAL ANTENNA

A. Type: Directional or omnidirectional based on manufacturer's recommendation and signal strength.

B. Gain: As required based on signal strength.

C. Designed for all weather applications.

D. Lightning protection: DC Ground.

2.03 INTERIOR ANTENNA

A. Type: Panel or Dome.

B. Quantity: As required by signal booster manufacturer.

2.04 COAXIAL CABLE

A. Type: As recommended by signal booster manufacturer.

PART 3 EXECUTION

3.01 INSTALLATION

A. In accordance with manufacturer's recommendations.

B. Provide all mounting hardware and accessories to install signal booster and interior antennas.

C. Provide external antenna per including anchoring and mounting material.

D. Locate external antenna per specification requirements and as directed by COR.

E. Provide all cable and raceways required to connect Contractor supplied equipment and to provide fully functional system that meets requirements in these specifications. Cable and raceways shall be in accordance with Section 26 05 10 – Conductors and Cables and 26 05 33 – Raceways and Boxes.

F. Make cable and conduit connections. Use watertight conduit hubs for conduit connections.

G. Make power connections to equipment utilizing station service 120 VAC.

H. Gradual and uniform pulling stresses only will be permitted on cable.

I. Where lubricant is needed as an aid to pulling cable, only soapstone or other approved material not injurious to cable outer sheath shall be used.

J. Make grounding and bonding connections in accordance with Section 26 05 20 – Grounding and Bonding.

K. Cable and Wire Identification: In accordance with Section 26 05 10 – Conductors and Cables.

3.02 FIELD CHECKOUT

A. Notify Government in writing at least 15-days prior to beginning of tests.

B. Testing will be witnessed by Government personnel. Government personnel will actively oversee testing.

C. All material and equipment furnished under this contract shall be subjected to tests by Contractor, and at expense of Contractor at times and location specified.

D. Furnish all necessary testing equipment and pay all cost of tests.

E. Tests shall verify that system functional, performance, and design requirements as set forth in this solicitation are met.

F. Perform Field Operational Checkout Test (FOC):

1. Verify signal strength of installed signal booster hardware and carrier switching applications.

G. Perform field wiring checkout and test in accordance with Section 26 05 10 – Conductors and Cables.

H. Acceptance of equipment, or waiving of tests thereof, shall in no way relieve Contractor of responsibility of furnishing equipment and materials meeting specification requirements.

END OF SECTION

SECTION 28 10 01
SECURITY AND SURVEILLANCE SYSTEM

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Security and Surveillance System:

1. Payment: Lump sum prices offered in Price Schedules.

1.02 REFERENCE STANDARDS

A. National Electrical Manufacturer's Association (NEMA)

1.	NEMA 250-2020	Enclosures for Electrical Equipment (1,000-Volts Maximum)

B. National Fire Protection Association, Inc. (NFPA)

1.	NFPA 70-2023	National Electrical Code (NEC)

1.03 SUBMITTALS

A. Submit the following in accordance with Sections 01 33 00 – Submittals and 01 33 26 – Electrical Drawings and Data.

B. RSN 28 10 01-1, Approval Drawings and Data:

1. Data:
   a. Manufacturer's product data.
   b. Electrical characteristics.
   c. Bill of materials.
2. Drawings in government format:
   a. Detailed installation drawings and instructions of system being provided.
   b. Device, conduit, and circuit plan layouts.
   c. Schematic and wiring diagrams.

C. RSN 28 10 01-2, Test Procedure and Test Schedule.

1. Detailed test procedure, identifying:
   a. Equipment and systems to be tested.
   b. Methods of testing equipment and systems:
      1) Strategy.

2) Tools and test equipment to be used.

3) Metrics to be collected.

4) Number of equipment and system configurations to be tested.

5) Special requirements or procedures for testing, if any.

c. Test criteria:

1) Test data to be collected.

2) Pass/Fail criteria.

3) Proposed test result logs.

2. Test schedule.

D. RSN 28 10 01-3, Test Report:

1. Certification of performance and results of test.

E. RSN 28 10 01-4, Final Drawings and Operations and Maintenance Manuals:

1. Final Drawings including "As-built" changes.

2. Operations and Maintenance Manuals.

3. List of recommended spare parts and components.

4. Final bill of materials.

1.04 DELIVERY, STORAGE, AND HANDLING

A. Handle and store equipment in accordance with manufacturer's instructions. Include copy of these instructions with equipment at time of shipment.

1.05 PERFORMANCE REQUIRMENTS

A. The Turnout Buildings shall each have an identical Security and Surveillance System with identical components and equipment. This Specification describes performance requirements for one Turnout Building but all provisions within shall apply to each Turnout Buildings individually.

B. Provide a video surveillance system for each of the following Turnout Buildings:

1. Sanostee Turnout.

2. Newcomb Turnout.

3. Naschitti Turnout.

4. BBN9 Turnout.

5. Coyote Canyon Turnout.

6. Rock Springs Turnout.

7. Window Rock Turnout.

C. Design video surveillance system to accommodate the following:

1. IP based digital video surveillance system.
2. Allows simultaneous live viewing and recording.
3. Allows full resolution or low-resolution video, based on bandwidth limitations.
4. Supports streaming to allow users from remote (off-site) locations to stream video. Supports streaming full resolution, low resolution, and still shots to remote locations.
5. Pan/Tilt/Zoom (PTZ) camera control, with full remote control via IP network. Supports continuous scan of project site.
6. Digital Video Recorder:
    a. Automatic control and recording of each camera in the system.
    b. Manual control of each camera in the system via keyboard, mouse and monitor.
7. Cameras:
    a. Limit stops shall be programmed to limit pan range of camera to wall adjacent to camera.
    b. Turnout Building interior:
        1) Camera shall power up upon entry indication into building and pan to opened door.
        2) Camera shall begin continuous scan of entire room once initial preset is reached.
    c. Turnout Building yard:
        1) Allow coverage of entire treatment facility yard from camera(s) mounted in facility yard.
        2) Camera(s) shall provide continuous scan of entire facility yard.

D. Additional equipment not specified in these specifications but necessary to make system complete and operational shall be provided by Contractor.

E. Provide software essential for system to meet functional requirements.

F. Placement:

1. Camera Coverage:
    a. Interior cameras shall be placed so that the intruder represents at least 10 percent of the image height (no less than 70-pixels) at normal camera image resolution.

b. All cameras shall be placed to monitor potential intrusion points, unless larger images are required.

c. Cameras shall:

1) Snap to intrusion point and remain on for 60 seconds upon detection of intrusion.

2. Locations:

a. Interior area:

1) PTZ camera.

2) Coverage for entire interior area.

3) Coverage for all building entrances.

a) Camera shall be tied to door magnetic contacts.

b. Turnout Building exterior area:

1) PTZ cameras.

2) Coverage for entire building perimeter and ability to monitor entrance gate:

a) Camera shall be tied to magnetic contact located on gate hinge.

G. Provide Door/Gate/Hatch Intrusion Monitoring:

1. Utilize balanced magnetic contacts to provide door, gate, control panel, network interface panel, and security and communication network rack intrusion monitoring unless the installation arrangement requires alternative method to monitor position. Video analyzation alone is not sufficient.

2. Provides specific indication of which entry point opened.

3. Minimum Locations:

a. Doors or entry points for entering the building. This includes personnel doors, equipment doors, and roll-up doors.

b. Equipment control panels and network racks.

c. Gates allowing entry to the site or through fences.

d. Access hatches for vaults.

4. Alerts: Provide hardwired (not network or communication based) alarms to Control System for system alarm or unauthorized entry.

H. Provide Emergency Services Access:

1. Emergency services access, including fire department, via key locked access box for key storage, appropriately keyed for emergency services access.

2. Accessible outside of building.

3. Acceptable to the local fire marshal.

I. Design Security and Surveillance System to accommodate the following:

1. Provide security surveillance equipment protection by installing equipment in a locked security and communication network rack that has limited access by authorized personnel only.

PART 2 PRODUCTS

2.01 DIGITAL VIDEO RECORDER

A. On-Site Storage Requirements: 30 days at 10 frames per second, minimum.

B. PTZ Control, with full remote control via TCP/IP network.

C. Ports:

1. 10/100BaseT Ethernet.
2. Video Inputs: 4, minimum.
3. Mouse and keyboard.
4. USB ports: 3, minimum.

D. Archive Devices:

1. For long term storage, greater than 30-days.
2. DVDRW.
3. USB.

E. Remote/Web-Client capable.

F. Features:

1. 720p, minimum.
2. H.264 compression.
3. HDMI.

G. Monitor:

1. Screen dimension, 24-inch minimum.
2. Resolution: 1080p.
3. Screen Type: Flat panel, LCD.
4. Video input: HDMI.

2.02 PAN/TILT/ZOOM (PTZ) DOME CAMERAS

A. Type: Heavy Duty.

B. Construction:

1. Type: Pendant.
2. Back box: Aluminum.

C. Auto-focus.

D. Zoom: 18x optical, minimum.

E. Resolution: 720p, minimum.

F. Power over Ethernet (PoE) capable.

G. Outdoor camera to include the following:

1. Enclosure: NEMA 4X.
2. Dome: Smoked polycarbonate.
3. Protective cage: Stainless steel.
4. Day/Night mode.
5. Operating temperature: -40 to 50 degrees C.
6. High Powered PoE injector for heater element.
7. Corner mounting for 270-degree view around building.

2.03 INDUSTRIAL NETWORK SWITCH

A. Rack-mounted, industrial, managed Ethernet switch.

B. Physical Features:

1. Copper ports: Ten 10/100BaseTX, RJ-45, PoE, minimum.
2. Up to 30-watts per PoE port.
3. Status indicators on each port.
4. Power: Redundant 24 VDC inputs.
5. Enclosure: Metal case.

C. Network Features:

1. IEEE 802.3 compliance.
2. IEEE 802.3af/at compliance.
3. Full/Half Duplex operation on all ports.

4. 100-Mbps throughput, minimum.

2.04 POWER

1. Primary Power Supply:
   a. Supply primary power for Security and Surveillance System and access monitoring components from facility Control System UPS power system.
2. Backup Power Supply:
   a. Backup power for Security and Surveillance System and access monitoring components to allow its continued operation upon loss of power.

B. Uninterruptible Power Supply (UPS) from Section 25 00 01 - Control and Monitoring System to supply normal and backup power as required by Security and Surveillance System and access monitoring components for Turnout Building.

2.05 BALANCED MAGNETIC CONTACT

A. Designation: Balanced magnetic contact.
1. Separate magnet housing and contact.
2. All metal housing.

2.06 SURVEILLANCE SIGNS

A. Type:
1. Aluminum construction.
2. Size: 10- by 14-inches, minimum.
3. Sign shall read “RESTRICTED AREA MONITORED BY VIDEO CAMERA”.

2.07 SURGE SUPRESSION DEVICES

A. Type: Designed to protect outdoor cameras.

2.08 INSULATED CONDUCTORS

A. Insulated conductors used to wire security and surveillance system shall be in accordance with requirements of Section 26 05 10 – Conductors and Cables.

PART 3 EXECUTION

3.01 INSTALLATION

A. In accordance with manufacturer’s recommendations.

B. All connections shall be made to place security and surveillance system into operation.

C. Provide all cable and raceways required to connect equipment.

1. Provide fully functional system which meets requirements in these specifications.

2. Cable and raceways shall be provided in accordance with Sections 26 05 10 – Conductors and Cables and 26 05 33 – Raceways and Boxes.

D. Make grounding and bonding connections in accordance with Section 26 05 20 – Grounding and Bonding and NFPA 70.

E. Mount exterior security cameras on elevated standoffs on corners of structures at diagonally opposite corners to get optical coverage of all sides of the structure.

F. Contractor is to complete the surveillance design and determine the quantity and location of cameras on approval drawings.

G. Locate surveillance system hardware in Communication and Security Rack cabinet specified in Section 25 00 01 – System Control and Monitoring.

H. Place DVR monitor, keyboard, and mouse in Turnout Building, where space is available.

3.02 CONTRACTOR FIELD QUALITY TESTING

A. Perform field wiring checkout and test in accordance with Section 26 05 90 – Wiring Checkout and Tests.

B. Government will witness Security and Surveillance System test. Notify COR in writing at least 20-days prior to beginning of Security and Surveillance system testing.

C. Purpose of test is to demonstrate functionality, performance, and stability of hardware and software.

D. Test shall verify that all hardware and software meet requirements of specifications and drawings.

END OF SECTION

SECTION 31 02 10
WATER FOR DUST ABATEMENT

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Water for Dust Abatement:

1. Measurement: Volume of water applied for dust abatement as directed by COR:
    a. Only water used for dust abatement as directed by COR will be included.
    b. Quantity will be measured by calibrated water meter approved by COR.
2. Payment: M (1,000) gallons prices offered in Price Schedules.

1.02 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals.

B. RSN 31 02 10-1, Meter Calibration.

PART 2 PRODUCTS

2.01 WATER

A. Required for application.

PART 3 EXECUTION

3.01 APPLYING WATER FOR DUST ABATEMENT

A. Furnish calibrated water meter for use in measuring water usage.

B. Provide water in accordance with Section 01 51 00 – Temporary Utilities.

C. Provide means of conveying water to point of use and applying water.

D. Install water meter at source.

E. Use pressure spray or distributor bar to apply water evenly.

F. Do not use water for dust abatement on temporary equipment roads, completed backfill areas, and other construction activities unless directed by COR.

G. Apply water for dust abatement as directed by COR.

END OF SECTION

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SECTION 31 02 30
DUST PALLIATIVE

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Dust Palliative:

1. Measurement: Area covered as approved by COR:
   a. Only palliative applied in areas approved by COR will be included.
2. Payment: Acre prices offered in Price Schedules.

1.02 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals.

B. RSN 31 02 30-1, Dust Palliative:

1. Proposed application area.
2. Include dust palliative composition, equipment, and application.
3. Certification that dust palliative does not inhibit revegetation.

PART 2 PRODUCTS

2.01 DUST PALLITIVE

A. Dirtglue, as manufactured by Dirtglue Enterprises, www.dirtglue.com; or equal, with the following essential characteristics:

1. Biodegradable.
2. Water based.
3. Will not inhibit revegetation.

2.02 WATER

A. Required for mixing and application.

PART 3 EXECUTION

3.01 PREPARATION

A. Surface Preparation:

1. Shape and grade surface.
2. Loosen surface with rock rake or blade grader to depth of 1- to 2-inches.
3. Water loose surface so it is visibly moist.
4. In areas without surfacing materials: Do not disturb subgrade, moisten only.

3.02 APPLICATION

A. Apply in areas approved by COR immediately after clearing, grubbing, and topsoil stockpiling have been completed.

B. Apply palliative to cleared and grubbed areas not disturbed by excavation, stockpiled topsoil materials, roads, and access ramps.

C. Do not apply to areas to be excavated unless directed by COR.

D. After backfill and topsoil has been replaced.

E. Areas where application of water for dust abatement is ineffective as determined by COR.

F. Perform maintenance applications to control dust for duration of contract as recommended by manufacturer and as directed by COR.

G. Apply within 24-hours after completion of surface preparation.

H. Equipment: Designed for uniform application on variable widths at predetermined rates.

I. Location: Access road, stockpiles, and right-of-way (ROW) as directed by COR.

J. Area: Surface width as necessary with designated ROW.

K. Reseeded area: Suitable for reseeding, as recommended by manufacturer.

L. Limitations:

1. Not during rain.
2. Not if rain is anticipated within 4 hours following treatment.
3. Not on frozen ground.

M. Application Rate:

1. As recommended by manufacturer.
2. If material runs off during application, make repeated passes at lesser rates.

N. Compaction:

1. By equipment travel.

2. Keep traffic off treated surface until dust palliative has penetrated and cured enough to prevent excessive pickup under traffic.

END OF SECTION

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SECTION 31 03 33
REMOVAL AND CONTROL OF WATER

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in applicable prices offered in Price Schedules for items of work requiring removal and control of water from excavations.

1.02 DEFINITIONS

A. Dewatering: Removal and control of groundwater from pores or other open spaces in soil or rock formations to allow construction activities to proceed as intended through use of constructed wells or other systems, includes relief of groundwater pressure and control and discharge of effluent water.

B. Dewatering System: May consist of, but is not limited to deep wells, vacuum well points, and/or jet eductors/ejectors.

C. Unwatering: Removal and control of ponded, seeping, or flowing surface water or emerging subsurface water from excavated surfaces and from precipitation within and adjacent to excavations and construction zones using channels, ditches, gravel drains, gravel blankets, pipes, sumps, and discharge lines includes control and discharge of effluent water.

D. Unwatering System(s): May consist of, but is not limited to sumps, trash pumps, power supply, piping/plumbing, gravel blankets/drains, ditches, channels, drainage trenches, and/or sediment control.

E. Removal and Control of Water System: Any combination of dewatering and unwatering systems used to remove and control water.

F. Abandon:To grout or fill removal and control of water system(s) to prevent movement of surface water into aquifers penetrated by a well/sump/ditch/trench and to prevent vertical movement of water between aquifers after the work is completed.

1.03 PROJECT CONDITIONS

A. Conditions which may influence removal and control of water include:

1. Frequency and rate of precipitation at site(s).
2. Water levels in nearby arroyos.
3. Subsurface conditions consisting of layered deposits with variable thickness, permeability, storativity, and groundwater levels.
4. Efficiency of pumps, collectors, and discharge systems.

B. Water content and water levels in subsurface materials vary with location, depth, and material.

C. Refer to Section 53 10 00 – Geologic Investigations for additional information regarding subsurface conditions. Additional geotechnical field investigations shall be performed in accordance with Section 13 34 00 – Prefabricated Engineered Structures.

1. Removal and control of water methods shall be designed based upon the results from the geotechnical field investigations.

1.04 GOVERNMENT RESPONSIBILITY

A. Government may evaluate the effectiveness of removal and control of water system(s).

1. If COR determines water in the excavation is impacting construction, COR will request changes to removal and control of water system(s) in writing. COR may halt construction if water in excavation continues to impact construction.

1.05 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 31 03 33-1, Removal and Control of Water Plan:

1. Include narrative describing removal and control of water plan and methods:

a. Include proposed equipment for removal of water, including pump types and capacities.

b. Include sketches, figures, and/or site maps showing proposed methods and discharge locations.

2. Narrative describing abandonment and removal plan and methods.

C. RSN 31 03 33-2, Permits:

1. Include required Federal, State, Tribal, and local permits for water discharge and other activities associated with removal and control of water.

1.06 CONTRACTOR RESPONSIBILITY

A. Design removal and control of water system(s) in work areas as needed to collect and control ponding water, seepage from excavated slopes, runoff in open excavations and on cut or excavated slopes.

1. Use information obtained from the geotechnical field investigations to design the removal and control of water system(s).

B. Furnish, install, operate, and maintain removal and control of water system(s) during excavation, construction, and backfilling operations, in accordance with approved RSN 31 03 33-1 submittal.

1. Provide power for operation of removal and control of water system(s) on 24-hour, 7-day a week during construction.
2. Provide reserve equipment onsite at all times for the entire duration that the removal and control of water system(s) is operating to reduce downtime due to equipment failure.
3. Locate removal and control of water system(s) to maximize water removal and minimize construction interference and protect removal and control of water system(s) from construction operations.
4. Design and layout removal and control of water system(s) to collect and convey water to designated approved discharge points.
5. Maintain excavations, foundations, and other parts of work free from water as required for constructing each part of the work.

C. Remove equipment and materials associated with removal and control of water system(s) from site(s) at the end of the project, including related power systems. Removed equipment and materials shall remain property of the Contractor.

1. Damaged or destroyed removal and control of water system(s) shall be repaired or replaced by Contractor at Contractor's expense.
2. Backfill excavations for removal and control of water system(s) with materials conforming to gradation and compaction requirements specified for adjacent materials, or as approved by COR.

D. Removal and control of water system(s) may be placed in operation upon approval of RSN 31 03 33-1, but nothing in this section shall relieve Contractor from full responsibility for adequacy of water removal installation and operation.

1.07 REGULATORY REQUIREMENTS

A. Obtain required Federal, State, Tribal, and local permits for water discharge and other activities associated with removal and control of water.

PART 2 PRODUCTS

A. Not used

PART 3 EXECUTION

A. Not used

END OF SECTION

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SECTION 31 11 00
CLEARING AND GRUBBING

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in applicable lump sum prices offered in Price Schedules for Turnout Buildings.

1. Includes disposal of materials.

1.02 DEFINITIONS

A. Vegetation: Trees, shrubs, brush, stumps, exposed roots, down timber, branches, grass, and weeds.

B. Clearing: Removal of all debris and vegetation with a trunk diameter smaller than 3-inches.

C. Grubbing: Cutting and removal of all large vegetation with a trunk diameter greater than or equal to 3-inches.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 GENERAL

A. Obtain clearances and permits before clearing and grubbing.

B. Obtain COR approval before clearing and grubbing.

3.02 CLEARING AND GRUBBING

A. Clear only areas to be stripped of topsoil.

B. Clear rights-of-way (ROW) to be occupied by permanent construction and temporary construction ROW to facilitate work as approved by COR.

C. Clear area adjacent to excavation/cut or fill sections to a minimum distance of 3 feet outside of slope lines.

D. Obtain COR approval before removing trees.

E. Remove vegetation, rubbish, and objectionable material as determined by COR.

F. Where vegetation cutting is required:

1. Usable trees:

a. Size: 3-inch diameter or larger.

b. Remove branches and leave them on roadside for local residents to collect and use as firewood.

c. Remove and dispose of tree root balls.

d. Remove and dispose of stumps.

2. Smaller woody vegetation:

a. Size: Less than 3-inch diameter.

b. Chip and spread on ROW during revegetation process.

G. Clear areas in accordance with Sections 01 57 50 – Tree and Plant Protection and 01 57 60 – Protected Species.

3.03 DISPOSAL OF CLEARED MATERIAL

A. Dispose of material in accordance with Section 01 74 00 – Cleaning and Waste Management, or as directed by COR.

END OF SECTION

SECTION 31 14 10
TOPSOIL STRIPPING, STOCKPILING AND PLACEMENT

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include cost in applicable lump sum price offered in Price Schedules for Turnout Buildings.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 GENERAL

A. Obtain clearances and permits before stripping topsoil.

B. Obtain COR approval before stripping topsoil.

C. Strip all areas to be disturbed by construction activities to expose unknown cultural resources immediately after mobilization. Refer to Section 01 57 90 – Preservation of Historical and Archeological Data if cultural resources are discovered.

D. Vegetation removal in accordance with Section 31 11 00 – Clearing and Grubbing.

3.02 STRIPPING

A. Remove topsoil to a depth of 6 inches, and to additional depths where directed by COR, only in areas to be disturbed by construction activities:

1. If a true topsoil layer does not exist, consider top 6 inches of surface soils removed to be topsoil for purpose of this section.
2. If surface is rock, stripping is not required.

3.03 STOCKPILING

A. Transport and stockpile topsoil for future use, prior to placing.

B. Protect quality of topsoil. Stockpile topsoil separately from other excavated materials to avoid mixing during construction and reclamation activities.

C. Do not compact, drive on, place equipment on, or otherwise disturb stockpiled topsoil.

D. Cover stockpile with non-permeable tarp or dust palliative as approved by COR.

E. Haul excess topsoil offsite in accordance with Section 31 23 39 – Disposal of Excavated Materials as approved by COR.

3.04 PLACEMENT

A. Prior to topsoil placement, remove trash, debris, weeds, stones larger than 6 inches, and large pieces of vegetative materials.

B. Place topsoil in areas of earthen surface disturbed by construction.

C. Place topsoil to a minimum thickness of 6 inches.

D. Equipment travel over replaced topsoil is prohibited unless associated with reseeding/reclamation activities to avoid compaction and improve chances for successful reclamation, unless otherwise authorized by COR.

END OF SECTION

SECTION 31 23 02
COMPACTING EARTH MATERIALS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include cost of compacting earth materials in prices offered in Price Schedules for items of work where earth materials are required to be compacted:

a. Includes:

1) Furnishing water and moistening materials.

2) Contractor Quality Testing.

1.02 DEFINITIONS

A. Use definitions from ASTM D653.

B. Control Fraction: Portion of soil sample consisting of particles smaller than designated sieve size. Control fraction is used to compare in-place unit weight with standard laboratory unit weight. Control sieve size depends on laboratory test used to determine laboratory maximum density.

C. C-Value: Ratio expressed as a percentage of (1) in-place unit weight at fill moisture content to (2) wet unit weight of laboratory-compacted specimen prepared at fill moisture content as determined by rapid method of construction control in accordance with ASTM D5080. C-Value is a comparison of compactive effort of field compaction equipment to standard laboratory compactive effort.

D. D-value: Ratio expressed as a percentage of (1) in-place wet unit weight at fill moisture content to (2) laboratory maximum wet unit weight as determined from compaction curve constructed at fill moisture content as determined by rapid method of construction control in accordance with ASTM D5080. D-value is approximately the equivalent of percent compaction in accordance with ASTM D698.

E. Percent Compaction: Ratio expressed as a percentage of field in-place dry unit weight to laboratory maximum dry unit weight in accordance with ASTM D698 or ASTM D7382.

F. Special compaction: Compaction close to structures or in spaces not accessible by standard width mechanical rollers.

1.03 REFERENCE STANDARDS

A. ASTM International (ASTM)

1.	ASTM D422-63(2007)e2	Particle-Size Analysis of Soils
2.	ASTM D653-26	Terminology Relating to Soil, Rock, and Contained Fluids
3.	ASTM D698-12(2021)	Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/ft^3 (600 kN-m/m^3))
4.	ASTM D1140-25	Amount of Material in Soils Finer than the No. 200 (75-µm) Sieve
5.	ASTM D1556/D1556M-24	Density and Unit Weight of Soil in Place by the Sand-Cone Method (Withdrawn)
6.	ASTM D2216-19	Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass
7.	ASTM D2487-17(2025)	Classification of Soils for Engineering Purposes (Unified Soil Classification System)
8.	ASTM D2488-26	Description and Identification of Soils (Visual-Manual Procedure)
9.	ASTM D4318-17e1	Liquid Limit, Plastic Limit, and Plasticity Index of Soils
10.	ASTM D4546-25	One-Dimensional Swell or Collapse of Cohesive Soils
11.	ASTM D4564-08e1	Standard Test Method for Density of Soil in Place by the Sleeve Method (Withdrawn)
12.	ASTM D4647/D4647M-13(2020)	Identification and Classification of Dispersive Clay Soils by the Pinhole Test
13.	ASTM D4718/D4718M-15(2023)	Correction of Unit Weight and Water Content for Soils Containing Oversize Particles
14.	ASTM D4914/D4914M-24	Density of Soil and Rock in Place by the Sand Replacement Method in a Test Pit
15.	ASTM D5030/D5030M-21	Density of Soil and Rock in Place by the Water Replacement Method in a Test Pit
16.	ASTM D5080-20	Rapid Determination of Percent Compaction
17.	ASTM D6572-21	Determining Dispersive Characteristics of Clayey Soils by the Crumb Test
18.	ASTM D6913/D6913M-17(2025)	Particle-Size Distribution (Gradation) of Soils Using Sieve Analysis

19.	ASTM D6938-23	In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depths)
20.	ASTM D7382-25	Determination of Maximum Dry Unit Weight and Water Content Range for Effective Compaction of Granular Soils Using a Vibrating Hammer
21.	ASTM D7928-21e1	Standard Test Method for Particle-Size Distribution (Gradation) of Fine-Grained Soils Using the Sedimentation (Hydrometer) Analysis

B. Bureau of Reclamation (Reclamation)

1. *Guidelines for Earthwork Construction Control Testing of Gravelly Soils*, Earth Sciences and Research Laboratory, Technical Service Center, Bureau of Reclamation, Denver, Colorado, September 2006.

1.04 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals.

B. RSN 31 23 02-1, Test Results:

1. Results of tests listed in Table 31 23 02A – Contractor Materials Testing Requirements and Frequency.
2. Include test date, time, location, and name of person performing tests.
3. Notify COR within 2 hours if testing does not meet specified requirements.

1.05 AMBIENT CONDITIONS

A. Do not place or compact soil under the following conditions:

1. Rain or seepage that creates puddles in clayey or silty materials.
2. Heat or wind that dries material below specified moisture conditions.
3. Ice, frost, or snow pockets are visible in soil being placed or on working surface (underlying materials).

B. Ambient air temperature below freezing may make placement and compaction operations difficult:

1. Cease operations when placed materials are at or below freezing.
2. Obtain COR approval before performing operations when ambient air temperatures are below freezing.
3. Minimize interruptions of operations.
4. Continuous operations may be required to eliminate freezing of materials.

PART 2 PRODUCTS

2.01 WATER

A. Required for soil mixing and moisture control.

B. Water requirements per Section 01 51 00 – Temporary Utilities.

2.02 SOIL CLASSIFICATION

A. When required, classify earth materials using the Unified Soil Classification System (USCS) according to ASTM D2487:

1. Gradation tests for classification: ASTM D422, ASTM D6913 and or ASTM D1140.
2. Atterberg limits testing for classification: ASTM D4318.

2.03 SOIL TYPES

A. Clean Fill:

1. Any soil except soils with classifications of Peat (PT), Organic Silt and Organic Clay (OL and OH), or Elastic Silt (MH), per ASTM D2487.
2. Do not use soils that exhibit expansive or dispersive characteristics in accordance with ASTM D4546, ASTM D4647, or ASTM D6572.
3. Free of roots, stumps, limbs, vegetation, organic matter, and snow/ice.
4. Does not contain construction debris, scrap materials, refuse, man-made wastes, or chemical or hydro-carbon contamination.

B. Do not use frozen soils.

C. Special Gradations and Plasticity.

1. In some cases, such as compacted backfill under foundations and bedding and embedment for buried pipe, special gradations, plasticity index requirements, and/or swell characteristics may be required. Refer to Sections Section 31 23 10 - Earthwork for Site and Structures and 31 23 22 – Pipe Trench Earthwork for details.

2.04 DESIGNATION OF SOILS FOR COMPACTION

A. Requirements for lift thickness, method of compaction, and method of determining degree of compaction depend on whether soil is silty or clayey, coarse-grained, or coarse-grained containing some silt and clay.

B. Silty or Clayey Soils:

1. Contains appreciable amounts of fines (more than 15 percent plastic fines or 30 percent non-plastic fines).

C. Coarse-Grained Soils:

1. Contains few fines (less than 5 percent fines).
2. Coarse-Grained Free-Draining Material: Water does not pond on surface.
3. Coarse-Grained Non-Free-Draining Material: Water does pond on surface.

D. Coarse-Grained Soils Containing Some Silt or Clay:

1. Contains some silt or clay (between 5 percent fines and 15 percent plastic fines or 30 percent non-plastic fines).

2.05 MAXIMUM PARTICLE SIZE

A. Soil for compacted backfill:

1. Maximum particle size limitations are described in Sections 31 23 10 - Earthwork for Site and Structures and 31 23 22 – Pipe Trench Earthwork. Refer to the appropriate Sections for compacted backfill, pipe bedding, and pipe embedment.
2. Otherwise, no particles larger than 3-inches.

PART 3 EXECUTION

3.01 SURFACE PREPARATION

A. Clear, grub, and strip surface area.

B. Prepare in-situ horizontal surface so first compacted lift is placed on a firm and stable foundation base.

1. Building and structure foundations: if excavated foundation base is soil: scarify, moisten and proof-roll horizontal surface (excavated footprint) with a 10-ton vibratory steel drum (or equivalent compaction equipment for fine grained soils, see below) until surface does not settle (yield).
2. Excavate areas that settle excessively (yield) or rut and replace with compacted backfill.
3. Use woven geotextile, geogrid, or similar product at base of the excavation, if development of a stable base is causing difficulties during proof roll.
4. Minimum number of compaction equipment passes as approved by COR.

C. Scarify and moisten in-situ surface to provide satisfactory bonding surface before placing layer of material to be compacted.

D. COR will approve surface before the first lift is placed.

3.02 SOIL MOISTURE CONTENT

A. Moisten or aerate (dry) material, as necessary, to provide moisture content that will readily facilitate obtaining specified compaction. Add water to soil only in increments that will permit moisture content to be uniform and homogenous throughout each layer after mixing.

B. Silty or Clayey Soils and Coarse-Grained Soils Containing Some Silt or Clay:

1. Add no more than 2 percent water to fill by sprinkling just prior to compaction when fill is clayey and contains dry clods of clay:

a. If clayey borrow soil is more than 2 percent below optimum moisture, pre-conditioning and curing may be required to obtain uniform and homogenous distribution of moisture in the clods.

b. Use of disks, harrows, rakes, or other equipment may be required to blend moisture in borrow area.

2. Moisture content during compaction:

a. Soils controlled by impact compaction:

1) Within 2 percent dry to 2 percent wet of optimum moisture content determined in accordance with ASTM D698.

b. Soils controlled by vibrating hammer:

1) Applicable only to Coarse-Grained soils for which consistent results are not obtained using ASTM D698.

2) Within water content range for effective compaction as determined by ASTM D7382.

c. Compare field compaction moisture content with moisture contents determined in accordance with:

1) ASTM D2216, or

2) ASTM D6938. Correct moisture from nuclear gauge for gauge error for specific soils tested, moisture content of total material may require adjustment for control fraction (ASTM D4718). If nuclear gauge is utilized, rod shall test entire lift being tested.

C. Coarse-Grained Soils:

1. Add water during compaction as necessary because these soils are typically free-draining.

3.03 PLACEMENT

A. Placed soils to be compacted in horizontal layers.

B. If necessary, blend materials prior to placement so compacted backfill is homogenous and free from lenses, pockets, streaks, voids, laminations, or other imperfections.

3.04 COMPACTION

A. Compact earth materials according to the following methods and techniques appropriate to the type of soil.

B. Density, based on Percent Compaction of Maximum Dry Unit Weight using ASTM D698:

1. Compacted backfill:
    a. Minimum: 98 percent beneath structures, extending down and away from outside bottom edge of structure (foundation) at 1H:1V slope.
    b. Minimum: 95 percent outside of structures and 1H:1V slope.
2. Pipe Bedding Minimum: 95 percent.
3. Pipe Embedment Minimum: 95 percent.

C. Silty or Clayey Soils:

1. Compact with mechanical impact tampers, tamping rollers, vibrating pad foot rollers (not smooth), other suitable compaction equipment, or equipment travel:
    a. Uniformly distribute equipment passes.
    b. Compact in horizontal layers to compacted thickness of 6-inches or less.
2. Special compaction: Compact with walk-behind roller, handheld impact tampers, or small tamping equipment:
    a. Uniformly distribute effort.
    b. Compact in horizontal layers to compacted thickness of 4-inches.
    c. Required within 3.0 feet horizontally of:
        1) Structures.
        2) Pipe.
        3) Buried utilities.

D. Coarse-Grained Soils:

1. Compact with crawler-type tractors, vibrating drum rollers, surface plate vibrator, or similar equipment:
    a. Uniformly distribute equipment passes.
    b. Compact in horizontal layers to compacted thickness of 12-inches or less.
2. Special compaction: Compact with walk-behind vibrating roller, handheld impact tampers, vibrating plate tampers, or small tamping equipment:
    a. Uniformly distribute effort.
    b. Compact in horizontal layers to compacted thickness of 6-inches.

c. Required within 3.0 feet horizontally of:

1) Structures.

2) Pipe.

3) Buried utilities.

E. Coarse-Grained Soils Containing Some Silt or Clay:

1. Compact in accordance with either procedure above.

F. Adjustment:

1. Silty or clayey soils containing 5 to 30 percent oversized particles:

a. In-place unit weight of minus No. 4 size control fraction determined by screening gravel, washing, and determining mass and volume by assuming surface saturated dried moisture as outlined in ASTM D4718.

2. Silty and clayey soils containing more than 30 percent oversize particles:

a. Required D-ratio or Percent Compaction may be adjusted in accordance with appropriate curve on Figure 3 in Reclamation *Guidelines for Earthwork Construction Control Testing of Gravelly Soils*.

G. Demonstration:

1. Lift thicknesses may vary depending on equipment and methods.

2. Before changing requirements in this Section, demonstrate that required density shall be obtained.

3.05 MEASURE OF COMPACTION

A. Determine unit weight of in-place soils by using one of the following methods appropriate to the type of soil:

1. Silty or Clayey Soils and Coarse-Grained Soils Containing Some Silt or Clay:

a. ASTM D1556, or

b. ASTM D4914, or

c. ASTM D5030, or

d. ASTM D6938.

2. Coarse-Grained Soils:

a. ASTM D4564, or

b. ASTM D6938.

B. Determine Percent Compaction by one of the following techniques appropriate to the type of soil:

1. Silty or Clayey Soils:

a. Laboratory Compaction Test: Comparison of in-place dry unit weight of minus No. 4 sieve size control fraction to laboratory maximum dry unit weight as determined by ASTM D698, Procedure A.

b. Rapid Method: ASTM D5080.

2. Coarse-Grained Soils:

a. In-place dry unit weight of minus 2-inch control fraction compared to the maximum dry unit weight determined by ASTM D7382.

3. Coarse-Grained Soils Containing Some Silt or Clay:

a. Use ASTM D698 except for soils for which consistent results are not obtained using ASTM D698. In this case, use ASTM D7382.

4. Adjustment:

a. For soils containing 5 to 30 percent oversized particles:

1) In-place unit weight of minus No. 4 size control fraction determined by screening gravel, washing, and determining mass and volume by assuming surface saturated dried moisture as outlined in ASTM D4718.

b. For soils containing more than 30 percent oversized particles:

1) Required D-ratio or Percent Compaction may be adjusted in accordance with appropriate curve on Figure 3 in Reclamation *Guidelines for Earthwork Construction Control Testing of Gravelly Soils*.

3.06 CONTRACTOR FIELD QUALITY TESTING

A. Testing:

1. Independent testing laboratory shall perform sampling, testing, and reporting as required in Table 31 23 02A - Contractor Materials Testing Requirements and Frequency:

a. Independent testing laboratory shall meet requirements specified in Section 01 46 20 – Testing Agency Services.

2. Notify independent testing laboratory and COR 24-hours before compaction work begins and 24-hours before significant change in compaction operations (major change in equipment or procedure used).

3. Notify independent testing laboratory and COR immediately of equipment change due to breakdown, or re-deployment.

4. Testing Frequency:

a. At minimum, perform test at frequencies specified in Table 31 23 02A - Contractor Materials Testing Requirements and Frequency.

b. Greater frequency of testing shall be performed at beginning of new work, new work crew, or new equipment.

c. After successful work operation pattern is established, testing frequency may be performed at minimum guidelines.

d. Perform additional tests at sites considered questionable by COR.

1) Such as suspected incomplete compaction, surfaces that have become excessively wet or dry since compaction, compacted surfaces torn up by subsequent equipment travel, or other similar circumstances.

2) Frequency of additional testing is at discretion of COR.

e. Perform compaction test at every road crossing to verify compaction.

Table 31 23 02A – Contractor Materials Testing Requirements and Frequency

PROCEDURE	TEST STANDARD	STANDARD TITLE	MINIMUM FREQUENCY OF TESTING
Soil Classification	ASTM D2487	Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)	As necessary to classify material to meet specification requirements or to index material for determining density by nuclear method.
	ASTM D2488	Standard Practice for Description and Identification of Soils (Visual-Manual Procedure)	
Moisture Content	ASTM D2216	Standard Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass	With in-place density or as required to index material for determining density by nuclear method.
Sand Cone	ASTM D1556	Standard Test Method for Density and Unit Weight of Soil in Place by Sand-Cone Method	One test every 500 yd^3 Minimum 1 test per day per compacted backfill operation.

Table 31 23 02A – Contractor Materials Testing Requirements and Frequency

PROCEDURE	TEST STANDARD	STANDARD TITLE	MINIMUM FREQUENCY OF TESTING
Rapid Construction Control	ASTM D5080	Standard Test Method for Rapid Determination of Percent Compaction	One test every 500 yd^3 Minimum 1 test per day per compacted backfill operation.
Laboratory Compaction (Maximum Dry Unit Weight)	ASTM D 698, Method A	Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/ft^3)	One test every 500 yd^3 Minimum 1 test per day per compacted backfill operation.
Laboratory Compaction with Vibratory Hammer	ASTM D7382	Standard Test Methods for Determination of Maximum Dry Unit Weight and Water Content Range for Effective Compaction of Granular Soils Using a Vibrating Hammer	One test every 500 yd^3 Minimum 1 test per day per compacted backfill operation.
Nuclear Method	ASTM D6938	Standard Test Methods for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth)	One test every 100 yd^3 Minimum 1 test per day per compacted backfill operation. Minimum 1 test per foundation prior to concrete placement.
Oversized Correction	ASTM D4718	Standard Practice for Correction of Unit Weight and Water Content for Soils Containing Oversize Particles	When oversize material is between 5 and 30 percent
Rock Factor		Reclamation Guidelines for Earthwork Construction Control Testing of Gravelly Soils	When oversize material is greater than 30 percent

B. Contractor Support:

1. Provide timely access to areas for density testing:
    a. Excavate and level an area in compacted material to provide surface for testing.
    b. Test fills compacted by sheepsfoot rollers one or two lifts below surface.
2. COR may select location of testing.
3. When density is being measured by:
    a. Sand-cone device in accordance with ASTM D1556, cease construction activity in immediate vicinity of testing.
    b. Nuclear gauge reading shall be taken with the rod extended to full depth.
4. Dig test pits as requested to examine compacted soil against structures or pipe.
5. Replace compacted backfill within test pits to original requirements.
6. Provide warning lights, flags, or other safety devices as needed by testing personnel.
7. Provide adequate lighting for performing test if required because of darkness.

3.07 FIELD QUALITY ASSURANCE

A. COR may perform testing as required to verify contractor quality control testing.

B. Contractor shall provide access and support (at Contractor's cost) as required for COR to perform quality assurance testing.

END OF SECTION

SECTION 31 23 10
EARTHWORK FOR SITE AND STRUCTURES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Includes cost of earthwork for sites and structures in price offered in applicable Price Schedules for Turnout Buildings.

1.02 DEFINITIONS

A. Additional Excavation: Excavation beyond specified lines as directed by COR to remove unsuitable foundation material.

B. Over-excavation: Excavation performed for convenience, fault, or operation of Contractor beyond specified or directed additional excavation lines.

C. Rock Excavation:

1. Material that cannot be ripped using conventional excavation equipment and requires a 400 horsepower or larger dozer tractor equipped with rock tooth ripper rated for hard rock.
2. Material that cannot be excavated with conventional excavation equipment and requires a 300 horsepower or larger hydraulic excavator, with rock bucket and teeth, specifically manufactured for rock excavation, and/or a hydraulic hammer (breaker, hoe-ram), in areas where it is not feasible for ripping with a dozer tractor.

D. Additional Backfill: Backfill of additional excavation as directed by COR.

E. Cover: Distance between top surfaces of buried cables, conductors, or conduits and finished grade.

F. Earthwork for Structures: Excavation, placing backfill, and compacting backfill for:

1. Buildings.
2. Vaults.
3. Thrust blocks.
4. Other structures not covered in other Sections.
5. In areas as directed by COR.

1.03 REFERENCE STANDARDS

A. ASTM International (ASTM)

1.	ASTM D698-12(2021)	Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/ft^3 (600 kN-m/m^3))
2.	ASTM D4546-25	One-Dimensional Swell or Collapse of Soils

1.04 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals.

B. RSN 31 23 10-1, Processing Plan for Soils under Structures:

1. Describe equipment and methods for removing, stockpiling, processing, crushing, screening, moistening, placing, and compacting material to achieve requirements in Parts 2.02 and 3.03.
2. Provide schedule and sequence.
3. Include working and stockpiling areas and haul route plans.

C. RSN 31 23 10-2, Test Results:

1. Include test date, time, location, name of person collecting soil samples, and correlating in-situ soil density and moisture content according to Section 31 23 02 - Compacting Earth Materials.
2. Provide laboratory test results for swell and collapse potential in accordance with ASTM D4546, and as indicated in Parts 3.05 and 3.09.

1.05 PROJECT CONDITIONS

A. Refer to Section 53 10 00 – Geologic Investigations for additional information regarding subsurface soil and rock conditions.

B. Soils requiring remediation may be encountered on site (e.g. expansive, collapsible, or dispersive materials).

C. Excavate and process in-situ soils and bedrock, and place compacted backfill to provide a stable foundation for buildings, vaults, thrust blocks, and other structures.

PART 2 PRODUCTS

2.01 EXCAVATED MATERIALS

A. Operations shall be such that excavations will yield as much suitable material for use in permanent construction required under these specifications as practicable.

B. Place excavated materials which are too wet for immediate compaction temporarily in stockpiles, and process until moisture content is reduced sufficiently to permit placement.

2.02 MATERIAL FOR COMPACTED BACKFILL

A. Type and amount of material used for compacted backfill, and manner of depositing material shall be subject to approval by COR.

B. Obtain compacted backfill material from required excavations. If sufficient suitable material is not available from this source, obtain additional material from other sources approved by COR:

1. Government makes no guarantee that specified materials for compacted backfill are contained in or can be processed from excavated materials.

2. All reasonable effort shall be expended to obtain suitable material for compacted backfill from required excavation prior to obtaining commercial sources.

C. Do not use material removed during topsoil stripping.

D. Do not use organic material or other deleterious material.

E. Material shall be thoroughly mixed and evenly moistened (processed) to achieve a uniform material and to avoid pockets or lenses of variable material.

F. Maximum particle size: 3-inch.

G. Minimum of 60 percent particles passing the No. 4 sieve size, by weight.

H. Plasticity Index (PI): Less than 15.

I. Swell potential: Less than one percent under an inundation/saturated load of 1,000 pounds per square-foot (psf) (ASTM D4546).

J. Where special compaction is required, in accordance with 31 23 02 – Compacting Earth Materials, the gradation requirements are the same.

PART 3 EXECUTION

3.01 EXCAVATION, GENERAL

A. Excavate structure foundations to elevations and lines shown on drawings, as approved, or as directed by COR.

B. Government reserves the right, during progress of work, to vary slopes, grades, and dimensions of excavations from those specified.

C. Government does not represent that excavation performed under these specifications can be made to or maintained at paylines shown on the drawings or described in these specifications.

D. Perform excavation in the dry.

E. Do not excavate in frozen materials without written approval of COR.

F. Blasting: Not allowed.

G. Preserve material below and beyond established lines of excavation in soundest possible condition:

1. Damage to work due to Contractor's operations shall be repaired by and at expense of Contractor.

2. Material beyond required or prescribed excavation lines which is loosened by Contractor's operations shall be removed at expense of Contractor.

3.02 OVER-EXCAVATION

A. If foundation material is excavated beyond lines required to receive structure, fill over-excavation with suitable materials as directed by COR and compact in accordance with Section 31 23 02 – Compacting Earth Materials.

B. If foundation material is disturbed or loosened during excavation or otherwise, compact foundation in place or remove and replace it with suitable material as directed by COR and compact in accordance with Section 31 23 02 – Compacting Earth Materials.

3.03 PROCESSING AND STOCKPILING EXCAVATED MATERIAL TO BE USED AS COMPACTED BACKFILL

A. Stockpile material in maximum 12-inch lifts as it is removed from excavation.

B. Do not compact stockpile material.

C. Wet material as it is stockpiled to minimum moisture content of optimum as determined by ASTM D698.

D. Test moisture content in accordance with Section 31 23 02 – Compacting Earth Materials.

E. Excavated materials may require crushing, screening, and/or other processing to meet requirements of Part 2.02. Provide processing equipment and perform work necessary to process excavated materials to produce materials meeting the requirements of these specifications.

F. Compacted backfill material shall be thoroughly mixed and moistened to achieve a uniform material and to avoid pockets or lenses of variable material.

G. Earthwork equipment should be used to uniformly blend, moisture condition, aerate if necessary, and stockpile material to achieve a uniform fill material before it is placed as compacted backfill material.

3.04 SURFACES OF EXCAVATION

A. Vertical offsets within rock surfaces are not to extend more than 2 inches unless surfaces are to be covered with concrete or surfacing materials.

3.05 PREPARATION OF STRUCTURE FOUNDATIONS

A. Prepare foundations at structure sites by methods which will provide firm foundations for structures:

1. Finish bottom and side slopes of excavation, upon or against which the structure is to be placed, to prescribed dimensions.
2. Moisten and compact prepared surfaces with suitable tools to form firm foundations upon or against which to place structure.

B. Where unsuitable material is encountered in the foundation for a structure, COR will direct performance of additional excavation to remove unsuitable material: Fill additional excavation with compacted backfill in accordance with Section 31 23 02 - Compacting Earth Materials.

C. Keep surfaces of shale, claystone, and/or mudstone (if encountered) that are exposed by excavation moist until concrete or earth material are placed on them.

D. If potentially expansive materials, such as clay, shale, claystone, or mudstone are encountered below structures, in-place material shall be tested for swell potential as directed by COR.

1. Test in accordance with ASTM D4546.
2. Include information required in RSN 31 23 11-2 Test Results.

E. Test moisture content of foundation subgrade no more than 12 hours prior to placing concrete in accordance with 31 23 02 – Compacting Earth Materials.

3.06 FOUNDATION INSPECTION

A. Inspection will be performed by Government to obtain a geologic record of the final foundation surface:

1. Provide safe access for Government personnel during inspection.
2. Notify COR at least 7 days before reaching final excavation elevation for foundation.
3. Once an adequate foundation has been obtained, and before backfilling, Government will require 3-days to inspect, map and survey foundation for structures:
    a. Clean surface with straight edge of equipment bucket to expose foundation as directed by COR.

4. Maintain prepared surfaces free of debris and standing water until inspection and geologic mapping are completed.

5. Do not operate equipment in area being inspected until inspection and survey are complete.

3.07 PLACING COMPACTED BACKFILL

A. Place compacted backfill to lines and grades shown on drawings, as approved, or as directed by COR.

B. Place material carefully and spread in uniform layers so compacted backfill is homogeneous and free from lenses, pockets, voids, laminations, or other imperfections.

C. Do not place compacted backfill material when either material or surfaces on which it is to be placed are frozen.

D. Place backfill to the same elevation on both sides of the structure to prevent unequal loading.

E. Subsequent layers of compacted backfill may be placed as soon as compaction of previous layers is completed. Placing of additional materials may be delayed by COR at designated locations for procurement of samples for testing.

F. Additional information is provided in 31 23 02 – Compacting Earth Materials.

3.08 CONTRACTOR FIELD QUALITY TESTING

A. Notify COR within 48 hours of completion of excavation but before compaction. Do not compact until approved by COR.

3.09 COMPACTING BACKFILL MATERIAL

A. Place, moisten, and compact materials of the compacted backfill as provided in Section 31 23 02 – Compacting Earth Materials.

B. Testing of swell potential of in-place compacted backfill material shall be performed. Testing shall be in accordance with ASTM D4546, as directed by COR, and as follows:

1. Take one test per 5,000 square feet per lift, or as directed by COR. Take a minimum of one test per day.

2. Swell Potential shall be less than one percent under load of 1,000 psf (ASTM D4546, inundated/saturated at 1,000 psf).

3. Include details listed in RSN 31 23 11-2, Test Results.

3.10 PROTECTION

A. To provide adequate protection for compacted backfill around structure, Government reserves the right to direct Contractor to place enough uncompacted backfill material over compacted backfill within 72 hours after compaction of compacted backfill.

3.11 DISPOSAL OF EXCAVATED MATERIALS

A. Dispose of excavated materials unsuitable for or in excess of backfill or other earthwork requirements, as determined by COR, as provided in Section 31 23 39 - Disposal of Excavated Materials.

END OF SECTION

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SECTION 31 23 22
PIPE TRENCH EARTHWORK

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include cost of common excavation for pipeline trenches in prices offered in Price Schedules for applicable Steel Manifold Piping and Line Pipe.
2. Include cost of rock excavation for pipeline trenches in prices offered in Price Schedule for applicable Steel Manifold Piping and Line Pipe.
3. Include cost of processing pipeline backfill materials in prices offered in Price Schedules for applicable Steel Manifold Piping and Line Pipe.
4. Include cost of compacted backfill pipeline trenches in prices offered in Price Schedules for applicable Steel Manifold Piping and Line Pipe.
5. Include cost of controlled low strength material (CLSM) in prices offered in Price Schedules for Steel Manifold Piping and applicable Line Pipe.

1.02 DEFINITIONS

A. Additional Excavation: Excavation beyond specified lines as directed by COR to remove unsuitable foundation material.

B. Additional Backfill: Backfill of additional excavation when directed by COR.

C. Additional Compacted Backfill: Compacted backfill of additional excavation when directed by COR.

D. Over-excavation: Excavation performed for the convenience, fault, or operation of Contractor beyond specified or directed additional excavation lines.

E. Rock Excavation:

1. Material that cannot be ripped using conventional excavation equipment and requires a 400 horsepower or larger dozer tractor equipped with rock tooth ripper rated for hard rock.
2. Material that cannot be excavated with conventional excavation equipment and requires a 300 horsepower or larger hydraulic excavator, with rock bucket and teeth, specifically manufactured for rock excavation, and/or a hydraulic hammer (breaker, hoe-ram), in areas where it is not feasible for ripping with a dozer tractor.

F. Cover: Distance between top surfaces of buried cables, conductors, or conduits and finished grade.

1.03 REFERENCE STANDARDS

A. American Association of State Highway and Transportation Officials (AASHTO)

1. AASHTO SSHB-02 Standard Specifications for Highway Bridges, 17th Edition

1.04 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals.

B. RSN 31 23 22-1, Processing Plan:

1. Describe equipment and methods for removing, stockpiling, processing, crushing, screening, moistening, placing, and compacting backfill to achieve requirements in Parts 2.03 and 3.05.
2. Provide schedule and sequence.
3. Include working and stockpiling areas and haul route plans.

C. RSN 31 23 22-2, Test Results:

1. Include test date, time, location, person collecting soil samples, and correlating in-situ soil density and moisture content information in accordance with Section 31 23 02 – Compacting Earth Materials.

1.05 PROJECT CONDITIONS

A. Refer to Section 53 10 00 – Geologic Investigations for additional information regarding subsurface (soil and rock) conditions.

B. Soils requiring remediation may be encountered on site (e.g. expansive, collapsible, or dispersive materials).

C. Excavate and process in-situ soils and bedrock.

PART 2 PRODUCTS

2.01 EXCAVATED MATERIALS

A. Operations in excavations shall be such that excavations yield as much suitable material for use in permanent construction required under these Specifications as practicable.

B. Place excavated materials too wet for immediate placement and compaction temporarily in stockpiles, and process until moisture content is reduced sufficiently to permit placement and compaction.

2.02 CONTROLLED LOW STRENGTH MATERIAL (CLSM) FOR BEDDING AND EMBEDMENT

A. Use for bedding.

B. Use for embedment as detailed on Drawing 1695-D-62554.

C. Controlled low strength materials (CLSM) in accordance with Section 31 23 70 - Controlled Low Strength Material (CLSM).

2.03 SOIL MATERIAL FOR PIPE EMBEDMENT AND BACKFILL

A. Contractor shall provide processing equipment and perform work necessary to process excavated materials to produce materials meeting requirements of these Specifications.

B. Type and amount of material used for pipe embedment, backfill or compacted backfill, and manner of depositing material, shall be subject to approval by COR.

C. Obtain pipe embedment and backfill material from required excavations. If sufficient suitable material is not available from this source, obtain additional material from other sources approved by COR:

1. Government makes no guarantee that specified pipe embedment and backfill materials are contained in or can be processed from materials excavated.

2. All reasonable effort shall be expended to obtain suitable pipe embedment and backfill materials from required excavation prior to obtaining commercial sources.

D. Do not use material removed during topsoil stripping for pipe embedment and backfill materials.

E. Do not use organic material, frozen materials, or other deleterious material.

F. Do not use unprocessed expansive or dispersive materials (if encountered) for pipe embedment.

G. Material shall be thoroughly mixed and evenly moistened to achieve a uniform material and to avoid pockets or lenses of variable material.

H. Select material in accordance with Drawing 1695-D-62554.

I. Material gradation and plasticity index (PI)l:

1. Embedment:

a. As shown on Drawing 1695-D-62554.

b. PI of processed and thoroughly mixed materials should be less than 15.

2.

3. Compacted Backfill: Maximum particle size:
   a. 3/4-inches within 12-inches of the pipe.
   b. All other locations, 3-inches.
4. Uncompacted Backfill: Maximum particle size:
   a. 3/4-inches within 12-inches of the pipe.
   b. All other locations, 5-inches.
5. Backfill placed where pipelines cross under open irrigation ditches and other watercourses: Use Silty or Clayey material with a Plasticity Index (PI) of 8 or more.
6. Where compaction or special compaction is required, perform in accordance with Section 31 23 02 – Compacting Earth Materials.
   a. In these instances, the backfill gradational requirements are identical.

2.04 WARNING TAPE

A. Polyethylene tape.

B. Thickness: 4 mils.

C. Non-detectable.

D. Width, minimum: 6-inches.

E. Color: Blue with silver lettering.

F. Legend repeated at least every 3 feet: "CAUTION: WATERLINE BURIED BELOW – FOR MORE INFORMATION CONTACT THE NAVAJO TRIBAL UTILITIES AUTHORITY"

PART 3 EXECUTION

3.01 GENERAL

A. Perform operations so that land can be restored to original conditions.

B. If backfill operations fall behind pipelaying operations, stop excavation and pipelaying operations until sufficient progress has been made as directed by COR.

C. Place temporary protective backfill over pipe, as approved by COR, if backfilling operations are interrupted for more than 24-hours.

D. Fence open trench with temporary safety fencing in accordance with Section 01 56 32 – Temporary Safety Fence.

E. Install warning tape over center of pipe, at least 18 inches below ground, and at least 18 inches above pipe.

3.02 EXCAVATION

A. Excavate pipe trenches for pipelines, accessories, and appurtenances to lines, grades, and dimensions shown on drawings.

B. Finish bottom of excavated trench to lines and grades shown on drawings. Trench foundation shall be approved by COR.

C. Government may, during progress of work, vary slopes, grades, and dimensions of excavations from those specified for safety.

D. Government does not represent that excavation performed under these Specifications can be made to or maintained as shown on the drawings or described in these Specifications.

E. Perform excavation in the dry.

F. Do not excavate in frozen materials without written approval of COR.

G. Blasting: Not allowed.

H. Preserve material below and beyond established lines of excavation in soundest possible condition:

1. Damage to work due to Contractor's operations shall be repaired by and at expense of Contractor.
2. Material beyond required or prescribed excavation lines which is loosened by Contractor's operations shall be removed at expense of Contractor.

3.03 ADDITIONAL EXCAVATION

A. Perform additional excavation within pipe trench, beyond specified lines, as directed by COR to remove unsuitable foundation material.

3.04 OVER-EXCAVATION

A. If foundation material is excavated beyond specified or directed lines, fill over-excavation with suitable materials as directed by COR and compact in accordance with Section 31 23 02 – Compacting Earth Materials.

B. If foundation material is disturbed or loosened during excavation or otherwise, either compact material in-place or remove and replace with suitable material as directed by COR and compact in accordance with Section 31 23 02 – Compacting Earth Materials.

3.05 PROCESSING AND STOCKPILING EXCAVATED MATERIAL FOR TO BE USED AS EMBEDEMENT AND BACKFILL

A. Stockpile excavated materials which meet, or will be processed to meet, material requirements for backfill in pipe trenches.

B. Stockpile material in maximum 12-inch lifts as it is removed from excavation.

C. Do not compact stockpile material.

D. Wet material as it is stockpiled to minimum of optimum moisture content.

E. Test moisture content in accordance with Section 31 23 02 – Compacting Earth Materials prior to placement.

F. Excavated materials may require crushing, screening, and/or other processing to meet requirements of this Section. Contractor shall provide processing equipment and perform work necessary to process excavated materials to produce materials meeting the requirements of these Specifications.

G. All pipe embedment and backfill material shall be thoroughly mixed and moistened to achieve uniform material and avoid pockets or lenses of variable material.

H. Earthwork equipment should be used to uniformly blend, moisture condition, aerate if necessary, and stockpile material in order to achieve uniform fill material before it is placed as pipe embedment or backfill material.

3.06 EXCAVATION SURFACES

A. Vertical offsets within rock surfaces shall not extend more than 2-inches unless surfaces are to be covered with concrete or surfacing materials.

3.07 PREPARATION OF PIPE TRENCH FOUNDATION

A. Prepare pipe trench foundation by methods which will provide firm foundations for pipe:

1. Finish bottom and side slopes of excavation, upon or against which pipe is to be placed, to prescribed dimensions shown on drawings.
2. Moisten and compact prepared surfaces with suitable tools to form firm foundations upon or against which to place pipe.

B. Where unsuitable material is encountered in pipe trench, COR will direct performance of additional excavation to remove unsuitable material:

1. Replacement of materials where additional excavation is performed will be directed by COR in accordance with Section 31 23 02 – Compacting Earth Materials and/or Section 31 23 70 – Controlled Low Strength Material (CLSM).

C. Keep exposed surfaces of shale, claystone, and/or mudstone (if encountered) exposed by excavation moist until CLSM or earthen backfill materials are placed on or against them.

3.08 PLACING CONTROLLED LOW STRENGTH MATERIAL (CLSM) FOR PIPE BEDDING AND/OR EMBEDMENT

A. Place CLSM in accordance with Section 31 23 70 – Controlled Low Strength Material (CLSM) and as shown on drawings.

B. Allow CLSM to obtain required strength in accordance with Section 31 23 70 – Controlled Low Strength Material (CLSM) before placing pipe backfill above the CLSM bedding and embedment.

3.09 PIPE INSTALLATION

A. Install pipe in accordance with Section 35 21 95 – Steel Manifold Piping.

3.10 PLACING SOIL MATERIAL FOR PIPE EMBEDMENT AND BACKFILL

A. Place to lines, grades, widths, and depths shown on drawings, unless otherwise directed by COR, except:

1. Place CLSM encasement about Reinforced Concrete Pipe Culvert to a height not less than a factor of 0.375 of the outside diameter (0.375*OD) of pipe.

B. Do not drop soil materials directly on pipe at heights exceeding 3.0-feet.

C. Do not place CLSM bedding or embedment, or earthen backfill materials when either material or surfaces on which it is to be placed are frozen or contain ponded water.

D. Place embedment material at same elevation on both sides of pipe to prevent unequal loading and displacement of pipe. Elevation difference on sides of pipe shall not exceed 6-inches.

E. Placing of backfill materials may be delayed by COR at designated locations for procurement of bedding or embedment CLSM samples for testing.

F. Place backfill carefully and spread in uniform layers so that backfill, once compacted, is homogeneous and free from lenses, pockets, voids, laminations, or other imperfections.

Place temporary protective uncompacted backfill over pipe as approved by COR, if backfilling operations are interrupted for more than 24 hours.

G. Uncompacted backfill: Maximum lift height: 1.0-foot.

3.11 COMPACTION

A. Compact to lines, grades, widths, and depths shown on drawings, unless otherwise directed by COR.

B. Place, moisten, and compact materials for backfill as provided in Section 31 23 02 – Compacting Earth Materials.

C. Compacted backfill layers that do not meet the requirements of these Specifications, based on field quality test results, shall be removed and replaced at Contractor's expense.

D. For special compaction, there should be at least 1.0-foot of cover over pipe before using hand-held impact tampers or walk-behind small tamping equipment.

1. The first 2.0-feet of compacted backfill material shall be considered special compaction as provided in Section 31 23 02 – Compacting Earth Materials.

E. Location of Compacted Backfill:

1. As shown on drawings.

2. Where utility pipelines cross pipeline: Compact to horizontal centerline of utility pipeline, or as required by utility owner, whichever is more stringent.

3. At tees with horizontal outlets, pipe bends, encasements, collars including collar-type blocking, and pipe plugs as shown on drawings.

4. Around valve boxes, vaults and manholes as shown on drawings.

3.12 PROTECTION

A. To provide adequate protection for CLSM bedding and embedment, compacted backfill, and pipe within the pipe trenches, Government may direct the Contractor to place uncompacted backfill material over CLSM or compacted backfill within 72-hours after CLSM placement or compaction of backfill.

B. Place uncompacted backfill to a minimum thickness of 3.0-feet or compacted backfill to a minimum thickness of 2.0-feet above top of pipe before allowing construction equipment to travel over pipe.

1. After these minimum earth covers are in-place, maximum equipment loading allowed over the pipe shall be HS-20 loading (16,000-pound wheel load) in accordance with AASHTO SSHB.

2. If construction equipment that exerts a larger wheel load is proposed to be used, submit construction equipment loadings to COR for analysis and determination of required backfill depths or other protective measures.

3. Limit construction equipment passing over pipe to 5 mph until at least 3.5-feet of cover is achieved.

3.13 DISPOSAL OF EXCAVATED WASTE MATERIALS

A. Dispose of excavated waste materials, as determined by COR, as provided in Section 31 23 39 - Disposal of Excavated Materials.

END OF SECTION

SECTION 31 23 39
DISPOSAL OF EXCAVATED MATERIALS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for items of work requiring excavation.

1.02 DEFINITIONS

A. Waste material: Material from required excavation not suitable or required for compacted backfill, miscellaneous backfill, embedment, bedding, or topsoil.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 DISPOSAL OF EXCAVATED MATERIALS

A. Dispose of excavated waste material within right-of-way (ROW) unless prior approvals of cultural and environmental clearance have been obtained outside of ROW.

B. Place excavated waste material as directed by COR, in layers of 6-inches or less.

C. Do not place excavated waste material in wetlands, within 12 feet of drainage channels, within 12 feet of edge of prescribed or actual cuts for laterals, wasteways, or drains.

D. Do not waste material by dumping from top of slope.

E. Grade waste banks to drain, with reasonably even and uniform surfaces that blend with natural terrain:

1. Minimum slope: 2 percent.
2. Maximum slope: 4H:1V (horizontal:vertical).

F. Seed surface in accordance with Section 32 92 20 - Seeding.

END OF SECTION

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SECTION 31 23 50
EARTHWORK FOR ROADS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Access Road:

1. Payment: Lump sum price offered in applicable Price Schedules.

   a. Includes all materials and labor to make existing haul road route safe and usable for construction and public traffic.

   b. No payment will be made for excavation and replacement or refill of excavation beyond prescribed excavation lines.

   c. Includes cost of rip rap and concrete.

B. BBN9 Low Water Crossing:

1. Payment: Lump sum price offered in applicable Price Schedules.

   a. Includes all materials and labor to make existing haul road route safe and usable for construction and public traffic.

   b. No payment will be made for excavation and replacement or refill of excavation beyond prescribed excavation lines.

   c. Includes cost of rip rap, concrete, and reinforcement.

1.02 DEFINITIONS

A. Refer to Section 31 23 10 – Earthwork for Site and Structures.

1.03 PROJECT CONDITIONS

A. Government may vary lines, grades, and dimensions of excavation or subgrade fill from those shown on drawings based on actual subsurface materials encountered during excavation.

B. Subsurface information is provided in Section 53 10 00 – Geologic Investigations.

PART 2 PRODUCTS

2.01 EARTHEN FILL

A. Clean fill materials in accordance with Section 31 23 02 – Compacting Earth Materials.

B. Maximum Particle Size: 3 inches.

C. Source: Obtain and process materials from excavated materials or other sources, as approved by COR.

PART 3 EXECUTION

3.01 EXCAVATION

A. Excavate to lines, grades, and dimensions shown on drawings or prescribed by COR.

B. Excavate in dry conditions.

C. Backfill excavation beyond prescribed excavation lines with compacted backfill in accordance with Section 31 23 02 – Compacting Earth Materials.

D. Remove foundation material adjacent to excavation which is disturbed or loosened during excavation operations or other work.

1.04 PREPARATION

E. Grade and prepare subgrade free from depressions and soft spots:

1. Additional excavation required for soft soils as directed by COR. Replace with compacted backfill in accordance with 31 23 02 – Compacting Earth Materials.

3.02 PROTECTION

A. Preserve materials below and beyond prescribed excavation lines in soundest possible condition. Repair damage to work caused by construction operations.

B. Provide temporary construction, pumping, bailing, draining, shoring, sheeting, bracing, and other work necessary to maintain excavations during construction operations.

3.03 PLACING

A. Use suitable excavated materials for required backfill. Locations and procedures for stockpiling will be subject to approval by COR.

B. Placing and Compacting:

1. Place in layers and compact in accordance with Section 31 23 02 – Compacting Earth Materials.

2. Obtain COR approval of subgrade before placing surfacing material.

3.04 DISPOSAL

1. Dispose of excavated materials which are not used or suitable for backfill in accordance with 31 23 39 – Disposal of Excavated Waste Materials.

END OF SECTION

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SECTION 31 23 70
CONTROLLED LOW STRENGTH MATERIALS (CLSM)

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for Steel Manifold Piping:

1. Includes:

a. Pipe bedding and embedment zones.

b. Including road and utility crossing requirements.

1.02 REFERENCE STANDARDS

A. American Concrete Institute (ACI)

1.	ACI 306.1-90 (2002)	Cold Weather Concreting
2.	ACI 318-19	Building Code Requirements for Structural Concrete

B. ASTM International (ASTM)

1.	ASTM C33/C33M-24a	Concrete Aggregate
2.	ASTM C94/C94M-25	Ready-Mixed Concrete
3.	ASTM C143/C143M-20	Slump of Hydraulic-Cement Concrete
4.	ASTM C1602/C1602M-22	Mixing Water Used in the Production of Hydraulic Cement Concrete
5.	ASTM D4832-23	Preparation and Testing of Controlled Low Strength Material (CLSM) Test Cylinders
6.	ASTM D6024/ASTM D6024M-24	Ball Drop on Control Low Strength Material (CLSM) to determine Suitability for Load Application

1.03 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals.

B. RSN 31 23 70-1, Approval Data for CLSM Mix:

1. Mix Design: For each CLSM mix:

a. Mixture proportions.

b. Material sources:

1) Name and manufacturer of each cementitious material.

2) Name of aggregate sources.

3) Product name and manufacturer of admixtures to be used in mix.

4) Government reserves right to require submission of samples of CLSM materials for testing before or during use in concrete.

c. Physical properties:

1) Trial mixtures:

a) Results from trial batches made within past 6 months.

b) Trial mix test results, three 6-inch-diameter cylinders, each at 7 and 28 days.

c) Average compressive strength of trial batch cylinders at specified design age.

d. Resubmit mix design for change in material source or type.

e. Cementitious materials manufacturer's certifications and test reports within last 3 months.

f. Aggregate test reports for gradation, less than 12 months old.

g. Method to determine moisture and consistency of materials to maintain specified strengths. Provide testing frequency.

2. NRMCA Certification of Production Facilities:

a. Automatic digital recording of cementitious materials, aggregate, water, and chemical admixtures.

b. Verification that facility meets requirements for furnishing concrete in subfreezing weather.

c. ACI Personnel Certification(s):

1) Aggregate Testing Technician.

2) Concrete Field-Testing Technician.

3) Concrete Strength Testing Technician.

3. If Quality Control test results show CLSM does not meet requirements, submit revised mix design.

C. RSN 31 23 70-2, Cold Weather Protection Plan

1. In accordance with ACI 306.

2. Detailed procedures for the production, transportation, placement, protection, curing, and temperature monitoring of CLSM during cold weather. Include procedures to be implemented upon abrupt changes in weather conditions or equipment failures. Do not begin cold weather placements until these procedures have been reviewed and accepted.

D. RSN 31 23 70-3, Quality Control Test Results:

1. Notify COR within 2-hours if testing does not meet specified requirements.
2. If Quality Control Test Results show CLSM does not meet specified requirements, submit a revised mix design.

1.04 DELIVERY, STORAGE, AND HANDLING

A. Furnish complete and legible batch ticket with each batch of CLSM in accordance with ASTM C94. Deliver ticket to COR at jobsite during batch delivery.

1.05 QUALIFICATIONS

A. References for foreman in charge of CLSM placement:

1. Minimum three projects within last 3 years.

PART 2 PRODUCTS

2.01 BACKFILL FOR PIPE TRENCHES

A. Refer to Sections 31 23 22 – Pipe Trench Earthwork and 31 23 02 – Compacting Earth Materials.

2.02 STEEL PIPE

A. Refer to Section 35 21 95 – Steel Manifold Piping.

2.03 CEMENTITIOUS MATERIALS

A. Cementitious Materials Options: Specified portland cement plus 20 to 80 percent by weight of total cementitious (cement plus pozzolan) specified pozzolan, in accordance with ACI 318.

B. Portland Cement: Meet requirements listed in Section 03 30 00 – Cast-In-Place Concrete.

C. Pozzolan: Meet requirements listed in Section 03 30 00 – Cast-In-Place Concrete.

D. Blended hydraulic cement: Meet requirements listed in Section 03 30 00 – Cast-In-Place Concrete.

2.04 WATER

A. ASTM C1602.

2.05 AGGREGATE

A. Native material is not allowed.

B. Aggregate:

1. ASTM C33.

2. Maximum particle size: 1-1/2-inches or 1/8 of open distance between conduit and trench wall, whichever is less.

2.06 MIX

A. Mixture of Aggregate, Cementitious Materials, Water, and Admixtures:

1. Cementitious material content: Percent by dry weight of aggregate to obtain specified compressive strength.

2. Make trial mixes prior to placing CLSM to determine mixture adequacy:

a. Determine compressive strength in accordance with ASTM D4832.

b. Determine slump in accordance with ASTM C143.

B. Water Content: Not to exceed that required to provide mix that will flow and can be pumped.

C. Seven-day Compressive Strength, ASTM D4832: Not less than 50 pounds per square inch (lbs/in^2) and not more than 150 lbs/in^2.

D. Consistency:

1. Except, when a stiffer mix required to prevent CLSM from flowing down trenches on a steep slope:

a. Slump, ASTM C143: 8- to 10-inches.

2.07 CLSM TEMPERATURE

A. CLSM Temperature at time of placement: 50 to 85 degrees F (10 to 30 degrees C).

PART 3 EXECUTION

3.01 BATCHING EQUIPMENT

A. Manufacture and deliver in accordance with ASTM C94. In addition to requirements of ASTM C94, use water meter approved by COR to measure and record mix water for each batch.

3.02 PREPARATION

A. Place pipe on soil pads or other approved material such as extruded polystyrene foam insulation or equal.

B. Soil pads shall maintain horizontal and vertical alignment during backfilling operations.

C. Do not create point loads on pipe: Soil pads shall have lower compressible strength than surrounding CLSM.

3.03 PLACING

A. Notify COR at least 24-hours before batching CLSM. Include quantity of CLSM required for each daily placement. Unless inspection is waived, perform batching in presence of Government inspector.

B. Do not place CLSM during rain, snow, hail, or if excessive moisture is expected during placement.

C. Do not mix or place CLSM when ambient temperature is below 40 degrees F. Except when, ambient temperature is 35 degrees F and rising and daily high is predicted to be above 40 degrees F, CLSM may be placed, if approved by COR.

1. CLSM may be placed when ambient temperatures are below 40 degrees F if a Cold Weather Protection Plan has been approved prior to placement in cold weather.

D. Place CLSM to lines, grade, and dimensions shown on drawings:

1. Initially, place CLSM from one side of pipe. Where necessary, rod or vibrate CLSM so that CLSM flows under pipe and appears on other side.
2. Add CLSM evenly to both sides of pipe and rod or vibrate until CLSM completely fills space between pipe and trench.
3. Do not disturb pipe trench or allow foreign material to become mixed with CLSM.

E. Restrain pipe to prevent flotation during placement of CLSM per approved plan. Do not point load pipe.

F. Do not place backfill material over CLSM until CLSM has reached initial set:

1. As determined by ASTM D6024 (ball drop test) or ASTM D1558 in presence of COR.
2. Do not place backfill over pipe until CLSM has compressive strength of 50 lbs/in^2 or greater.

G. Discharge CLSM within 2 hours of batching.

3.04 CONTRACTOR FIELD QUALITY TESTING

A. Testing:

1. Independent testing laboratory shall perform sampling, testing, and reporting:
   a. Independent testing laboratory shall meet requirements specified in Section 01 46 20 – Testing Agency Services.

b. Government may inspect testing laboratory facilities. COR will notify testing facilities 72-hours before inspection. Periodic inspection may be required by COR.

2. Obtain samples and test to determine compressive strength in accordance with ASTM D4832 and slump in accordance with ASTM C143:

a. Testing frequency:

1) At least once for each shift when placing CLSM.

2) Once every 100 cubic yards or,

3) If consistency of materials change.

b. Acceptance Criteria:

1) 7-day compressive strength, ASTM D4832:

a) Not less than 50 lb/in^2 and not more than 150 lb/in^2.

b) 70 percent of test cylinders shall exceed 75 lb/in^2.

c. Make adjustments to mixture to comply with strength requirements.

d. Stop work if specified requirements are not met.

3.05 FIELD QUALITY ASSURANCE

A. Government may obtain samples and test for compressive strength in accordance with ASTM D4832 and slump in accordance with ASTM C143.

3.06 PROTECTION

A. When subsequent lifts of CLSM are to be placed, maintain surface of CLSM in moist condition by use of tarps or water mist until subsequent lift of CLSM is placed.

B. If backfill will not be placed over CLSM within 8-hours, place 6-inch minimum cover of moist backfill over CLSM. Maintain moisture in 6-inch soil cover until additional backfill is placed.

C. If ambient temperature is forecast to be 50 degrees F or less within 8-hours of placing CLSM, place 12-inch minimum additional cover of loose backfill over 6-inch moist backfill cover before end of workday.

1. Do not allow CLSM to freeze.

END OF SECTION

SECTION 31 31 30
SOIL-APPLIED HERBICIDE

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in the Price Schedules for other items of work.

1.02 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals.

B. RSN 31 31 30-1, Use Plan.

C. RSN 31 31 30-2, Applicator Certification.

1.03 QUALIFICATIONS

A. Applicators: Certified by appropriate Federal, State, Tribal, or local agencies, as required.

1.04 REGULATORY REQUIREMENTS

A. Comply with label directions, and applicable EPA, Federal, State, Tribal, or local laws and regulations.

PART 2 PRODUCTS

2.01 HERBICIDES

A. Label: EPA (Environmental Protection Agency) pesticide registration number.

B. Provide soil-applied herbicide conforming to one of the following:

1. Atrazine:
    a. Composed of a finely divided, wettable powder containing a minimum of 80 percent chloro-4-(ethylamino)-6-(isopropylamino)-s-triazine and related compounds.
    b. Mix: Minimum of 2 gallons of water per pound of dry product.
2. Bromacil:
    a. Composed of a finely divided, wettable powder containing a minimum of 80 percent bromo-3-sec-butyl-6-methyluracil.
    b. Mix: Minimum of 2 gallons of water per pound of dry product.

3. Prometon:

   a. Composed of a liquid emulsifiable solution containing a minimum of 25 percent 2-methoxy-4, 6-bis isopropylamino-s-triazine.

   b. Mix with water:

      1) For small areas, add sufficient water to provide thorough and uniform coverage.

      2) For large areas, add concentrate to 100 gallons of water per acre.

4. Commercial mixtures meeting requirements may be used.

PART 3 EXECUTION

3.01 PREPARATION

A. Apply herbicide after gravel surfacing has been placed.

3.02 APPLICATION

A. Apply herbicide to gravel surfacing placed under this section as directed by COR.

1. Select one herbicide from those specified.

2. Apply soil-applied herbicide at following rate:

   a. Atrazine: Equivalent of 0.035 pound of dry product per 100 square feet (about 15 pounds per acre).

   b. Bromacil: Equivalent of 0.035 pound of dry product per 100 square feet (about 15 pounds per acre).

   c. Prometone: Equivalent of 0.017 gallon of herbicide concentrate per 100 square feet (about 7 1/2-gallons per acre).

B. Apply herbicide only with approval of COR.

C. Apply uniformly.

D. During application, continuously agitate suspension.

END OF SECTION

SECTION 31 37 00
RIPRAP

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Riprap:

1. Measurement: Volume measured to outlines placed and to thickness shown on drawings or as directed by COR.
2. Payment: Cubic yard price offered in Price Schedules.
   a. Includes:
      1) Rock spalls and gravel to fill voids in riprap.
      2) Geotextile.

1.02 REFERENCE STANDARDS

A. ASTM International (ASTM)

1.	ASTM C88/C88M-24	Soundness of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate
2.	ASTM C97/C97M-25	Absorption and Bulk Specific Gravity of Dimension Stone
3.	ASTM C535-16(2024)	Resistance to Degradation of Large-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine

1.03 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals.

B. RSN 31 37 00-1, Approval Data and Test Results:

1. Name and location of commercial rock source.
2. Gradation of rock used for riprap.
3. Certified laboratory test results to demonstrate rock meets specified material requirements.

PART 2 PRODUCTS

2.01 ROCK

A. Rock from approved sources shall be excavated, selected, and processed to meet specified quality and grading requirements at the time rock is installed.

B. Rock shall consist of hard, dense, durable stone, angular in shape and resistant to weathering to ensure permanence in structure and climate in which it is used.

C. Limestone or crushed/recycled concrete material sources are not allowed.

D. Riprap physical properties:

1. Specific Gravity, ASTM C97, Minimum: 2.50.
2. Absorption, ASTM C97, Maximum: 2 percent.
3. Loss, Sulfate Soundness, ASTM C88, Maximum: 10 percent.
4. Percent wear of not more than 60 as determined by ASTM C535.

E. Rounded stone or boulders shall not be used as riprap material.

F. Individual rock shape:

1. Maximum dimension not greater than 3 times minimum dimension.

G. Conform to gradation requirements given in Table 31 37 00A – Riprap Gradation Requirements. Riprap type and locations as shown on drawings. Control of gradation will be by visual inspection.

Table 31 37 00A – Riprap Gradation Requirements

Riprap Type	Stone Size D_{50} (Inches)	Typical Stone Dimensions (Inches)	Percent of Material Smaller than Typical Stone Size by Weight
Type L	9	15 12 9 3	100 50 -70 35 - 50 2 - 10

2.02 GEOTEXTILE

A. Geotextile material in accordance with Section 32 15 10 – Gravel Surfacing.

B. Place geotextile as shown on drawings.

PART 3 EXECUTION

3.01 AREAS TO RECEIVE RIPRAP

A. Place riprap to outlines and thicknesses as shown on drawings.

3.02 PLACING

A. Bed riprap on geotextile as shown on drawings.

B. Rock in riprap need not be compacted.

C. Begin riprap placement at toe of slope and proceed up slope.

D. Rock drop height, maximum: 1 foot.

E. Dump/drop rocks and smooth by moving rocks into position so placed material is stable.

F. Place rocks so larger rocks are evenly distributed, and small rock fragments fill spaces. Include rock spalls or gravel in an amount to fill voids in riprap, but not more than an amount to be determined by COR.

G. Before placing riprap, demonstrate that placing technique will not damage geotextile and concrete features:

1. If demonstration does not show riprap can be installed without damaging geotextile materials and concrete features, modify riprap placing technique.
2. Possible modifications to placing technique include reducing drop height, installing layer of sacrificial geotextile, hand placing rocks, or installing gravel cushion.

H. Leave no unreasonably large unfilled spaces within riprap.

END OF SECTION

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SECTION 32 15 10
GRAVEL SURFACING

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Gravel Surfacing:

1. Measurement: Volume of in-place compacted material to outlines of area covered and to specified thickness.
2. Payment: Cubic yard price offered in Price Schedules.
   a. Includes furnishing and installing geotextile.
   b. Includes removal and reinstallation of gravel surfacing for areas disturbed by construction activities at existing roads.

1.02 REFERENCE STANDARDS

A. American Association of State Highway Transportation (AASHTO):

1.	AASHTO T 26-08	Test for Quality of Water to Be Used in Concrete

B. ASTM International (ASTM)

1.	ASTM D4632/D4632M-15a (2023) Geotextiles	Grab Breaking Load and Elongation of
2.	ASTM D4751-21a	Determining Apparent Opening Size of a Geotextile
3.	ASTM D6241-22a	Static Puncture Strength of Geotextiles and Geotextile-related Products Using a 50 mm Probe

C. New Mexico State Department of Transportation (NMDOT)

4.	NMDOT SS 2019	Standard Specifications for Highway and Bridge Construction

1.03 GENERAL

A. Place gravel surfacing at the following locations as shown on drawings:

1. Turnout Building Yard(s).
2. Access Road(s).
3. Restoring existing gravel roads after pipe installation.

1.04 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals.

B. RSN 32 15 10-1, Gravel Certification:

1. Gravel gradation range with percent passing by weight of respective sieve sizes.
2. Test results from representative 300-lb (150 kg) sample.

C. RSN 32 15 10-2, Geotextile Installation:

1. Manufacturer's recommendation.

D. RSN 32 15 10-3, Geotextile Certification:

1. Geotextile meets specified requirements.

PART 2 PRODUCTS

2.01 GRAVEL

A. NMDOT SS Table 303.2.2:2 – Type II Base Course Gradation Band.

B. NMDOT SS Table 303.2.2:3 – Base Course Physical Properties.

C. Limestone or crushed/recycled concrete material sources are not allowed.

2.02 WATER

A. In accordance with AASHTO T 26.

B. Do not use residual water, wash water, or recycled water generated by Equipment, mixer trucks, or central mixers.

2.03 GEOTEXTILE

A. 170N non-woven geotextile, manufactured by Mirafi, www.tencate.com or equal with the following essential characteristics:

1. Grab Tensile Strength (ASTM D4632): 180 pounds (lbs).
2. Elongation (ASTM D4632): 50 percent.
3. Puncture Strength (ASTM D6241): 400 lbs.
4. Apparent Opening Size (ASTM D4751): 70 US Sieve size.

PART 3 EXECUTION

3.01 PREPARATION AND EXCAVATION

A. Clear surface of loose and deleterious material.

B. Ensure top 6 inches of roadbed is free from depressions and soft spots.

C. Excavate limits shown on the drawings or as determined by COR.

1. Excavate in dry conditions.
2. Remove unsuitable or unstable material below finished subgrade.

D. Finish subgrade to smooth and uniform surface.

E. Obtain Government approval of subgrade prior to placing gravel surfacing.

3.02 GEOTEXTILE

A. As directed at locations shown on drawings.

B. In accordance with manufacturer's recommendations.

C. Do not tear geotextile during placement and compaction of gravel surfacing.

3.03 PLACING AND COMPACTING

A. At locations shown on drawings.

B. Place in layers and compact in accordance with Section 31 23 02 – Compacting Earth Materials.

C. Route hauling equipment uniformly over full width of surface to minimize rutting or uneven compaction.

3.04 DISPOSAL

A. Dispose of excess and unsuitable materials in accordance with Section 31 23 39 – Disposal of Excavated Materials.

END OF SECTION

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SECTION 32 31 19
STEEL ORNAMENTAL FENCE

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Ornamental Fence:

1. Measurement: Sloping length along top rail from centerline to centerline of posts, including gates.
2. Payment: Linear foot price offered in the Price Schedule.

1.02 REFERENCE STANDARDS

A. ASTM International (ASTM)

1.	ASTM A653/A653M-25a	Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
2.	ASTM F2408-16(2023)	Ornamental Fences Employing Galvanized Steel Tubular Pickets

1.03 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 - Submittals.

B. RSN 32 31 19-1, Certifications:

1. Manufacturers' certification that fence materials, fittings, and accessories meet specified requirements.
2. Include manufacturers' names and product designations and specified product standards in the certification.

C. RSN 32 31 19-2, Installation Plans

1. Drawings indicating the following:
   a. Locations and configurations of ornamental fence, including gates, to be installed based on verified field dimensions. Show location of footings in relation to yard pipe thrust blocks.
   b. Include details for fence and fence and gate post footings.

PART 2 PRODUCTS

2.01 ORNAMENTAL FENCE SYSTEM

A. Montage Commercial – Invincible Style as manufactured by Ameristar Perimeter Security, 1555 N Mingo Rd, Tulsa, OK 74116; Centurian Ornamental Steel Fence System – 3-Rail Enforcer Series as manufactured by Stephen's Pipe and Steel LLC, 2224 E Hwy 619, Russel Springs, KY 42642; or equal, having the following essential characteristics:

1. All furnished ornamental fence system components shall be from one manufacturer.
2. Fence Panels:
    a. 3/4-inch x 16 gauge pickets, minimum.
    b. Pickets spaced such that air spacing does not exceed 4".
    c. Pickets above top rail are pointed and angle outward.
    d. 2.5-inch x 14 gauge square posts.
    e. 1.4375-inch w x 1.5-inch h x 14 gauge rails.
    f. Minimum 3-rails.
    g. Fence panel height: 8-feet from grade.
3. Vehicle Gates:
    a. Vehicle gates panels to match fence panels.
    b. 3-inch x 12-gauge square gate posts, minimum.
    c. Dimensions are as shown on drawings.
4. Steel material for fence panels, gates, posts, and hardware shall conform to the requirements of ASTM A653, with a minimum yield strength of 45,000 psi and minimum coating designation G-60.

2.02 CONCRETE

A. Manufacture and delivery: In accordance with Section 03 30 00 - Cast-In-Place Concrete.

2.03 FABRICATION

A. The manufactured fence system shall be capable of meeting the vertical load, horizontal load, and infill performance requirements for industrial weight fences under ASTM F2408.

B. Coating applied to each component prior to assembly to ensure complete coverage and shall meet or exceed requirements show in in ASTM F2408 Table 3.

PART 3 EXECUTION

3.01 PREPARATION

A. Clear and remove trees, brush, ground surface irregularities, and other obstacles which interfere with proper erection of fence in advance of starting fencing work.

B. Dispose of removed material in accordance with Section 31 11 00 - Clearing and Grubbing.

3.02 INSTALLATION

A. Install fence and gates as shown on drawings.

B. Install fence and gates in accordance with manufacturer recommendations.

C. Install fence and gate post footings per manufacturers' recommendations. Ensure minimum footing depths are fully within compacted backfill.

3.03 ELECTRICAL GROUNDING

A. Bond all metallic fence components to the site grounding electrode system in accordance with Section 26 05 20 – Grounding and Bonding.

3.04 REPAIR

A. Repair damage to coatings per manufacturers' recommendations.

3.05 TESTING AND ADJUSTING

A. Test and adjust gates for proper operation after installation.

END OF SECTION

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SECTION 32 92 20
SEEDING

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Seeding:

1. Measurement: Area of surface disturbance seeded, or as determined by COR.
2. Payment: Acre price offered in the Price Schedule.

1.02 DEFINITIONS

A. Pure live seed content: Weight of seed times percent purity times percent germination.

1.03 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 32 92 20-1, Seeding Plan:

1. Method and equipment used for seeding.
2. Date for seeding.
3. Mulch type and application equipment.
4. Qualifying experience of person responsible for supervision of seeding, and approval.
5. Names, addresses, and telephone numbers of references.
6. Name and address of seed suppliers.

C. RSN 32 92 20-2, Certifications:

1. Origin of seed.
2. Percent purity and germination.
3. Prohibited and restricted weed seed content.

1.04 DELIVERY STORAGE AND HANDLING

A. Seed Containers:

1. Sealed.
2. Labeled:
   a. Identify seed origin on label.

1) Intrastate shipping: In accordance with New Mexico State Seed Laws and Regulations.

2) Interstate shipping: In accordance with U.S. Department of Agriculture Rules and Regulations under the Federal Seed Act.

1.05 AMBIENT CONDITIONS

A. Do not seed when ambient temperature is below 38 degrees F without approval of COR.

B. Do not seed when ground is snow covered or frozen.

C. Do not seed, mulch, or hydroseed when wind velocities prevent uniform application of materials or would drift materials.

PART 2 PRODUCTS

2.01 SEED

A. Weed seeds classified by State Seed Department:

1. Prohibited noxious weeds: None allowed.
2. Restricted noxious weeds: 0.5 percent maximum, by weight.

B. Seed Mixture:

1. Refer to Table 32 92 20A – Seed Mixture below:

Table 32 90 20A-Seed Mixture

Common Name	Scientific Name	Variety	Season	Form	Seeding Rate* (Pounds pure live seed per acre)
Sterile triticale	*Triticum aestivum X Secale cereale 'Quickguard'*	VNS	-		2.0
Fourwing saltbush	*Atriplex confertifolia*	VNS	Cool	Shrub	3.0
Shadscale	*Krascheninnikovia lanata*	VNS	Cool	Subshrub	2.0
Indian ricegrass	*Achnatherum hymenoides*	Paloma or Rimrock	Cool	Bunch	3.0
Western wheatgrass	*Pascopyrum smithii*	Arriba	Cool	Sod	2.0
Galleta	*Pleuraphis jamesii*	Viva florets	Warm	Bunch/Sod	2.0

Purple threeawn	*Aristida purpurea*	VNS	Warm	Bunch	2.0
Sand dropseed	*Sporobolus cryptandrus*	VNS	Warm	Bunch	0.25
Alkali sacaton	*Sporobolus airoides*	VNS	Warm	Bunch	0.25
Scarlet globemallow	*Sphaeralcea coccinea*	VNS	Warm	Forb	0.25
Narrowleaf penstemon	*Penstemon angustifolius*	VNS	Cool	Forb	0.25
Rocky mountain beeplant	*Cleome serrulate*	VNS	Warm	Forb	0.25
Hairy false goldenaster	*Heterotheca villosa*	VNS	Warm	Forb	0.25
*Based on 60 pure live seeds (PLS) per square foot, drill seeded. Double this rate (120 PLS per square foot) if broadcast or hydroseeded					

2. Purity, minimum: 85 percent.
3. Germination, minimum: 85 percent.
    a. Germination test: Less than 1-year old at time of seeding.

2.02 STRAW MULCH

A. Wheat or barley straw.

B. Free of mold or other evidence of decomposition.

C. Free from weed seed.

2.03 HYDROMULCH

A. Mat-Fiber Plus, manufactured by Mat Inc.,12402 Hwy 2, Floodwood, MN 55736, 1 888 477-3028; Spray Mulch X-80 manufactured by Pacific Wood Fibers, PO Box 2109, Redmond WA 98052; or equal, having the following essential characteristics:
    1. Wood cellulose fiber.
    2. No germination or growth inhibiting factors.
    3. Dyed appropriate color to allow visual metering of application.
    4. Evenly dispersed and suspended when agitated in water.
    5. Forms blotter like ground cover that readily absorbs water and allows infiltration to underlying soil.

2.04 TACKIFIER

A. Mixture of at least three specially blended compatible hydrocolloids:

1. One hydrocolloid will act as a slippery agent during suspension.
2. Will form loose, long-chain-like film on drying.
3. No growth or germination inhibiting factors.
4. Hydrates and disperses in circulating water to form homogeneous slurry.
5. Equilibrium air dry moisture content at time of manufacture of 8 percent, plus or minus 2 percent.
6. Minimum water holding capacity: 6 1/2 times weight of dry material.

PART 3 EXECUTION

3.01 SEEDBED PREPARATION

A. Complete prior to seeding and mulching or hydromulching.

B. Rip, till, disk compacted areas to create suitable medium for reseeding prior to spreading topsoil.

C. Respread topsoil and scarify or harrow and rake topsoil to minimum depth of 3 inches.

D. Remove stiff clods, lumps, roots, litter, stones, and other foreign material greater than 6-inches in size from surface. Dispose of removed materials in accordance with Section 01 74 00 – Cleaning and Waste Management.

E. Fill or smooth topsoil surface to remove rills, gullies, and depressions.

F. Protect prepared topsoil surfaces from erosion and washouts. Repair damaged surfaces as required.

3.02 SEEDING APPLICATION

A. Seeding in Project Areas:

1. Apply seed and mulch by one of the following methods:
   a. Broadcast seeding followed by mulching or hydromulching.
   b. Drill seeding followed by mulching or hydromulching.
   c. Hydroseeding.
2. Apply seed mixture at rate specified in applicable Table 32 92 20A – Seed Mixture. Double rate of seed application if broadcast seeding.
3. Seed between July 15 and November 15.

B. Seeding Under Erosion Control Blanket:

1. Prior to placing erosion control blankets apply seed mixture at rate specified in Table 32 92 20A. Double rate of seed application if broadcast seeding.

2. Seed between July 15 and November 15.

3.03 BROADCAST SEEDING

A. Broadcast seed only in areas not accessible for drilling or hydroseeding.

B. Mechanical Broadcasting:

1. Equipment:

a. Centrifugal type.

2. Designed and regulated to apply seed uniformly at proper rate per acre.

C. Hand Broadcasting:

1. By hand broadcaster.

2. By hand.

3. Uniformly applied.

D. Cover seed with soil to depth of 1/4-inch to 1/2-inch immediately after broadcasting:

1. Use hand rake or float.

2. Do not use log chain or similar devices.

3.04 DRILLING SEED

A. Drill:

1. Disc-type seed drill.

2. Space rows 1 foot or less apart.

3. Perform on contour, perpendicular to slopes to minimize runoff, rilling, and erosion.

B. Regulate drill to uniformly distribute seed at rate specified in Table 32 92 20A – Seed Mixture.

3.05 MULCHING

A. Certified weed-free straw or native grass hay.

B. Spread and crimp within 24 hours of seeding.

C. Rate: 1 to 2 tons/acre uniformly spread.

D. Anchor with threader.

1. Operate crosswise to slope.

2. Depth: 3 to 4 inches.

3. Interval: 6 to 12 inches across slope.

3.06 HYDROSEEDING AND HYDROMULCHING

A. Slurry:

1. Mix to keep homogenous.
2. Ingredients.
    a. Water.
    b. Tackifier.
    c. Seed.
    d. Wood cellulose fiber mulch:
        1) Rate: 4,000 pounds per acre at 10 percent moisture content.
        2) Add to water slurry after seed.
3. Maximum time between batching slurry and application: 1 hour.

B. Spray apply slurry mix uniformly.

C. Use mulch coloring as metering agent.

END OF SECTION

SECTION 33 11 12
STEEL LINE PIPE

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Line Pipe for BBN9 Turnout:

1. Payment: Lump sum price offered in Price Schedule 4.

a. Includes pipes, valves, and associated appurtenances shown on Drawings 1695-D-62548.

b. Includes cost of the following:

1) Pipe trench excavation in accordance with Section 31 23 22 – Pipe Trench Earthwork.

2) Furnishing and placing materials, including but not limited to the following:

a) Fittings, bedding, embedment, utility markers, concrete, reinforcing steel, blocking and collars, not included elsewhere.

b) Laying pipe.

B. Line Pipe for Coyote Canyon

1. Payment: Lump sum price offered in Price Schedule 5.

a. Includes cost of the following:

1) Pipe trench excavation in accordance with Section 31 23 22 – Pipe Trench Earthwork.

2) Furnishing and installing Coyote Canyon Turnout as shown on Drawing 1695-D-62550, including but not limited to the following:

a) Fittings, bedding, embedment, utility markers, concrete, reinforcing steel, blocking and collars, not included elsewhere.

b) Laying pipe.

C. Does not include pipe specified in Section 35 21 95 – Steel Manifold Piping.

1.02 PROJECT CONDITIONS

A. Refer to Drawing 1695-D-62548, 1695-D-62549, 1695-D-62550, 1695-D-62551, 1695-D-62552, 1695-D-62553, and 1695-D-62554.

1.03 REFERENCE STANDARDS

A. ASTM International (ASTM)

1.	ASTM A139/A139M-16	Electric-Fusion (Arc) - Welded Steel Pipe (NPS 4 and Over)
2.	ASTM C150/C150M-17	Portland Cement
3.	ASTM A283/A283M-18	Low and Intermediate Tensile Strength Carbon Steel Plates
4.	ASTM A1011/A1011M-15	Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Allow and High-Strength Low-Allow with Improved Formability
5.	ASTM A1018/A1018M-16a5	Steel, Sheet and Strip, Heavy Thickness Coils, Hot Rolled, Carbon, Structural, High-Strength Low-Alloy, Columbium or Vanadium, and High-Strength Low-Alloy with Improved Formability
6.	ASTM E165/E165M-12	Liquid Penetrant Examination for General Industry
7.	ASTM E709/E709M-15	Standard Guide for Magnetic Particle Testing

B. American Welding Society, Inc. (AWS)

1.	AWS D1.1/D1.1M-15	Structural Welding Code – Steel

C. American Water Works Association (AWWA)

1.	AWWA C200-12	Steel Water Pipe - 6 Inch (150mm) and Larger
2.	AWWA C205-12	Cement-Mortar Protective Lining and Coating for Steel Water Pipe - 4 Inch (100mm) and Larger - Shop Applied
3.	AWWA C206-17	Field Welding of Steel Water Pipe
4.	AWWA C208-12	Dimensions for Fabricated Steel Water Pipe Fittings
5.	AWWA C227-11	Bolted, Split-Sleeve Restrained and Non-restrained Couplings for Plain-End Pipe
6.	AWWA C604-11	Installation of Buried Steel Water Pipe 4 Inch (100 mm) and Larger

7.	AWWA M11-04	Steel Pipe: A Guide for Design and Installation, Fourth Edition

1.04 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals.

B. RSN 33 11 12-1, Shop Drawings:

1. Show pipe and fitting fabrication details.
2. Show exact dimensions of joints, position, marking of pipe sections and proposed joint restraint.
3. Show exact dimensions of welding lead holes and repair method.
4. Show details of factory epoxy lining of joint cement mortar hold back area if alternate epoxy joint recess lining is proposed.
5. Include centerline and invert stationing and elevations at horizontal and vertical changes in alignment, and subgrade elevation for each pipe segment and fitting.
6. Include pipe thickness or pressure class of pipe segment or fitting.

C. RSN 33 11 12-2, Ventilation Plan:

1. Method for ventilating inside of pipe when placing mortar lining for joints.

D. RSN 33 11 12-3, Field Welder Qualifications

1. Welder certifications

E. RSN 33 11 12-4, Filling and Testing Plan:

1. Proposed rate, time, and procedure for:
   a. Cleaning.
   b. Filling.
   c. Field and pressure testing.
   d. Draining pipeline.
2. Method for disposing of water drained from pipeline to enable repair of leaks.
3. Leak testing equipment and data sampling plan.

F. RSN 33 11 12-5, Flotation Prevention Plan:

1. Show location and describe method of preventing pipe from floating.
2. Provide supporting calculations.

G. RSN 33 11 12-6, Pipe Deflection Measurements.

1.05 DELIVERY, STORAGE, AND HANDLING

A. Ship pipe after receiving COR approval as per Section 2.03 – Source Quality Assurance.

B. Prevent damage to pipe and fittings during loading, transporting, unloading, storing, and laying.

C. Transport metallic pipe and fittings on padded bolsters curved to fit outside of pipes. Use heavy padding under ties.

D. Tightly close open ends of shop-applied, cement-mortar-lined pipe with plastic wrap for protection of cement-mortar lining during shipment:

1. Plastic Wrap:

a. At least 2 thicknesses of 6-mil sheet polyethylene plastic.

b. Remain on pipe until installation.

E. Support and store pipe above ground surface. Do not allow bells and spigots to contact each other or the ground.

F. Government will inspect pipe once it is delivered.

G. Replace or repair, as approved by COR, any pipe that is damaged during shipment, storage, or installation at Contractor's expense. Include linings and coatings.

1.06 QUALIFICATIONS

A. Pipe manufacturers shall be certified in accordance with ISO:9001 or similar nationally certified program approved by COR.

B. Line pipe and specials manufacturers:

1. 10-years successful experience producing products as specified.

2. Provide references for at least 3 completed projects with similar pipe diameters (within 12-inches) and pressures.

C. Pipe Layers: Skilled and experienced in laying pipe with joints as needed for each material type of pipe used. Minimum 2-years installing pipe with similar pipe diameters (within 12-inches), pipe material, and pressures:

1. In the absence of specific pipe material experience, substitution of onsite pipe manufacturer training and quality control is allowed:

a. Provide manufacturer's representative certified in pipeline installation, including all applicable joints, for training pipe layers and pipe foreman prior to installation.

b. In lieu of specific pipe material installation experience, pipe layers and pipe foreman shall still have experience with installing similar pipe diameters and pressures.

D. Qualify welding procedures and welders in accordance with code under which welding is specified.

PART 2 PRODUCTS

2.01 STEEL LINE PIPE

A. Pipe:

1. Electric fusion (arc) welded helical-seam steel pipe: ASTM A139, Grade C, D, or E.
2. Fabricated in accordance with AWWA C200:
   a. Except:
      1) Steel plate: ASTM A283, Grade C or D.
      2) Steel Sheet or coil: ASTM A1011, designation SS, Grade 40, 45, or 50; or ASTM A1018, designation SS, Grade 40 or designation HSLAS-F, Grade 50.

B. Inside diameter measured to inside of lining shall not be less than nominal diameter shown on drawings.

C. Minimum Steel Wall Thickness:

1. Greater thickness from the following requirements:
   a. Steel Wall Thickness for Handling:
      1) Equal to inside diameter (inches) of pipe steel shell divided by 240 for mortar lined and flexible coated steel pipe.
   b. Steel Wall Thickness for Internal Pressure:
      1) Hoop stress of pipe shell. Not to exceed:
         a) 50 percent yield strength as defined by steel grade of steel using the Barlow formula. With pressure defined by Hydraulic Grade Line as shown on pipeline plan and profile drawings.
         b) 75 percent yield strength as defined by grade of steel using the Barlow formula. With pressure defined by pipe head class as shown on pipeline plan and profile drawings.
      2) Barlow formula as defined by AWWA M11 Equation 4-1.

D. Steel shall be fully killed and conform to fine grain practice.

E. Rubber gaskets: In accordance with AWWA C200.

F. Materials and chemicals that may come into contact with drinking water shall be certified by NSF 61.

2.02 FITTINGS

A. Steel: ASTM A283, Grade C or D or ASTM A1011, designation SS, Grade 40, 45, or 50; or ASTM A1018, designation SS, Grade 40 or designation HSLAS-F, Grade 50.

B. Minimum steel wall thickness and fitting dimensions: In accordance with AWWA C200, AWWA C208 and AWWA M-11.

C. Welding: In accordance with AWS D1.1.

D. Lifting eyes and other handling devices: Made part of fitting before lining and coating are applied.

E. Bolts and Nuts: In accordance with ASME B18.2.1 and FS FF-N-836.

F. Coating and lining in accordance with Section 09 96 20 – Coatings.

G. Cement-Mortar Lining for Specials and Bends Larger than 24-inches:

1. In accordance with AWWA C205.
2. Apply lining with a resultant surface finish, including joints, equal to finish of adjacent pipe.

H. Joints between fittings and ductile iron pipe:

1. Rubber gasket joints.
2. Joint dimensions and tolerances: Same as pipe manufacturer's joint design.

I. Miter Bends: Fabricate steel bends in accordance with AWWA C208, AWWA M-11 and as shown on drawings.

J. Pipe Couplings:

1. Coupling types as shown on drawings.
2. Suitable for line pipe materials used.
3. Suitable for cold water pressures of pipe head classes on drawings.
4. Refer to Section 35 21 95 – Steel Manifold Piping for coupling requirements.

K. Closure Section Joints

1. Restraint requirements shown on drawings.
2. Field welded butt strap joints.

3. Bolted, Split-Sleeve Restrained and Nonrestrained Couplings for Plain – End:
   a. In accordance with AWWA C227.
   b. Refer to Section 35 21 95 – Metal Piping for Line Pipe for coupling requirements.

L. Tees:
1. Tees for air valves, blowoffs, manholes, and temporary construction line as shown on drawings.
2. Tee length, minimum: As shown on drawings, required for blocking, or in accordance with AWWA C208 whichever is greater

2.03 LININGS AND COATINGS

A. In accordance with Section 09 96 20 – Coatings.

B. Do not field apply mortar lining without OGR approval.

2.04 CONCRETE

A. Concrete in Thrust Blocks and Collars: Refer to Section 03 30 00 – Cast-in-Place Concrete.

2.05 FLANGES:

A. In accordance with AWWA C207.

B. Class: Working pressure meets or exceeds head class of attached pipe

2.06 FLANGE GASKETS:

A. BLUE-GARD Style 3000 manufactured by Garlock Sealing Technologies, 1666 Division Street, Palmyra NY 14522, or equal, having the following essential characteristics:
1. Sized in accordance with AWWA C207, Table 1.
2. For AWWA C207 flanged joints.
3. Compressed, Non-Asbestos (CNA) Gasketing with Aramid Fibers and a NGR Binder.
4. For potable cold-water service.

2.07 PIPE ACCESSORIES

A. Sectionalizing Valve, Air Valve, and Blowoff Assemblies: In accordance with Section 35 22 14 – Valves and Equipment.

B. Buried Manhole Assemblies:
1. In accordance with Section 03 48 00 – Precast Concrete Structures.

2. Cement-mortar lining: AWWA C205, in accordance with shop applied mortar for lining of specials, except:
   a. Reinforcement required for pipe larger than 24-inch diameter.
   b. Cement: ASTM C150, Type V.
3. Required at locations shown on drawings.

C. Warning Tape:
1. Polyethylene tape.
2. Thickness: 4 mils.
3. Magnetically detectable.
4. Width, minimum: 6-inches.
5. Color: Blue with silver lettering.
6. Legend repeated at least every 3-feet: "CAUTION: WATERLINE BURIED BELOW - FOR MORE INFORMATION CONTACT THE NAVAJO TRIBAL UTILITY AUTHORITY".

2.08 CONTRACTOR SOURCE QUALITY TESTING

A. Pipe and Fittings:
1. Hydrostatic Test:
   a. Perform shop hydrostatic test on pipe which stresses steel to 75 percent of minimum yield point of the steel.
   b. Hold pressure long enough to allow thorough inspection of welded joints.
   c. Repair leaks by rewelding and retesting joints.
   d. Test sections prior to forming bell and spigot joints.
   e. Repair defects and retest section before applying lining and coating.
   f. Government may witness hydrostatic testing and calibration of pressure gauges. Notify Government 30 days prior to performing hydrostatic test.
2. Fittings fabricated from tested steel pipe do not require hydrostatic testing if welds are tested.
   a. Weld test: Dye penetrant test in accordance with ASTM E165 or Magnetic Particle in accordance with ASTM E709.
   b. Weld defect repair. Chip, flame gouge, or grind to sound metal; reweld; and test repaired weld.

B. Joint Testing: Assemble one joint for each pipe diameter to check bell and spigot to check fit prior to coating or lining.

PART 3 EXECUTION

3.01 GENERAL

A. Perform cleaning, filling, and testing after backfill has been placed to finished grade or as approved by OGR.

B. Responsible Surveyor licensed in State of New Mexico or a certified Construction Surveyor, as defined in and in accordance with Section 01 71 20 – Surveying, shall be onsite during excavation and pipelaying.

C. Protect pipe from contamination, damage, and elements during storage on-site by covering with tarp as directed by OGR.

D. Install pipe based on these specifications or pipe manufacturer's recommendation, whichever is more stringent.

E. Isolate dissimilar metals.

3.02 INSTALLATION

A. Keep openings to installed pipe closed with watertight inflatable plugs during work stoppage, including end of work day, breaks, and work delays.

B. If pipe is flooded during construction, clear floodwater by draining and flushing with water, or other approved method, until pipe is clean.

C. Lubricant:

1. Keep clean.
2. Apply with dedicated, clean applicator brushes.
3. Apply lubricant as specified by pipe manufacturer:
    a. Use only lubricant supplied by pipe manufacturer.
    b. Lubricate spigot end of pipe as recommended by manufacturer.
    c. Do not lubricate either gasket or gasket groove in bells where gaskets are field installed.
    d. Lubricate factory-installed, non-removable-type gaskets as recommended by pipe manufacturer.

D. Coat buried manholes with cement mortar as shown on drawings.

E. Install warning tape over center of pipe at least 18-inches below ground and at least 18-inches above pipe.

3.03 LAYING PIPE

A. Excavate pipe trench in accordance with Section 31 23 22 – Pipe Trench Earthwork.

1. Grade pipe trenches to provide uniform slope along bottom of pipe.
2. At joints involving bells or collars, provide holes at joint of ample size to prevent bells or collars from coming in contact with subgrade and to allow for bedding placement.

B. Keep pipe trenches free of water during pipelaying operations.

C. Lower pipe into trench and place pipe in position such that no soil gets inside pipe and pipe is not damaged.

D. Install pipe to line and grade as shown on drawings.

E. On grades exceeding 10 percent, lay pipe uphill.

F. When pipelaying is not in progress, keep ends of pipelines closed with watertight inflatable plugs.

G. Joining Pipe:

1. In accordance with AWWA C604.
2. Verify inside joint gap meets manufacturer's tolerances as work progresses.

H. Joints for Pipe with Welded Joints:

1. In accordance with AWWA C206. Welded joints after backfill not allowed.
2. Perform dye penetrant test on welds in accordance with ASTM E165.
3. Welded Butt Joints:
   a. As required for pipe on steep slopes as shown on drawings.
4. Double Lap Weld Joints:
   a. Provide joint with full fillet welds.
   b. Air test double lap welds in accordance with AWWA C206.
   c. Required on thrust restrained joints greater than 30 degree bend deflection angles, unless butt welded joints are required.
5. Single Lap Weld Joints:
   a. May be used on thrust restrained joints with 30 degree bend deflection angles or less.
   b. Road and wash crossings unless butt or double-lap welded joints are required.

I. Connections at Thrust Blocks and Structures: As shown on drawings

J. Closure Sections:

1. Where necessary as determined by Contractor and subject to approval of OGR.
2. Ambient temperature when closure section is welded, maximum: 60 degrees F.

3. Assemble push-on or mechanical joints in accordance with applicable AWWA standard for pipe type and manufacturer's recommendations.
4. If adjustment of the position of length of pipe is required after installation, remove and relay length of pipe as new pipe.
5. Maintain pipe firmly in final position.
6. Placing Elastomeric Gasket:
    a. Clean gasket, bell, especially the groove, and spigot with rag, brush, or paper towel to remove any dirt or foreign material.
    b. Use only gaskets which are designed for and supplied with pipe.
    c. Lubricate pipe as recommended by manufacturer.
    d. Spigot Groove Method:
        1) Follow manufacturer's recommended practices for gasket installation.
        2) After placing elastomeric gasket in spigot groove, equalize elastomeric gasket cross section by inserting tool such as a large screwdriver under elastomeric gasket and moving it around periphery of pipe spigot.
    e. Coupling Groove Method: Follow manufacturer's recommended practices for gasket installation. Typically, factory installed.
7. Fit pipe unit's together, spigot to bell or coupling, and draw joints together so that bells or couplings and spigots are fully engaged with uniform contact to gasket.
8. Do not swing or "stab" joint and do not suspend pipe and swing into bell or coupling.
9. Fit pipe units together in a manner to avoid twisting or otherwise displacing or damaging elastomeric gasket.
10. After joining pipe units, insert feeler gauge between pipe bell and pipe spigot, move it around periphery of pipe to determine that position of elastomeric gasket is correct and there are no fish-mouth problems. If adjustment of position of a length of pipe is required after installation, remove and relay length of pipe as new pipe.

K. Changes in Alignment and Grade:
1. Make changes in alignment and grade with miter bends as shown on drawings.
2. Make other minor changes in alignment and grade by providing small deflections between adjacent pipe:
    a. Do not exceed manufacturer's deflection tolerances.
    b. Pipelaying diagrams shall only deflect joints no more than 50-percent of manufacturer's maximum deflection tolerances.

3. Provide restrained joint lengths upstream and downstream of bends or thrust blocks as shown on drawings. Ductile iron fittings with restrained joint lengths as limited by drawings may be used if Contractor submits revised plan and profile drawings for approval.

4. Do not encase rubber gasket joints in concrete.

5. Refer to individual pipe Sections for additional requirements. If requirements conflict, use most stringent requirement.

L. After pipelaying and joining operations are completed, clean inside of pipe and remove debris.

M. Schedule work so that at no time pipe remains in trench more than 7-days before backfill is placed to original ground surface or to other specified backfill limits shown on drawings.

3.04 THRUST RESTRAINT

A. Restrained joint couplings or concrete blocking and collars as shown on drawings. Refer to Section 03 30 00 – Cast-in-Place Concrete and in accordance with applicable sections listed in 2.01.A.

B. Plan, profile, and restraint system shown on drawings require modification if alternate methods are used.

C. Alternate methods of restraint may not be used unless submitted and approved by COR.

3.05 TOLERANCES

A. Lay pipe to lines and grades shown on drawings or established by OGR to following tolerances:

1. Total departure from established alignment and grade, maximum: 1-inch.

3.06 ADJUSTMENTS IN ALIGNMENT AND GRADE FOR PULLED JOINTS

A. Per approved shop drawings.

B. Make small minor changes in alignment and grade by providing small deflections between adjacent pipe sections as follows:

1. Pulling rubber gasket bell-and-spigot joint:

a. A total 1-inch joint deflection may be permitted by reducing normal 1/2-inch inside mortar space to 1/4-inch and pulling opposite side of pipe 3/4-inch from normal closure.

b. Provide unsymmetrical closure at welded lap joint up to maximum pullout of 1-inch.

2. Maximum deflection angle between adjacent pipe sections: Manufacture's recommendations but shall not exceed 5 degrees.

C. Lay ends of each section of steel line pipe on theoretical centerline of curve and to grade shown on drawings within laying tolerances prescribed above.

3.07 JOINT LINING AND COATING

A. Field joints for pipe with shop applied cement-mortar lining.

1. Line field joints with cement-mortar in accordance with AWWA C205.
2. Coat exterior joints as specified in Section 09 96 20 – Coatings.
3. Do not backfill field joints until coating is approved by OGR.

B. Apply shrink sleeves to field joints in accordance with Section 09 96 20 – Coatings.

3.08 BACKFILL

A. Keep internal supports in place until embedment has been placed and compacted above bottom of pipe to minimum height of 0.7 times diameter.

3.09 PIPE DIAMETER DEFLECTION

A. Allowable short-term vertical pipe diameter deflection after backfilling is complete.

1. Decrease, maximum: 2 percent of nominal pipe diameter.
2. Elongation, maximum: 3 percent of nominal pipe diameter as measured when backfill reaches pipe crown.
3. Short term is within 2 weeks after backfilling is complete.

B. Allowable long-term vertical pipe diameter deflection, at end of warranty period:

1. Decrease, maximum: 3 percent of nominal pipe diameter.

END OF SECTION

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SECTION 33 13 00
DISINFECTION OF WATER UTILITY DISTRIBUTION

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Disinfection of Water Utility Distribution:

1. Payment: Lump sum price offered in Price Schedules.
    a. Include the cost of disinfecting booster pump stations and pipes.

1.02 REFERENCE STANDARDS

A. American Water Works Association (AWWA):

1.	AWWA B300-24	Hypochlorites
2.	AWWA C604-17	Installation of Buried Steel Water Pipe-4 In. (100 mm) and Larger
3.	AWWA C651-23	Disinfecting Water Mains
4.	AWWA C655-25	Field Dechlorination

B. National Sanitation Foundation International / American National Standards Institute / National Standard of Canada (NSF/ANSI/CAN)

1.	NSF/ANSI/CAN 60-2025	Drinking Water Treatment Chemicals – Health Effects
2.	NSF/ANSI/CAN 61-2025	Drinking Water System Components – Health Effects

1.03 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals:

B. RSN 33 13 00-1, Flushing, Testing, Disinfection, and Verification Plan:

1. Demonstrate compliance with AWWA C604, AWWA C651 and AWWA C655.
2. Show product compliance with NSF/ANSI/CAN 60 and/or NSF/ANSI/CAN 61.
3. Permits and approvals:
    a. Identify required permits and approvals.
    b. Plan, procedures, and schedule.
4. Initial Flushing:
    a. Plan, procedures, and schedule.

b. Include flushing locations, water source, water volume, and flow rate.

5. Hydrostatic Pressure and Leakage Testing:

a. Plan, procedures, and schedule.

b. Include pipe segment materials, pipe diameters, pipe stations, water source, water volumes, pressure gauge measurement locations, test pressure calculations, acceptable makeup water volume calculations, and pass/fail determinations.

6. Disinfection:

a. Plan, procedures, and schedule.

b. Include disinfection method, disinfectant type and form, disinfectant dose, chlorine residual sample locations, disinfectant storage, disinfectant disposal, water source, water volume, and flow rate.

c. If disinfection will be performed in multiple segments, identify and delineate each segment.

7. Final Flushing:

a. Plan, procedures, and schedule.

b. Include flushing locations, water source, water volume, and flow rate.

c. Include dechlorination plan and procedures.

8. Verification:

a. Plan, procedures, and schedule.

b. Include proposed sampling locations.

c. Identify independent testing laboratory, including certifications/qualifications to perform bacteriological testing.

9. Plan and procedures for prevention of recontamination of system components/segments once they are disinfected.

10. Safety:

a. Chemical spill response plan for any chemicals to be used.

b. Job Hazard Analysis (JHA), including, but not limited to:

1) Personal protective equipment required for handling disinfection and dechlorination chemicals.

2) Emergency eyewash and shower locations.

3) Local emergency responder communication information.

C. RSN 33 13 00-2, Flushing, Testing, Disinfection, and Verification Report:

1. Detailed report demonstrating flushing and disinfection activities were accomplished in accordance with AWWA C651 and AWWA C655.

2. Examination
    a. Name of field person in charge of examination.
    b. Date and time of examination.
    c. Include verification of existing conditions.
    d. Include verification of pipe and associated equipment cleaning.
    e. Include verification of equipment operations.
3. Initial Flushing:
    a. Name of field person in charge of initial flushing.
    b. Date and time for start and completion.
    c. Flow rate and velocity achieved.
    d. Any data collected or readings taken during flushing activities.
4. Hydrostatic Pressure and Leakage Testing:
    a. Name of field person in charge of initial flushing.
    b. Date and time for start and completed.
    c. Include initial and final pressure readings for each measurement location, test pressures achieved, makeup water volumes required, pressure gauge serial numbers and calibration certification dates, duration of conditioning period and formal test periods, and signatures of responsible parties certifying the testing results.
    d. Test records for each test performed, including failures.
5. Disinfection:
    a. Name of field person in charge of disinfection and sampling.
    b. Type and form of disinfectant used.
    c. Date and time for disinfection start and completion.
    d. Chlorine residual sample location names identified on drawing or map.
    e. Table with the following for each chlorine residual sample location:
        1) Location name.
        2) Date and time of samples taken.
        3) Free chlorine residual concentration at each sample time.
    f. Any other data collected or readings taken during disinfection to verify compliance with AWWA C651 and AWWA C655.
6. Final Flushing:
    a. Name of field person in charge of final flushing.
    b. Date and time for start and completion.

c. Chlorine residual sample locations and discharge location identified on a drawing or map.

d. Any data collected or readings taken during final flushing activities, including, but not limited to, free chlorine concentrations at identified sample locations.

7. Bacteriological Test Results:

a. Bacteriological sample location names identified on drawing or map.

b. Table with the following for each bacteriological sample location:

1) Location name.

2) Name of sample collector.

3) Date and time of each sample.

4) Free chlorine residual test results.

5) Total coliform bacteria test results.

6) Escherichia coli bacteria test results.

7) Heterotrophic plate count (HPC) bacteria test results.

8) Certify water meets or does not meet bacterial test requirements.

c. Provide data summary and necessary graphs and plots of tests performed.

d. Testing laboratory: name, address, and telephone number.

e. Testing laboratory report(s) attached, including sample chain(s) of custody.

f. Water Quality Certificate: Certified, signed, dated and attested to having met the requirements of AWWA C651.

8. Issues and Resolutions Log:

a. Include any deviations from the approved Flushing, Testing, Disinfection, and Verification Plan, as approved by the COR.

9. Field Check Prints: Marked field check prints with drawing modifications in accordance with Sections 01 33 26 – Electrical Drawings and Data and 01 78 30 – Project Record Documents.

a. Use in preparing Final Drawings, including, but not limited to, the addition of any taps for sampling.

1.04 QUALIFICATIONS

A. Entities preparing disinfection documentation or performing disinfection and related activities shall have at least 10 years' experience performing similar services.

B. The laboratory performing bacteriological testing to confirm successful disinfection shall be certified by the State of New Mexico to perform the required analyses for drinking water compliance.

PART 2 PRODUCTS

2.01 GENERAL

1. In accordance with NSF/ANSI/CAN 60.
2. In accordance with NSF/ANSI/CAN 61.

PART 3 EXECUTION

3.01 GENERAL

A. In accordance with AWWA C604, AWWA C651 and AWWA C655, including, but not limited to, flushing, testing, disinfection, and bacteriological testing.

1. Optional sections of AWWA C651 that are required:
    a. 5.2.9 Backflow protection.
    b. Bacteriological testing at all sampling locations shall include total coliform, E. coli, and standard heterotrophic plate count (HPC) analyses.

B. In accordance with the approved version of RSN 33 13 00-1, Flushing, Testing, Disinfection, and Verification Plan. Any deviations from the approved Flushing, Testing, Disinfection, and Verification Plan must be approved by the COR.

C. Flushing, testing, disinfection, and verification activities shall be performed after completion and approval of all construction testing in accordance with specifications.

D. Provide water, chemicals, equipment, instrumentation, testing, and personnel necessary for performing flushing, testing, disinfection, and verification activities.

E. Potable water shall be used for all flushing, testing, disinfection, and verification activities.

F. Once disinfected, pipeline, booster pump stations, and appurtenances shall remain isolated and not be exposed to any potential or actual recontamination. A newly disinfected portion of the system shall only be connected to existing or previously disinfected portions of the system once disinfection is confirmed via bacteriological testing and written permission is obtained from COR.

3.02 COORDINATION

A. Coordinate work activities in accordance with Section 01 31 19 – Project Management and Coordination.

B. Government may witness any flushing, testing, disinfection, and verification activities.

C. Contractor shall:

1. Not schedule flushing, testing, disinfection, and verification activities until Flushing, Testing, Disinfection, and Verification Plan is approved.
2. Notify COR in writing a minimum of 60 days prior to beginning any flushing, testing, disinfection, and verification activities.
3. Complete all construction testing and request written confirmation of completion prior to beginning flushing, testing, disinfection, and verification activities.
4. Notify COR in writing of scheduled activities a minimum of 14 days prior to performing activity to allow Government opportunity to witness.
5. Coordinate with COR, NTUA, and water distribution system operators prior to commencing flushing and disinfection and during flushing and disinfection.
6. Ensure work is coordinated with commissioning, startup and ongoing operations activities along with ongoing distribution system operations.
7. Hold daily coordination meetings one week prior to Initial Flushing with COR, NTUA, and water distribution system operators.

3.03 EXAMINATION

A. Verify existing conditions before starting work.

1. Designated flush water discharge location(s) is acceptable and capable of receiving the volume and flow rate of the flush without causing:
   a. Flooding or erosion at the discharge location(s).
   b. Damage to adjacent infrastructure.
   c. Environmental violation.
2. All connections to the existing live system are closed and confirmed closed.
3. Pressure differentials are calculated and documented to ensure flow in intended direction upon valve openings.
4. No cross-connections exist between the new untested, unflushed system and the potable distribution system.

B. Verify pipe and associated equipment has been cleaned and are operational.

1. Visual inspection of accessible pipe interiors shall be performed, including:
   a. Inspection of all open pipe ends before jointing the connections.
   b. Confirmation that no visible debris, standing water, mud, rocks, tools, rags, or construction materials remain inside the pipe.
   c. Inspection of all appurtenance vaults and valve boxes for debris.

2. All gate and butterfly valves have been operated through full open and close cycles.
3. All air release/vacuum valves are installed, operational, and will function during the disinfection fill to allow air purging.
4. Meters are in place and ready to receive flow.

3.04 INSTALLATION

A. Authority Having Jurisdiction (AHJ) (e.g., NMED):

1. Flushing, disinfection, and verification activities shall be performed in accordance with AHJ requirements.
2. Obtain AHJ approvals and permits before commencing flushing, testing, disinfection, and verification activities.

B. Initial Flushing.

1. Prior to Initial Flushing, complete Examination.
2. Obtain and use potable water for flushing.
3. Minimum velocity in accordance with AWWA C651.

C. Hydrostatic Pressure Test and Leakage Test.

1. In accordance with AWWA C651, AWWA C604, and the AHJ.

   In the event of conflicting standards, the more stringent requirement governs the testing procedures.
2. Appurtenances to be included in the hydrostatic pressure test.
3. Obtain and use potable water for hydrostatic pressure testing.
4. Both sustained pressuring testing and leakage testing required, performed simultaneously, if possible.
5. Both sustained pressure testing and leakage testing shall have a minimum duration of 2 hours, excluding any conditioning period of the system.
6. Testing pressure: 150 psi or 1.5x the system operating pressure, whichever is greater.
   a. Testing pressure to be achieved at the highest point of the piping segment, unless maximum pressure at the lowest point exceeds pipe pressure class rating.
7. All equipment information shall be provided to and approved by the COR prior to testing.
8. Calibrated flow meter for measuring makeup water volume required.
   a. Makeup water allowances shall meet the requirements of AWWA C604.
   b. Verify and record allowance calculations.

c. Verify and record makeup water volumes required.

9. Calibrated pressure gauges used during hydrostatic pressure testing to verify pressure throughout pipe.
   a. Locations:
      1) Test pump.
      2) High point in piping system.
   b. Calibrated for accuracy and dial range within 12 months of hydrostatic pressure testing.
10. Verify and record successful weld, joint, and gauge operations during hydrostatic pressure testing.
11. Test failures include:
   a. Pressure drop acceptable tolerances include pressure changes within instrument resolution.
   b. Leakage exceeding allowance calculations.
   c. Weld, joint, gauge, and/or appurtenance failing during testing.
12. Retesting:
   a. Retesting of the hydrostatic pressure and leakage tests is required upon any test failures.
   b. Retesting shall be done until tests are successfully completed.

D. Post-Repair Flush:
1. Required only if repairs to piping segments are necessary.
2. Obtain and use potable water for flushing.
3. Minimum velocity in accordance with AWWA C651.

E. Disinfection.
1. Prior to Disinfection, complete Initial Flushing.
2. Obtain and use potable water for disinfection.
3. In accordance with AWWA C651.
   a. Introduce water into pipeline system including backflow prevention.
   b. Perform disinfection procedures.
   c. Minimum initial free chlorine concentration in accordance with AWWA C651 throughout system.
   d. Operate valves and hydrants to ensure these are disinfected.
   e. Maintain minimum free chlorine concentration throughout the system for a contact time in accordance with AWWA C651 based on method chosen.

F. Final Flushing.

1. Prior to Final Flushing, complete Disinfection.
2. Obtain and use potable water for flushing.
3. Flush system as soon as possible after the completion of disinfection.
4. Dechlorinate and discharge water in accordance with AWWA C655.
5. Confirm the system is fully flushed by either:
    a. Measuring feed and discharge free chlorine concentrations and ensuring they are the same (within 0.30 mg/L of each other).
    b. Measuring a discharge free chlorine concentration less than 1.0 mg/L.

G. Verification.

1. Bacteriological Testing.
    a. In accordance with AWWA C651.
    b. Total coliform, Escherichia coli, and heterotrophic plate count (HPC) bacteria testing required.
    c. Use the approved testing laboratory.
    d. Samples to be collected within 24 hours of completing Final Flushing.
    e. Samples to be stored correctly prior to delivery to testing laboratory to prevent inaccurate analyses.
    f. Samples shall be delivered to testing laboratory within 24 hours of collection.
2. Sampling Locations.
    a. In accordance with AWWA C651.
    b. If sampling location required spacing (e.g. every 1200 ft.) cannot be obtained through existing infrastructure, install testing saddles.
3. Bacteriological Testing Results.
    a. Passing test.
        1) In accordance with AWWA C651.
        2) Meet required chlorine residual at sampling point.

H. Completion:

1. Replace permanent system devices removed for flushing, testing, disinfection, and verification activities.
2. If sampling taps installed, seal taps and include in Final Drawings.
3. Obtain written permission from COR prior to connecting to existing system.

4. Final acceptance will be approval of RSN 33 13 00-2, Flushing, Testing, Disinfection, and Verification Report.

3.05 REDISINFECTION

A. Repeat disinfection, final flushing, and verification steps if:

1. Water quality testing fails to meet requirements.
2. A disinfected portion of the system was exposed to any potential or actual contamination.
3. Water is in disinfected pipe for greater than 30 days before the start of operations and potable water delivery to customers.
    a. COR shall be notified once pipeline is disinfected to begin tracking of 30-day timeline.

B. If more than three consecutive test failures occur, COR shall be notified, and a corrective action plan shall be submitted.

END OF SECTION

SECTION 33 13 00
DISINFECTION OF WATER UTILITY DISTRIBUTION

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Disinfection of Water Utility Distribution:

1. Payment: Lump sum price offered in Price Schedules.
    a. Include the cost of disinfecting pipes.

1.02 REFERENCE STANDARDS

A. American Water Works Association (AWWA):

1.	AWWA B300-24	Hypochlorites
2.	AWWA C604-17	Installation of Buried Steel Water Pipe-4 In. (100 mm) and Larger
3.	AWWA C651-23	Disinfecting Water Mains
4.	AWWA C655-25	Field Dechlorination

B. National Sanitation Foundation International / American National Standards Institute / National Standard of Canada (NSF/ANSI/CAN)

1.	NSF/ANSI/CAN 60-2025	Drinking Water Treatment Chemicals – Health Effects
2.	NSF/ANSI/CAN 61-2025	Drinking Water System Components – Health Effects

1.03 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals:

B. RSN 33 13 00-1, Flushing, Testing, Disinfection, and Verification Plan:

1. Demonstrate compliance with AWWA C604, AWWA C651 and AWWA C655.
2. Show product compliance with NSF/ANSI/CAN 60 and/or NSF/ANSI/CAN 61.
3. Permits and approvals:
    a. Identify required permits and approvals.
    b. Plan, procedures, and schedule.
4. Initial Flushing:
    a. Plan, procedures, and schedule.

b. Include flushing locations, water source, water volume, and flow rate.

5. Hydrostatic Pressure and Leakage Testing:

a. Plan, procedures, and schedule.

b. Include pipe segment materials, pipe diameters, pipe stations, water source, water volumes, pressure gauge measurement locations, test pressure calculations, acceptable makeup water volume calculations, and pass/fail determinations.

6. Disinfection:

a. Plan, procedures, and schedule.

b. Include disinfection method, disinfectant type and form, disinfectant dose, chlorine residual sample locations, disinfectant storage, disinfectant disposal, water source, water volume, and flow rate.

c. If disinfection will be performed in multiple segments, identify and delineate each segment.

7. Final Flushing:

a. Plan, procedures, and schedule.

b. Include flushing locations, water source, water volume, and flow rate.

c. Include dechlorination plan and procedures.

8. Verification:

a. Plan, procedures, and schedule.

b. Include proposed sampling locations.

c. Identify independent testing laboratory, including certifications/qualifications to perform bacteriological testing.

9. Plan and procedures for prevention of recontamination of system components/segments once they are disinfected.

10. Safety:

a. Chemical spill response plan for any chemicals to be used.

b. Job Hazard Analysis (JHA), including, but not limited to:

1) Personal protective equipment required for handling disinfection and dechlorination chemicals.

2) Emergency eyewash and shower locations.

3) Local emergency responder communication information.

C. RSN 33 13 00-2, Flushing, Testing, Disinfection, and Verification Report:

1. Detailed report demonstrating flushing and disinfection activities were accomplished in accordance with AWWA C651 and AWWA C655.

2. Examination
    a. Name of field person in charge of examination.
    b. Date and time of examination.
    c. Include verification of existing conditions.
    d. Include verification of pipe and associated equipment cleaning.
    e. Include verification of equipment operations.
3. Initial Flushing:
    a. Name of field person in charge of initial flushing.
    b. Date and time for start and completion.
    c. Flow rate and velocity achieved.
    d. Any data collected or readings taken during flushing activities.
4. Hydrostatic Pressure and Leakage Testing:
    a. Name of field person in charge of initial flushing.
    b. Date and time for start and completed.
    c. Include initial and final pressure readings for each measurement location, test pressures achieved, makeup water volumes required, pressure gauge serial numbers and calibration certification dates, duration of conditioning period and formal test periods, and signatures of responsible parties certifying the testing results.
    d. Test records for each test performed, including failures.
5. Disinfection:
    a. Name of field person in charge of disinfection and sampling.
    b. Type and form of disinfectant used.
    c. Date and time for disinfection start and completion.
    d. Chlorine residual sample location names identified on drawing or map.
    e. Table with the following for each chlorine residual sample location:
        1) Location name.
        2) Date and time of samples taken.
        3) Free chlorine residual concentration at each sample time.
    f. Any other data collected or readings taken during disinfection to verify compliance with AWWA C651 and AWWA C655.
6. Final Flushing:
    a. Name of field person in charge of final flushing.
    b. Date and time for start and completion.

c. Chlorine residual sample locations and discharge location identified on a drawing or map.

d. Any data collected or readings taken during final flushing activities, including, but not limited to, free chlorine concentrations at identified sample locations.

7. Bacteriological Test Results:

a. Bacteriological sample location names identified on drawing or map.

b. Table with the following for each bacteriological sample location:

1) Location name.

2) Name of sample collector.

3) Date and time of each sample.

4) Free chlorine residual test results.

5) Total coliform bacteria test results.

6) Escherichia coli bacteria test results.

7) Heterotrophic plate count (HPC) bacteria test results.

8) Certify water meets or does not meet bacterial test requirements.

c. Provide data summary and necessary graphs and plots of tests performed.

d. Testing laboratory: name, address, and telephone number.

e. Testing laboratory report(s) attached, including sample chain(s) of custody.

f. Water Quality Certificate: Certified, signed, dated and attested to having met the requirements of AWWA C651.

8. Issues and Resolutions Log:

a. Include any deviations from the approved Flushing, Testing, Disinfection, and Verification Plan, as approved by the COR.

9. Field Check Prints: Marked field check prints with drawing modifications in accordance with Sections 01 33 26 – Electrical Drawings and Data and 01 78 30 – Project Record Documents.

a. Use in preparing Final Drawings, including, but not limited to, the addition of any taps for sampling.

1.04 QUALIFICATIONS

A. Entities preparing disinfection documentation or performing disinfection and related activities shall have at least 10 years' experience performing similar services.

B. The laboratory performing bacteriological testing to confirm successful disinfection shall be certified by the State of New Mexico to perform the required analyses for drinking water compliance.

PART 2 PRODUCTS

2.01 GENERAL

1. In accordance with NSF/ANSI/CAN 60.
2. In accordance with NSF/ANSI/CAN 61.

PART 3 EXECUTION

3.01 GENERAL

A. In accordance with AWWA C604, AWWA C651 and AWWA C655, including, but not limited to, flushing, testing, disinfection, and bacteriological testing.

1. Optional sections of AWWA C651 that are required:
    a. 5.2.9 Backflow protection.
    b. Bacteriological testing at all sampling locations shall include total coliform, E. coli, and standard heterotrophic plate count (HPC) analyses.

B. In accordance with the approved version of RSN 33 13 00-1, Flushing, Testing, Disinfection, and Verification Plan. Any deviations from the approved Flushing, Testing, Disinfection, and Verification Plan must be approved by the COR.

C. Flushing, testing, disinfection, and verification activities shall be performed after completion and approval of all construction testing in accordance with specifications.

D. Provide water, chemicals, equipment, instrumentation, testing, and personnel necessary for performing flushing, testing, disinfection, and verification activities.

E. Potable water shall be used for all flushing, testing, disinfection, and verification activities.

F. Once disinfected, pipeline, booster pump stations, and appurtenances shall remain isolated and not be exposed to any potential or actual recontamination. A newly disinfected portion of the system shall only be connected to existing or previously disinfected portions of the system once disinfection is confirmed via bacteriological testing and written permission is obtained from COR.

3.02 COORDINATION

A. Coordinate work activities in accordance with Section 01 31 19 – Project Management and Coordination.

B. Government may witness any flushing, testing, disinfection, and verification activities.

C. Contractor shall:

1. Not schedule flushing, testing, disinfection, and verification activities until Flushing, Testing, Disinfection, and Verification Plan is approved.
2. Notify COR in writing a minimum of 60 days prior to beginning any flushing, testing, disinfection, and verification activities.
3. Complete all construction testing and request written confirmation of completion prior to beginning flushing, testing, disinfection, and verification activities.
4. Notify COR in writing of scheduled activities a minimum of 14 days prior to performing activity to allow Government opportunity to witness.
5. Coordinate with COR, NTUA, and water distribution system operators prior to commencing flushing and disinfection and during flushing and disinfection.
6. Ensure work is coordinated with commissioning, startup and ongoing operations activities along with ongoing distribution system operations.
7. Hold daily coordination meetings one week prior to Initial Flushing with COR, NTUA, and water distribution system operators.

3.03 EXAMINATION

A. Verify existing conditions before starting work.

1. Designated flush water discharge location(s) is acceptable and capable of receiving the volume and flow rate of the flush without causing:
   a. Flooding or erosion at the discharge location(s).
   b. Damage to adjacent infrastructure.
   c. Environmental violation.
2. All connections to the existing live system are closed and confirmed closed.
3. Pressure differentials are calculated and documented to ensure flow in intended direction upon valve openings.
4. No cross-connections exist between the new untested, unflushed system and the potable distribution system.

B. Verify pipe and associated equipment has been cleaned and are operational.

1. Visual inspection of accessible pipe interiors shall be performed, including:
   a. Inspection of all open pipe ends before jointing the connections.
   b. Confirmation that no visible debris, standing water, mud, rocks, tools, rags, or construction materials remain inside the pipe.
   c. Inspection of all appurtenance vaults and valve boxes for debris.

2. All gate and butterfly valves have been operated through full open and close cycles.
3. All air release/vacuum valves are installed, operational, and will function during the disinfection fill to allow air purging.
4. Meters are in place and ready to receive flow.

3.04 INSTALLATION

A. Authority Having Jurisdiction (AHJ) (e.g., NMED):

1. Flushing, disinfection, and verification activities shall be performed in accordance with AHJ requirements.
2. Obtain AHJ approvals and permits before commencing flushing, testing, disinfection, and verification activities.

B. Initial Flushing.

1. Prior to Initial Flushing, complete Examination.
2. Obtain and use potable water for flushing.
3. Minimum velocity in accordance with AWWA C651.

C. Hydrostatic Pressure Test and Leakage Test.

1. In accordance with AWWA C651, AWWA C604, and the AHJ.

   In the event of conflicting standards, the more stringent requirement governs the testing procedures.
2. Appurtenances to be included in the hydrostatic pressure test.
3. Obtain and use potable water for hydrostatic pressure testing.
4. Both sustained pressuring testing and leakage testing required, performed simultaneously, if possible.
5. Both sustained pressure testing and leakage testing shall have a minimum duration of 2 hours, excluding any conditioning period of the system.
6. Testing pressure: 150 psi or 1.5x the system operating pressure, whichever is greater.
   a. Testing pressure to be achieved at the highest point of the piping segment, unless maximum pressure at the lowest point exceeds pipe pressure class rating.
7. All equipment information shall be provided to and approved by the COR prior to testing.
8. Calibrated flow meter for measuring makeup water volume required.
   a. Makeup water allowances shall meet the requirements of AWWA C604.
   b. Verify and record allowance calculations.

c. Verify and record makeup water volumes required.

9. Calibrated pressure gauges used during hydrostatic pressure testing to verify pressure throughout pipe.
   a. Locations:
      1) Test pump.
      2) High point in piping system.
   b. Calibrated for accuracy and dial range within 12 months of hydrostatic pressure testing.
10. Verify and record successful weld, joint, and gauge operations during hydrostatic pressure testing.
11. Test failures include:
   a. Pressure drop acceptable tolerances include pressure changes within instrument resolution.
   b. Leakage exceeding allowance calculations.
   c. Weld, joint, gauge, and/or appurtenance failing during testing.
12. Retesting:
   a. Retesting of the hydrostatic pressure and leakage tests is required upon any test failures.
   b. Retesting shall be done until tests are successfully completed.

D. Post-Repair Flush:
1. Required only if repairs to piping segments are necessary.
2. Obtain and use potable water for flushing.
3. Minimum velocity in accordance with AWWA C651.

E. Disinfection.
1. Prior to Disinfection, complete Initial Flushing.
2. Obtain and use potable water for disinfection.
3. In accordance with AWWA C651.
   a. Introduce water into pipeline system including backflow prevention.
   b. Perform disinfection procedures.
   c. Minimum initial free chlorine concentration in accordance with AWWA C651 throughout system.
   d. Operate valves and hydrants to ensure these are disinfected.
   e. Maintain minimum free chlorine concentration throughout the system for a contact time in accordance with AWWA C651 based on method chosen.

F. Final Flushing.

1. Prior to Final Flushing, complete Disinfection.
2. Obtain and use potable water for flushing.
3. Flush system as soon as possible after the completion of disinfection.
4. Dechlorinate and discharge water in accordance with AWWA C655.
5. Confirm the system is fully flushed by either:
    a. Measuring feed and discharge free chlorine concentrations and ensuring they are the same (within 0.30 mg/L of each other).
    b. Measuring a discharge free chlorine concentration less than 1.0 mg/L.

G. Verification.

1. Bacteriological Testing.
    a. In accordance with AWWA C651.
    b. Total coliform, Escherichia coli, and heterotrophic plate count (HPC) bacteria testing required.
    c. Use the approved testing laboratory.
    d. Samples to be collected within 24 hours of completing Final Flushing.
    e. Samples to be stored correctly prior to delivery to testing laboratory to prevent inaccurate analyses.
    f. Samples shall be delivered to testing laboratory within 24 hours of collection.
2. Sampling Locations.
    a. In accordance with AWWA C651.
    b. If sampling location required spacing (e.g. every 1200 ft.) cannot be obtained through existing infrastructure, install testing saddles.
3. Bacteriological Testing Results.
    a. Passing test.
        1) In accordance with AWWA C651.
        2) Meet required chlorine residual at sampling point.

H. Completion:

1. Replace permanent system devices removed for flushing, testing, disinfection, and verification activities.
2. If sampling taps installed, seal taps and include in Final Drawings.
3. Obtain written permission from COR prior to connecting to existing system.

4. Final acceptance will be approval of RSN 33 13 00-2, Flushing, Testing, Disinfection, and Verification Report.

3.05 REDISINFECTION

A. Repeat disinfection, final flushing, and verification steps if:

1. Water quality testing fails to meet requirements.
2. A disinfected portion of the system was exposed to any potential or actual contamination.
3. Water is in disinfected pipe for greater than 30 days before the start of operations and potable water delivery to customers.
    a. COR shall be notified once pipeline is disinfected to begin tracking of 30-day timeline.

B. If more than three consecutive test failures occur, COR shall be notified, and a corrective action plan shall be submitted.

END OF SECTION

SECTION 35 21 95
STEEL MANIFOLD PIPING

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Steel Manifold Piping:

1. Payment: Lump sum price offered in Price Schedules.
    a. Includes cost of pipe and fittings within the turnout buildings.
    b. Includes cost of pipe and fittings for yard pipe located outside of the turnout buildings.

1.02 DEFINITIONS

A. Steel Manifold Piping:

1. Details of steel manifolds are shown on drawings 1695-D-62558 through 1695-D-62568.

1.03 REFERENCE STANDARDS

A. American Society of Mechanical Engineers (ASME)

1.	ASME B1.20.1-2013(R2018)	Pipe Threads, General Purpose (Inch)
2.	ASME B16.9-2024	Factory Made Wrought Buttwelding Fittings
3.	ASME B16.11-2021	Forged Fittings, Socket-Welding and Threaded

B. ASTM International (ASTM)

1.	ASTM A36/A36M-19	Carbon Structural Steel
2.	ASTM A53/A53M-24	Pipe, Steel, Black and Hot-Dipped, Zinc-Coated Welded and Seamless
3.	ASTM A105/A105M-24	Carbon Steel Forgings for Piping Applications
4.	ASTM A106/A106M-26	Seamless Carbon Steel Pipe for High-Temperature Service
5.	ASTM A139/A139M-22	Electric-Fusion (Arc)-Welded Steel Pipe (NPS 4 and Over)
6.	ASTM A194/A194M-24	Carbon Steel, Alloy Steel, and Stainless-Steel Nuts for Bolts for High Pressure or High Temperature Service, or Both

7.	ASTM A234/A234M-25	Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service
8.	ASTM A420/A420M-25	Piping Fittings of Wrought Carbon Steel and Alloy Steel for Low Temperature Service
9.	ASTM A1011/A1011M-25	Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength
10.	ASTM A1018/A1018M-25	Steel, Sheet and Strip, Heavy Thickness Coils, Hot Rolled, Carbon, Commercial, Drawing, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength
11.	ASTM E165/E165M-23	Liquid Penetrant Examination for General Industry
12.	ASTM E709-21	Magnetic Particle Testing

C. American Welding Society (AWS)

1.	AWS D1.1/D1.1M:2025	Structural Welding Code - Steel

D. American Water Works Association (AWWA)

1.	AWWA C200-23	Steel Water Pipe - 6 In (150 mm) and Larger
2.	AWWA C206-23	Field Welding of Steel Water Pipe
3.	AWWA C207-23	Steel Pipe Flanges for Waterworks Service, Sizes 4 In. through 144 In. (100 mm Through 3,600 mm)
4.	AWWA C208-22	Dimensions for Fabricated Steel Water Pipe Fittings
5.	AWWA C604-24	Installation of Buried Steel Water Pipe – 4 In. (100 mm) and Larger

E. Manufacturer’s Standardization Society (MSS)

1.	MSS SP-97-25	Integrally Reinforced Forged Branch Outlet Fittings: Socket Welding, Threaded, and Buttwelding Ends

F. National Sanitation Foundation International / American National Standards Institute (NSF/ANSI)

1.	NSF/ANSI/CAN 61-24	Drinking Water System Components – Health Effects
2.	NSF/ANSI/CAN 372-24	Drinking Water System Components – Lead Content

1.04 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals.

B. RSN 35 21 95-1, Shop Drawings:

1. Checked, detailed shop drawings including dimensions and tolerances, material data, joint details, and welding requirements:
    a. Provide checked box on each drawing indicating date drawing was checked and by whom. Drawings must be stamped by a PE (Professional Engineer).
    b. When revised drawings and data are resubmitted, clearly identify changes from previous submittals.

C. RSN 35 21 95-2, Commercial Products Data:

1. Items in Part 2 Products.
2. Manufacturer's technical product data, including the applicable sizes, standards, material, and pressure rating.

D. RSN 35 21 95-3, Final Data:

1. Weld nondestructive testing (NDT) reports including retests after weld repair.
2. Results of air and soap tests on field welded double lap and butt strap joints.
3. Hydrostatic test reports, including time-pressure record.
4. Affidavit of compliance to AWWA C200 and C206 as applicable.

E. RSN 35 21 95-4, Final Drawings:

1. As-Built drawings
    a. Clearly mark drawings 1695-D-62558 through 1695-D-62568 to show all field modifications of the final installation.
    b. Submit drawings in accordance with Section 01 33 00 – Submittals article 1.04 paragraph C.6 and Section 01 78 30 –Project Record Documents.

1.05 QUALIFICATIONS

A. Pipe Installers: Installed similar manifold on 3 projects within the last 5 years.

B. Steel Pipe Welders: Welders and Welding Operators to be qualified in accordance with AWS D1.1.

1.06 DELIVERY, STORAGE, AND HANDLING

A. During loading, transporting, unloading, storing, and laying, prevent damage to steel manifold piping linings and coating. Transport and handle in accordance with fabricator's instructions:

1. Provide padded bolsters curved to fit under outside of pipe.
2. Provide heavy padding under ties during transportation and storage.

B. Maintain circularity of pipe sections during coating, lining, transporting, and installing operations:

1. Provide stulls or adjustable spiders.
2. Measured deviation between maximum and minimum diameters at stulls or spiders shall not exceed 0.5 percent of nominal pipe diameter.

C. Do not store directly on the ground.

D. Store so that lifting straps can be passed beneath pipe sections.

E. Provide space to allow personnel to move between pipe sections.

F. Cover ends with tarps and tie down securely.

G. Repair damage to lining and coating as directed if, in the opinion of COR, a satisfactory repair can be made; otherwise, replace damaged section at expense of Contractor and in accordance with Section 09 96 20 – Coatings.

PART 2 PRODUCTS

2.01 GENERAL REQUIREMENTS

A. Piping and equipment materials:

1. ANSI/NSF/CAN 61 certified.
2. ANSI/NSF/CAN 372 certified lead-free.
3. Suitable for potable water.
4. Materials suitable for free chlorine residual exposure.
    a. Typical free chlorine residual concentration: Maximum 4 milligrams per liter (mg/l).

2.02 EQUIPMENT

A. Steel plate: ASTM A36, ASTM A1011, SS, Grade 36, 40, or 45; or ASTM A1018, SS, Grade 36 or 40.

B. Standard and Schedule Steel Pipe:

1. ASTM A53, Type E or S, Grade B:

a. Pipe 2-inches in diameter: Fusion Bonded Epoxy.

1) Acceptable alternative ASTM A106.

b. Pipe 3-inches and larger in diameter: In accordance with Section 09 96 20 – Coatings.

C. Electric–Fusion– (arc–) Welded, Spiral–Seam Steel Pipe: ASTM A139, Grade C, D, or E.

D. Corrosion Collar:

1. Steel plate: ASTM A36, ASTM A1011, Grade 36, 40, or 45; or ASTM A1018, Grade 36 or 40.

E. Flanges:

1. Steel: Slip-on type, flat faced, AWWA C207.

2. Flat face with finish suitable for gasket: Provide spiral or concentric serrated grooved flange face gasket surface finishes that are suitable for flat gasket material provided and for operating pressures.

3. Do not remove machine marks.

F. Gaskets:

1. Suitable for cold water service.

2. Do not reuse compressed gaskets.

3. Use Multi-Swell Style 3760 manufactured by Garlock, www.garlock.com; or equal, having the following essential characteristics:

a. Dimensional requirements of AWWA C207 as shown on drawings.

b. Flat, full face type.

c. Minimum 300 psig cold water pressure rating.

d. Minimum 1/16-inch thick.

G. Flange bolts, Nuts, and Washers:

1. AWWA C207 for AWWA flanges.

2. Bolts: ASTM A193/A193M, Grade B7.

3. Nuts:

a. Greater than 1-inch diameter: ASTM A194/A194M, Grade 2H

b. 1-inch diameter or less: ASTM A563

4. Washers: ASTM F436.

H. Pipe Flange Supports:

1. Trumbull Flange Style Adjustable Pipe Support manufactured by Trumbull Manufacturing, www.trumbull-mfg.com, or equal, having the following essential characteristics:

a. Adjustable height.

b. Material: Galvanized steel.

c. Suitable for pipe sizes as shown on the drawings.

d. Provides vertical restraint to pipe flange.

I. Threaded Fittings:

1. Threaded NPT: ASME B1.20.1.

a. Steel: ASME B1.20.1, ASME B16.11, ASTM A105 ASTM A106 and MSS SP-97.

2.

J. Butt-Welding Steel Fittings: ASME B16.9 and ASTM A234 or ASTM A420 - WPL6.

K. Concentric Steel Pipe Reducers: Designed, fabricated and dimensioned in accordance with AWWA C208.

L. Steel Pipe Miter Bend Elbows: Designed, fabricated, and dimensioned in accordance with AWWA C208.

M. Pipe couplings:

1. Grooved-end: Style 77 manufactured by Victaulic, www.victaulic.com; or equal, having the following essential characteristics:

a. Flexible joint type with rolled grooves.

b. Minimum cold water working pressure rating: 300 psig.

c. Minimum angular deflection

1) 6-inch pipe diameter: 1-5’ degrees

2) 8-inch pipe diameter: 0-50’ degrees

3) 12-inch pipe diameter: 0-34’ degrees

N. Anchor bolts: In accordance with Section 03 20 60 – Concrete Anchors.

O. Flange insulation:

1. Flange isolating gasket kit complete with the “Trojan” type of gasket manufactured by Advanced Products and Systems, Inc., www.apsonline.com, or equal with the following essential characteristics:

a. Provide full-faced gaskets as shown on drawings. Outside diameter of gasket is same as outside diameter of flange.

b. Provide retainer with integral Nitrile sealing element ("Trojan" type).

c. Retainer material to be suitable for cold water service as recommended by gasket manufacturer.

d. One insulating sleeve and 2 insulating washers for each bolt. Material suitable for cold water service as recommended by gasket manufacturer.

e. Asbestos gaskets are not acceptable.

f. Prevent damage to gasket, gasket supplier to provide required bolt torques, recommended bolt tightening sequence, and flange face finish for insulating flanged joints.

g. Suitable for 300 psig cold water service.

2.03 FABRICATION

A. Fabricate steel manifold piping in accordance with these specifications, as indicated on drawings, and in accordance with AWWA C200.

B. Longitudinal, girth, and spiral joints, other than field welds:

1. Double-welded butt joints with complete penetration.
2. Stagger longitudinal joints.
3. Longitudinal, girth, and spiral joints shall not intersect at outlet connections.

C. Cut plates accurately to size and shape.

D. Bend or roll plates to true circular with curvature continuous from edges of plates.

E. Properly form edges to be joined by welding to suit type of welding and to allow complete penetration.

F. Perform welding with a process that protects molten metal from atmosphere:

1. Where practicable, use automatic machines.
2. Where weld metal is deposited in successive layers, clean each layer thoroughly before subsequent layer is deposited.
3. Take particular care in aligning and separating edges of plates to be joined by butt welding so that complete penetration and fusion of welds will be ensured.
4. After welding is complete, remove weld spatter.

G. Protect work and operator from wind, rain, and snow during welding operations:

1. Welding is not permitted on wet surfaces or when temperature of steel is lower than 0 degrees F.

2. Apply heat to welding areas as needed during field installation. At pipe surface temperatures between 0 and 32 degrees F, heat surface of areas within 3-inches of the point where a weld is to be started to at least 60 degrees F.

H. Ends to be joined by a pipe coupling: Exterior surface of pipe end to be connected with sleeve-type coupling shall be free from weld reinforcement (bead), seams, scars, dents, flats, or other defects likely to interfere with joint-sealing process in accordance with coupling manufacturer instructions.

I. Flanges and flanged connections:

1. During attachment of flanges to associated steel manifold pipe, take adequate steps to ensure that face of flanges remains flat and perpendicular to centerline of attached piping.

2. Remove by machining, initial, residual, or latent irregularities or warping that remains in face of flange after its attachment to steel manifold piping.

3. After machining, flange thickness shall not be reduced below minimum thickness specified in AWWA C207, or as shown on drawings.

4. Weld pipe flanges to the piping so that bolt holes straddle vertical centerline when in the installed position.

5. Provide pipe flanges that connect to valve body flanges with same number of bolt holes, bolt pattern, bolt circle diameter, and face type, flat or raised, as valves to which they are connected.

J. Continuously weld joints of flange supports and pipe supports.

2.04 CONTRACTOR SOURCE QUALITY TESTING

A. Flange Weld Testing:

1. Test flange welds by magnetic particle or liquid penetrant method in accordance with AWS D1.1.

2. Repair defects and retest weld.

B. Hydrostatic Pressure Testing:

1. Straight sections of steel piping manufactured and hydrostatically tested in accordance with ASTM A53 or ASTM A139 need not be retested.

2. Notify Government a minimum of 30-days before test.

3. Government may witness hydrostatic test and calibration of pressure gauges.

4. Use 2 or more pressure gauges with one being a recording type:

a. Upper limit of pressure gauges shall be between 1.5 and 3 times required test pressure. Calibrate pressure gauges with a deadweight tester.

b. Calibrate at least 1 gauge, which is used for acceptance of test, within 30-days prior to use.

c. Make a time-pressure record of hydrostatic pressure test.

5. Furnish temporary supports, bulkheads, air vents, dead weight tester, and other equipment necessary to perform hydrostatic tests.

6. If flange is attached to piping section after piping section has been hydrostatically tested, piping section will not require retesting.

7. Water temperature for tests: Greater than 45 degrees F.

8. Vent air from test sections.

9. Fill each section with water and uniformly increase pressure until required test pressure is reached:

   a. Hold test pressure for a minimum of 15-minutes.

   b. Apply and release test pressure 3 successive times.

   c. After each application, hold pressure to 2/3 of required pressure until welded joints and seams can be examined.

   d. Repair defects. After defects are repaired repeat complete test procedure.

10. After hydrostatic testing is completed, cut sections for handling purposes on planes normal to pipe axis.

PART 3 EXECUTION

3.01 PREPARATION

A. Locations of existing lateral turnout tie-ins are approximate, confirm locations prior to fabrication of steel manifold piping.

B. Construction of cPWS distribution pipes is ongoing and may be constructed at some sites. Coordinate with COR to verify consecutive distribution entry points.

3.02 INSTALLATION

A. Install steel manifold piping to line and grade as shown on drawings and in accordance with AWWA C604:

B. Prevent steel pipe from being displaced or overstressed during concrete and backfill placement.

C. Contact between pipe and rebar will not be allowed.

D. Install pipe spools with flanged connections so that bolt holes straddle vertical centerline when in installed position.

E. Dielectrically insulate between dissimilar metals. Provide insulating type unions at threaded valve, pipe or fitting connections between dissimilar metals.

F. Install pipe couplings, gaskets, and other appurtenances in accordance with manufacturer instructions:

1. Maximum allowable misalignment (lateral offset) between pipe ends at split sleeve type couplings: 0.125-inch:

a. Government may witness installation of pipe couplings.

b. If misalignment exceeds 0.125-inch, piping shall be repaired at the expense of the Contractor to correct misalignment. Piping repair method subject to COR approval.

c. Do not deflect pipe between coupling pairs at couplings for dynamic joint deflection to correct misalignment.

G. Install anchor bolts in accordance with manufacturer instructions.

H. Field joints:

1. Field joints shown on drawings are required to facilitate lining or installation of steel piping.

2. Allowable field joints for steel piping:

a. AWWA C207, AWWA C228 flanged joints per ASME PCC-1 respectively:

1) Flanges shall be suitable for working pressure of adjacent pipe.

2) Flanges shall not be permitted underneath buildings, , or in concrete encasements.

b. Full-penetration butt welds.

c. Double welded lap joints, or double welded butt strap joints.

d. Threaded couplings for pipe less than 4-inches in diameter.

3. Welded field joints shall be in accordance with AWWA C206 or AWWA C231.

4. Prepare field welded surfaces and recoat in accordance with Section 09 96 20 – Coatings.

I. Steel manifold piping earthwork: In accordance with Section 31 23 22 – Pipe Trench Earthwork.

3.03 CONTRACTOR FIELD QUALITY TESTING

A. Nondestructive Testing of Field Welds of Steel Pipe:

1. Test all welds in accordance with AWWA C206 and this specification.

2. Complete inspections and tests as work progresses in presence of COR and in accordance with AWS D1.1.

3. When welds are found to be defective by the standards of AWS, repair welds in accordance with AWS D1.1 and reexamine welds to ensure the adequacy of repairs.
4. Visual Inspection: Visually inspect welds following procedures, techniques, and standards in accordance with AWS D1.1.
5. Testing of full penetration joints:
    a. Ultrasonic methods in accordance with AWS D1.1.
6. Testing of partial penetration joint field welds and fillet field welds, including testing of double welded butt strap joints or double welded lap joints:
    a. Magnetic particle testing: ASTM E709.
    b. Dye penetrant: ASTM E165.
    c. Standards of acceptance for both types: AWS D1.1.
    d. Caulking or peening of defective welds is not permitted.
    e. Repair and retest unsatisfactory welds.
7. Inspect all double lap welded and butt strap joints by air and soap test:
    a. AWWA C206.
    b. Test the joints by shop drilling and tapping for a 1/8-inch or 1/4-inch national pipe thread in the lap end of the pipe.
    c. Apply 40 psig of air into the connection between the two fillet welds and maintain for a period of 5-minutes.
    d. Paint the welds with a soap solution.
    e. Mark any leaks indicated by the escaping air bubbles.
    f. Repair welds at leak locations in accordance with AWS D1.1 and AWWA C206 and retest.
    g. Close the threaded openings with pipe plugs.
    h. Seal weld the pipe plugs and grind the weld smooth.
    i. Submit test results to COR within one week after testing.
    j. Prepare field welded surfaces and recoat in accordance with Section 09 96 20 – Coatings.

3.04 LINING AND COATING

A. Refer to Section 09 96 20 – Coatings.

END OF SECTION

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SECTION 35 22 14
VALVES AND EQUIPMENT

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Valves and Equipment:

1. Payment: Lump sum prices offered in Price Schedules for various types and sizes of valves and equipment provided as shown on drawings.

1.02 DEFINITIONS

A. Details of valves and equipment are shown on drawings: 1695-D-62548 through 1695-D-62552 and 1695-D-62558 through 1695-D-62568.

1.03 REFERENCE STANDARDS

A. American Society of Mechanical Engineers (ASME)

1.	ASME B16.5-2025	Pipe Flanges and Flanged Fittings: NPS 1/2 through NPS 24 Metric/Inch

B. ASTM International (ASTM)

1.	ASTM A216/A216M-21	Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service
2.	ASTM A536-24	Ductile Iron Castings

C. American Society of Safety Engineers (ASSE)

1.	ASSE 1063-21	Air Valve and Vent Inflow Preventer

D. American Water Works Association (AWWA)

1.	AWWA C207-23	Steel Pipe Flanges for Waterworks Service, Sizes 4 In. through 144 In.
2.	AWWA C504-23	Rubber-Seated Butterfly Valves
3.	AWWA C507-23	Ball Valves, 6 In. Through 60 In. (150 mm Through 1,500 mm)
4.	AWWA C508-25	Swing-Check Valves for Waterworks Service, 2-In. through 48-In. (50-mm Through 1200-mm) NPS
5.	AWWA C509-23	Resilient-Seated Gate Valves for Water Supply Service

6.	AWWA C512-25	Air-Release, Air/Vacuum, and Combination Air Valves for Water and Wastewater Service
7.	AWWA C514-21	Air Valve and Vent Inflow Preventer Assemblies for Potable Water Distribution System and Storage Facilities
8.	AWWA C515-20	Reduced-Wall, Resilient-Seated Gate Valves for Water Supply Service
9.	AWWA C530-22	Pilot Operated Control Valves

E. Manufacturers Standardization Society of the Valves and Fittings Industry (MSS)

1.	MSS SP-110-2010	Ball Valves Threaded, Socket–Welding, Solder Joint, Grooved, and Flared Ends

F. National Sanitation Foundation International / American National Standards Institute (NSF/ANSI)

1.	NSF/ANSI/CAN 61-24	Drinking Water System Components – Health Effects
2.	NSF/ANSI/CAN 372-24	Drinking Water System Components – Lead Content

1.04 SUBMITTALS

A. Submit the following in accordance with the requirements of Sections 01 33 00 – Submittals.

B. RSN 35 22 14–1, Commercial Products Data:

1. For items in Part 2 Products.
2. Manufacturer's technical product data, including the applicable size, standards, bill of materials, and pressure rating:
    a. For pilot-operated control valves, include the following for each type and size of valve:
        1) Parts identification lists.
        2) Lists of special tools and accessories.
        3) Schematics showing pilot-operated control system, connections and manufacturer recommended componentry:
        4) Instructions for installing, adjusting, operating, lubricating, and maintaining equipment.
        5) Cavitation chart showing flow rate, differential pressure, valve characteristic curve (Cv versus percent open), and system velocity.

C. RSN 35 22 14-2, Field Installation Procedures.

1. For items listed in Part 2 Products.
2. Pilot-operated control valves: Include circuit schematic of pilot-operating system.

D. RSN 35 22 14–3, Shop Testing Certifications, Reports, and Data:

1. Hydrostatic pressure testing certifications and pressures used in testing valves:
   a. AWWA gate valves tested in accordance with AWWA C509, or AWWA C515.
   b. Pilot-operated control valves tested in accordance with AWWA C530.
   c. Silent check valves tested in accordance with AWWA C508.
2. Affidavit of compliance to applicable standards:
   a. AWWA gate valves.
   b. AWWA silent check valves.
   c. Pilot-operated control valves.
3. Include pertinent observations and notable occurrences during testing.

E. RSN 35 22 14–4, Service Manuals and Final Data:

1. Items in Part 2 Products.
2. Complete parts identification lists including clearly identified recommended spare parts, lists of special tools and accessories, detailed instructions for installing, adjusting, operating, lubricating, and maintaining the equipment.

1.05 DELIVERY, STORAGE, AND HANDLING

A. Take precautions during loading, transportation, unloading, storage, and laying, to prevent damage to the valves and equipment:

1. Repair damage to the valves and equipment as directed if, in the opinion of COR, a satisfactory repair can be made; otherwise replace damaged valves and equipment.
2. Ship, handle, store, and position according to manufacturer’s instructions.
3. Cover valves until installed.

1.06 LOCATION DATA

A. Refer to table as shown on drawing 1695-D-62558 for flow control valve sizes.

PART 2 PRODUCTS

2.01 GENERAL REQUIREMENTS

A. Valves and equipment materials:

1. ANSI/NSF/CAN 61 certified.
2. ANSI/NSF/CAN 372 certified lead-free.
3. Suitable for potable water.

B. Valve Body Flange Faces and Flanges:

1. Valve manufacturers are responsible for providing the correct flange face gasket surface finish.
2. Provide spiral or concentric serrated grooved flange face gasket surface finishes that are suitable for the gasket material provided and for the operating pressures.
3. Valve flange bolt patterns to match pipe flanges.

C. Furnish valves and actuators of same type and size from same manufacturer.

D. Spare parts:

1. Provide one extra set of seats for each size and type of gate valve.
2. parts for each size and type of air valve including:
    1) Seats, lever, and gasket.
    2) Additional parts recommended by manufacturer.
    3) Box and clearly label box: Include list of spare parts inside box.
3. Store spare parts in location as directed by COR.
4.

2.02 VALVES AND OPERATOR

A. Pilot Operated Control Valves with Solenoid Shut-off:

1. General:
    a. AWWA C530.
    b. Full, globe style.
    c. Anti-cavitation trim as recommended by manufacturer.
    d. During valve operation, water vented from upper pressure chamber of main valve (when main valve is opening) shall be directed into pilot piping so that it discharges into downstream side of main control valve.
    e. Diaphragm operated.

f. Pressure gauges integral to the main control valve to measure upstream and downstream pressures.

g. Drip tight shut-off.

h. Sealing disk: Suitable for potable water.

i. ASTM A536 ductile iron or ASTM A216 WCB cast steel.

j. Stainless steel trim.

k. Control system components to be manufactured by same company as main valve.

l. Pilot piping for the valves:

1) Design, furnish, and test per manufacturer's recommendations.

2) Hydraulic (water) operated control pilots with seamless copper or stainless steel tubing.

3) Heavy-duty, ball-type shut off valves.

4) Opening and closing speed control valves.

5) Pilot "Y" strainers with blowdown valves.

6) Check valves.

7) Atmospheric drain.

8) Pressure gauges.

9) Provide automatic vent valves.

m. Valve visual position indication:

1) Stainless steel position indicator rod.

2) Brass or stainless steel protective housing which allows viewing of valve position from two sides.

3) Manual air purge valve at top of assembly.

4) Pyrex tube or covers to enclose stem inside housing.

n. Furnish and install pilot piping and associated valving and fittings as recommended by the valve manufacturer.

2. Model 106-SC manufactured by Singer Valve, www.singervalve.com, Model No. 43-01, manufactured by Cla-Val Company, www.cla-val.com, Model A115 Series manufactured by Apollo Valves, www.apollovalves.com, Model 120-01 manufactured by OCV Control Valves, www.controlvalves.com; or equal, having the following essential characteristics:

a. Automatically controls flow through valve to specified flow rate.

b. Flow control settings shall be field adjustable.

c. Orifice plates:

1) Install orifice plates and valves as shown on the drawings. Actual valve and orifice plate locations may vary depending on control valve manufacturer's recommendations.

2) Provide orifice plate bore size in accordance with manufacturer's recommendations to meet required flow rates.

d. Solenoid shut-off valve:

1) 120 vac.

2) Control valve to close when supplied power.

3) Control valve will remain open on loss of power.

e. Limit switch:

1) Adjustable.

2) Integral to control valve.

3) Actuates when valve is fully closed.

4) Double pole double pole switches rated at 1 amp at 125 vdc and 120 vac.

5) UL rated with NEMA 4 enclosure.

B. Ball Valves:

1. Carbon Steel, Cast Iron or Bronze:

2. 3-inch and smaller:

a. 70LF-100 Series manufactured by Apollo Valves, www.apollovalves.com; or equal, having the following essential characteristics:

1) Cold water pressure rating of not less than 300 psig

2) Body: Full ported carbon steel or lead-free bronze.

3) Ball: Stainless steel or lead -free brass.

4) Valve stem: Stainless steel or lead-free brass.

5) Packing gland: Steel or brass.

6) Install where shown on drawings. Bronze or carbon steel ball valves shall be installed on carbon steel pipe.

7) Bubble tight at the rated pressure in either direction and suitable for operation after long periods of inactivity.

8) MSS SP–110.

9) Female threaded ends.

3. 4-inch and larger:

a. Val-Matic Rubber Seated Ball Valves, manufactured by Val-Matic, www.valmativ.com; or equal, having the following essential characteristics;

1) Flanged ends:

a) AWWA Class 150 valves: Flanges suitable for connection to AWWA Class D flanges or ANSI Class 150 flanges.

2) Valves suitable for manual service.

3) AWWA C507.

4) Pressure ratings suitable for steel pipe pressure.

4. Provide dielectric insulating connections as necessary to prevent galvanic corrosion between dissimilar metals.

C. Gate Valves:

1. Series A-2361 manufactured by Mueller Co. LLC, www.muellercompany.com, High Pressure Resilient Wedge Gate Valve manufactured by Kennedy Valve, www.kennedyvalve.com; or equal, having the following essential characteristics:

a. AWWA C509 or AWWA C515.

b. OS&Y or NRS

c. Ductile iron body and wedge shall be made of ductile iron.

d. Bonnet: Ductile iron or cast iron.

e. Handwheel.

f. Valve ends shall be flat faced flanges and shall match the drilling dimensions and pattern of AWWA C207 Class F flanges.

g. Wedges shall be fully encapsulated with EPDM.

h. Minimum cold-water working pressure rating of 300 psig.

i. Zero leakage.

D. Check Valves:

1. Series 581 Silent Check Valve manufactured by Cla-Val, www.cla-val.com, or equal, having following essential characteristics:

a. Silent operating type.

b. Globe style.

c. Center guided, spring loaded disc.

d. All parts to be field replaceable.

e. Resilient seal.

f. Zero leakage.

g. Minimum working pressure rating of 300 psig, cold water service.

h. Suitable for bolting to AWWA Class F steel pipe flanges.

i. Valve body: ASTM A536 grade 65-45-12 ductile iron or ASTM A216 grade WCB cast steel.

j. Disc: ASTM A351 grade stainless-steel.

k. Seat and compression spring, type 316 stainless-steel.

l. Valve body with integral flanges (not wafer style).

m. Flat face flanges: Provide spiral or concentric serrated grooved flange face gasket surface finishes that are suitable for flat gasket material.

E. Combination Air Valves:

1. Manufactured by DeZurik/Apco, www.dezurik.com; Cla-Val, www.cla-val.com; Val-Matic Valve and Manufacturing Corporation, www.valmatic.com; or equal having the following essential characteristics:

a. Cold water service.

b. Cold-water pressure rating: 300 psig, minimum.

c. Single or dual body.

d. Body material: Ductile iron or cast steel.

e. AWWA C512.

f. Actuated by a float or floats.

g. Stainless steel internal parts such as guides, bushings, and screws.

h. Stainless steel floats.

i. Float-pivot supports: Brass, bronze, stainless steel, or cast iron.

j. Inlet: Threaded

k. Provide combination air-inlet and air-release type or air and vacuum type actuated by a float as shown on the drawings.

l. Remains open for filling line until water has displaced air at point of attachment of valve assembly after which it closes.

m. Opens when pressure in pipeline drops sufficiently to create a vacuum.

n. Releases trapped air when the pipeline is under pressure.

o. 2-inch Air release minimum orifice size: 1/16-inch.

p. 3-inch Air release minimum orifice size: 5/64-inch

q. Operating pressure for seat materials: Resilient seat suitable for 0 to 300 psig.

r. Furnish with ball valves in accordance with paragraph 2.01 F:

1) One ball valve on top, permits checking effectiveness of air valve.

2) One ball valve on bottom, allows valve to be drained.

3) For dual body valves, provide full port isolation ball valve between main body and air release valve.

F. Inflow Preventer Assembly: Val-Matic model 1302/1303 with optional bracket SPK-1302/1303-80, and field test kit SPK-1302/1303 as applicable per air valve size, manufactured by Val-Matic Valve and Manufacturing Corporation, 905 Riverside Drive, Elmhurst, IL 60126; or equal, having the following essential characteristics:

1. Manufactured in accordance with AWWA C514 and ASSE 1063.
2. Stainless steel internal parts such as guides, bushings, and screws.
3. Upper and lower chambers: Ductile iron, ASTM A536 Grade 65-45-12.
4. Basket strainer:
   a. Type 304 stainless steel.
   b. 24# mesh opening (0.0277 inch).
   c. Quick disconnect joint.
   d. Total flow area of at least four times circular cross-sectional area having a diameter equal to nominal size of air valve outlet.
5. Float checks and trim: Type 316 stainless steel.
6. Cold-water pressure rating for body: Minimum of 25 psi.
7. Resilient seat: EPDM with fiberglass reinforcement.
8. Inlet connection: Internally threaded NPT.
9. Piping manifold and wall bracket as shown on drawings.
10. Supply mounting bracket and field test kit with all appurtenant piping and valves per air valve as shown on drawings.
11.

G. Butterfly Valves:

1. Cold water pressure up to 150 psig: AWWA C504, Class 150B.
2. Cold water pressures from 151 psig up to 250 psig: AWWA C504, Class 250B.
3. Bubble tight at rated pressure in either direction and suitable for operation after long periods of inactivity.
4. Tested in accordance with AWWA C504 including performance, leakage, and hydrostatic testing.
5. Axis of valve leaf: Horizontal when valve is in installed position except where shown on drawings.
6. Flanged ends:
   a. Class 150B valves: Body flanges suitable for connection to AWWA Class D flanges.

b. Class 250B valves: Body flanges suitable for connection to AWWA Class F flanges.

7. Valve Seat:

a. Rubber seat: Clamped, or mechanically secured to body or disc.

b. Material: AWWA C504, except bonded rubber seats will not be accepted.

c. Field replaceable seat.

d. Located either in the valve body, mating with a stainless steel seating surface, located on valve disc or in valve disc, mating with a stainless steel seating surface located in valve body.

8. Manual Operators:

e. Manually operated butterfly valves with worm-gear or traveling nut type manual operator in accordance with AWWA C504.

f. Ensure rim pull and component strength meets AWWA C504.

g. Operators that open and close valves at full unbalances pressure rating of valve.

h. Gears of self-locking type; holds valve in all positions without fluttering or creeping.

i. Handwheel where shown on drawings.

9. Buried butterfly valves:

a. Valve boxes.

b. Square nut operator: 2-inch.

c. Provide a total of 8 steel tee-handle wrenches, with a length of 4-feet.

1) Torque rating Match or exceed maximum torque input of valve actuator.

2) Tee wrench:

a) Capable of withstanding 2 times the torque rating.

b) Painted with primer and topcoat to manufacturer's standard.

d. Steel extension stems:

1) Required to provide operating nut to within 6-inches of ground surface for buried valves:

2) Rating to match or exceed maximum torque input of valve actuator.

3) Steel centering ring with 1/8-inch minimum thickness and welded all around to top of extension stem just below the operating nut.

4) Bottom socket with mud vents, welded or pinned to provide a secure connection to the square steel extension stem.

5) Connect top nut to square steel stem with a pin or two cap screws. Capable of withstanding 2 times the torque rating.

6) Painted with primer and topcoat to manufacturer's standard.

H.

I. Valve Boxes:

1. Model 6855 series, manufactured by Tyler Union, www.tylerunion.com; or equal, having the following characteristics:

a. Cast iron locking lid with pentagon shaped bolt head.

b. Cast iron extension from valve nut operator to ground surface.

c. Cast iron collar with lid marked "WATER".

d. 5 1/4-inch minimum inside diameter cast iron.

e. 2-piece or 3-piece and slip type.

2. Base and adequate extension items to extend from valve nut operator to ground surface.

3. Refer to Detail A on drawing 1695-D-62549.

J. Air Vent Damper:

1. Model No. 1512, FrostSafe two-way damper manufactured by Val-Matic Valve and Manufacturing Corporation, 905 S. Riverside Drive, Elmhurst IL 60126; or equal having the following essential characteristics:

a. For 6-inch air vent as shown on drawing 1695-D-62551.

b. Suitable for connection between AWWA Class D flanges.

c. Allows 2-way flow.

d. Minimizes thermal exchange of cold and warm air in and out of valve vault.

e. Allows vault to vent at full air valve capacity.

K. Pressure Reducing Valve

1. Single stage and suitable for water.

2. End connections: Threaded for regulators NPS 2-inch and smaller

3. Minimum working pressure: 300 psi.

L. Hose fittings

1. Hose Bibb:

a. Operation: 1/4 turn.

b. End connections: Threaded.

c. Suitable for working pressure of pipe.

d. Hose adapter;End connections: Threaded to hose barb.

e. Suitable for working pressure of pipe.

2.

M. Magnetic Flowmeter:

1. Electromagnetic Flowmeter, Ultra Mag, as manufactured by McCrometer,; or equal, having the following essential characteristics;

a. Accuracy: Plus, or minus, 0.5 percent of all flowrates, over range of measurement.

b. +0.5% Accuracy over specified flow range:

c. Straight pipe installation clearances:

1) 3 diameters upstream.

2) 2 diameters downstream.

d. Suitable for operation in ambient temperatures ranging from -4 to 140 degrees F.

e. Flanged, cylindrical flow-tube with polyurethane liner, grounding rings and gaskets.

f. Flanges:

1) Suitable for bolding to AWWA Class F.

g. Sensor Tubes:

1) Stainless steel electrode material

2) Bonded epoxy liner.

h. Transmitter:

1) Modbus or 4-20mA HART

2.03 APPURTENANCES:

A. Gaskets:

1. Multi-Swell Style 3760 manufactured by Garlock, www.garlock.com; or equal with the following essential characteristics:

a. Minimum 300 psig cold water pressure rating.

b. AWWA C207 or ASME B16.5.

c. Flat, full face type.

d. Suitable for cold water service.

2. Do not use compressed gaskets.

B. Delivery Point Signage Tags

1. As shown on Drawing 1695-D-62568
2. Engraved metal tag (blue with white lettering).
3. Wire Rope (1/8" diameter).
   a. Provide 6 spare tags, wire rope and crimps.

2.04 CONTRACTOR SOURCE QUALITY TESTING

A. Shop Testing:

1. Test valves at manufacturer's facility or an alternate facility approved by COR.
2. Tests will be Government witnessed. Notify COR 30-days prior to valve testing.
3. Minimize number of Government trips by testing valves from same manufacturer during same Government trip.
4. Prior to shop leakage test and in presence of Government, Contractor shall verify that when valve is fully closed, valve position indicator indicates, "CLOSED".
5. If indicator is not in closed position (exactly at closed), remove operator from valve, fully close valve, reinstall operator so that it indicates, "CLOSED".
6. Shop leakage test acceptance:
   a. Spray or high velocity leakage will not be acceptable.
   b. If valve leaks, adjustments shall be made until valve is drip tight with no leakage. Adjustments may include adjusting valve stops, seats, actuators, or other components.
   c. If valve is not drip tight, valve is unacceptable.
   d. After any adjustments are made, valve position indicator shall be inspected again and corrected if necessary.
7. Manually operated gate valves:
   a. Leakage test in both directions in accordance with AWWA C509 or AWWA C515, as applicable. Record results.
   b. Hydrostatic test in accordance with AWWA C509 or AWWA C515, as applicable. Record pressures used.
   c. Pressure test valves and their seats for leaks with water in 100 percent closed position for a timed period of 5-minutes with pressure equal to rated design pressure. Valves shall be drip tight with no leakage.
   d. Downstream position of valve shall be visible to Government inspector.
8. Check valves (globe style) shop testing:

a. AWWA C508 seat leakage and valve body hydrostatic test.

b. In accordance with manufacturer's recommendations.

9. Pilot-operated control valves with solenoid shutoff shop testing:

a. AWWA C530 seat leakage test and valve body hydrostatic test.

b. Manufacturers' requirements.

PART 3 EXECUTION

3.01 INSTALLATION

A. Dielectrically insulate between dissimilar metals:

1. Provide insulating type unions at threaded valve, pipe or fitting connections between dissimilar metals.

B. Provide a sufficient quantity of unions for removal, replacement and maintenance of valves and equipment.

C. Install the valves and equipment to line and grade as shown on drawings.

D. Install valves with flanged connections so that the bolt holes straddle the vertical centerline when in the installed position.

E. Install valves and gaskets in accordance with the manufacturer's instructions.

F. Valve Tags:

1. Provide brass or stainless steel tags for all valves identified in the Piping and Instrumentation Diagram drawings 1695-D-T0152 through 1695-D-T0161.

2. Stamp or engrave tag with the valve identification code designated in the Piping and Instrumentation Diagram drawings 1695-D-T0152 through 1695-D-T0161.

3. Permanently attach tag to valve with wire and metal crimp-type connector.

3.02 CONTRACTOR FIELD QUALITY TESTING

A. Field Pressure Testing:

1. After each valve has been completely installed, test valve in place by opening and closing valve through full range of operation 3 times.

2. Furnish testing equipment, as approved by COR, which provides observable and accurate measurements of leakage under specified conditions.

3. Maximum Filling Velocity: 0.25 feet per second calculated based on full area of pipe.

4. Expel air from piping system during filling.

5. Test Pressure:
   a. 150 percent of steel manifold pipe normal operating pressure.
   b. Maximum test pressure not to exceed 300 psi.
   c. Test pressure for some segments may be reduced subject to COR approval.
6. Apply and maintain specified test pressure with hydraulic force pump. Valve off piping system when test pressure is reached.
7. Maintain hydrostatic test pressure continuously for 2 hours minimum.
8. Maximum Allowable Leakage: No leakage.
9. Dispose of water used in testing.

B. Test pilot operated control valve per manufacturer's recommendation.

C. Test Flow control valve and Flowmeter per specification Section 25 08 10 – Automation System Testing and Acceptance.

3.03 COATINGS

A. Coat valves and equipment in accordance with Section 09 96 20 – Coatings.

END OF SECTION

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SECTION 51 00 00

INFORMATION AVAILABLE TO OFFERORS

PART 1 GENERAL

1.01 ORDER OF PRECEDENCE

A. In case of differences between sections in Division 51 – Information Available to Offerors and sections in other divisions of the specifications, requirements in other divisions will govern.

B. Government assumes no responsibility for deductions, interpretations, or conclusions made by Contractor based on information made available by Government, in accordance with clause at FAR 52.236-3, Site Investigations and Conditions Affecting the Work.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 51 00 40
NTUA - MARKERS

PART 1 GENERAL

1.01 MARKERS

A. Above Ground Markers as referenced in 10 14 16 – Utility Markers.



PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

SECTION 51 00 50

ELECTRONIC MEDIA RELEASE

U.S. Bureau of Reclamation
Electronic Media Release

U.S. Bureau of Reclamation, Contract Number/Solicitation Number: _____

This release for electronic media is dated the ___ day of _____ (month), ____ (year), between _____ (RECIPIENT) and U.S. Bureau of Reclamation, for the exchange of electronic media (CD ROMs, DVD, Flash Drives, External Hard Disk Drives, etc.) containing information on the _____ project (hereinafter referred to as the PROJECT) for use by the RECIPIENT.

Therefore, RECIPIENT and U.S Bureau of Reclamation (Reclamation), agree as follows:

1. All data and intellectual property contained on the electronic files remains the property of U.S. Bureau of Reclamation and may not be copied or distributed to any other party without the written permission of U.S. Bureau of Reclamation.
2. The electronic files provided to RECIPIENT by U.S. Bureau of Reclamation, for the PROJECT may be used or adapted by RECIPIENT for their own use. If RECIPIENT chooses to alter in any way, in whole or in part, the electronic files provided for the PROJECT or any future project(s), RECIPIENT agrees that the use of the altered files shall be without liability or legal exposure to U.S. Bureau of Reclamation.
3. Because information and data provided electronically may be altered, whether inadvertently or otherwise, U.S. Bureau of Reclamation, reserves the right to retain copies of the electronic file(s) and to remove from the electronic files provided to RECIPIENT all identification (such as logo, surveyor's seal, etc.) reflecting the involvement of U.S. Bureau of Reclamation, in their preparation.
4. The electronic files are provided solely as a convenience to RECIPIENT by U.S. Bureau of Reclamation and shall not be considered "Drawings of Record" or "Construction Documents." All documents considered "Drawings of Record" or "Construction Documents" will be hardcopy drawing sheets with signatures or scanned files of the hardcopy drawings with signatures.
5. RECIPIENT is advised to check all electronic media for viruses and malware before loading the files. RECIPIENT is solely responsible for intercepting and disabling viruses, if any, that may be transmitted with the electronic files and hereby agrees to indemnify and hold U.S. Bureau of Reclamation harmless from and against all claims of any type or nature asserted by RECIPIENT or any party as a result of viruses transmitted with the electronic files.
6. Files distributed electronically are subject to data erosion, erasure, and/or alteration, and computer systems and software become obsolete in time. By accepting these electronic files, RECIPIENT acknowledges these risks and agrees to waive all claims against U.S. Bureau of Reclamation, should data erosion, erasure, and/or alteration of these electronic files occur.

7. RECIPIENT agrees to defend, indemnify, and hold U.S. Bureau of Reclamation, harmless from all claims, injuries, damages, losses, expenses, and costs, including attorney's fees, arising out of breach of this agreement, or the modification or reuse of the electronic files.

ACCEPTED FOR RECIPIENT:	RELEASED BY U.S. BUREAU OF RECLAMATION:
By: _____	_____
Title: _____	_____
Date: _____	_____

END OF SECTION

SECTION 51 00 80
BLM HERBICIDES

PART 1 GENERAL

	Herbicides Formulations Approved for Use on Lands		
	The BLM Administers in the 17 Western States		
			Update: April 4, 2019
Restrictions associated with exisitng Environmental Impact Statements and individual Environmental Assessments (EA)			
at the present time, may restrict the use of individual herbicide active ingredients allowed for a particular			
project within that state. Refer to current EAs prior to selecting the active ingredient(s) and subsequent formulation(s).			
Refer to the complete label prior to considering the use of any herbicide formulation. Just because it has a Federal registration,			
it may not be registered in a particular State, for example California. Label changes can also impact the intended use			
through, such things as, creation or elimination of Special Local Need (SLN) or 24 (C) registrations, changes in application sites,			
rates and timing of application, county restrictions, etc.			
ACTIVE			EPA REG.
INGREDIENT	TRADE NAME	MANUFACTURER	NUMBER
Aminopyralid	Milestone	Dow AgroSciences	62719-519
Aminopyralid + 2,4-D	ForeFront HL	Dow AgroSciences	62719-630
	GrazonNext HL	Dow AgroSciences	62719-628
Aminopyralid + Clopyralid	Sendero	Dow AgroSciences	62719-645
Aminopyralid + Metsulfuron methyl	Chaparral	Dow AgroSciences	62719-597
	Opensight	Dow AgroSciences	62719-597
Aminopyralid + Triclopyr	Capstone	Dow AgroSciences	62719-572
Bromacil	Alligare Bromacil 80	Alligare, LLC	81927-4
	Ceannard Bromacil 80DF	Ceannard, Inc.	58035-19
	Hyvar X	Bayer Environmental Science	432-1546
	Hyvar X	DuPont Crop Protection	352-287
	Hyvar X-L	Bayer Environmental Science	432-1548
	Hyvar X-L	DuPont Crop Protection	352-346
Bromacil + Diuron	Alligare Bromacil/Diuron 40/40	Alligare, LLC	81927-3
	Ceannard Diuron/Bromacil 80 DF	Ceannard, Inc.	58035-18
	DiBro 2+2	Nufarm Americas Inc.	228-227
	DiBro 4+2	Nufarm Americas Inc.	228-386
	DiBro 4+4	Nufarm Americas Inc.	228-235
	Krovar I DF	Bayer Environmental Science	432-1551
	Krovar I DF	DuPont Crop Protection	352-505
	Weed Blast 4G	SSI Maxim	34913-19
	Weed Blast Res. Weed Cont.	Loveland Products Inc.	34704-576

ACTIVE INGREDIENT	TRADE NAME	MANUFACTURER	EPA REG. NUMBER
Chlorsulfuron	Alligare Chlorsulfuron 75	Alligare, LLC	81927-43
	Chlorsulfuron E-Pro 75 WDG	Nufarm Americas Inc.	79676-72
	Nufarm Chlorsulf SPC 75 WDG Herbicide	Nufarm Americas Inc.	228-672
	Telar XP	Bayer Environmental Science	432-1561
	Telar XP	DuPont Crop Protection	352-654
Clopyralid	Alligare Clopyralid 3	Alligare, LLC	81927-14
	CleanSlate	Nufarm Americas Inc.	228-491
	Pyramid R&P	Albaugh, LLC (Albuagh, Inc/Agri Star)	42750-94
	Reclaim	Dow AgroSciences	62719-83
	Spur	Albaugh, LLC (Albuagh, Inc/Agri Star)	42750-89
	Stinger	Dow AgroSciences	62719-73
	Transline	Dow AgroSciences	62719-259
Clopyralid + 2, 4-D	Alligare Cody Herbicide	Alligare, LLC	81927-28
	Commando	Albaugh, LLC (Albuagh, Inc/Agri Star)	42750-92
	Curtail	Dow AgroSciences	62719-48
	Cutback	Nufarm Americas Inc.	71368-72
2, 4-D	2,4-D 4# Amine Weed Killer	UAP-Platte Chem. Co.	34704-120
	2,4-D Amine	Helena Agri-Enterprises, LLC (Helena Chemical Company)	5905-72
	2,4-D Amine 4	Albaugh, LLC (Albuagh, Inc/Agri Star)	42750-19
	2,4-D Amine 4	Helena Agri-Enterprises, LLC (Helena Chemical Company)	42750-19-5905
	2,4-D LV 4	Albaugh, LLC (Albuagh, Inc/Agri Star)	42750-15
	2,4-D LV 6 Ester	Nufarm Americas Inc.	228-95
	2,4-D LV4	Helena Agri-Enterprises, LLC (Helena Chemical Company)	5905-90
	2,4-D LV 6	Albaugh, LLC (Albuagh, Inc/Agri Star)	42750-20
	2,4-D LV6	Helena Agri-Enterprises, LLC (Helena Chemical Company)	42750-20-5905
	2,4-D LV6	Helena Agri-Enterprises, LLC (Helena Chemical Company)	5905-93
	Alliagre 2,4-D Amine	Alligare, LLC	81927-38
	Alligare 2,4-D LV 6	Alligare, LLC	81927-39
	Aqua-Kleen	Nufarm Americas Inc.	71368-4
	Aqua-Kleen	Nufarm Americas Inc.	228-378
	Barrage HF	Helena Agri-Enterprises, LLC (Helena Chemical Company)	5905-529
	Barrage LV Ester	Helena Agri-Enterprises, LLC (Helena Chemical Company)	5905-504
	Base Camp Amine 4	Wilbur-Ellis Co., LLC (Wilbur-Ellis Co.)	71368-1-2935
	Base Camp LV6	Wilbur-Ellis Co., LLC (Wilbur-Ellis Co.)	2935-553
	Broadrange 55	Wilbur-Ellis Co., LLC (Wilbur-Ellis Co.)	2217-813-2935

ACTIVE INGREDIENT	TRADE NAME	MANUFACTURER	EPA REG. NUMBER
2,4-D - continued	Clean Amine	Loveland Products Inc.	34704-120
	Clean Crop Amine 4	UAP-Platte Chem. Co.	34704-5 CA
	Clean Crop Low Vol 6 Ester	UAP-Platte Chem. Co.	34704-125
	Clean Crop LV-4 ES	UAP-Platte Chem. Co.	34704-124
	Cornbelt 4 lb. Amine	Van Diest Supply Co.	11773-2
	Cornbelt 4# LoVol Ester	Van Diest Supply Co.	11773-3
	Cornbelt 6# LoVol Ester	Van Diest Supply Co.	11773-4
	D-638	Albaugh, LLC (Albuagh, Inc/Agri Star)	42750-36
	Esteron 99C	Nufarm Americas Inc.	62719-9-71368
	Five Star	Albaugh, LLC (Albuagh, Inc/Agri Star)	42750-49
	Formula 40	Nufarm Americas Inc.	228-357
	Freelexx	Dow AgroSciences	62719-634
	HardBall	Helena Agri-Enterprises, LLC (Helena Chemical Company)	5905-549
	Hi-Dep	PBI Gordon Corp.	2217-703
	Low Vol 4 Ester Weed Killer	Loveland Products Inc.	34704-124
	Low Vol 6 Ester Weed Killer	Loveland Products Inc.	34704-125
	Opti-Amine	Helena Agri-Enterprises, LLC (Helena Chemical Company)	5905-501
	Platoon	Nufarm Americas Inc.	228-145
	Rugged	WinField-United (WinField Solutions, LLC)	1381-247
	Saber	Loveland Products Inc.	34704-803
	Salvo	Loveland Products Inc.	34704-609
	Salvo LV Ester	UAP-Platte Chem. Co.	34704-609
	Savage DS	Loveland Products Inc.	34704-606
	Savage DS	UAP-Platte Chem. Co.	34704-606
	Shredder 2,4-D LV4	WinField-United (WinField Solutions, LLC)	1381-102
	Shredder Amine 4	WinField-United (WinField Solutions, LLC)	1381-103
	Solution Water Soluble	Nufarm Americas Inc.	228-260
	Solve 2,4-D	Albaugh, LLC (Albuagh, Inc/Agri Star)	42750-22
	Unison	Helena Agri-Enterprises, LLC (Helena Chemical Company)	5905-542
	Weedar 64	Nufarm Americas Inc.	71368-1
	WEEDestroy AM-40	Nufarm Americas Inc.	228-145
	Weedone LV-4	Nufarm Americas Inc.	228-139-71368
	Weedone LV-4 Solventless	Nufarm Americas Inc.	71368-14
	Weedone LV-6	Nufarm Americas Inc.	71368-11
	Whiteout 2,4-D	Loveland Products, Inc.	34704-1032
Dicamba	Alligare Cruise Control	Alligare, LLC	42750-40-81927
	Alligare Dicamba 4 Herbicide	Alligare, LLC	81927-55
	Banvel	Arysta LifeScience N.A. Corp.	66330-276
	Clarity	BASF Corporation	7969-137
	Diablo	Nufarm Americas Inc.	228-379
	Dicamba DMA	Albaugh, LLC (Albuagh, Inc/Agri Star)	42750-40

ACTIVE INGREDIENT	TRADE NAME	MANUFACTURER	EPA REG. NUMBER
Dicamba - continued	Kam-Ba	Drexel Chemical Company	19713-624
	Rifle	Loveland Products Inc.	34704-861
	Sterling Blue	WinField-United (WinField Solutions, LLC)	7969-137-1381
	Topeka	Rotam North America, Inc.	83100-34-83979
	Vanquish	Syngenta	100-884
	Vanquish Herbicide	Nufarm Americas Inc.	228-397
	Vision	Helena Agri-Enterprises, LLC (Helena Chemical Company)	5905-576
Dicamba + 2, 4-D	Alligare Dicamba + 2,4-D DMA	Alligare, LLC	81927-42
	Brash	WinField-United (WinField Solutions, LLC)	1381-202
	Brush-Rhap	Helena Agri-Enterprises, LLC (Helena Chemical Company)	5905-568
	Cimarron MAX - Part B	Bayer Environmental Science	432-1555
	Cimarron MAX - Part B	DuPont Crop Protection	352-615
	KambaMaster	Nufarm Americas Inc.	71368-34
	Latigo	Helena Agri-Enterprises, LLC (Helena Chemical Company)	5905-564
	Outlaw	Helena Agri-Enterprises, LLC (Helena Chemical Company)	5905-574
	Range Star	Albaugh, LLC (Albuagh, Inc/Agri Star)	42750-55
	Rifle-D	Loveland Products Inc.	34704-869
	Veteran 720	Nufarm Americas Inc.	228-295
	Weedmaster	Nufarm Americas Inc.	71368-34
Dicamba + Diflufenzopyr	Distinct	BASF Corporation	7969-150
	Overdrive	BASF Corporation	7969-150
NOTE: In accordance with the Record of Decision for the *Vegetation Treatments Using Herbicides on Bureau of Land Management* Lands in 17 Western *States Programmatic Environmental Impact Statement (PEIS), the aerial application of this herbicide* is prohibited.			
Diquat	Alligare Diquat Herbicide	Alligare, LLC	81927-43
	Diquat E-AG 2L	Nufarm Americas Inc.	79676-75
	Diquat E-Pro 2L	Nufarm Americas Inc.	79676-75
	Diquat SPC 2L Herbicide	Nufarm Americas Inc.	228-675
	Nufarm Diquat 2L Herbicide	Nufarm Americas Inc.	228-675
	Reward	Syngenta Professional Products	100-1091
Diuron	Alligare Diuron 4L	Alligare, LLC	81927-44
	Alligare Diuron 80DF	Alligare, LLC	81927-12
	Ceannard Diuron 80DF	Ceannard, Inc.	58035-16
	Direx 4L	DuPont Crop Protection	352-678
	Direx 4L	Makhteshim Agan of N. A. (ADAMA)	66222-54
	Diuron 4L	Drexel Chemical Company	19713-36

ACTIVE INGREDIENT	TRADE NAME	MANUFACTURER	EPA REG. NUMBER
Diuron - continued	Diuron 4L	Loveland Products Inc.	34704-854
	Diuron 4L	Makhteshim Agan of N. A. (ADAMA)	66222-54
	Diuron 80	Drexel Chemical Company	19713-274
	Diuron 80 WDG	Loveland Products Inc.	34704-648
	Diuron 80DF	WinField-United (WinField Solutions, LLC)	9779-318
	Diuron 80WDG	UAP-Platte Chem. Co.	34704-648
	Karmex DF	DuPont Crop Protection	352-692
	Karmex DF	Makhteshim Agan of N. A. (ADAMA)	66222-51
	Karmex IWC	DuPont Crop Protection	352-692
	Karmex XP	DuPont Crop Protection	352-692
	Parrot 4L	Makhteshim Agan of N. A. (ADAMA)	66222-54
	Parrot DF	Makhteshim Agan of N. A. (ADAMA)	66222-51
Fluridone	Alligare Fluridone	Alligare, LLC	81927-45
	Avast!	SePRO	67690-30
	Fluridone 4L	Albaugh, LLC (Albuagh, Inc/Agri Star)	42750-280
	Sonar AS	SePRO	67690-4
	Sonar Precision Release	SePRO	67690-12
	Sonar Q	SePRO	67690-3
	Sonar SRP	SePRO	67690-3
Fluroxypyr	Alligare Flagstaff	Alligare, LLC	81927-61
	Alligare Fluroxypyr	Alligare, LLC	66330-385-81927
	Comet Selective	Nufarm Americas Inc.	71368-87
	Vista XRT	Dow AgroSciences	62719-586
Fluroxypyr + 2,4-D + Dicamba	E-2 Herbicide	Nufarm Americas Inc.	228-442
Fluroxypyr + Clopyralid	Truslate Selective Herbicide	Nufarm Americas Inc.	71368-86
Fluroxypyr + Picloram	Alligare Triumph XTR Herbicide	Alligare , LLC	81927-64
	Surmount	Dow AgroSciences	62719-480
	Trooper Pro	Nufarm Americas Inc.	228-599
Fluroxypyr + Triclopyr	Alligare Cleargraze Pasture Herbicide	Alligare, LLC	81927-65
	PastureGard	Dow AgroSciences	62719-637
Glyphosate	Accord Concentrate	Dow AgroSciences	62719-324
	Accord SP	Dow AgroSciences	62719-322
	Accord XRT	Dow AgroSciences	62719-517
	Accord XRT II	Dow AgroSciences	62719-556
	Alligare Dryphosate 75SG	Alligare, LLC	81927-60
	Alligare Glyphosate 4 PLUS	Alligare, LLC	81927-9
	Alligare Glyphosate 5.4	Alligare, LLC	81927-8
	Aqua Neat	Nufarm Americas Inc.	228-365

ACTIVE INGREDIENT	TRADE NAME	MANUFACTURER	EPA REG. NUMBER
Glyphosate - continued	Aqua Star	Albaugh, LLC (Albuagh, Inc/Agri Star)	42750-59
	Aquamaster	Monsanto	524-343
	AquaPro Aquatic Herbicide	SePRO Corporation	62719-324-67690
	Buccaneer	Tenkoz	55467-10
	Buccaneer Plus	Tenkoz	55467-9
	Credit Xtreme	Nufarm Americas Inc.	71368-81
	Foresters	Nufarm Americas Inc.	228-381
	Gly Star Gold	Albaugh, LLC (Albuagh, Inc/Agri Star)	42750-61
	Gly Star Original	Albaugh, LLC (Albuagh, Inc/Agri Star)	42750-60
	Gly Star Plus	Albaugh, LLC (Albuagh, Inc/Agri Star)	42750-61
	Gly Star Pro	Albaugh, LLC (Albuagh, Inc/Agri Star)	42750-61
	Gly-4	Universal Crop Protection Alliance	42750-60-72693
	Gly-4 Plus	Universal Crop Protection Alliance	72693-1
	Gly-4 Plus	Universal Crop Protection Alliance	42750-61-72693
	GlyphoMate 41	PBI/Gordon Corporation	2217-847
	Glypro	Dow AgroSciences	62719-324
	Glypro Plus	Dow AgroSciences	62719-322
	Honcho	Monsanto	524-445
	Honcho Plus	Monsanto	524-454
	Imitator Aquatic	Drexel Chemical Company	19713-623
	Imitator DA	Drexel Chemical Company	19713-586
	Imitator Plus	Drexel Chemical Company	19713-526
	KleenUp Pro	Loveland Products, Inc.	34704-890
	Mad Dog Plus	Loveland Products, Inc.	34704-890
	Makaze	Loveland Products, Inc.	34704-890
	Mirage	Loveland Products Inc.	34704-889
	Mirage Herbicide	UAP-Platte Chem. Co.	524-445-34704
	Mirage Plus	Loveland Products Inc.	34704-890
	Rattler	Helena Agri-Enterprises, LLC (Helena Chemical Company)	524-445-5905
	Razor	Nufarm Americas Inc.	228-366
	Razor Pro	Nufarm Americas Inc.	228-366
	Rodeo	Dow AgroSciences	62719-324
	Roundup Custom	Monsanto	524-343
	Roundup Original	Monsanto	524-445
	Roundup Original II	Monsanto	524-454
	Roundup Original II CA	Monsanto	524-475
	Roundup PROMAX	Monsanto	524-579
	Roundup PRO	Monsanto	524-475
	Roundup PRO Concentrate	Monsanto	524-529
	Roundup PRO Dry	Monsanto	524-505
	Showdown	Helena Agri-Enterprises, LLC (Helena Chemical Company)	71368-25-5905

ACTIVE INGREDIENT	TRADE NAME	MANUFACTURER	EPA REG. NUMBER
Glyphosate + 2, 4-D	Campaign	Monsanto	524-351
	Imitator + 2,4-D	Drexel Chemical Company	19713-635
	Landmaster BW	Albaugh, LLC (Albuagh, Inc/Agri Star)	42570-62
	Landmaster BW	Monsanto	524-351
Hexazinone	Pronone 10G	Pro-Serve	33560-21
	Pronone 25G	Pro-Serve	33560-45
	Pronone MG	Pro-Serve	33560-21
	Pronone Power Pellet	Pro-Serve	33560-41
	Velosa	Helena Agri-Enterprises, LLC (Helena Chemical Company)	5905-579
	Velpar DF	DuPont Crop Protection	352-581
	Velpar DF VU	Bayer Environmental Science	432-1576
	Velpar L	DuPont Crop Protection	352-392
	Velpar L VU	Bayer Environmental Science	432-1573
	Velpar ULW	DuPont Crop Protection	352-450
Hexazinone + Sulfometuron methyl	Oustar	Bayer Environmental Science	432-1553
	Oustar	DuPont Crop Protection	352-603
	Westar	Bayer Environmental Science	432-1558
	Westar	DuPont Crop Protection	352-626
NOTE: In accordance with the Record of Decision for the *Vegetation Treatments Using Herbicides on Bureau of Land Management Lands in 17 Western States Programmatic Environmental Impact Statement (PEIS), the aerial application of this herbicide* is prohibited.			
Imazapic	Alligare Panoramic 2SL	Alligare, LLC	66222-141-81927
	Nufarm Imazapic 2SL	Nufarm Americas Inc.	71368-99
	Open Range G	Wilbur-Ellis Co., LLC (Wilbur-Ellis Co.)	2935-557
	Plateau	BASF Corporation	241-365
Imazapyr	Alligare Ecomazapyr 2SL	Alligare, LLC	81927-22
	Alligare Imazapyr 4SL	Alligare, LLC	81927-24
	Alligare Rotary 2 SL	Alligare, LLC	
	Arsenal	BASF Corporation	241-346
	Arsenal Applicators Conc.	BASF Corporation	241-299
	Arsenal PowerLine	BASF Corporation	241-431
	Chopper	BASF Corporation	241-296
	EZ-JECT Copperhead Herbicide Shells	EZ-JECT, Inc.	83220-2
	Habitat	BASF Corporation	241-426
	Habitat Herbicide	SePRO	241-426-67690
	Polaris	Nufarm Americas Inc.	228-534
	Polaris AC	Nufarm Americas Inc.	241-299-228
	Polaris AC	Nufarm Americas Inc.	228-480
	Polaris AC Complete	Nufarm Americas Inc.	228-570

ACTIVE INGREDIENT	TRADE NAME	MANUFACTURER	EPA REG. NUMBER
Imazapyr - continued	Polaris AQ	Nufarm Americas Inc.	241-426-228
	Polaris Herbicide	Nufarm Americas Inc.	241-346-228
	Polaris RR	Nufarm Americas Inc.	241-273-228
	Polaris SP	Nufarm Americas Inc.	228-536
	Polaris SP	Nufarm Americas Inc.	241-296-228
	SSI Maxim Arsenal 0.5G	SSI Maxim Co., Inc.	34913-23
	SSI Maxim Arsenal 5.0 G	SSI Maxim Co., Inc.	34913-24
	Stalker	BASF Corporation	241-398
Imazapyr + Diuron	Alligare Mojave 70 EG	Alligare, LLC	81927-25
	Imazuron	Nufarm Americas Inc.	228-654
	Sahara DG	BASF Corporation	241-372
	SSI Maxim Topsite 2.5G	SSI Maxim Co., Inc.	34913-22
Imazapyr + Metsulfuron methyl	Lineage Clearstand	Bayer Environmental Science	432-1578
	Lineage Clearstand	DuPont Crop Protection	352-766
Imazapyr + Sulfometuron methyl +	Lineage HWC	Bayer Environmental Science	432-1577
Metsulfuron methyl	Lineage HWC	DuPont Crop Protection	352-765
	Lineage Prep	Bayer Environmental Science	432-1579
	Lineage Prep	DuPont Crop Protection	352-767
NOTE: In accordance with the Record of Decision for the *Vegetation Treatments Using Herbicides on Bureau of Land Management* Lands in 17 Western *States Programmatic Environmental Impact Statement (PEIS), the aerial application of this herbicide* is prohibited.			
Metsulfuron methyl	Alligare MSM 60	Alligare, LLC	81927-7
	AmTide MSM 60DF Herbicide	AmTide, LLC	83851-3
	Cimarron MAX - Part A	Bayer Environmental Science	432-1555
	Cimarron MAX - Part A	DuPont Crop Protection	352-615
	Escort XP	Bayer Environmental Science	432-1549
	Escort XP	DuPont Crop Protection	352-439
	Patriot	Nufarm Americas Inc.	228-391
	PureStand	Nufarm Americas Inc.	71368-38
	Rometsol	Rotam North America, Inc.	831000-2-83979
Metsulfuron methyl + Chlorsulfuron	Cimarron Plus	Bayer Environmental Science	432-1572
	Cimarron Plus	DuPont Crop Protection	352-670
	Cimarron X-tra	DuPont Crop Protection	352-669
Picloram	Alligare Picloram 22K	Alligare, LLC	81927-18
	Grazon PC	Dow AgroSciences	62719-181
	OutPost 22K	Dow AgroSciences	62719-6
	Tordon 22K	Dow AgroSciences	62719-6
	Tordon K	Dow AgroSciences	62719-17

ACTIVE INGREDIENT	TRADE NAME	MANUFACTURER	EPA REG. NUMBER
Picloram - continued	Triumph 22K	Albaugh, LLC (Albuagh, Inc/Agri Star)	42750-79
	Triumph K	Albaugh, LLC (Albuagh, Inc/Agri Star)	42750-81
	Trooper 22K	Nufarm Americas Inc.	228-535
Picloram + 2, 4-D	Alligare Picloram + D	Alligare, LLC	81927-16
	Graslan L	Dow AgroSciences	62719-655
	Grazon P+D	Dow AgroSciences	62719-182
	GunSlinger	Albaugh, LLC (Albuagh, Inc/Agri Star)	42750-80
	HiredHand P+D	Dow AgroSciences	62719-182
	Pathway	Dow AgroSciences	62719-31
	Tordon 101 Mixture	Dow AgroSciences	62719-5
	Tordon RTU	Dow AgroSciences	62719-31
	Trooper P + D	Nufarm Americas Inc.	228-530
Picloram + 2, 4-D + Dicamba	Trooper Extra	Nufarm Americas Inc.	228-586
Rimsulfuron	Alligare Laramie 25DF	Alligare, LLC	81927-57
	Hinge	Rotam Borth America, Inc.	83100-40-83979
	Matrix SG	Dupont Crop Protection	352-768
Sulfometuron methyl	Alligare SFM 75	Alligare, LLC	81927-26
	Oust XP	Bayer Environmenatl Science	432-1552
	Oust DF	DuPont Crop Protection	352-401
	Oust XP	DuPont Crop Protection	352-601
	Spyder	Nufarm Americas Inc.	228-408
NOTE: In accordance with the Record of Decision for the *Vegetation Treatments Using Herbicides on Bureau of Land Management Lands in 17 Western States Programmatic Environmental Impact Statement (PEIS), the aerial application of this herbicide is prohibited.*			
Sulfometuron methyl + Chlorsulfuron	Landmark XP	Bayer Environmental Science	432-1560
	Landmark XP	DuPont Crop Protection	352-645
NOTE: In accordance with the Record of Decision for the *Vegetation Treatments Using Herbicides on Bureau of Land Management Lands in 17 Western States Programmatic Environmental Impact Statement (PEIS), the aerial application of this herbicide is prohibited.*			

ACTIVE INGREDIENT	TRADE NAME	MANUFACTURER	EPA REG. NUMBER
Sulfometuron methyl + Metsulfuron methyl	Alligare SFM Extra	Alligare, LLC	81927-5
	Oust Extra	Bayer Environmental Science	432-1557
	Oust Extra	DuPont Crop Protection	352-622
	Spyder Extra Selective	Nufarm Americas Inc.	228-690
NOTE: In accordance with the Record of Decision for the *Vegetation Treatments Using Herbicides on Bureau of Land Management* Lands in 17 Western *States Programmatic Environmental Impact Statement (PEIS), the aerial application of this herbicide* is prohibited.			
Tebuthiuron	Alligare Tebuthiuron 20 P	Alligare, LLC	81927-41
	Alligare Tebuthiuron 80 WG	Alligare, LLC	81927-37
	Spike 20P	Dow AgroSciences	62719-121
	Spike 80DF	Dow AgroSciences	62719-107
	SpraKil S-5 Granules	SSI Maxim Co., Inc.	34913-10
Tebuthiuron + Diuron	SpraKil SK-13 Granular	SSI Maxim Co., Inc.	34913-15
	SpraKil SK-26 Granular	SSI Maxim Co., Inc.	34913-16
Triclopyr	Alligare Boulder 6.3	Alligare, LLC	81927-54
	Alligare Triclopry 4	Alligare, LLC	81927-11
	Alligare Triclopyr 3	Alligare, LLC	81927-13
	Element 3A	Dow AgroSciences	62719-37
	Element 4	Dow AgroSciences	62719-40
	Forestry Garlon XRT	Dow AgroSciences	62719-553
	Garlon 3A	Dow AgroSciences	62719-37
	Garlon 4	Dow AgroSciences	62719-40
	Garlon 4 Ultra	Dow AgroSciences	62719-527
	Pathfinder II	Dow AgroSciences	62719-176
	Relegate	Nufarm Americas Inc.	228-521
	Relegate RTU	Nufarm Americas Inc.	228-522
	Remedy	Dow AgroSciences	62719-70
	Remedy Ultra	Dow AgroSciences	62719-552
	Renovate 3	SePRO Corporation	62719-37-67690
	Renovate OTF	SePRO Corporation	67690-42
	Tahoe 3A	Nufarm Americas Inc.	228-520
	Tahoe 4E	Nufarm Americas Inc.	228-385
	Tahoe 4E Herbicide	Nufarm Americas Inc.	228-517
	Triclopyr RTU	Albaugh, LLC (Albuagh, Inc/Agri Star)	42750-173
	Trycera	Helena Agri-Enterprises, LLC (Helena Chemical Company)	5906-580
	Vastlan	Dow AgroSciences	62719-687

ACTIVE INGREDIENT	TRADE NAME	MANUFACTURER	EPA REG. NUMBER
Triclopyr + 2, 4-D	Alligare Everett	Alligare, LLC	81927-29
	Aquasweep	Nufarm Americas Inc.	228-316
	Candor	Nufarm Americas Inc.	228-565
	Crossbow	Dow AgroSciences	62719-260
Triclopyr + Clopyralid	Alligare Prescott Herbicide	Alligare, LLC	81927-30
	Brazen	Nufarm Americas Inc.	228-564
	Redeem R&P	Dow AgroSciences	62719-337

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 52 00 00
DRAWINGS

PART 1 GENERAL

1.01 DISCREPANCIES, ERRORS, OR OMISSIONS

A. Inform the CO of discrepancies discovered on drawings in accordance with clause at FAR 52.236-21, Specifications and Drawings for Construction.

1.02 PROJECT CONDITIONS

A. Where there are differences as determined by the CO between details and dimensions shown on drawings and details and dimensions of existing features at jobsite, use details and dimensions of existing features at jobsite.

1.03 INFORMATION DRAWINGS

A. Drawings marked "For Information Only" in the drawing list are included to show existing features about which knowledge is required to perform work under this contract. These drawings do not show work to be performed under this contract.

B. If there are differences as determined by the CO between details and dimensions shown on these drawings and those of existing features at jobsite, use details and dimensions of existing features at jobsite.]

1.04 STANDARD DRAWINGS

A. Standard drawings may show details which are not a part of work under this contract. Disregard details shown on these drawings which are not applicable to work under this contract.

1.05 DRAWING NUMBERS IN NUMERICAL ORDER

A. Specification drawings are listed in numerical order in Table 52 00 00B - Drawings in Numerical Order.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

Table 52 00 00B - Drawings in Numerical Order

Drawing No.	Sheet No.
Specification Drawings	
1695-D-T0001	1
1695-529-60481	2
1695-529-T0004	3
1695-529-T0005	4
1695-529-T0006	5
1695-529-T0007	6
1695-529-T0008	7
1695-529-T0009	8
1695-529-T0010	9
1695-D-T0041	10
1695-D-T0049	11
1695-D-T0050	12
1695-D-T0051	13
1695-D-T0052	14
1695-D-T0053	15
1695-D-T0054	16
1695-D-T0055	17
1695-D-T0056	18
1695-D-T0057	19
1695-D-T0058	20
1695-D-T0059	21
1695-D-T0060	22
1695-D-T0061	23
1695-D-T0062	24

Table 52 00 00B - Drawings in Numerical Order

Drawing No.	Sheet No.
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1695-D-T0082	26
1695-D-T0083	27
1695-D-T0084	28
1695-D-T0085	29
1695-D-T0086	30
1695-D-T106011	31
1695-D-T106012	32
1695-D-T106013	33
1695-D-T106014	34
1695-D-T106015	35
1695-D-T106016	36
1695-D-T106017	37
1695-D-T0097	38
1695-D-T0099	39
1695-D-T0092	40
1695-D-T0132	41
1695-D-T0133	42
1695-D-T0134	43
1695-D-T0135	44
1695-D-T0136	45
1695-D-T0137	46
1695-D-T0138	47
1695-D-T0142	48

Table 52 00 00B - Drawings in Numerical Order

Drawing No.	Sheet No.
1695-D-T0144	49
1695-D-T0149	50
1695-D-T0150	51
1695-D-T0151	52
1695-D-T0152	53
1695-D-T0153	54
1695-D-T0155	55
1695-D-T0156	56
1695-D-T0157	57
1695-D-T0158	58
1695-D-T0159	59
1695-D-T0160	60
1695-D-T0161	61
1695-D-T0167	62
1695-D-T0168	63
1695-D-T0169	64
1695-D-T0170	65
1695-D-T0171	66
1695-D-T0172	67
1695-D-T0173	68
1695-D-T0174	69
1695-D-T0176	70
1695-D-T0185	71
1695-D-T0188	72

Table 52 00 00B - Drawings in Numerical Order

Drawing No.	Sheet No.
1695-D-T0191	73
1695-D-T0192	74
Information Drawings	
1695-D-60119	75
1695-D-60129	76
1695-D-60150	77
1695-D-60157	78
1695-D-60253	79
1695-D-60260	80
1695-D-60296	81
1695-D-60309	82
1695-D-60322	83
1695-D-60324	84
1695-D-60325	85
1695-D-60326	86
1695-D-60327	87
1695-D-60684	88
1695-D-60695	89
1695-D-60713	90
1695-D-60720	91
1695-D-60722	92
1695-D-60724	93
1695-D-60727	94
1695-D-60728	95
1695-D-60731	96

Table 52 00 00B - Drawings in Numerical Order

Drawing No.	Sheet No.
085-N-1110	97
085-N-1111	98
085-N-1115	99
085-N-1116	100
085-N-1120	101
085-N-1121	102
085-N-1125	103
085-N-1126	104
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085-N-1131	106
085-N-1135	107
085-N-1136	108
085-N-1150	109
085-N-1151	110
Standard Drawings	
40-D-4334	111
40-D-4335	112
40-D-4753	113
40-D-5246	114
40-D-5247	115
40-D-6234	116
40-D-6551	117
40-D-6552	118
40-D-7102	119
40-D-60001	120

Table 52 00 00B - Drawings in Numerical Order

Drawing No.	Sheet No.
40-D-60002	121
40-D-60003	122
40-D-60004	123
104-D-254	124
104-D-757	125
104-D-1150	126
104-D-1165	127

END OF SECTION

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SECTION 53 10 00
GEOLOGIC INVESTIGATIONS

PART 1 GENERAL

1.01 GENERAL

A. General

1. Geologic descriptions, drawings, logs of subsurface explorations, water level data, and test data in these specifications include information and records of geologic investigations for the work, and are geologic data upon which the design of this work is based. These data supersede previous versions which may be available for examination by Offerors. These data are contained in Section 53 20 00 - Records of Geologic and Subsurface Investigations.

2. Selected samples recovered during investigations were tested by Reclamation's Four Corners Construction Office (FCCO) Laboratories in Farmington, New Mexico and the Geotechnical Laboratory and Field Support (GLFS) Laboratory in Denver, Colorado. Summaries of Physical Properties Test Results from laboratory tests performed on these samples are contained in Section 53 20 00 - Records of Geologic and Subsurface Investigations. Offerors are encouraged to inspect the site and to obtain their own samples and perform tests on the materials to evaluate properties which the Offeror believes to be significant. Offerors wishing to take samples at the site should make arrangements with the CO.

3. Copies of the following reports and memoranda on the area, site geology, and materials properties may be examined at Reclamation FCCO, 1235 LaPlata Highway, Farmington, NM]. Offerors wishing to inspect the reports and memoranda should make arrangements with the CO.

   a. Geologic Design Data Report San Juan Lateral Turnout Buildings, Navajo Gallup Water Supply Project, Department of the Interior, Bureau of Reclamation, Upper Colorado Basin, Four Corners Construction Office, Farmington, NM, September 2025.

   b. CONTROLLED Technical Memorandum No. 86-68320-2026-1: Geophysical Surveys: Seismic and Electrical Resistivity Surveys at Navajo Gallup Turnout Building Sites, Reaches 6 through 12.2., Navajo Gallup Water Supply Project, Upper Colorado Region, Department of the Interior, Bureau of Reclamation, Technical Service Center, Denver Colorado, January 2026.

4. The geologic logs of test pits, water-level data, and other available geologic information indicate the conditions encountered during the investigations. The water-level data show the conditions at the particular time or times the information was obtained and may not indicate variations such as those caused by drought, precipitation, changes in the surface elevation of nearby reservoirs, irrigation, or flooding.

5. The geologic drawings portray generalized geologic conditions. The geologic logs of explorations should be consulted for specific details. For General Geologic Legend, Explanation, and Notes, refer to drawing number 1695-529-60463.

6. General geologic unit definitions, explanations of geologic symbols, and geologic notes are shown on drawing number 1695-529-60463.

7. Surface geology and location of explorations, cross sections, and note drawings for the work are included in these specifications on Drawing Nos. 1695-529-60463 through 1695-529-60470.

8. Surficial materials recorded in geologic exploration logs included in these Specifications were classified under current Unified Soil Classification System procedures as described in USBR 5000 (laboratory classification) and USBR 5005 (visual classification). These two procedures are similar to ASTM D 2487 and ASTM D 2488, respectively.

9. Reclamation has established standard descriptors and descriptive criteria for rock and standard descriptors and descriptive criteria for discontinuities. These standards are available for review at https://www.usbr.gov/tsc/techreferences/mands/manuals.html. The standards provide the basis for classification and description of rock and discontinuities logged after September 1984.

1.02 REFERENCE STANDARDS

A. ASTM International (ATSM)

1.	ASTM D 2487-11	Classification of Soils for Engineering Purposes (Unified Soil Classification)
2.	ASTM D 2488-09a	Description and Identification of Soils (Visual-Manual Procedure)

B. Bureau of Reclamation (USBR)

1. USBR EM – Earth Manual, Part 2, Third Edition (1990)

2. Procedure No. and Title

a. USBR 5000-86 Determining Unified Soil Classification (Laboratory Method)

b. USBR 5005-86 Determining Unified Soil Classification (Visual Method)

c. Copies of the above Reclamation procedures are available for review at Reclamation FCCO, 1235 LaPlata Highway, Farmington, NM.

1.03 GEOLOGIC INVESTIGATIONS

A. Geologic investigations conducted at the NGWSP Turnout Buildings are listed below; and available logs of investigations are contained in Section 53 20 00 - Records of Geologic and Subsurface Investigations.

1. Geologic investigations for the NGWSP turnout buildings are comprised of back hoe test pits. Test pits include TPTB-25-1 through TPTB-25-12. In-place density tests using the sand cone method were conducted in all test pits except TPTB-25-11. Corrosivity samples were collected between 4 and 7 feet below ground surface. Fifty-pound bag samples were collected from the same intervals where in-place densities were performed.

2. Soil testing conducted by the Four Corners Construction Office Laboratory included in-place density, Proctor, Atterberg Limits, and gradation analysis. Corrosion testing was performed by the Geotechnical Laboratory and Field Support (GLFS) group at the Denver Technical Services Center (TSC).

1.04 REGIONAL GEOLOGY

A. Most of the San Juan Lateral lies within the San Juan Basin, which is characterized by young plateaus, mesas, and dry-wash canyons currently undergoing erosion in an arid climate. The San Juan Basin is a structural depression approximately 5,000 feet deep and contains Tertiary fill resting on Cretaceous-age rocks, which crop out around the basin margins. The Defiance Uplift marks the southern and western edges of the basin and is composed of Cretaceous and Jurassic bedrock formations, which dip steeply to the southeast near the New Mexico–Arizona border.

1.05 SITE GEOLOGY

A. Site geology is specific to each of the seven turnout building locations. The turnout buildings are dispersed along the San Juan Lateral at strategic locations and are within the construction Right of Way. Most of the turnout building locations are within previously disturbed and backfilled ground where the pipeline was installed.

B. Stratigraphy

1. Fill: Consists of varying native materials including intermixed, lensed, clay, silt, sand, gravel, and cobble material. Large particles are primarily subangular to angular, moderately hard (H4) to hard (H3) sandstone. Sporadic construction debris including garbage and concrete fragments are sometimes encountered.
2. Quaternary Alluvium (Qal): Derived from a variety of sources including slope wash, eolian and colluvial deposition as well as weathering and decomposition of in-place bedrock. Alluvium contains variable amounts of plastic to non-plastic fines and sand, and may contain gravel, cobbles, and boulders. Specific soil types are identified in the test pit logs.
3. Cretaceous Menefee Formation (Kmf): Consists primarily of light brown, fine-grained sandstone with interbedded claystone, shale, and siltstone, though color and grain size may vary. The sandstone is generally moderately weathered (W5), moderately soft (H5) to soft (H6), and slightly fractured at the surface. Below ground, the same sandstone may be moderately hard (H4) and slightly weathered (W3).
4. A hard, calcareous sandstone bed frequently caps the upper 1 to 3 feet in some areas and may also appear as interbeds or lenses within softer sandstone units. Intermittent hard (H3) concretionary lenses are also present throughout the formation. This caprock/interbedded sandstone is often hard (H3) and can only be broken with a heavy hammer blow.
5. The claystone is sandy, light brown to gray, generally laminated to thinly bedded, intensely to moderately weathered (W6), and moderately soft (H5) to soft (H6). It often grades into siltstone and may become fissile, transitioning into shale. Thin coal seams are also occasionally encountered. The coal is subbituminous to bituminous, friable, soft (H6), and ranges from moderately to intensely weathered (W6) to moderately to slightly weathered (W4).

C. Sanostee Turnout Building, 2 test pits were performed.

1. Test pit TPTB-25-1 encountered Lean Clay with Sand (CL)s from 0.0 to 2.2 feet, Poorly Graded Sand with Silt (SP-SM) from 2.2 to 5.5 feet, and sandstone from 5.5 to 8.5 feet, where refusal was met.
    a. An in-place density test was performed at 4.5 feet with the following results: 91.6 lb/ft^3.
    b. Moisture content: 2.6%, D value: 83.3%, Laboratory classification: Silty Sand (SM).
2. Test pit TPTB-25-2 encountered Lean Clay with Sand (CL)s from 0.0 to 3.3 feet, Poorly Graded Sand with Silt (SP-SM) from 3.3 to 6.0 feet, and Lean Clay with Sand (CL)s from 6.0 to 9.0 feet, where excavation was discontinued in very stiff soil.

a. An in-place density test was performed at 4.0 feet with the following results: 86.3 lb/ft³, Moisture content: 3.7%, D value: 73.8%, Laboratory classification: Silty Sand (SM).

D. Newcomb Turnout Building: 2 test pits were performed.

1. Test pit TPTB-25-3 encountered Clayey Sand (SC) from 0.0 to 8.5 feet, Poorly Graded Sand (SP) from 8.5 to 9.5 feet, and Clayey Sand (SC) from 9.5 to 11.0 feet, where the test pit was discontinued.

a. An in-place density test was performed at 5.0 feet with the following results: 90.4 lb/ft³, Moisture content: 6.3%, D value: 79.0%, Laboratory classification: Clayey Sand (SC).

2. Test pit TPTB-25-4 encountered Clayey Sand (SC) from 0.0 to 3.0 feet and Poorly Graded Sand with Silt (SP-SM) from 3.0 to 11.5 feet, where the test pit was discontinued.

a. An in-place density test was performed at 4.0 feet with the following results: 91.0 lb/ft³, Moisture content: 6.9%, D value: 79.1%, Laboratory classification: Silty Sand (SM).

E. Naschitti Turnout Building: 2 test pits were performed.

1. Test pit TPTB-25-5 encountered Clayey Sand (SC) from 0.0 to 1.6 feet, Silty Sand (SM) from 1.6 to 9.7 feet, and sandstone from 9.7 to 10.0 feet, where refusal was met.

a. An in-place density test was performed at 5.0 feet with the following results: 93.8 lb/ft³, Moisture content: 6.4%, D value: 87.2%, Laboratory classification: Silty Sand (SM).

2. Test pit TPTB-25-6 encountered Clayey Sand (SC) from 0.0 to 1.5 feet, Silty Sand (SM) from 1.5 to 4.5 feet, Clayey Sand (SC) from 4.5 to 5.5 feet, and sandstone from 5.5 to 6.5 feet, where refusal was met.

a. An in-place density test was performed at 4.5 feet with the following results: 99.4 lb/ft³, Moisture content: 6.0%, D value: 88.6%, Laboratory classification: Silty Sand (SM).

F. BBN9 Turnout Building: 1 test pit was performed.

1. Test pit TPTB-25-7 encountered Clayey Sand (SC) from 0.0 to 8.5 feet and Poorly Graded Sand with Silt (SP-SM) from 8.5 to 12.0 feet.

a. An in-place density test was performed at 4.5 feet with the following results: 84.1 lb/ft³, Moisture content: 14.8%, D value: 75.9%, Laboratory classification: Sandy Lean Clay (CL).

G. Coyote Canyon Turnout Building: 2 test pits were performed.

1. Test pit TPTB-25-8 encountered Clayey Sand (SC) from 0.0 to 0.8 feet, a mix of cement gravel and soil from 0.8 to 1.5 feet, and Clayey Sand (SC) from 1.5 to 10.0 feet, where the test pit was discontinued.

   a. An in-place density test was performed at 4.5 feet with the following results: 92.8 lb/ft³, Moisture content: 6.3%, D value: 82.5%, Laboratory classification: Sandy Lean Clay (CL).

   b. Note: The cement gravel mixture appears to be residue from a concrete truck washout, covered by a thin layer of native material.

2. Test pit TPTB-25-9 encountered Clayey Sand (SC) from 0.0 to 1.0 feet, Silty Sand with Gravel (SM)g from 1.0 to 3.5 feet, and Silty Sand (SM) from 3.5 to 11.5 feet where the test pit was discontinued.

   a. An in-place density test was performed at 4.5 feet with the following results: 94.3 lb/ft³, Moisture content: 6.3%, D value: 80.6%, Laboratory classification: Clayey Sand with Silt (SC-SM).

H. Rock Springs Turnout Building: 1 test pit was performed.

1. Test pit TPTB-25-10 encountered Clayey Sand (SC) from 0.0 to 2.0 feet and Clayey Sand with Gravel (SC) from 2.0 to 10.0 feet where the test pit was discontinued.

   a. An in-place density test was performed at 5.5 feet with the following results: 92.6 lb/ft³, Moisture content: 2.4%, D value: 84.3%, Laboratory classification: Sandy Lean Clay (CL).

I. Window Rock Turnout Building 2 test pits were performed.

1. Test pit TPTB-25-11 encountered Poorly Graded Sand (SP) from 0.0 to 4.0 feet, Poorly Graded Sand with Gravel (SP)g from 4.0 to 4.5 feet, and Poorly Graded Sand (SP) from 4.5 to 12.0 feet where the test pit was discontinued. Note: No in-place density was recorded for this test pit.

2. Test pit TPTB-25-12 encountered Poorly Graded Sand with Silt (SP-SM) from 0.0 to 12.0 feet where the test pit was discontinued.

   a. An in-place density test was performed at 7.0 feet with the following results: 92.3 lb/ft³, Moisture content: 2.4%, D value: 80.7%, Laboratory classification: Silty Sand (SM).

1.06 WATER OCCURRENCE

A. Sand washes and tributary streams in project area may have intermittent surface water flow particularly after rainfall. Surface flow would have an adverse effect on excavation

and construction activities. Surface flow is likely to increase groundwater elevation in surrounding areas and would likely increase groundwater elevations in areas that are dry when streams do not have flowing surface water.

B. Groundwater was not encountered in any geologic explorations at the San Juan Lateral Turnout Buildings. Minor amounts of groundwater are sometimes observed under perched conditions atop claystone or shale bedrock surfaces after rainfall or snowmelt along the San Juan Lateral.

1.07 ENGINEERING GEOLOGY AND GEOTECHNICAL CONSIDERATIONS

A. Excavation:

1. Unconsolidated soils and weathered bedrock can be excavated using common methods. The depth of weathering is generally less than 2 feet in the claystone and is less than 1 foot in sandstone units.

2. Degree of weathering varies from decomposed (W9) to slightly weathered to fresh (W2). Excavation of the bedrock may be slow and difficult and could locally require rock excavation methods particularly where the trench excavation becomes constricted in width or hard cap rock, interbeds or lenses are encountered.

B. Slope Stability

1. The stability of surficial deposits is dependent primarily on material composition. Natural slopes in unconsolidated alluvium are generally stable at approximately 1.5:1 or sometimes flatter in sandy materials. Bedrock exposures in the area are typically stable at about 1.5:1 or steeper in claystone and shale and 1:1 to vertical in sandstone.

2. Recommendations for cut slopes in surficial deposits are based on material type and texture. All cut slopes must comply with Reclamation Safety and Health Standards and OSHA regulations. These recommendations assume dry or adequately dewatered conditions; materials with excessive moisture will require flatter slopes for stability.

3. Type C soils (SP, SP-SM, SM): recommended cut slope = 1.5:1

4. Type B soils (SC/SM, SC): recommended cut slope = 1:1

5. Lean to fat clays (classified as Type B due to construction traffic): recommended cut slope = 1:1 or flatter

6. Cut slopes in bedrock depend on rock type and degree of weathering. Decomposed to intensely weathered, very soft bedrock is classified as Type B or C soil, depending on composition. Moderately weathered to fresh, moderately soft

to hard bedrock may be classified as Stable Rock if all Reclamation Safety and Health Standards are met.

C. Compaction

1. Low density soils were encountered in most of the test pits at the turnout building locations. In place density tests with D-values below 85% are considered low density.
2. The turnout building locations are mostly located within disturbed pipeline construction backfill and were likely not returned to natural compaction conditions.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

SECTION 53 20 00
RECORDS OF GEOLOGIC AND SUBSURFACE INVESTIGATIONS

PART 1 GENERAL

1.01 GEOLOGY DRAWINGS

A. Some drawings are in color to highlight geologic information. Geologic information shown on drawings is of equal importance regardless of whether or not it is highlighted.

Geology Drawings

Sheet No.	Drawing No.	Title
G1	103-D-347	Unified Soil Classification – Including Identification and Description
G2	40-D-7022	Geology for design & Specifications, Standard Descriptors and Descriptive Criteria for Rock
G3	40-D-7023	Geology for design & Specifications, Standard Descriptors and Descriptive Criteria for Discontinuities
G4	1695-529-60459	Navajo Gallup Water Supply Project, New Mexico, San Juan Lateral, Turnout Buildings, General Vicinity Map
G5	1695-529-60463	Navajo Gallup Water Supply Project, New Mexico, San Juan Lateral, Turnout Buildings, General Geologic Legend, Explanation and Notes
G6	1695-529-60464	Navajo Gallup Water Supply Project, New Mexico, San Juan Lateral, Turnout Buildings, Sanostee Turnout, Surface Geology & Location of Exploration
G7	1695-529-60465	Navajo Gallup Water Supply Project, New Mexico, San Juan Lateral, Turnout Buildings, Newcomb Turnout, Surface Geology & Location of Exploration
G8	1695-529-60466	Navajo Gallup Water Supply Project, New Mexico, San Juan Lateral, Turnout Buildings, Naschitti Turnout, Surface Geology & Location of Exploration

Geology Drawings

Sheet No.	Drawing No.	Title
G9	1695-529-60467	Navajo Gallup Water Supply Project, New Mexico, San Juan Lateral, Turnout Buildings, BBN9 Turnout, Surface Geology & Location of Exploration
G10	1695-529-60468	Navajo Gallup Water Supply Project, New Mexico, San Juan Lateral, Turnout Buildings, Coyote Canyon (BBN9) Turnout, Surface Geology & Location of Exploration
G11	1695-529-60469	Navajo Gallup Water Supply Project, New Mexico, San Juan Lateral, Turnout Buildings, Rock Springs Turnout, Surface Geology & Location of Exploration
G12	1695-529-60470	Navajo Gallup Water Supply Project, New Mexico, San Juan Lateral, Turnout Buildings, Window Rock Turnout, Surface Geology & Location of Exploration

1.02 GEOLOGIC LOGS

Geologic Logs

Sheet No.	Title
Test Pits	
G13	Log of Test Pit No. TPTB-25-1
G14	Log of Test Pit No. TPTB-25-2
G15	Log of Test Pit No. TPTB-25-3
G16	Log of Test Pit No. TPTB-25-4
G17	Log of Test Pit No. TPTB-25-5
G18	Log of Test Pit No. TPTB-25-6
G19	Log of Test Pit No. TPTB-25-7
G20	Log of Test Pit No. TPTB-25-8

Geologic Logs

Sheet No.	Title
G21	Log of Test Pit No. TPTB-25-9
G22	Log of Test Pit No. TPTB-25-10
G23	Log of Test Pit No. TPTB-25-11
G24	Log of Test Pit No. TPTB-25-12

1.03 LABORATORY TEST DATA

Laboratory Test Data

Sheet No.	Title
Laboratory Test Data for Test Pits	
G25	Summary of Physical Properties Test Results, Navajo Water Supply Project, Turnout Buildings

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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